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(54) **TETRADENTATE ORGANIC
ELECTROLUMINESCENT MATERIALS AND
DEVICES**

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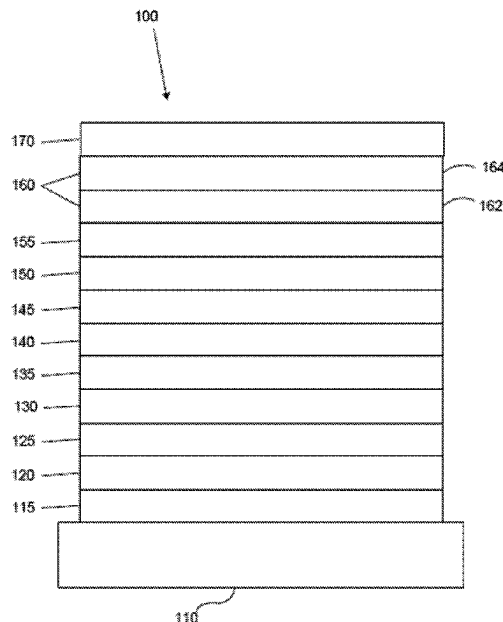
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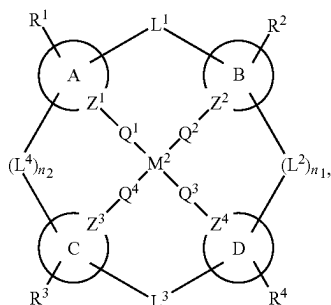
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(57) **ABSTRACT**

New organometallic complexes having bis- or tris-hetero-
leptic ligands and large aspect ratio in one direction and their
use in OLEDs to enhance the efficiency is disclosed. Such
compounds can have a structure of Formula III,
(Continued)





Formula III

where rings A, B, C, and D are ring; L^1 and L^3 each a direct bond or a linking group; n_1 and n_2 are 0 or 1; if n_1 or n_2 is 1, then L^2 or L^4 is a direct bond or a linking group, and if n_1 or n_2 is 0, L^2 or L^4 is not present; Q^1 , Q^2 , Q^3 and Q^4 are a direct bond or oxygen; ring A is trans to ring D, ring B is trans to ring C in a square-planar coordination configuration; M^2 is Pt, and the compound comprises at least one Pt-carbene bond.

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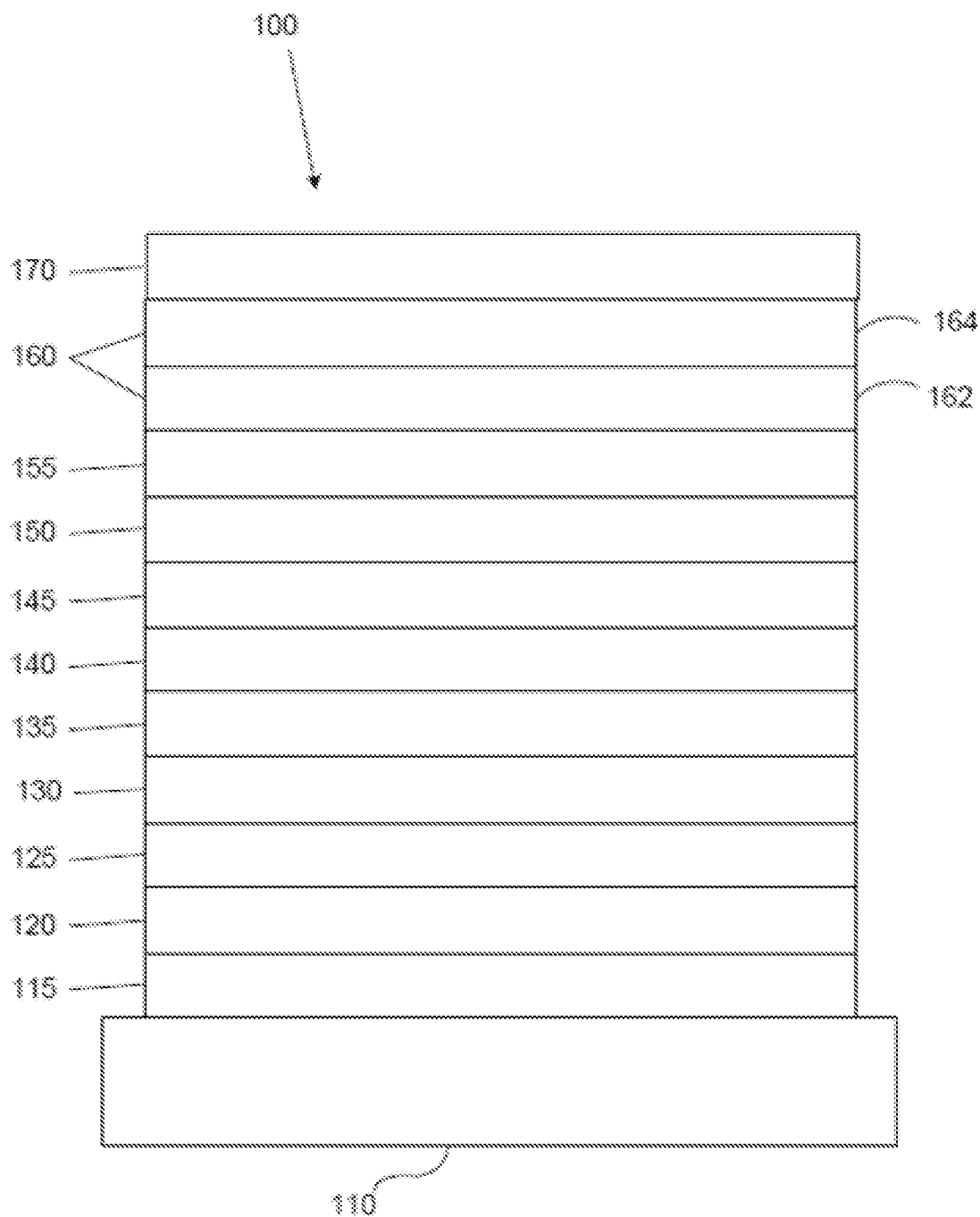


FIG. 1

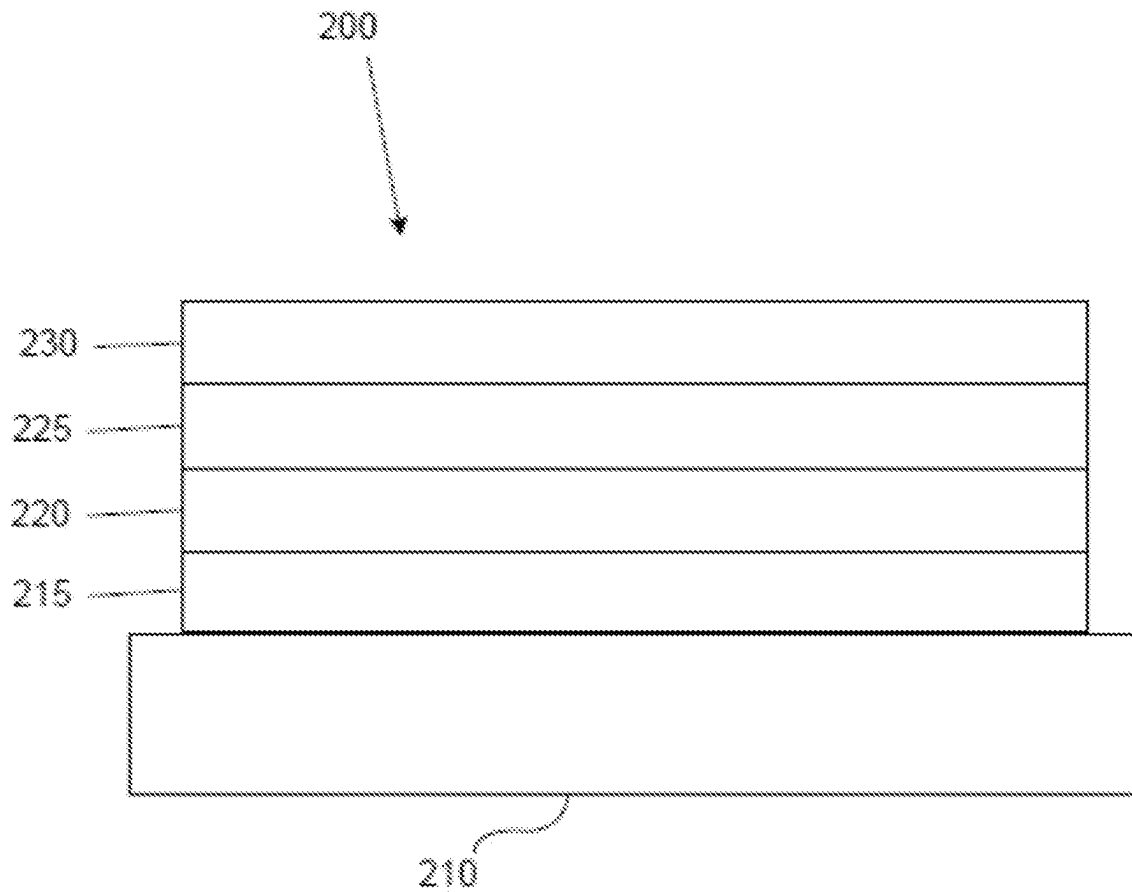


FIG. 2

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TETRADENTATE ORGANIC ELECTROLUMINESCENT MATERIALS AND DEVICES

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/069,016, filed Dec. 20, 2022, which is a continuation of U.S. application Ser. No. 17/075,989, filed Oct. 21, 2020, now U.S. Pat. No. 11,588,121, which is a divisional of U.S. application Ser. No. 15/619,170, filed Jun. 9, 2017, now U.S. Pat. No. 10,862,054, which claims priority under 35 U.S.C. § 119(e)(1) to U.S. Provisional Application Ser. No. 62/516,329, filed Jun. 7, 2017, U.S. Provisional Application Ser. No. 62/352,139, filed Jun. 20, 2016, U.S. Provisional Application Ser. No. 62/450,848, filed Jan. 26, 2017, U.S. Provisional Application Ser. No. 62/479,795, filed Mar. 31, 2017, and U.S. Provisional Application Ser. No. 62/480,746, filed Apr. 3, 2017, the entire contents of which are incorporated herein by reference.

FIELD

The present disclosure relates to compounds for use as phosphorescent emitters, and devices, such as organic light emitting diodes, including the same. More specifically, this disclosure relates to organometallic complexes having large aspect ratio in one direction and their use in OLEDs to enhance the efficiency.

BACKGROUND

Opto-electronic devices that make use of organic materials are becoming increasingly desirable for a number of reasons. Many of the materials used to make such devices are relatively inexpensive, so organic opto-electronic devices have the potential for cost advantages over inorganic devices. In addition, the inherent properties of organic materials, such as their flexibility, may make them well suited for particular applications such as fabrication on a flexible substrate. Examples of organic opto-electronic devices include organic light emitting diodes/devices (OLEDs), organic phototransistors, organic photovoltaic cells, and organic photodetectors. For OLEDs, the organic materials may have performance advantages over conventional materials. For example, the wavelength at which an organic emissive layer emits light may generally be readily tuned with appropriate dopants.

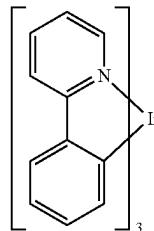
OLEDs make use of thin organic films that emit light when voltage is applied across the device. OLEDs are becoming an increasingly interesting technology for use in applications such as flat panel displays, illumination, and backlighting. Several OLED materials and configurations are described in U.S. Pat. Nos. 5,844,363, 6,303,238, and 5,707,745, which are incorporated herein by reference in their entirety.

One application for phosphorescent emissive molecules is a full color display. Industry standards for such a display call for pixels adapted to emit particular colors, referred to as “saturated” colors. In particular, these standards call for saturated red, green, and blue pixels. Alternatively the OLED can be designed to emit white light. In conventional liquid crystal displays emission from a white backlight is filtered using absorption filters to produce red, green and blue emission. The same technique can also be used with OLEDs. The white OLED can be either a single EML device

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or a stack structure. Color may be measured using CIE coordinates, which are well known to the art.

One example of a green emissive molecule is tris(2-phenylpyridine) iridium, denoted Ir(ppy)₃, which has the following structure:



In this, and later figures herein, we depict the dative bond from nitrogen to metal (here, Ir) as a straight line.

As used herein, the term “organic” includes polymeric materials as well as small molecule organic materials that may be used to fabricate organic opto-electronic devices. “Small molecule” refers to any organic material that is not a polymer, and “small molecules” may actually be quite large. Small molecules may include repeat units in some circumstances. For example, using a long chain alkyl group as a substituent does not remove a molecule from the “small molecule” class. Small molecules may also be incorporated into polymers, for example as a pendent group on a polymer backbone or as a part of the backbone. Small molecules may also serve as the core moiety of a dendrimer, which consists of a series of chemical shells built on the core moiety. The core moiety of a dendrimer may be a fluorescent or phosphorescent small molecule emitter. A dendrimer may be a “small molecule,” and it is believed that all dendrimers currently used in the field of OLEDs are small molecules.

As used herein, “top” means furthest away from the substrate, while “bottom” means closest to the substrate. Where a first layer is described as “disposed over” a second layer, the first layer is disposed further away from substrate. There may be other layers between the first and second layer, unless it is specified that the first layer is “in contact with” the second layer. For example, a cathode may be described as “disposed over” an anode, even though there are various organic layers in between.

As used herein, “solution processible” means capable of being dissolved, dispersed, or transported in and/or deposited from a liquid medium, either in solution or suspension form.

A ligand may be referred to as “photoactive” when it is believed that the ligand directly contributes to the photoactive properties of an emissive material. A ligand may be referred to as “ancillary” when it is believed that the ligand does not contribute to the photoactive properties of an emissive material, although an ancillary ligand may alter the properties of a photoactive ligand.

As used herein, and as would be generally understood by one skilled in the art, a first “Highest Occupied Molecular Orbital” (HOMO) or “Lowest Unoccupied Molecular Orbital” (LUMO) energy level is “greater than” or “higher than” a second HOMO or LUMO energy level if the first energy level is closer to the vacuum energy level. Since ionization potentials (IP) are measured as a negative energy relative to a vacuum level, a higher HOMO energy level corresponds to an IP having a smaller absolute value (an IP that is less negative). Similarly, a higher LUMO energy level

corresponds to an electron affinity (EA) having a smaller absolute value (an EA that is less negative). On a conventional energy level diagram, with the vacuum level at the top, the LUMO energy level of a material is higher than the HOMO energy level of the same material. A “higher” HOMO or LUMO energy level appears closer to the top of such a diagram than a “lower” HOMO or LUMO energy level.

As used herein, and as would be generally understood by one skilled in the art, a first work function is “greater than” or “higher than” a second work function if the first work function has a higher absolute value. Because work functions are generally measured as negative numbers relative to vacuum level, this means that a “higher” work function is more negative. On a conventional energy level diagram, with the vacuum level at the top, a “higher” work function is illustrated as further away from the vacuum level in the downward direction. Thus, the definitions of HOMO and LUMO energy levels follow a different convention than work functions.

More details on OLEDs, and the definitions described above, can be found in U.S. Pat. No. 7,279,704, which is incorporated herein by reference in its entirety.

SUMMARY

According to an aspect of the present disclosure, a compound comprising.

According to another aspect, an OLED is disclosed. The OLED comprises: an anode; a cathode; and an organic layer, disposed between the anode and the cathode, comprising the compound having Formula I.

According to another aspect, a consumer product comprising an OLED is disclosed, where the OLED comprises: an anode; a cathode; and an organic layer, disposed between the anode and the cathode, comprising the compound having Formula I.

A formulation comprising the compound having Formula I is also disclosed.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows an organic light emitting device.

FIG. 2 shows an inverted organic light emitting device that does not have a separate electron transport layer.

DETAILED DESCRIPTION

Generally, an OLED comprises at least one organic layer disposed between and electrically connected to an anode and a cathode. When a current is applied, the anode injects holes and the cathode injects electrons into the organic layer(s). The injected holes and electrons each migrate toward the oppositely charged electrode. When an electron and hole localize on the same molecule, an “exciton,” which is a localized electron-hole pair having an excited energy state, is formed. Light is emitted when the exciton relaxes via a photoemissive mechanism. In some cases, the exciton may be localized on an excimer or an exciplex. Non-radiative mechanisms, such as thermal relaxation, may also occur, but are generally considered undesirable.

The initial OLEDs used emissive molecules that emitted light from their singlet states (“fluorescence”) as disclosed, for example, in U.S. Pat. No. 4,769,292, which is incorporated by reference in its entirety. Fluorescent emission generally occurs in a time frame of less than 10 nanoseconds.

More recently, OLEDs having emissive materials that emit light from triplet states (“phosphorescence”) have been demonstrated. Baldo et al., “Highly Efficient Phosphorescent Emission from Organic Electroluminescent Devices,” *Nature*, vol. 395, 151-154, 1998; (“Baldo-I”) and Baldo et al., “Very high-efficiency green organic light-emitting devices based on electrophosphorescence,” *Appl. Phys. Lett.*, vol. 75, No. 3, 4-6 (1999) (“Baldo-II”), are incorporated by reference in their entirety. Phosphorescence is described in more detail in U.S. Pat. No. 7,279,704 at cols. 5-6, which are incorporated by reference.

FIG. 1 shows an organic light emitting device **100**. The figures are not necessarily drawn to scale. Device **100** may include a substrate **110**, an anode **115**, a hole injection layer **120**, a hole transport layer **125**, an electron blocking layer **130**, an emissive layer **135**, a hole blocking layer **140**, an electron transport layer **145**, an electron injection layer **150**, a protective layer **155**, a cathode **160**, and a barrier layer **170**. Cathode **160** is a compound cathode having a first conductive layer **162** and a second conductive layer **164**. Device **100** may be fabricated by depositing the layers described, in order. The properties and functions of these various layers, as well as example materials, are described in more detail in U.S. Pat. No. 7,279,704 at cols. 6-10, which are incorporated by reference.

More examples for each of these layers are available. For example, a flexible and transparent substrate-anode combination is disclosed in U.S. Pat. No. 5,844,363, which is incorporated by reference in its entirety. An example of a p-doped hole transport layer is m-MTDATA doped with F₄-TCNQ at a molar ratio of 50:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. Examples of emissive and host materials are disclosed in U.S. Pat. No. 6,303,238 to Thompson et al., which is incorporated by reference in its entirety. An example of an n-doped electron transport layer is BPhen doped with Li at a molar ratio of 1:1, as disclosed in U.S. Patent Application Publication No. 2003/0230980, which is incorporated by reference in its entirety. U.S. Pat. Nos. 5,703,436 and 5,707,745, which are incorporated by reference in their entirety, disclose examples of cathodes including compound cathodes having a thin layer of metal such as Mg:Ag with an overlying transparent, electrically-conductive, sputter-deposited ITO layer. The theory and use of blocking layers is described in more detail in U.S. Pat. No. 6,097,147 and U.S. Patent Application Publication No. 2003/0230980, which are incorporated by reference in their entirety. Examples of injection layers are provided in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its entirety. A description of protective layers may be found in U.S. Patent Application Publication No. 2004/0174116, which is incorporated by reference in its entirety.

FIG. 2 shows an inverted OLED **200**. The device includes a substrate **210**, a cathode **215**, an emissive layer **220**, a hole transport layer **225**, and an anode **230**. Device **200** may be fabricated by depositing the layers described, in order. Because the most common OLED configuration has a cathode disposed over the anode, and device **200** has cathode **215** disposed under anode **230**, device **200** may be referred to as an “inverted” OLED. Materials similar to those described with respect to device **100** may be used in the corresponding layers of device **200**. FIG. 2 provides one example of how some layers may be omitted from the structure of device **100**.

The simple layered structure illustrated in FIGS. 1 and 2 is provided by way of non-limiting example, and it is

understood that embodiments of the invention may be used in connection with a wide variety of other structures. The specific materials and structures described are exemplary in nature, and other materials and structures may be used. Functional OLEDs may be achieved by combining the various layers described in different ways, or layers may be omitted entirely, based on design, performance, and cost factors. Other layers not specifically described may also be included. Materials other than those specifically described may be used. Although many of the examples provided herein describe various layers as comprising a single material, it is understood that combinations of materials, such as a mixture of host and dopant, or more generally a mixture, may be used. Also, the layers may have various sublayers. The names given to the various layers herein are not intended to be strictly limiting. For example, in device 200, hole transport layer 225 transports holes and injects holes into emissive layer 220, and may be described as a hole transport layer or a hole injection layer. In one embodiment, an OLED may be described as having an "organic layer" disposed between a cathode and an anode. This organic layer may comprise a single layer, or may further comprise multiple layers of different organic materials as described, for example, with respect to FIGS. 1 and 2.

Structures and materials not specifically described may also be used, such as OLEDs comprised of polymeric materials (PLEDs) such as disclosed in U.S. Pat. No. 5,247,190 to Friend et al., which is incorporated by reference in its entirety. By way of further example, OLEDs having a single organic layer may be used. OLEDs may be stacked, for example as described in U.S. Pat. No. 5,707,745 to Forrest et al, which is incorporated by reference in its entirety. The OLED structure may deviate from the simple layered structure illustrated in FIGS. 1 and 2. For example, the substrate may include an angled reflective surface to improve out-coupling, such as a mesa structure as described in U.S. Pat. No. 6,091,195 to Forrest et al., and/or a pit structure as described in U.S. Pat. No. 5,834,893 to Bulovic et al., which are incorporated by reference in their entireties.

Unless otherwise specified, any of the layers of the various embodiments may be deposited by any suitable method. For the organic layers, preferred methods include thermal evaporation, ink-jet, such as described in U.S. Pat. Nos. 6,013,982 and 6,087,196, which are incorporated by reference in their entireties, organic vapor phase deposition (OVPD), such as described in U.S. Pat. No. 6,337,102 to Forrest et al., which is incorporated by reference in its entirety, and deposition by organic vapor jet printing (OVJP), such as described in U.S. Pat. No. 7,431,968, which is incorporated by reference in its entirety. Other suitable deposition methods include spin coating and other solution based processes. Solution based processes are preferably carried out in nitrogen or an inert atmosphere. For the other layers, preferred methods include thermal evaporation. Preferred patterning methods include deposition through a mask, cold welding such as described in U.S. Pat. Nos. 6,294,398 and 6,468,819, which are incorporated by reference in their entireties, and patterning associated with some of the deposition methods such as ink-jet and OVJP. Other methods may also be used. The materials to be deposited may be modified to make them compatible with a particular deposition method. For example, substituents such as alkyl and aryl groups, branched or unbranched, and preferably containing at least 3 carbons, may be used in small molecules to enhance their ability to undergo solution processing. Substituents having 20 carbons or more may be used, and 3-20 carbons is a preferred range. Materials with asym-

metric structures may have better solution processability than those having symmetric structures, because asymmetric materials may have a lower tendency to recrystallize. Dendrimer substituents may be used to enhance the ability of small molecules to undergo solution processing.

Devices fabricated in accordance with embodiments of the present invention may further optionally comprise a barrier layer. One purpose of the barrier layer is to protect the electrodes and organic layers from damaging exposure to harmful species in the environment including moisture, vapor and/or gases, etc. The barrier layer may be deposited over, under or next to a substrate, an electrode, or over any other parts of a device including an edge. The barrier layer may comprise a single layer, or multiple layers. The barrier layer may be formed by various known chemical vapor deposition techniques and may include compositions having a single phase as well as compositions having multiple phases. Any suitable material or combination of materials may be used for the barrier layer. The barrier layer may incorporate an inorganic or an organic compound or both. The preferred barrier layer comprises a mixture of a polymeric material and a non-polymeric material as described in U.S. Pat. No. 7,968,146, PCT Pat. Application Nos. PCT/US2007/023098 and PCT/US2009/042829, which are herein incorporated by reference in their entireties. To be considered a "mixture", the aforesaid polymeric and non-polymeric materials comprising the barrier layer should be deposited under the same reaction conditions and/or at the same time. The weight ratio of polymeric to non-polymeric material may be in the range of 95:5 to 5:95. The polymeric material and the non-polymeric material may be created from the same precursor material. In one example, the mixture of a polymeric material and a non-polymeric material consists essentially of polymeric silicon and inorganic silicon.

Devices fabricated in accordance with embodiments of the invention can be incorporated into a wide variety of electronic component modules (or units) that can be incorporated into a variety of electronic products or intermediate components. Examples of such electronic products or intermediate components include display screens, lighting devices such as discrete light source devices or lighting panels, etc. that can be utilized by the end-user product manufacturers. Such electronic component modules can optionally include the driving electronics and/or power source(s). Devices fabricated in accordance with embodiments of the invention can be incorporated into a wide variety of consumer products that have one or more of the electronic component modules (or units) incorporated therein. Such consumer products would include any kind of products that include one or more light source(s) and/or one or more of some type of visual displays. Some examples of such consumer products include flat panel displays, computer monitors, medical monitors, televisions, billboards, lights for interior or exterior illumination and/or signaling, heads-up displays, fully or partially transparent displays, flexible displays, laser printers, telephones, cell phones, tablets, phablets, personal digital assistants (PDAs), wearable devices, laptop computers, digital cameras, camcorders, viewfinders, micro-displays (displays that are less than 2 inches diagonal), 3-D displays, virtual reality or augmented reality displays, vehicles, video walls comprising multiple displays tiled together, theater or stadium screen, and a sign. Various control mechanisms may be used to control devices fabricated in accordance with the present invention, including passive matrix and active matrix. Many of the devices are intended for use in a temperature range comfortable to

humans, such as 18 degrees C. to 30 degrees C., and more preferably at room temperature (20-25 degrees C.), but could be used outside this temperature range, for example, from -40 degree C. to +80 degree C.

The materials and structures described herein may have applications in devices other than OLEDs. For example, other optoelectronic devices such as organic solar cells and organic photodetectors may employ the materials and structures. More generally, organic devices, such as organic transistors, may employ the materials and structures.

The term "halo," "halogen," or "halide" as used herein includes fluorine, chlorine, bromine, and iodine.

The term "alkyl" as used herein contemplates both straight and branched chain alkyl radicals. Preferred alkyl groups are those containing from one to fifteen carbon atoms and includes methyl, ethyl, propyl, 1-methylethyl, butyl, 1-methylpropyl, 2-methylpropyl, pentyl, 1-methylbutyl, 2-methylbutyl, 3-methylbutyl, 1,1-dimethylpropyl, 1,2-dimethylpropyl, 2,2-dimethylpropyl, and the like. Additionally, the alkyl group may be optionally substituted.

The term "cycloalkyl" as used herein contemplates cyclic alkyl radicals. Preferred cycloalkyl groups are those containing 3 to 10 ring carbon atoms and includes cyclopropyl, cyclopentyl, cyclohexyl, adamantyl, and the like. Additionally, the cycloalkyl group may be optionally substituted.

The term "alkenyl" as used herein contemplates both straight and branched chain alkene radicals. Preferred alkenyl groups are those containing two to fifteen carbon atoms. Additionally, the alkenyl group may be optionally substituted.

The term "alkynyl" as used herein contemplates both straight and branched chain alkyne radicals. Preferred alkynyl groups are those containing two to fifteen carbon atoms. Additionally, the alkynyl group may be optionally substituted.

The terms "aralkyl" or "arylalkyl" as used herein are used interchangeably and contemplate an alkyl group that has as a substituent an aromatic group. Additionally, the aralkyl group may be optionally substituted.

The term "heterocyclic group" as used herein contemplates aromatic and non-aromatic cyclic radicals. Hetero-aromatic cyclic radicals also means heteroaryl. Preferred hetero-non-aromatic cyclic groups are those containing 3 to 7 ring atoms which includes at least one hetero atom, and includes cyclic amines such as morpholino, piperidino, pyrrolidino, and the like, and cyclic ethers, such as tetrahydrofuran, tetrahydropyran, and the like. Additionally, the heterocyclic group may be optionally substituted.

The term "aryl" or "aromatic group" as used herein contemplates single-ring groups and polycyclic ring systems. The polycyclic rings may have two or more rings in which two carbons are common to two adjoining rings (the rings are "fused") wherein at least one of the rings is aromatic, e.g., the other rings can be cycloalkyls, cycloalkenyls, aryl, heterocycles, and/or heteroaryls. Preferred aryl groups are those containing six to thirty carbon atoms, preferably six to twenty carbon atoms, more preferably six to twelve carbon atoms. Especially preferred is an aryl group having six carbons, ten carbons or twelve carbons. Suitable aryl groups include phenyl, biphenyl, triphenyl, triphenylene, tetraphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, and azulene, preferably phenyl, biphenyl, triphenyl, triphenylene, fluorene, and naphthalene. Additionally, the aryl group may be optionally substituted.

The term "heteroaryl" as used herein contemplates single-ring hetero-aromatic groups that may include from one to

five heteroatoms. The term heteroaryl also includes polycyclic hetero-aromatic systems having two or more rings in which two atoms are common to two adjoining rings (the rings are "fused") wherein at least one of the rings is a heteroaryl, e.g., the other rings can be cycloalkyls, cycloalkenyls, aryl, heterocycles, and/or heteroaryls. Preferred heteroaryl groups are those containing three to thirty carbon atoms, preferably three to twenty carbon atoms, more preferably three to twelve carbon atoms. Suitable heteroaryl groups include dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofurofuran, furodipyridine, benzothienopyridine, thienodipyridine, benzoselenophenopyridine, and selenophenodipyridine, preferably dibenzothiophene, dibenzofuran, dibenzoselenophene, carbazole, indolocarbazole, imidazole, pyridine, triazine, benzimidazole, 1,2-azaborine, 1,3-azaborine, 1,4-azaborine, borazine, and aza-analogs thereof. Additionally, the heteroaryl group may be optionally substituted.

The alkyl, cycloalkyl, alkenyl, alkynyl, aralkyl, heterocyclic group, aryl, and heteroaryl may be unsubstituted or may be substituted with one or more substituents selected from the group consisting of deuterium, halogen, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, cyclic amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acid, ether, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

As used herein, "substituted" indicates that a substituent other than H is bonded to the relevant position, such as carbon. Thus, for example, where R¹ is mono-substituted, then one R¹ must be other than H. Similarly, where R¹ is di-substituted, then two of R¹ must be other than H. Similarly, where R¹ is unsubstituted, R¹ is hydrogen for all available positions.

The "aza" designation in the fragments described herein, i.e. aza-dibenzofuran, aza-dibenzothiophene, etc. means that one or more of the C—H groups in the respective fragment can be replaced by a nitrogen atom, for example, and without any limitation, azatriphenylene encompasses both dibenzo[f,h]quinoxaline and dibenzo[f,h]quinoline. One of ordinary skill in the art can readily envision other nitrogen analogs of the aza-derivatives described above, and all such analogs are intended to be encompassed by the terms as set forth herein.

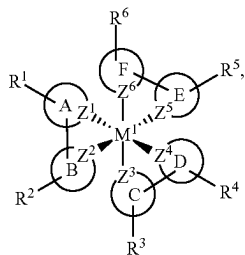
It is to be understood that when a molecular fragment is described as being a substituent or otherwise attached to another moiety, its name may be written as if it were a fragment (e.g. phenyl, phenylene, naphthyl, dibenzofuryl) or as if it were the whole molecule (e.g. benzene, naphthalene, dibenzofuran). As used herein, these different ways of designating a substituent or attached fragment are considered to be equivalent.

In this disclosure, organometallic complexes based on Ir, Os, Rh, Ru, Re, Pt, or Pd having bis- or tris-heteroleptic ligands and large aspect ratio in one direction are provided. The inventors have found that incorporating such compounds in OLEDs enhance the device efficiency. The ligands

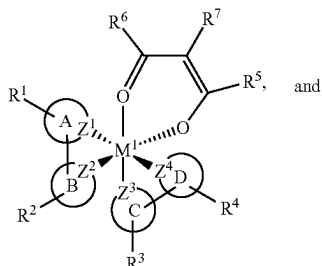
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are arranged in such a way that the length of the molecule in one direction is longer than in any other directions thus resulting in a large aspect ratio. These compounds with large aspect ratio when applied as emitters in PhOLED devices show enhanced external quantum efficiencies (EQEs) because they preferentially orient themselves in horizontal orientation to the plane of the substrate (i.e. parallel to the substrate) and therefore result in maximizing light extraction from the emitter compounds. The horizontal orientation maximizes the surface area of the light emitting molecules facing the light emitting façade of the device. Some examples of the organometallic compounds disclosed herein have three different bidentate cyclometalated ligands coordinating to an iridium metal center. Some other examples of the organometallic compounds have two different bidentate cyclometalated ligands coordinating to a platinum metal center.

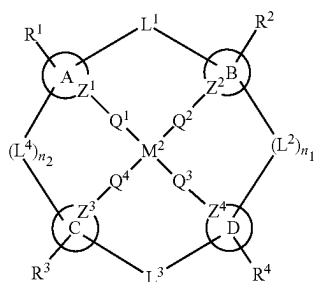
According to an aspect of the present disclosure, a compound having a formula selected from the group consisting of:



Formula I



Formula II



Formula III

is disclosed, wherein rings A, B, C, D, E, and F are each a 5 or 6-membered carbocyclic or heterocyclic ring;

wherein in Formula I: A-B, C-D, and E-F form three bidentate ligands coordinated to metal M¹; wherein A-B, C-D, and E-F are different from each other; wherein ring A is trans to ring D, ring B is trans to ring E, and ring C is trans to ring F in a octahedral coordination configuration;

wherein in Formula II: A-B, C-D, and one acetylacetonate ligand form three bidentate ligands coordinated to metal M¹; wherein A-B, and C-D are different from

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each other; wherein ring A is trans to ring D, ring B is trans to oxygen atom, and ring C is trans to oxygen atom in a octahedral coordination configuration;

wherein in Formula III: L¹ and L³ each independently selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof; n₁, n₂ each independently is 0 or 1; when n₁ or n₂ is 1, L² or L⁴ is selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof; when n₁ or n₂ is 0, L² or L⁴ is not present; Q¹, Q², Q³ and Q⁴ each independently selected from the group consisting of direct bond and oxygen; and when any of Z¹, Z², Z³ and Z⁴ is nitrogen, the Q¹, Q², Q³ and Q⁴ attached thereto is a direct bond;

ring A is trans to ring D, ring B is trans to ring C in a square-planar coordination configuration;

wherein R¹, R², R³, R⁴, R⁵, R⁶, and R⁷ each represents mono to the maximum possible number of substitution, or no substitution;

wherein Z¹, Z², Z³, Z⁴, Z⁵ and Z⁶ are each independently selected from the group consisting of carbon and nitrogen;

wherein M¹ is a metal selected from the group consisting of Ir, Os, Rh, Ru, and Re; M² is a metal selected from the group consisting of Pt and Pd;

wherein a first distance is the distance between the atom in R¹ that is the farthest away from M¹ to the atom in R⁴ that is the farthest away from M¹;

wherein a second distance is the distance between the atom in R² that is the farthest away from M¹ to the atom in R³ that is the farthest away from M¹;

wherein a third distance is the distance between the atom in R³ that is the farthest away from M¹ to the atom in R⁶ that is the farthest away from M¹;

wherein a fourth distance is the distance between the atom in R¹ that is the farthest away from M² to the atom in R⁴ that is the farthest atom away from M²;

wherein a fifth distance is the distance between the atom in R² that is the farthest atom away from M² to the atom in R³ that is the farthest atom away from M² in R³;

wherein the first distance is longer than the second distance and the third distance each by at least 1.5 Å; wherein the fourth distance is longer than the fifth distance by at least 1.5 Å;

wherein R, R', R¹, R², R³, R⁴, R⁵, R⁶, and R⁷ are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alk-enyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfonyl, sulfonyl, phosphino, and combinations thereof; and

wherein any two substituents among R, R', R¹, R², R³, R⁴, R⁵, R⁶, and R⁷ are optionally joined or fused into a ring. 1.5 Å mentioned above is the distance of a C—C bond (i.e., adding a methyl group) from calculation.

In other words, the above description defines the relationship between the molecular long axes defined by different pairs of substituent groups in each of the complexes represented by Formula I, Formula II, and Formula III. Each of the pairs of substituent groups identified above are substituent groups positioned substantially opposite from each other relative to the coordinating metal M¹ or M². The two end points of each of the molecular long axes defined are the

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atoms in each of the paired substituents that are the farthest away from the corresponding coordinating metal.

In some embodiments of the compound, any two substituents within each substituent groups R, R¹, R¹, R², R³, R⁴, R³, R⁶, and R⁷, when they are more than mono substitution, are optionally joined or fused into a ring.

In some embodiments of the compound, M¹ is Ir, M² is Pt.

In some embodiments of the compound, rings A, B, C, D, E, and F are each independently selected from the group consisting of phenyl, pyridine, and imidazole. In some embodiments of the compound, rings A, C, and E in Formula I and III, and rings A and D in Formula II are phenyl.

In some embodiments of the compound, rings B, D, and F in Formula I and III, and rings B and C in Formula II are selected from the group consisting of pyridine, pyrimidine, imidazole, and pyrazole.

In some embodiments of the compound, R¹, R², R³, R⁴, R³, R⁶, and R⁷ are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, silyl, aryl, heteroaryl, and combinations thereof.

In some embodiments of the compound, the first distance is longer than the second distance and the third distance each by at least 4.3 Å, and the fourth distance is longer than the fifth distance by at least 4.3 Å. The value 4.3 Å is representative of the diameter of a phenyl ring. Thus, the first distance in the compound is longer than the second distance and the third distance by at least a phenyl substitution and the fourth distance is longer than the fifth distance by at least a phenyl substitution.

In some embodiments of the compound, the first distance is longer than the second distance and the third distance each by at least 5.9 Å, and the fourth distance is longer than the fifth distance by at least 5.9 Å. The value 5.9 Å is representative of the distance spanning a para-tolyl group.

In some embodiments of the compound, at least one of the rings A, B, C, D, E, and F is fused by another 5- or

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6-membered ring. The another 5- or 6-membered ring can be an aromatic ring or a non-aromatic ring. The aromatic ring can be a phenyl ring.

In some embodiments where the compound is of Formula I, at least one of (i), (ii), and (iii) is true, wherein (i) one R¹ connects to one R², (ii) one R³ connects to one R⁴, (iii) one R³ connects to one R⁶. In some embodiments where the compound is of Formula II, at least one of (i) and (ii) is true, wherein (i) one R¹ connects to one R², (ii) one R³ connects to one R⁴.

In some embodiments, the first distance is longer than the second distance and the third distance each by at least 3.0 Å, and the fourth distance is longer than the fifth distance by at least 3.0 Å. The value 3.0 Å is representative of the distance spanning two methyl groups.

The following Table 1 lists the maximum linear length for various substituent groups defined along their long axis. This maximum linear length is defined as the distance between the two atoms that are the farthest apart along the long axis of the particular substituent group. The listed values can be used to estimate the difference in length between two molecular long axes defined above in connection with the structures of Formulas I, II, and III depending on the substituent group that is the differential between two molecular long axes being compared. For example, if the difference in length between two molecular long axes is the result of one molecular long axis being longer than the other by an extra phenyl substituent group, the fourth entry in Table 1 below provides that the difference in length between the two molecular long axes will be at least 4.3 Å (an extra C—C bond is required to make the connection). Any two or more of the following fragments can be linked together, and its distance can be calculated by simply adding up these numbers plus the total length of the single C—C bond distance used to connect them.

TABLE 1

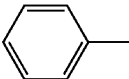
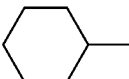
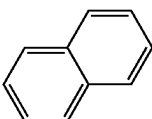
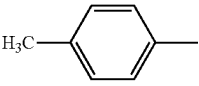
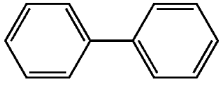
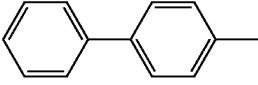
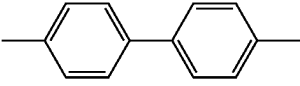
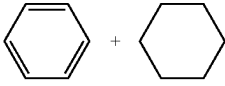
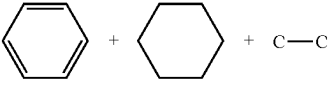
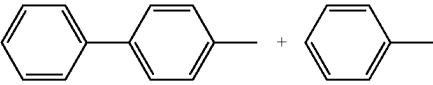
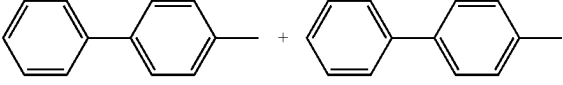
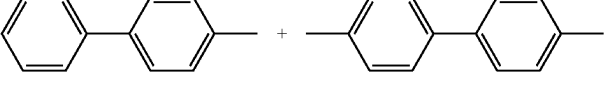
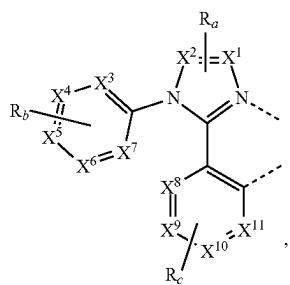
The difference between two directions	Longest Distance of the difference (Å)
C—C	1.5
	2.9
	3.0
	4.3
	4.4
	5.2

TABLE 1-continued

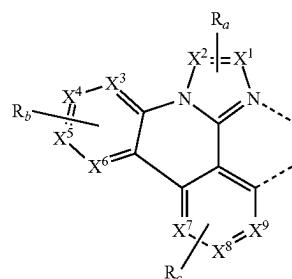
The difference between two directions	Longest Distance of the difference (Å)
	5.9
	7.3
	8.8
	10.3
two C—C	3.0
	7.3
	8.8
	13.1
	17.6
	19.1

In some embodiments of the compound, the bidentate ligand A-B, C-D, and E-F are each independently selected from the group consisting of:



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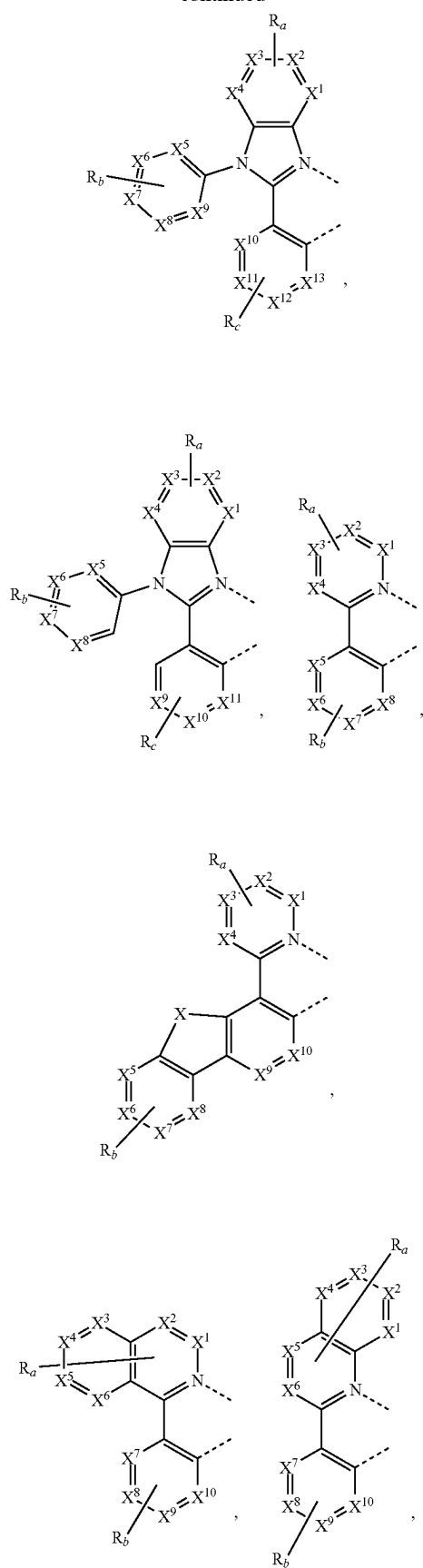
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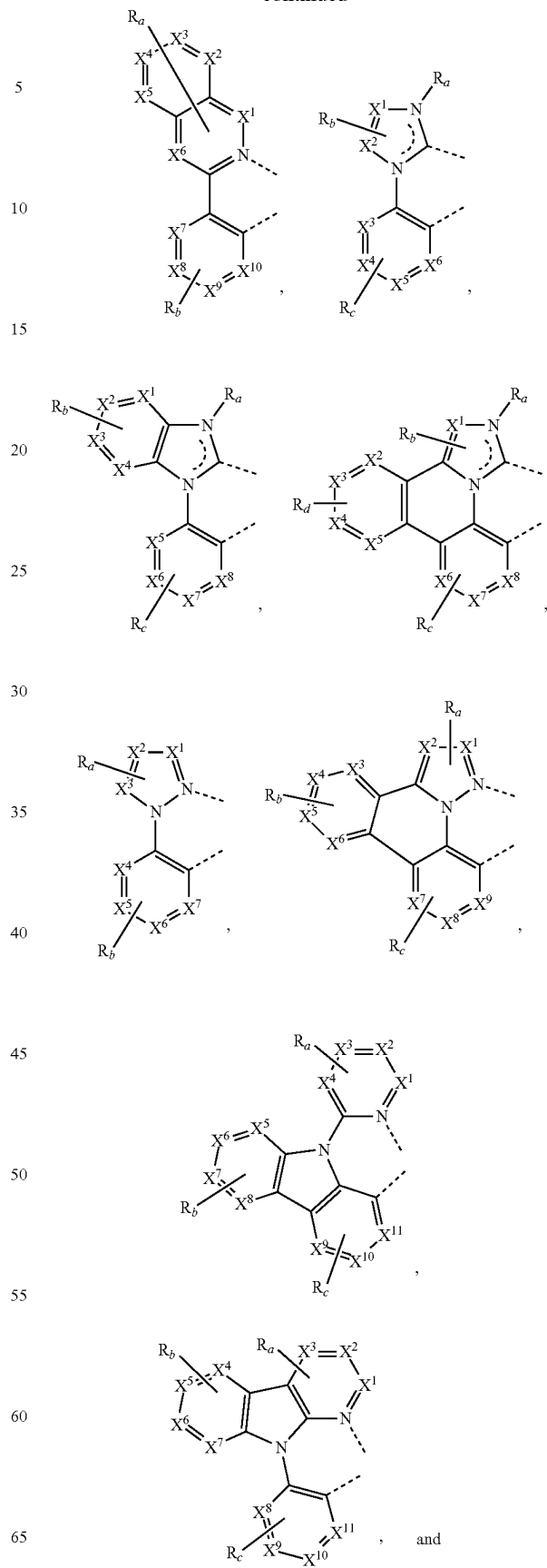
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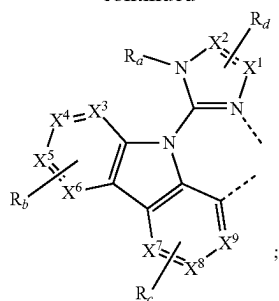
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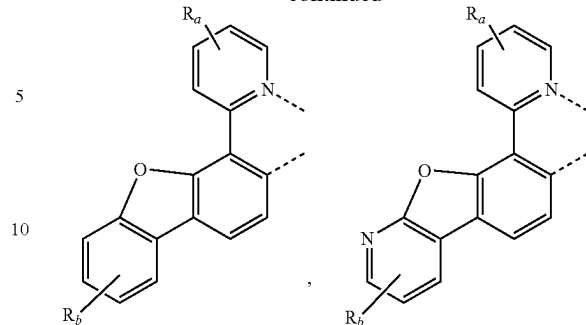
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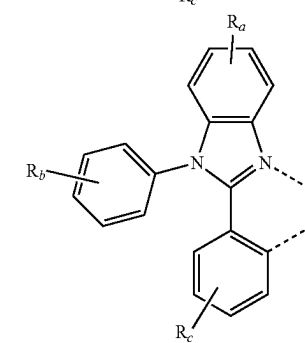
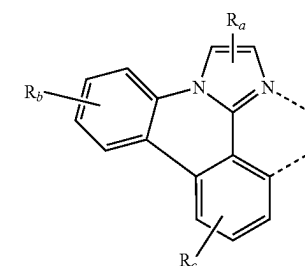
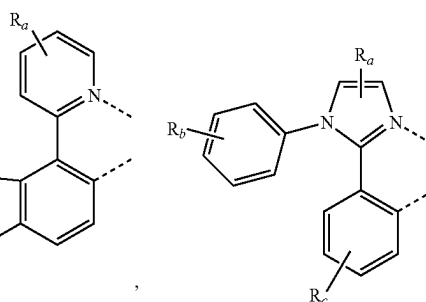
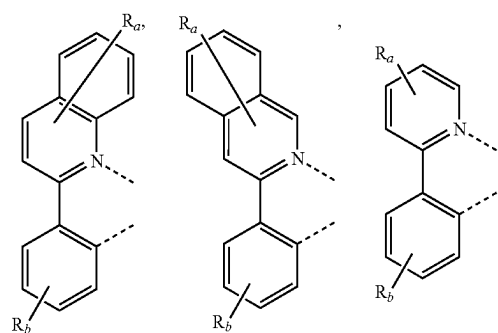
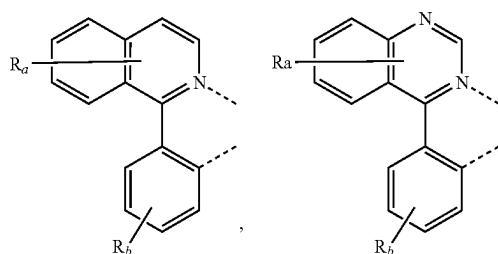
wherein each X^1 to X^{13} are independently selected from the group consisting of carbon and nitrogen; wherein X is selected from the group consisting of BR' , NR' , PR' , O, S, Se, $C=O$, $S=O$, SO_2 , $CR'R''$, $SiR'R''$, and $GeR'R''$;

wherein R' and R'' are optionally fused or joined to form a ring;

wherein R_a , R_b , R_c , and R_d each represents from mono substitution to the maximum possible number of substitution, or no substitution;

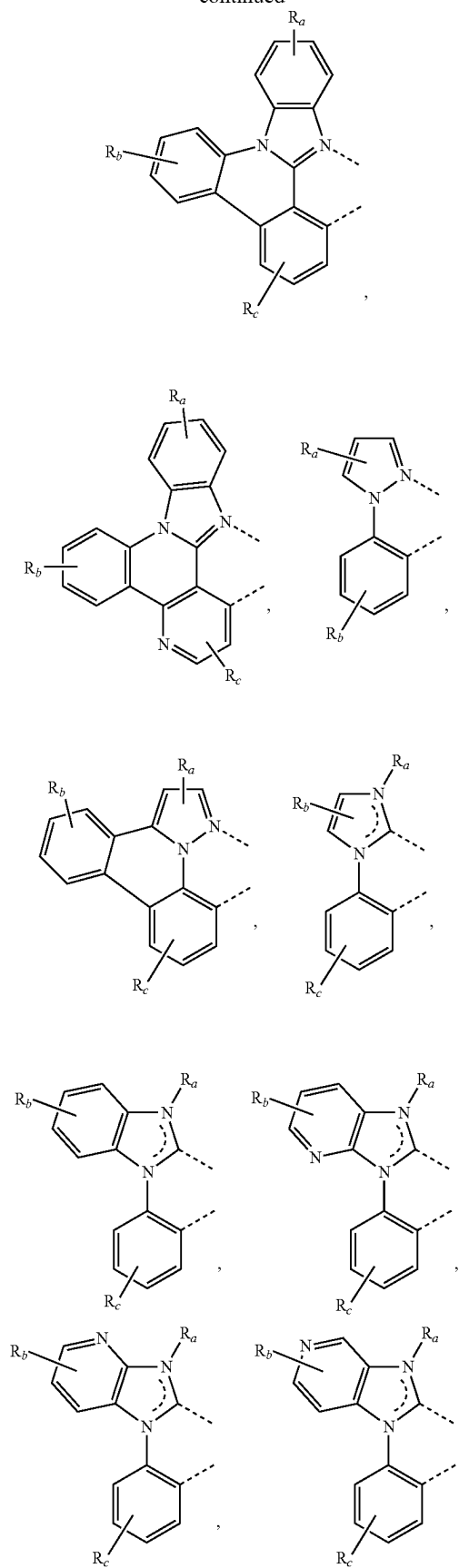
wherein R' , R'' , R_a , R_b , R_c , and R_d are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfonyl, sulfonyl, phosphino, and combinations thereof; and

wherein any two substituents among R_a , R_b , R_c , and R_d are optionally fused or joined to form a ring. In some embodiments of the compound, the bidentate ligand A-B, C-D, and E-F are each independently selected from the group consisting of:

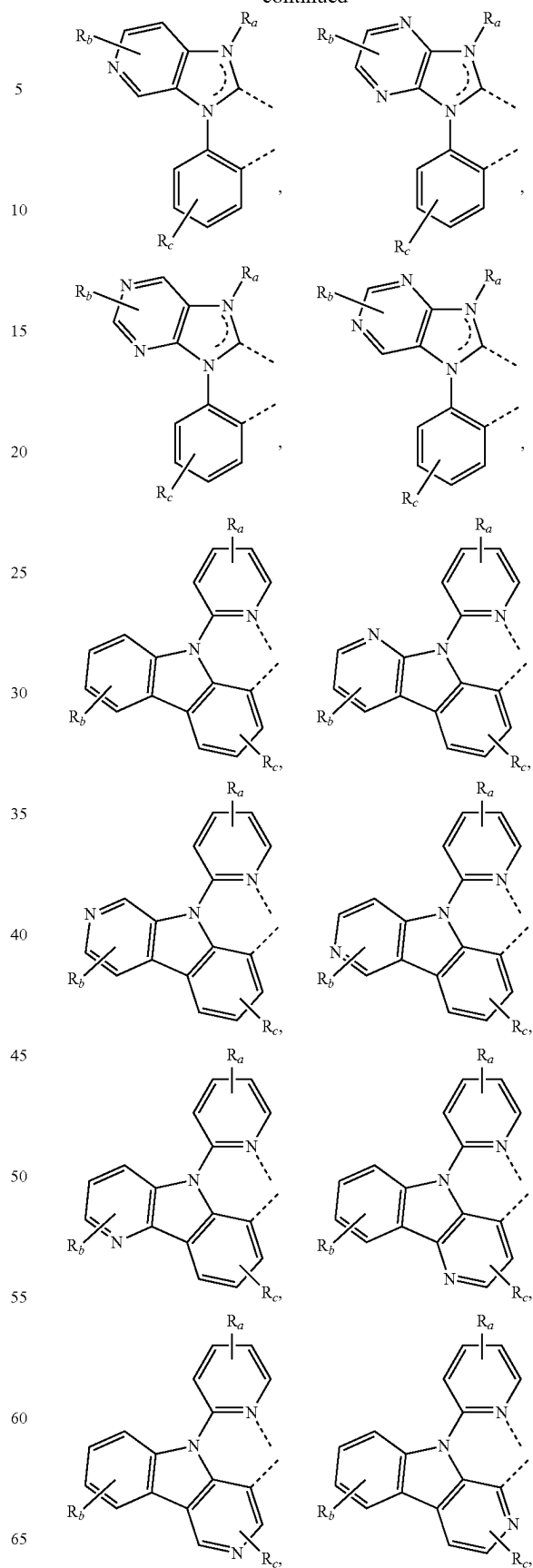


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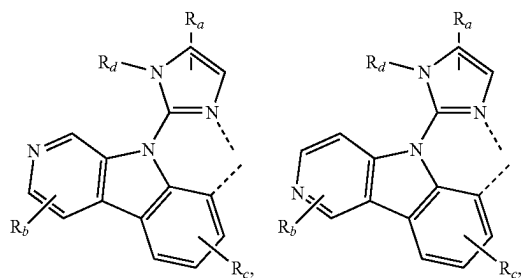
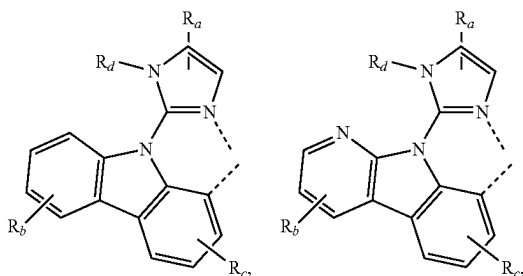
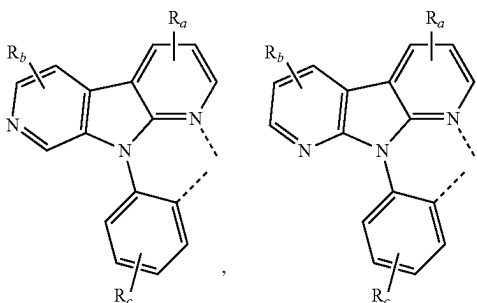
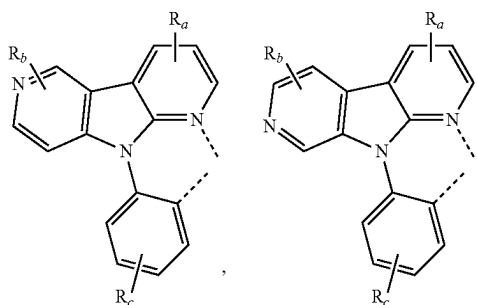
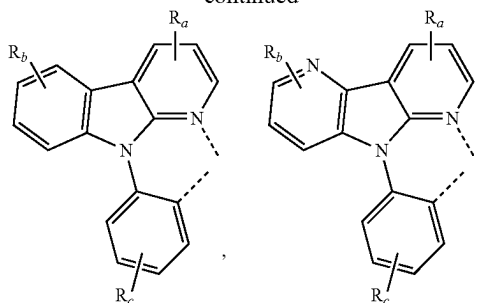
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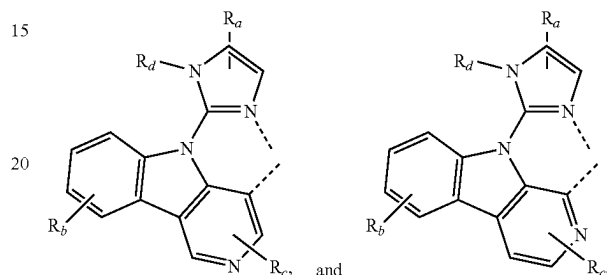
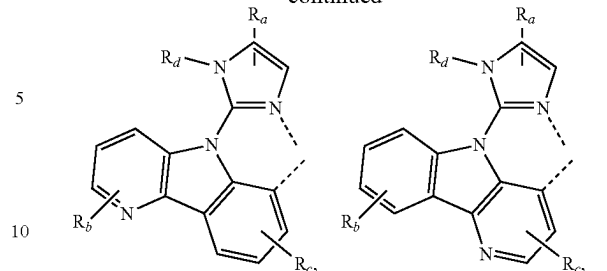
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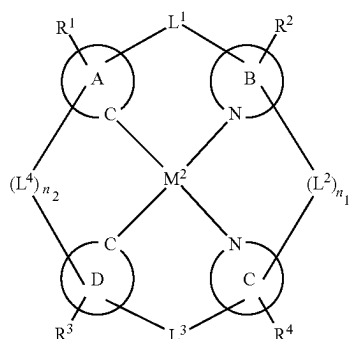
In some embodiments of the compound, n_1 is 1 and n_2 is 0. In some embodiments, n_1 is 1 and n_2 is 1. In some embodiments, n_1 is 0 and n_2 is 0.

In some embodiments of the compound, each of Q^1 , Q^2 , Q^3 , and Q^4 is a direct bond. In some embodiments, one of Q^1 , Q^2 , Q^3 , and Q^4 is oxygen, the remaining three of Q^1 , Q^2 , Q^3 , and Q^4 are direct bonds. In some embodiments, two of Q^1 , Q^2 , Q^3 , and Q^4 are oxygen, and the remaining two of Q^1 , Q^2 , Q^3 , and Q^4 are direct bonds.

In some embodiments of the compound, two of Z^1 , Z^2 , Z^3 , Z^4 are carbon atoms, and the remaining two of Z^1 , Z^2 , Z^3 , Z^4 are nitrogen atoms. In some embodiments, three of Z^1 , Z^2 , Z^3 , Z^4 are carbon atoms, and the remaining one of Z^1 , Z^2 , Z^3 , Z^4 is a nitrogen atom. In some embodiments, each of Z^1 , Z^2 , Z^3 , Z^4 is a carbon atom.

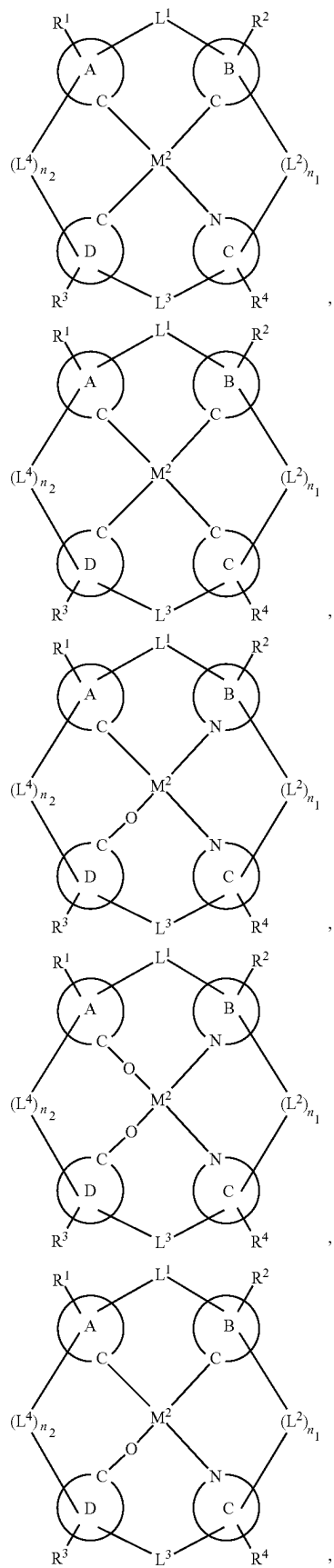
In some embodiments where each of Q^1 , Q^2 , Q^3 , and Q^4 is a direct bond, the compound is in cis configuration. In some embodiments, the compound has at least one Pt-carbene or Ir-carbene bond.

In some embodiments of the compound, the compound of Formula III is selected from the group consisting of:

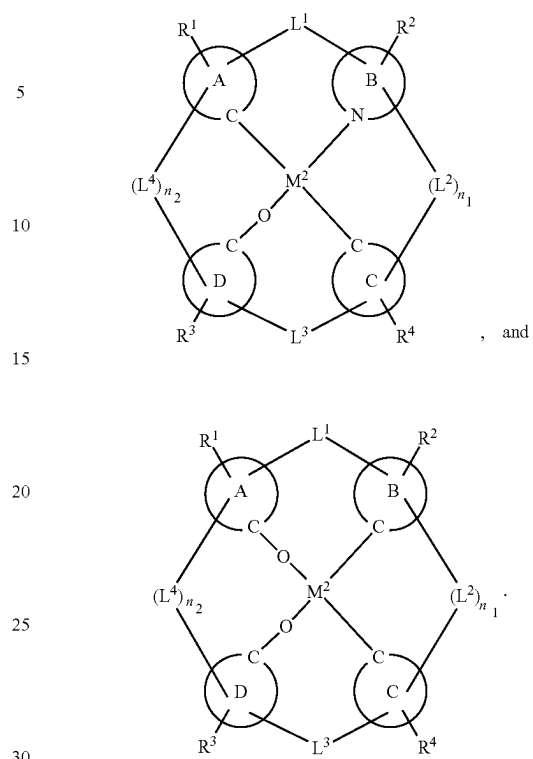


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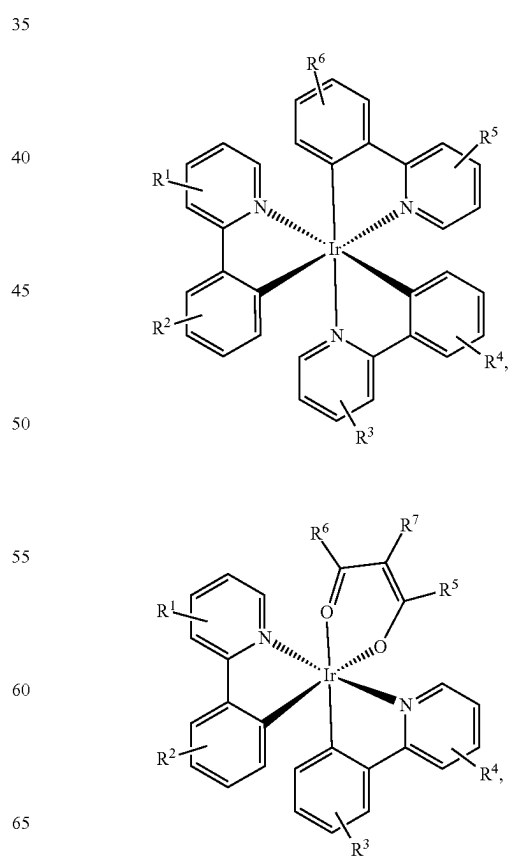
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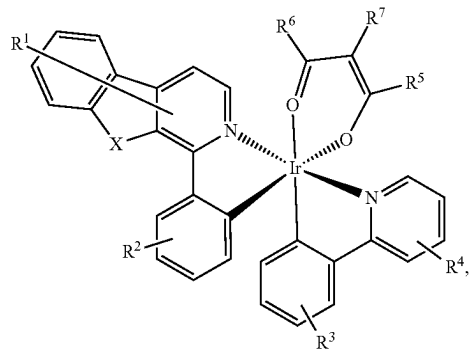
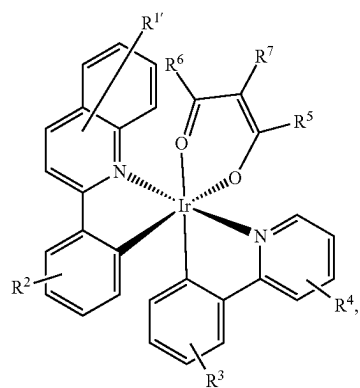
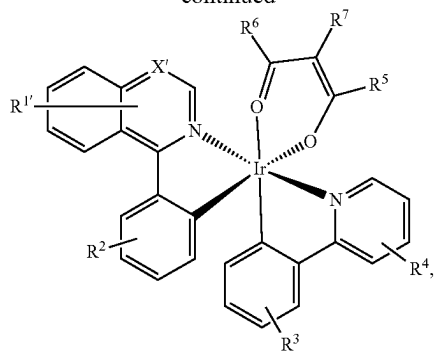
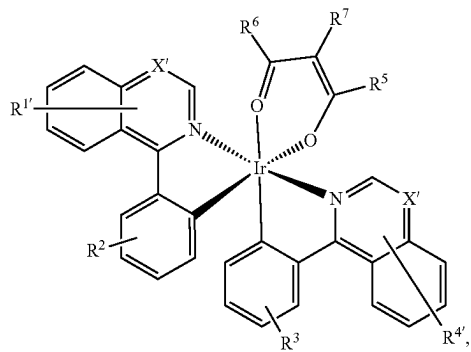


In some embodiments of the compound, the compound is selected from the group consisting of:

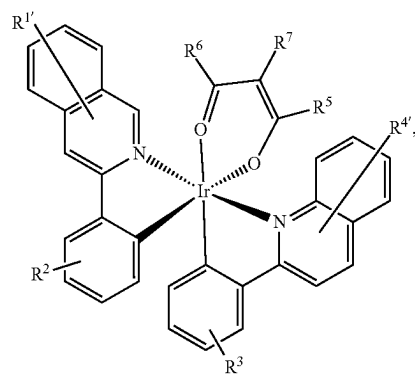
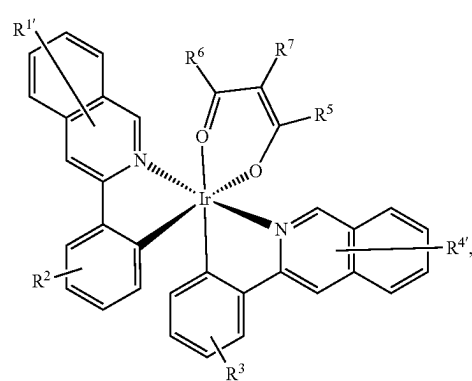
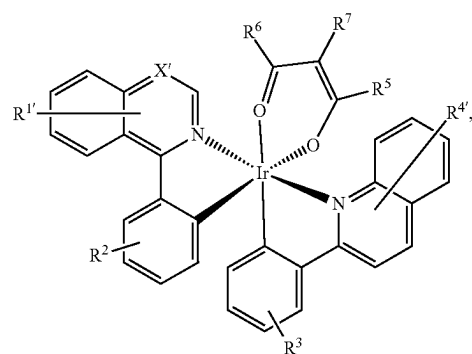
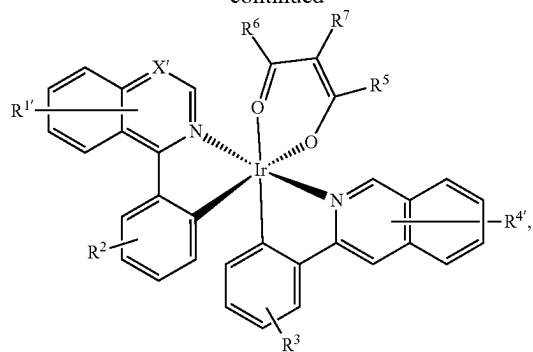


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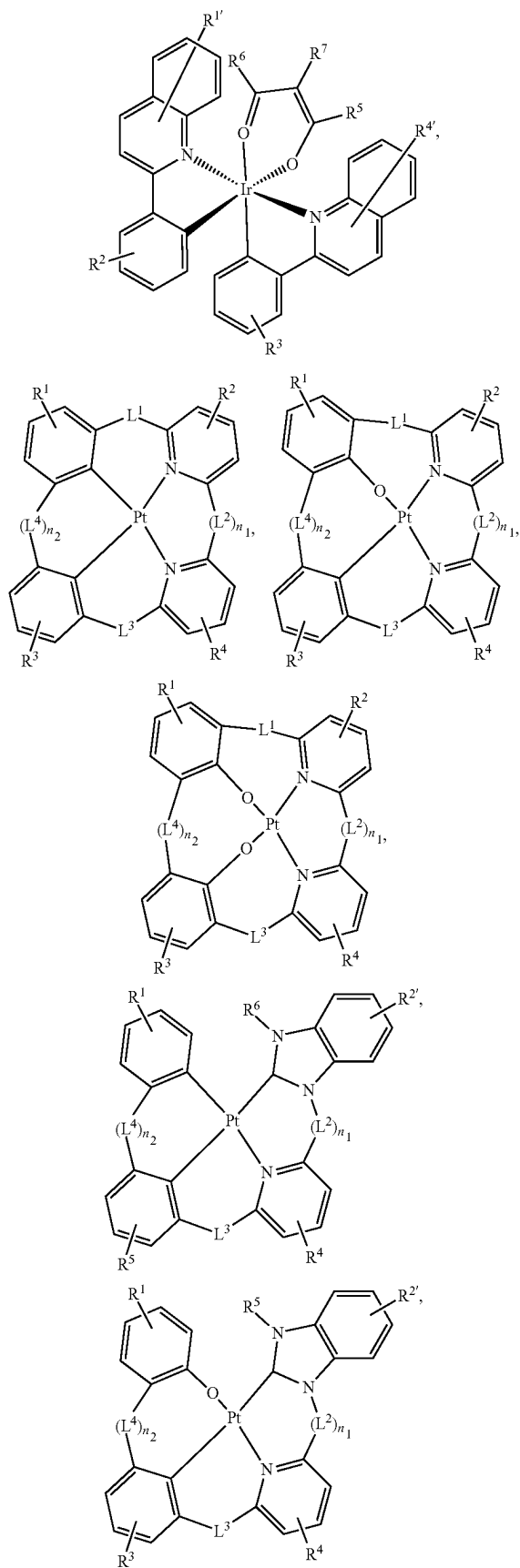
X = S, O, NR, CR₂**26**

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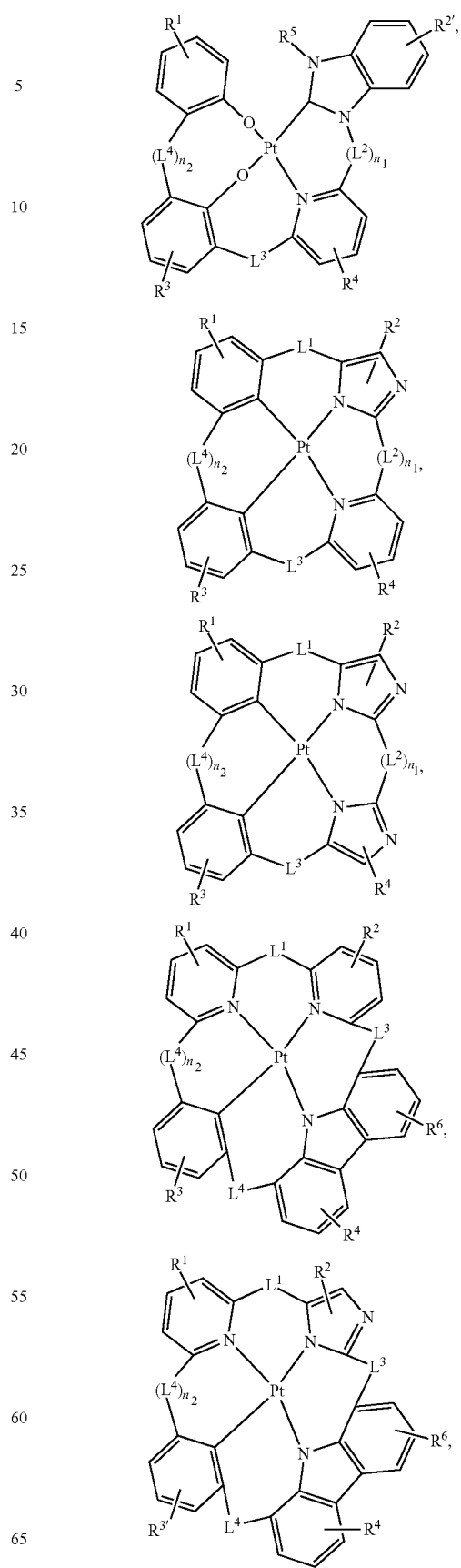


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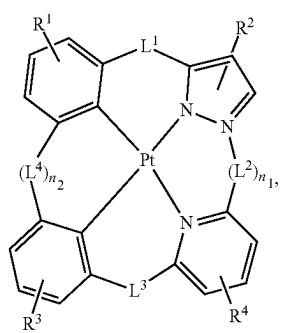
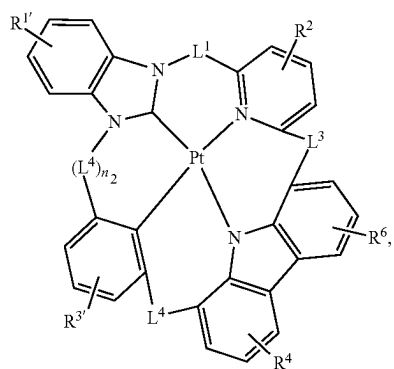
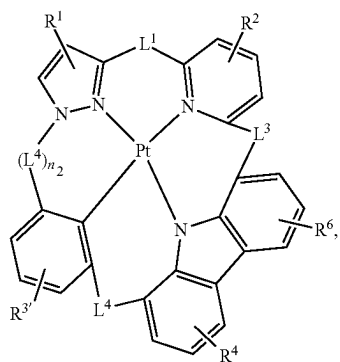
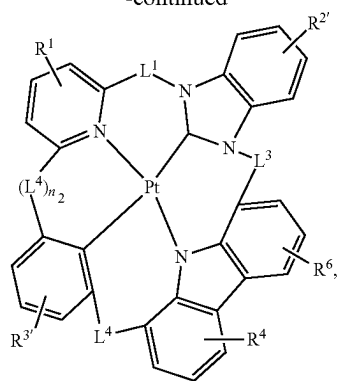
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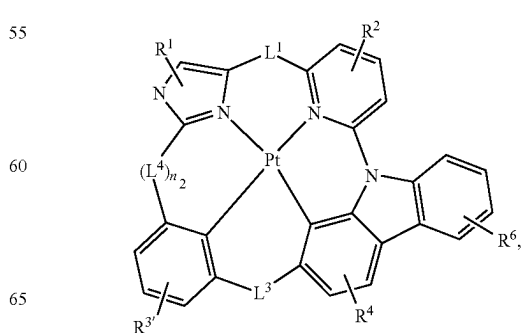
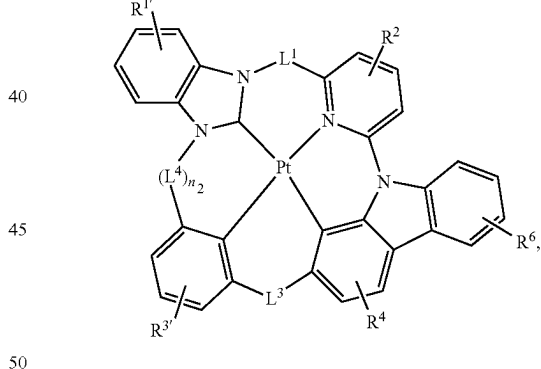
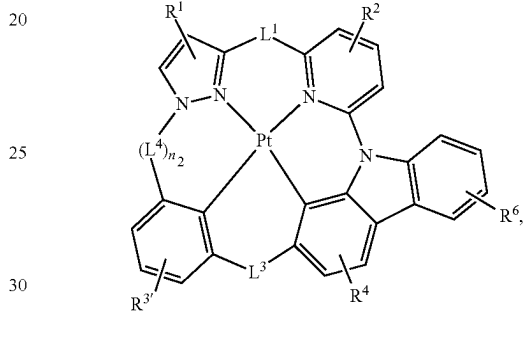
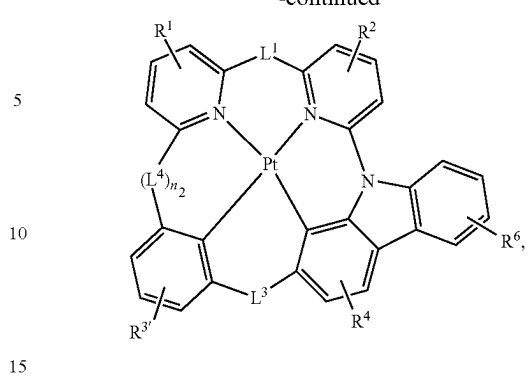


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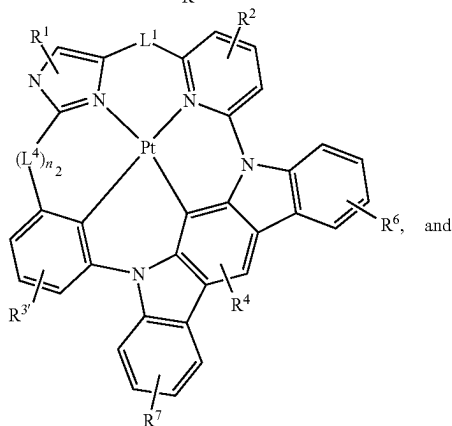
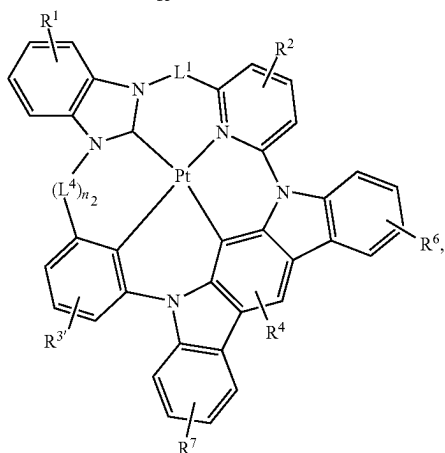
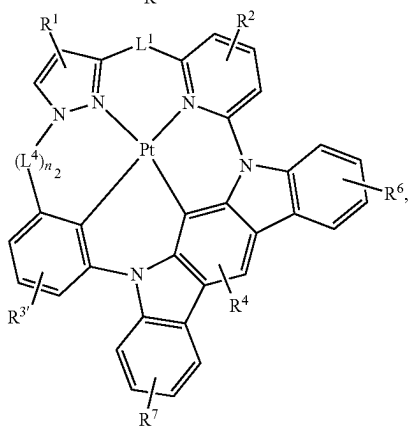
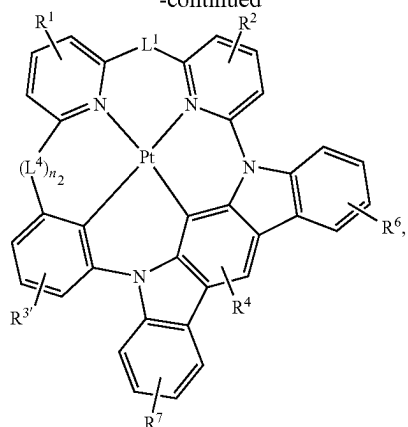
**30**

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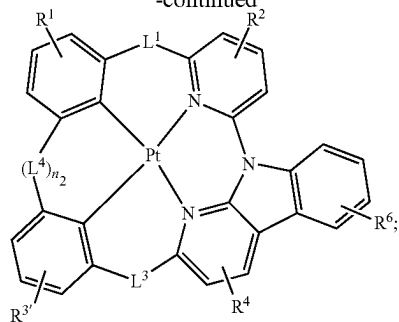


31

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**32**

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wherein X is selected from the group consisting of O, Se, and S;

wherein X' is carbon or nitrogen;

wherein R^{1'}, R^{2'}, R^{3'}, and R^{4'} each represents mono to the maximum possible number of substitution, or no substitution;

wherein each R^{1'}, R^{2'}, R^{3'}, and R^{4'} are independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfonyl, sulfonyl, phosphino, and combinations thereof; and

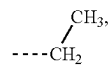
wherein any two substituents among R, R', R¹, R², R³, R⁴, R⁵, R⁶, and R⁷ are optionally joined or fused into a ring. In some embodiments of the compound, at least one R¹ is para to N coordinated to Ir, and at least one R⁴ is para to carbon coordinated to Ir. In some other embodiments of the compound, at least one of R¹ and R^{1'} and at least one R⁴ and R^{4'} is selected from the group consisting of:



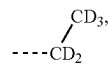
Substituent 1



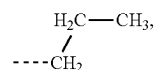
Substituent 2



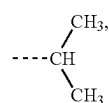
Substituent 3



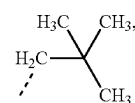
Substituent 4



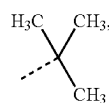
Substituent 5



Substituent 6



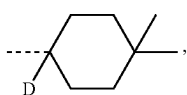
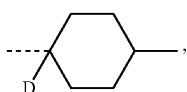
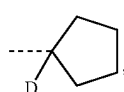
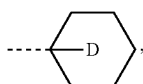
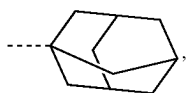
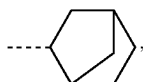
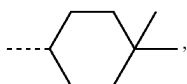
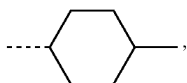
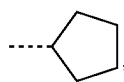
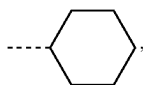
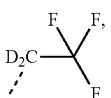
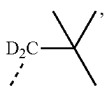
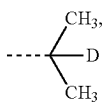
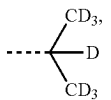
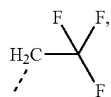
Substituent 7



Substituent 8

33

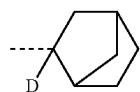
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**34**

-continued

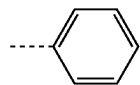
Substituent 9

5



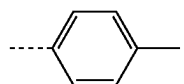
Substituent 10

10



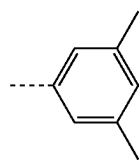
Substituent 11

15



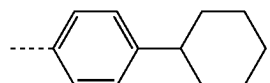
Substituent 12

20

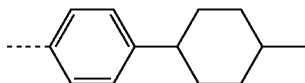


Substituent 13

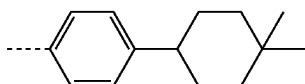
Substituent 14 25



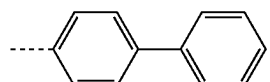
Substituent 15 30



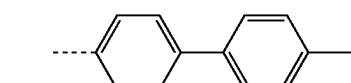
Substituent 16 35



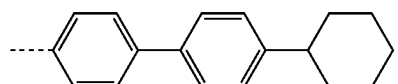
Substituent 17 40



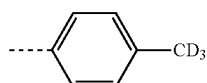
Substituent 18 45



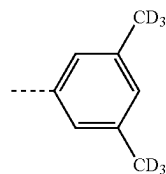
Substituent 19 50



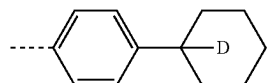
Substituent 20 55



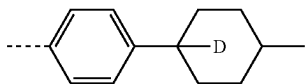
Substituent 21 60



Substituent 22 65



Substituent 23



Substituent 24

Substituent 25

Substituent 26

Substituent 27

Substituent 28

Substituent 29

Substituent 30

Substituent 31

Substituent 32

Substituent 33

Substituent 34

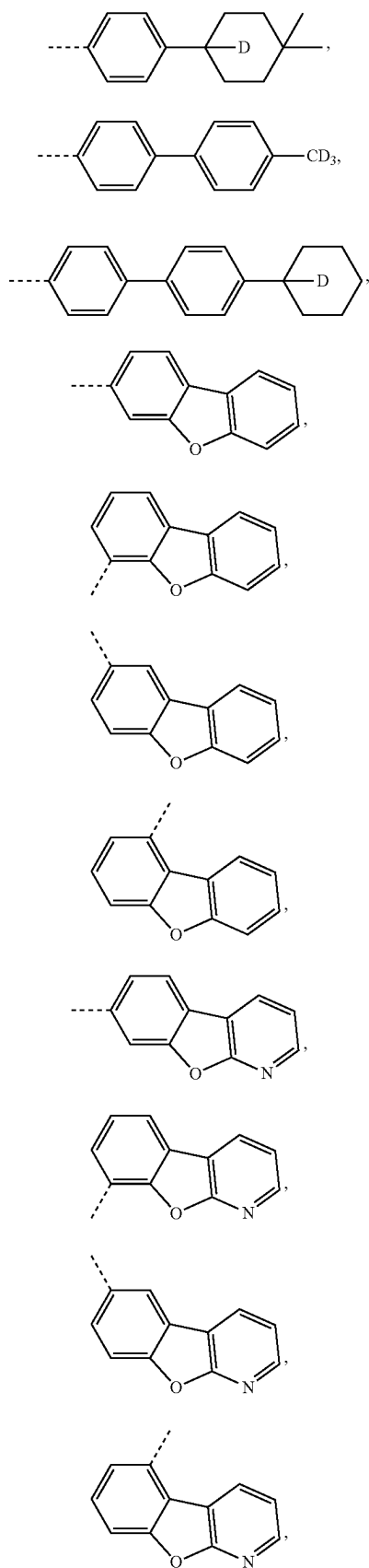
Substituent 35

Substituent 36

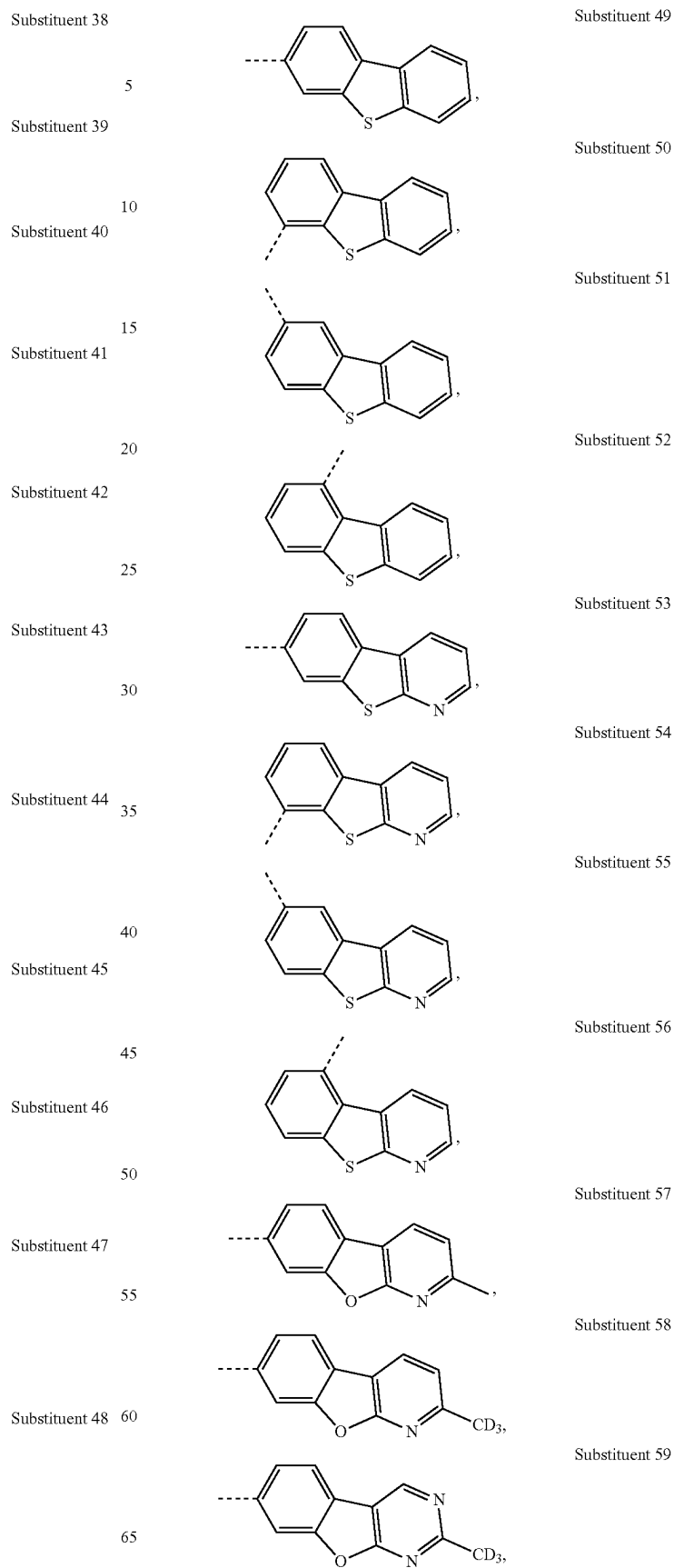
Substituent 37

35

-continued

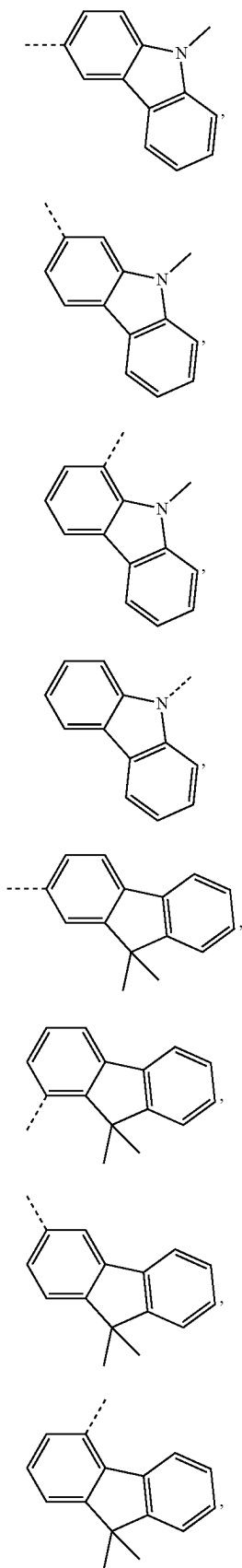
**36**

-continued

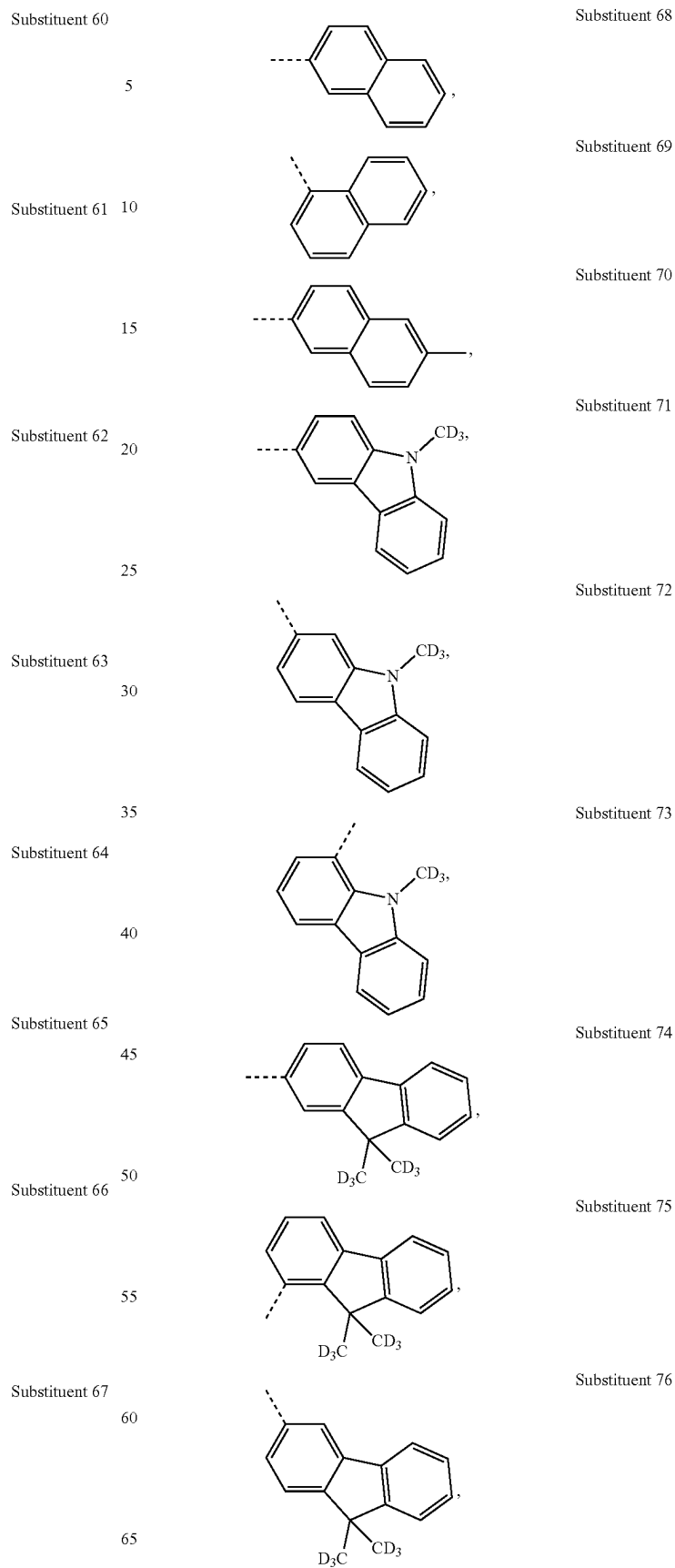


37

-continued

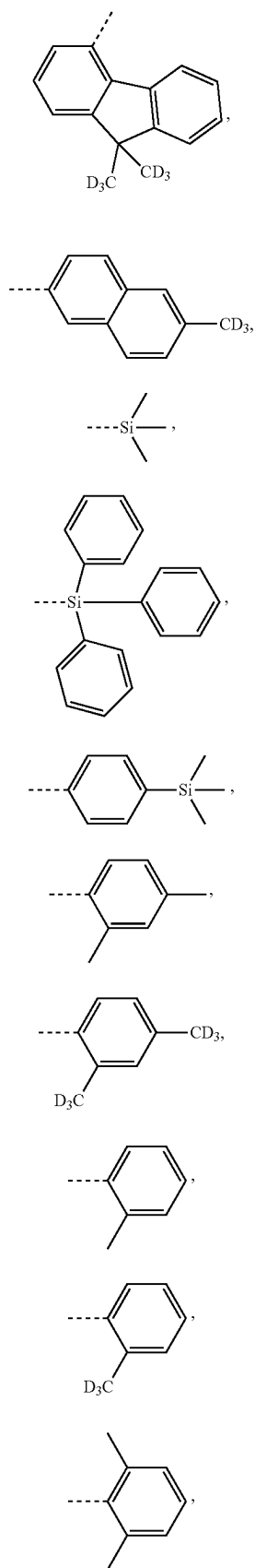
**38**

-continued

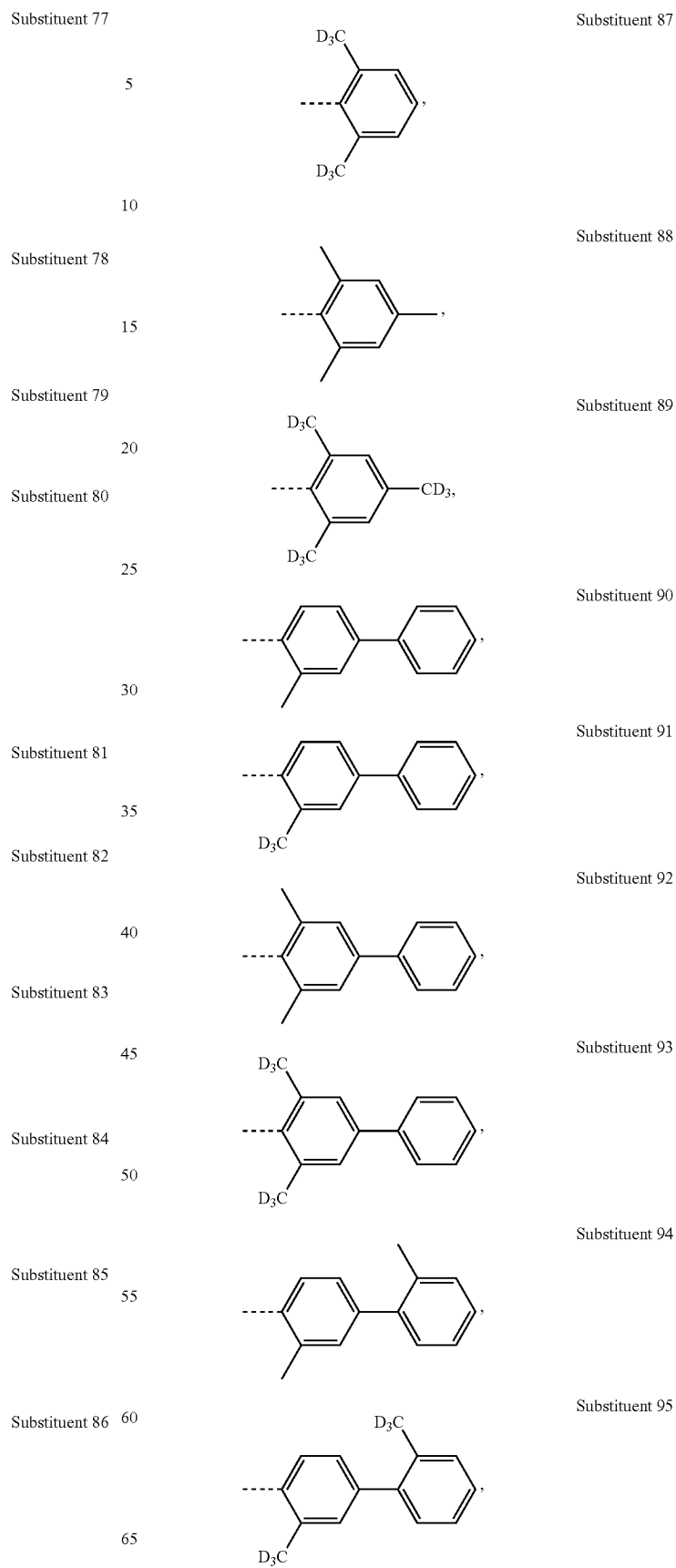


39

-continued

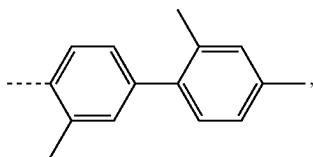
**40**

-continued



41

-continued

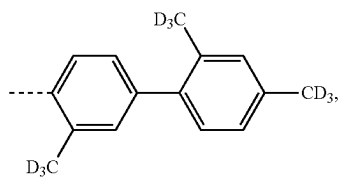


Substituent 96

5

10

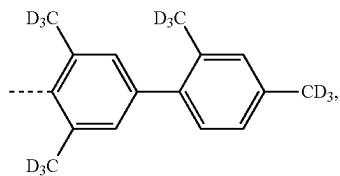
Substituent 97



15

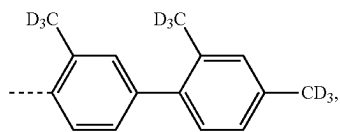
20

Substituent 98



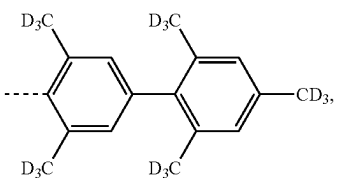
25

Substituent 99



35

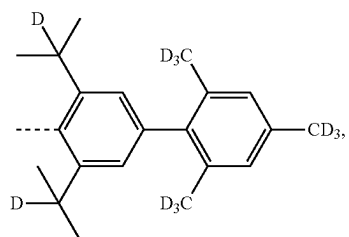
Substituent 100



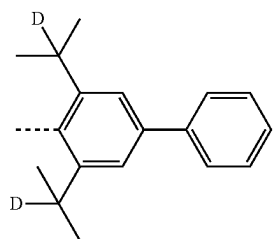
45

42

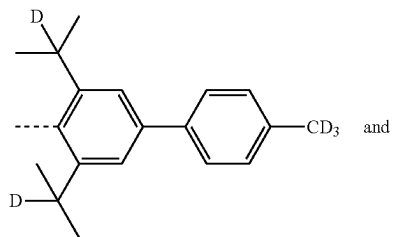
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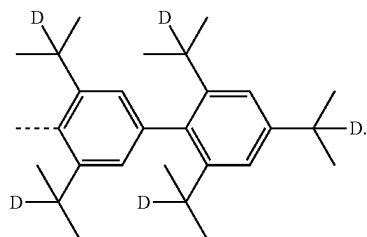
Substituent 101



Substituent 102



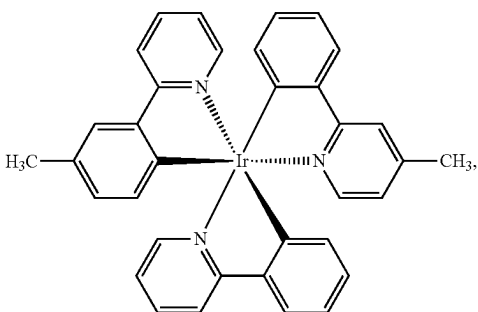
Substituent 103



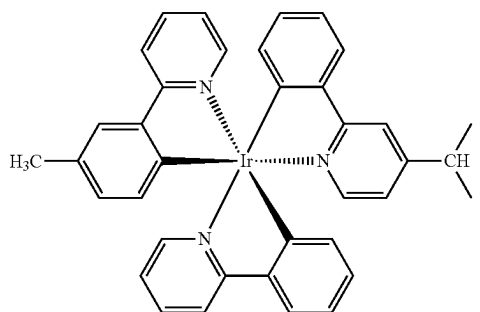
Substituent 104

In some embodiments of the compound, the compound is selected from the group consisting of:

Compound 1

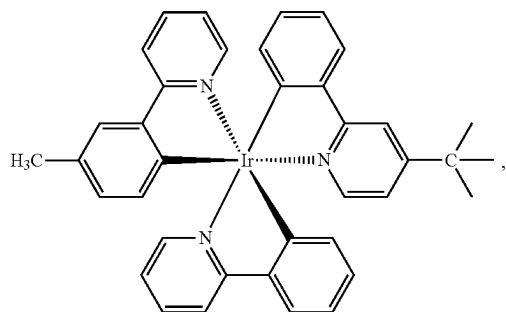


Compound 2



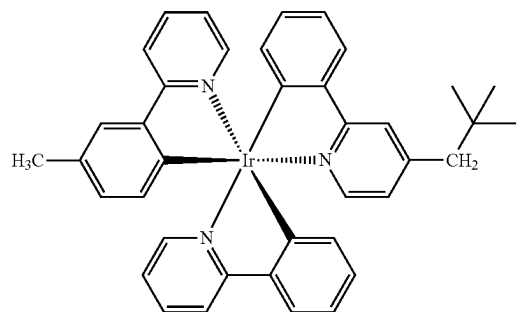
43

-continued
Compound 3

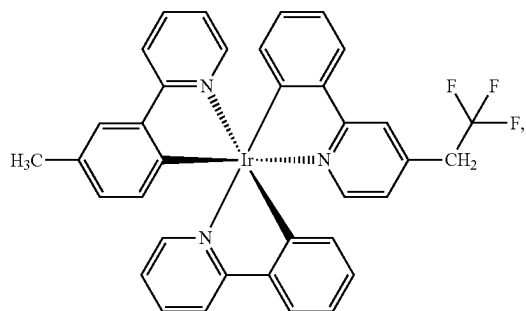


44

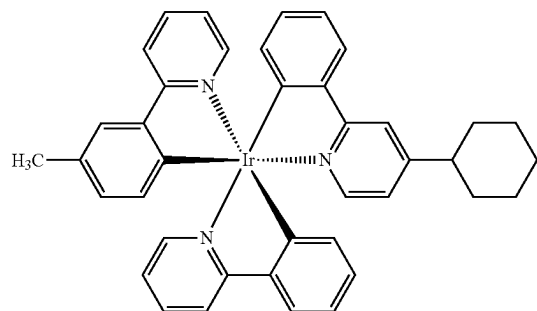
Compound 4



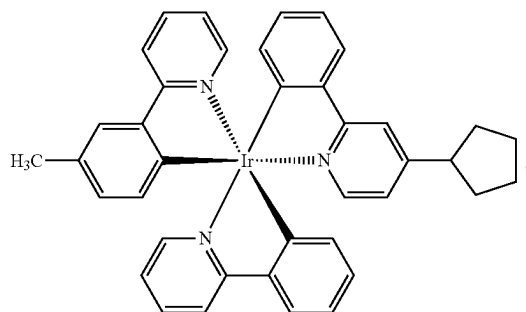
Compound 5



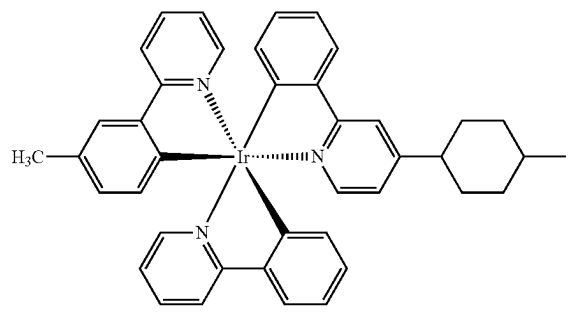
Compound 6



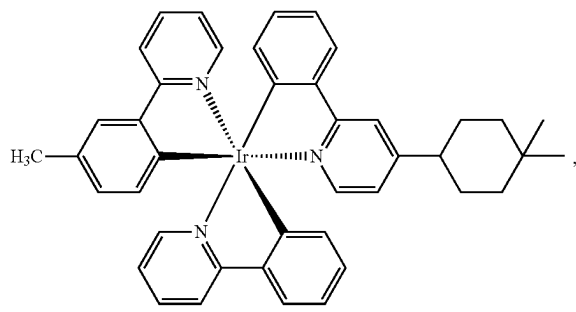
Compound 7



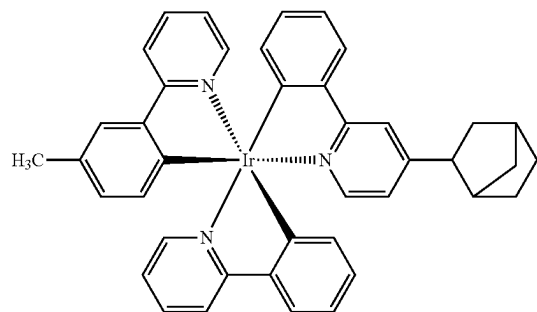
Compound 8



Compound 9



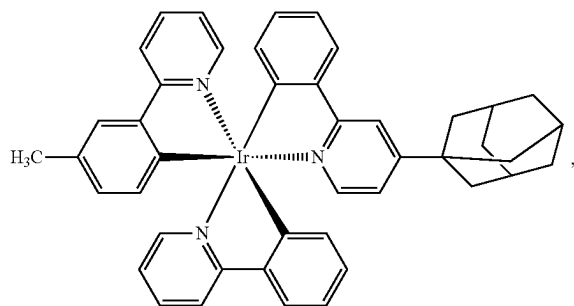
Compound 10



45

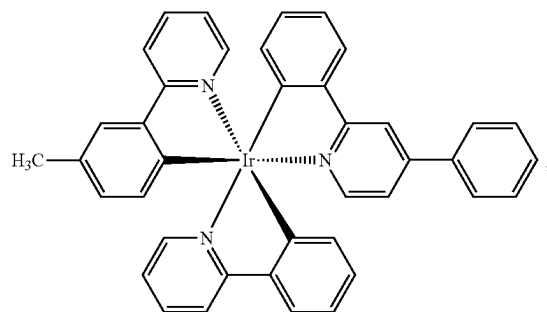
-continued

Compound 11

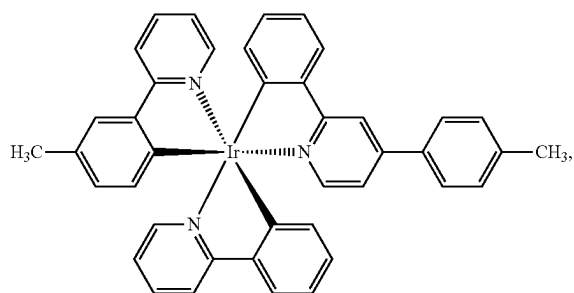


46

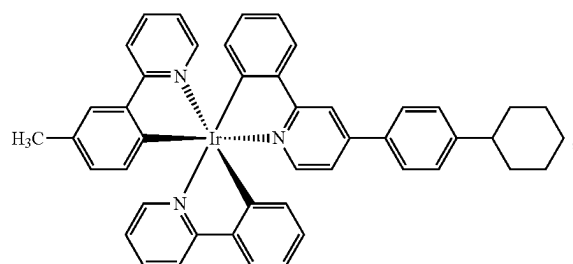
Compound 12



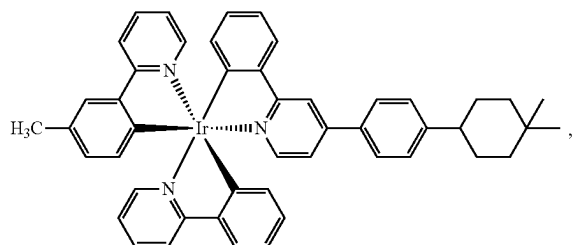
Compound 13



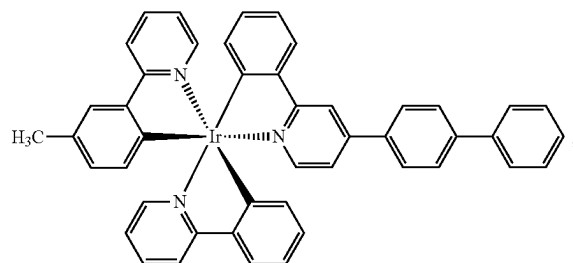
Compound 14



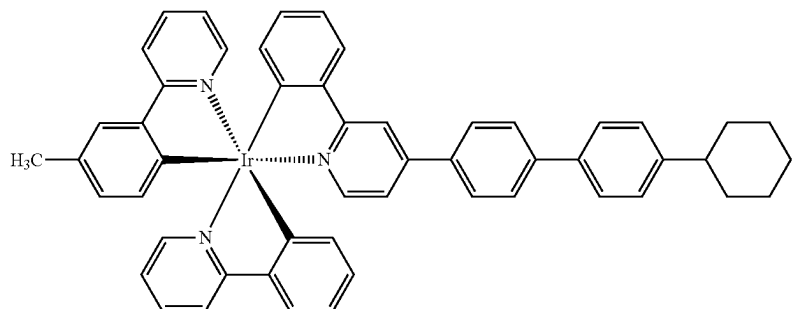
Compound 15



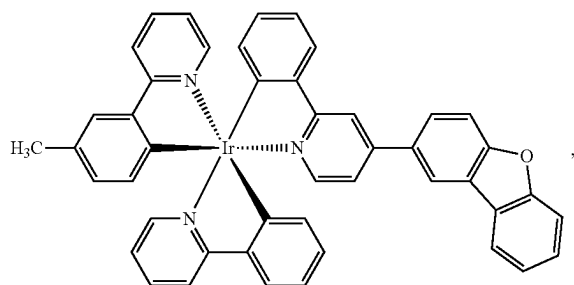
Compound 16



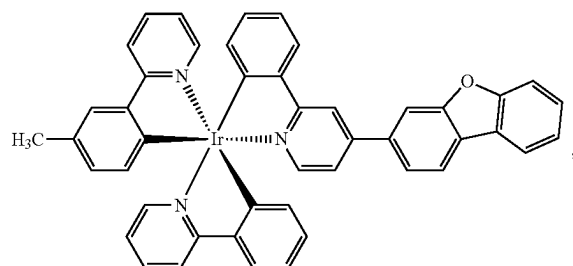
Compound 17



Compound 18



Compound 19

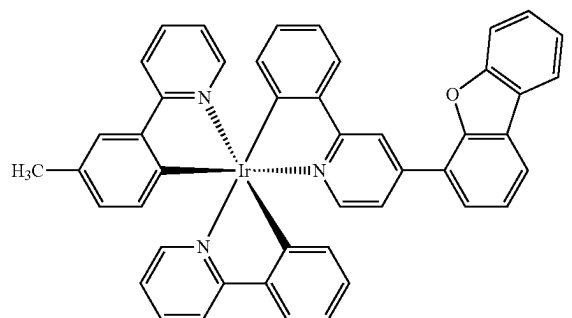


47

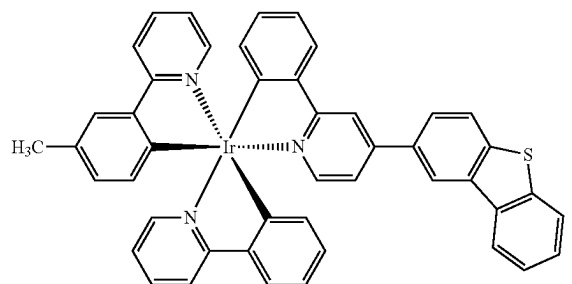
48

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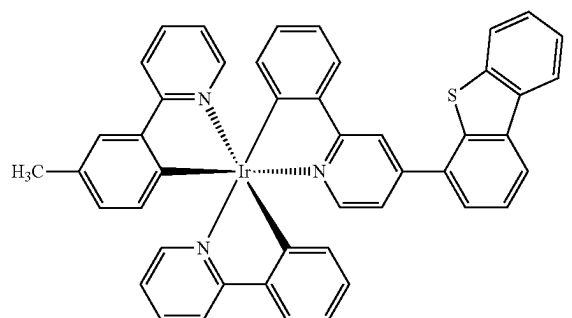
Compound 20



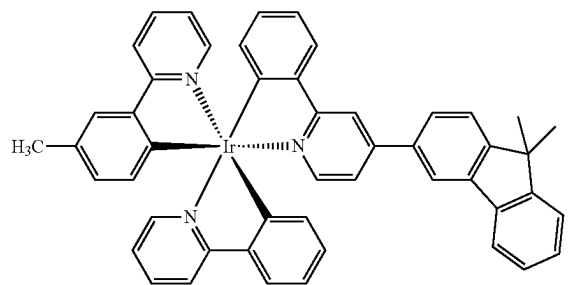
Compound 22



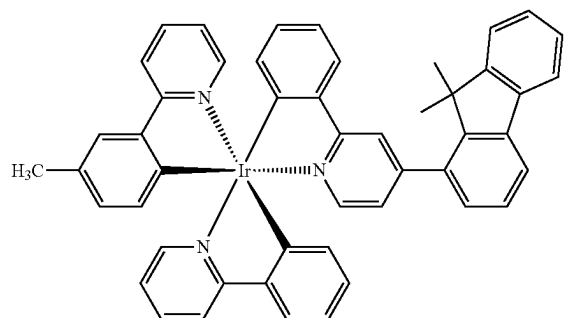
Compound 24



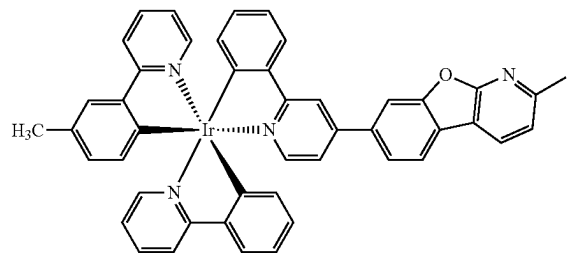
Compound 26



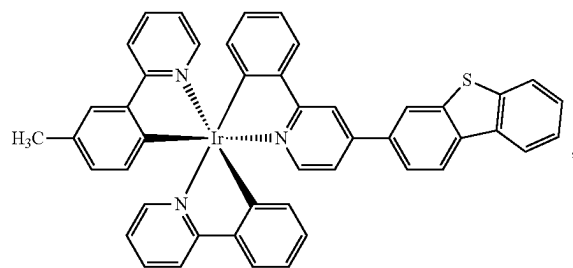
Compound 28



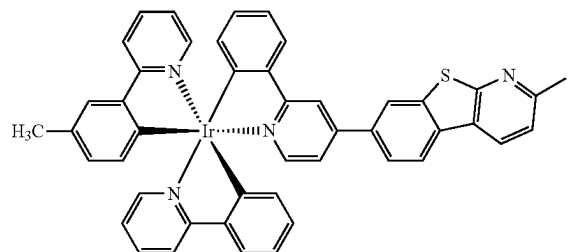
Compound 21



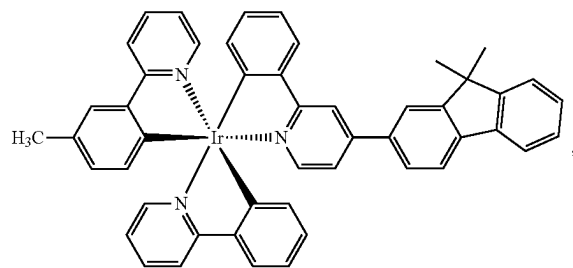
Compound 23



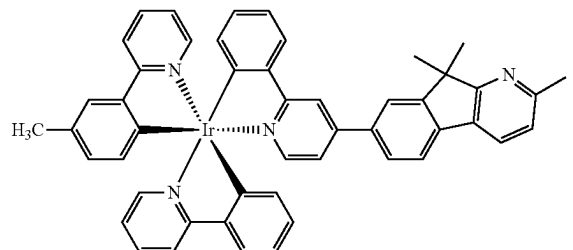
Compound 25



Compound 27



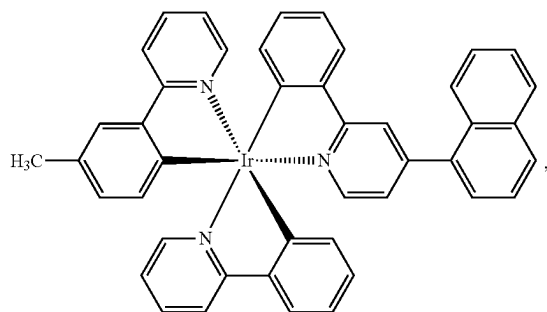
Compound 29



49

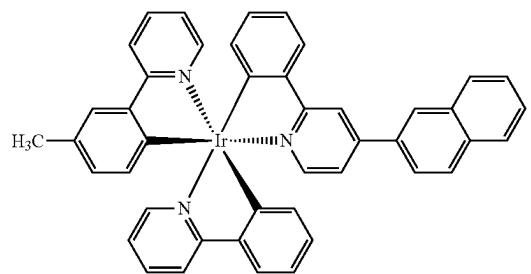
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Compound 30

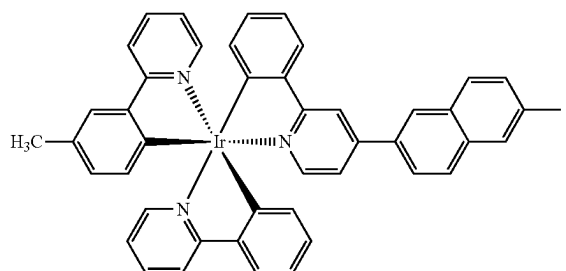


50

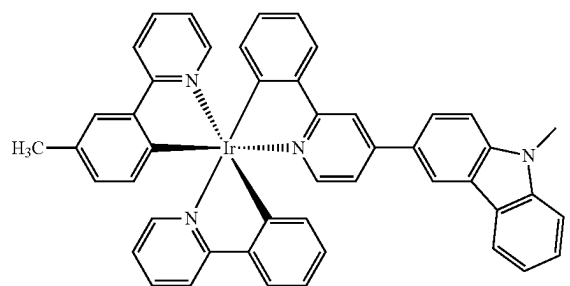
Compound 31



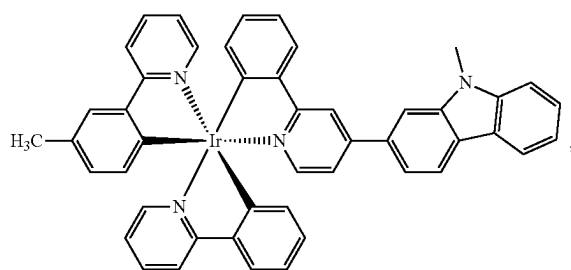
Compound 32



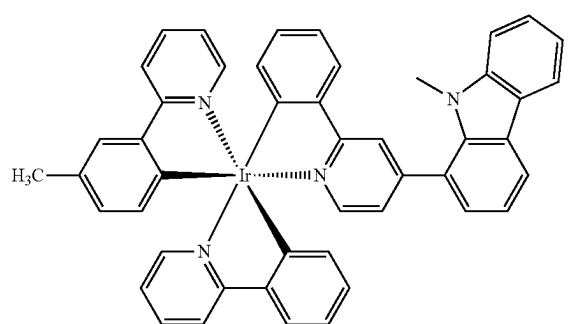
Compound 33



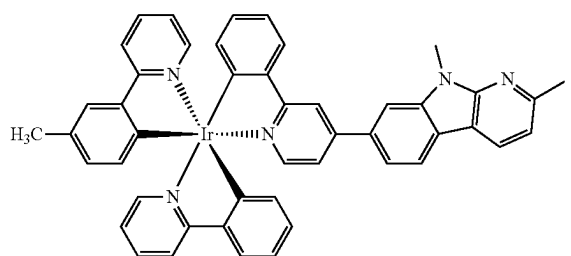
Compound 34



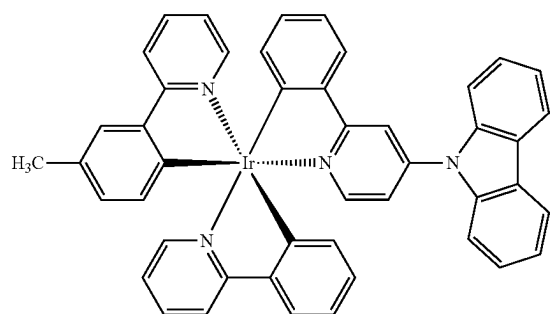
Compound 35



Compound 36



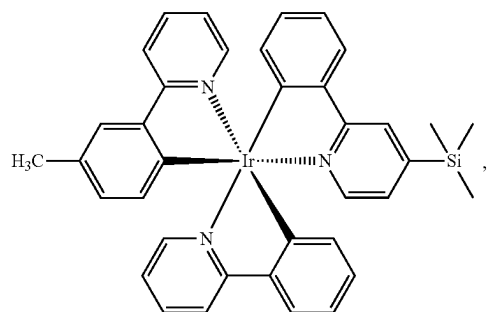
Compound 37



51

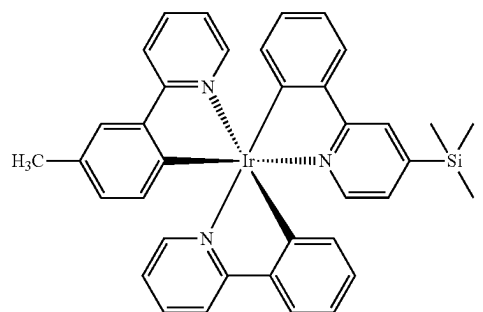
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Compound 38

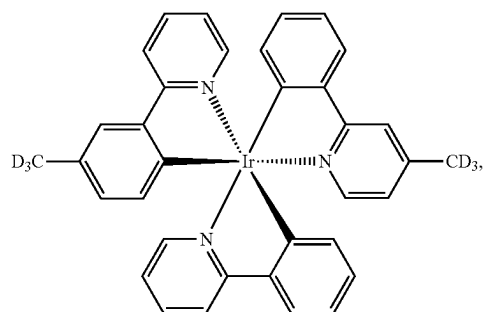


52

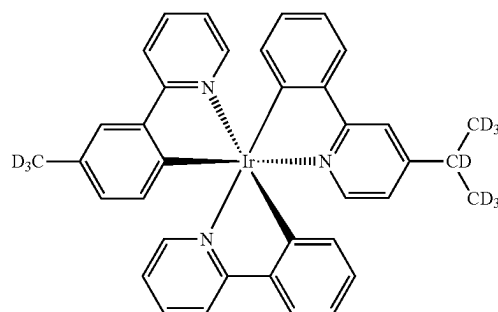
Compound 39



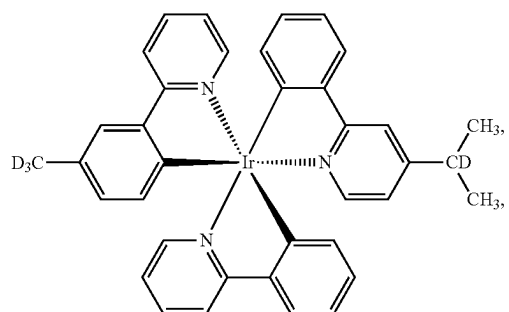
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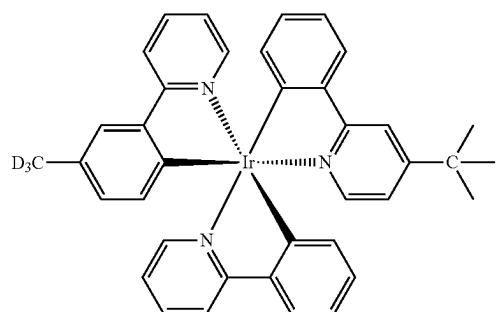
Compound 41



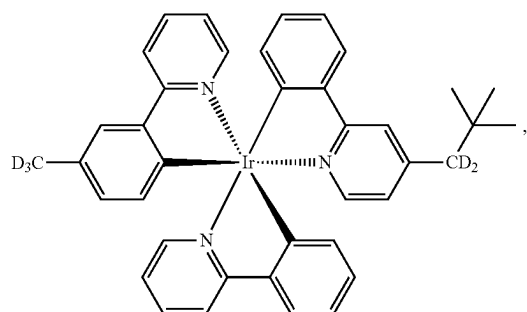
Compound 42



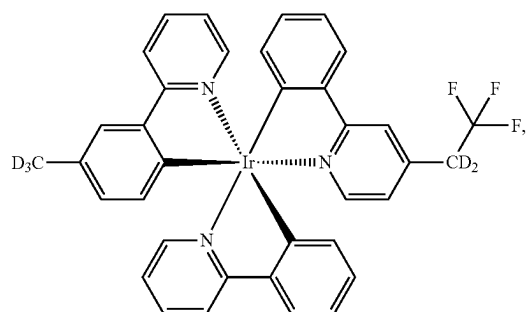
Compound 43



Compound 44



Compound 45



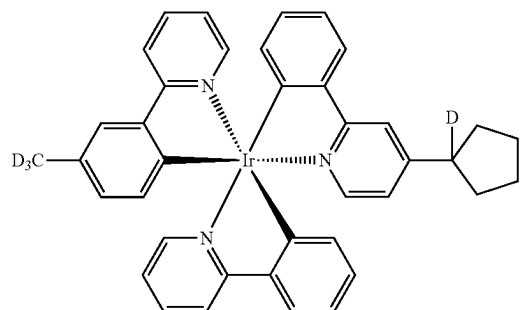
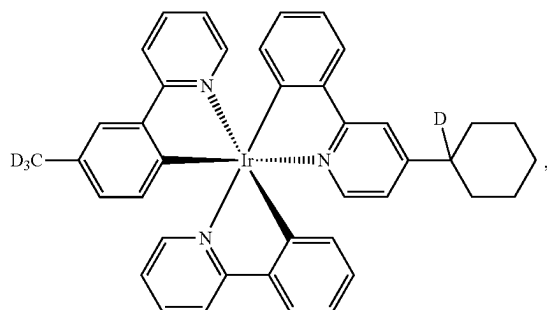
53

54

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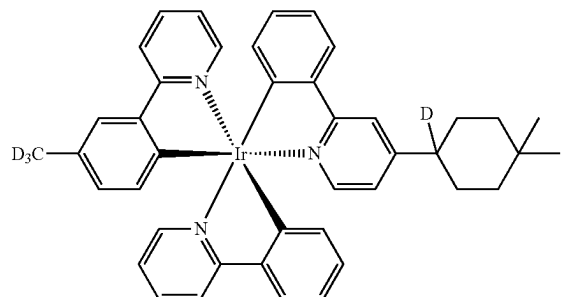
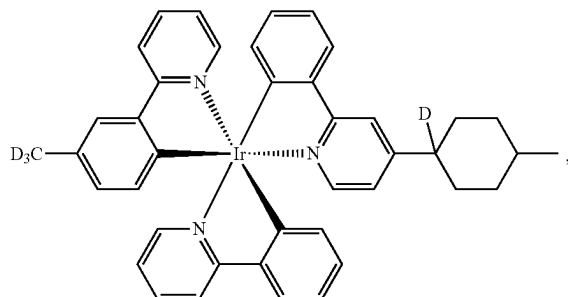
Compound 46

Compound 47



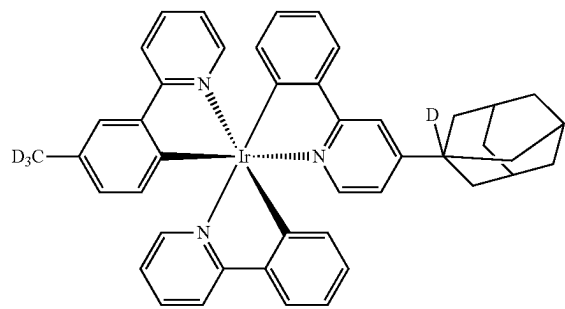
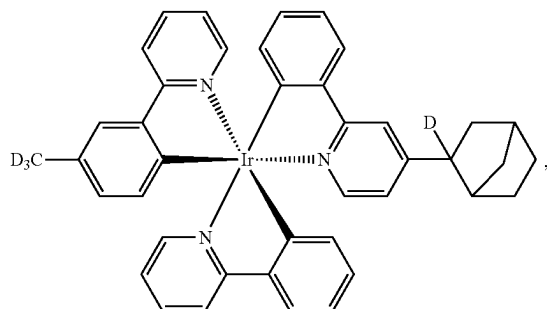
Compound 48

Compound 49



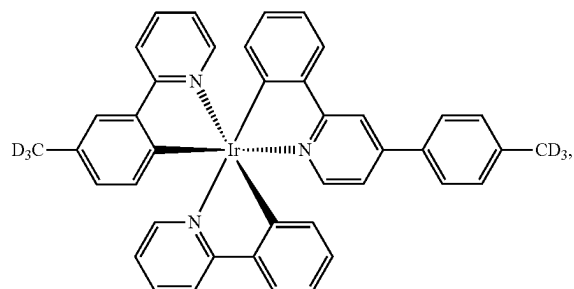
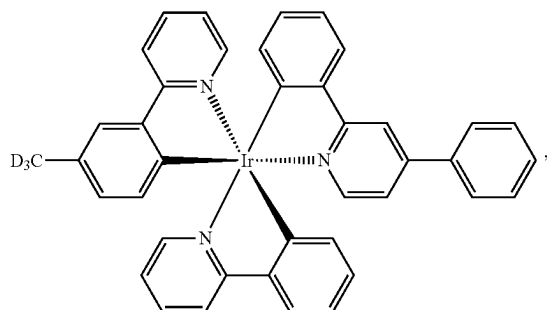
Compound 50

Compound 51



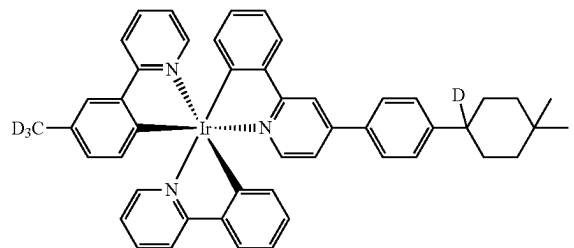
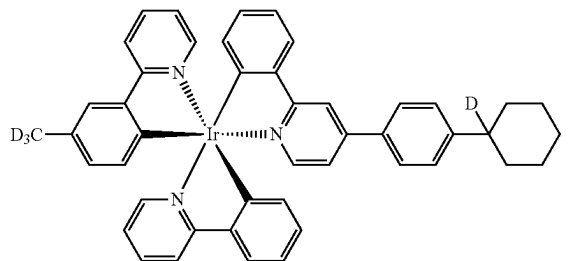
Compound 52

Compound 53



Compound 54

Compound 55

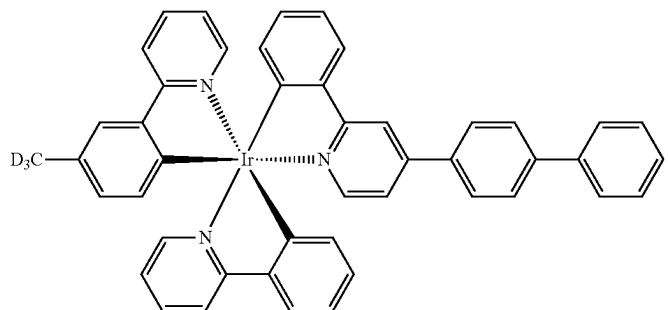


55

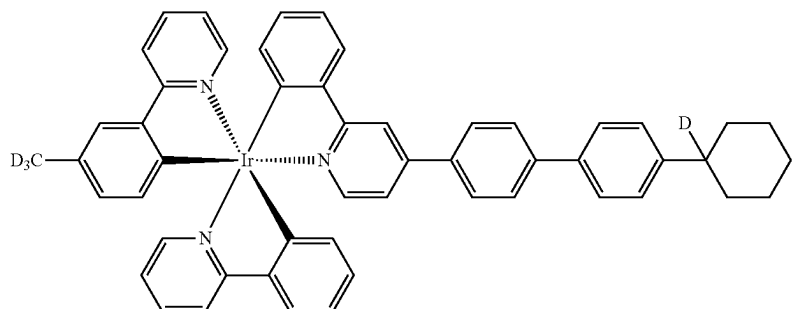
56

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Compound 56

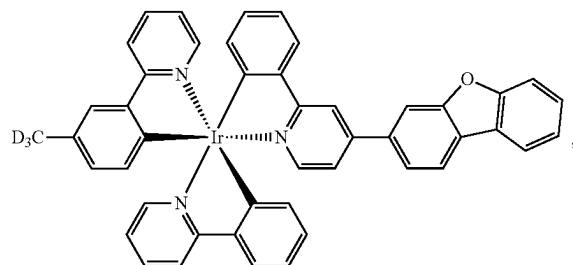
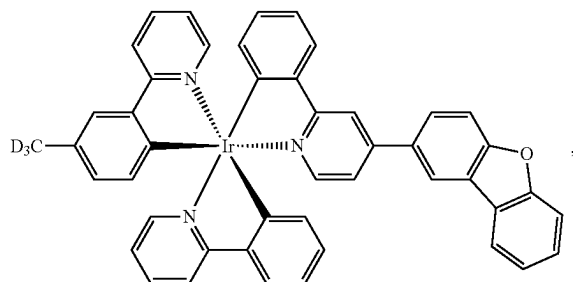


Compound 57



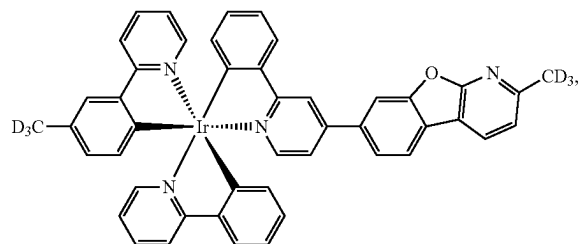
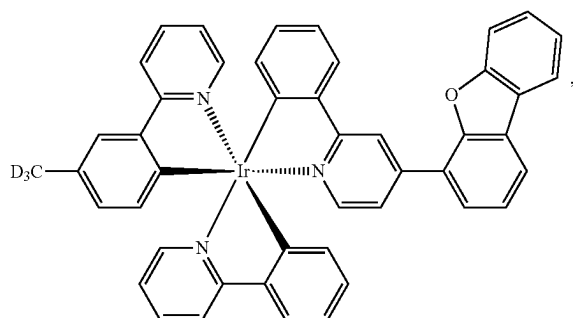
Compound 58

Compound 59



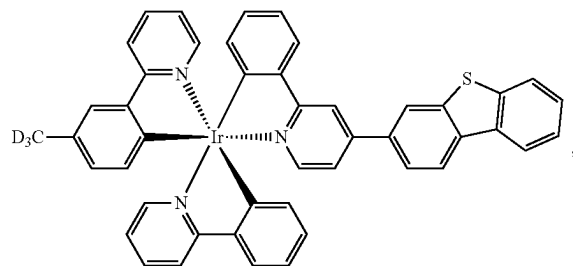
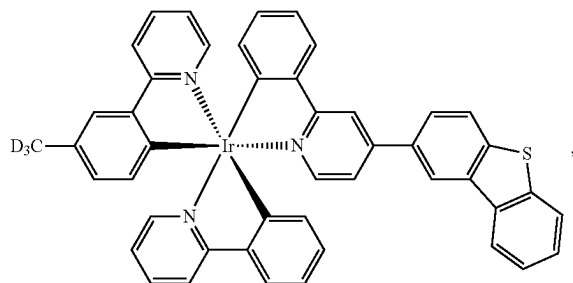
Compound 60

Compound 61



Compound 62

Compound 63

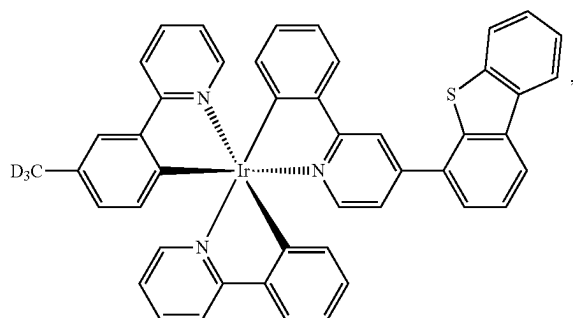


57

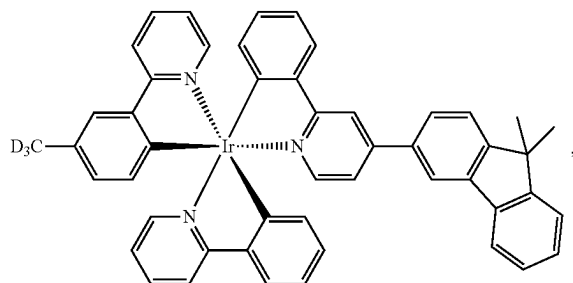
58

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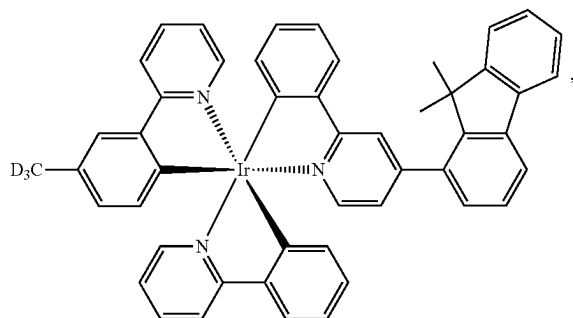
Compound 64



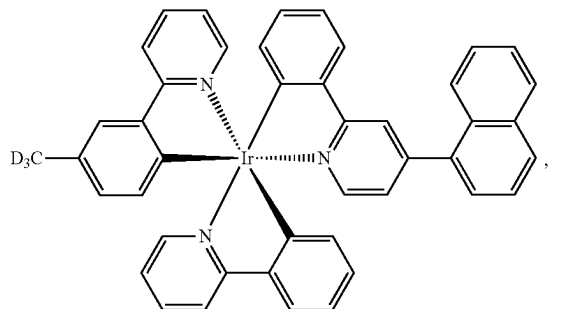
Compound 66



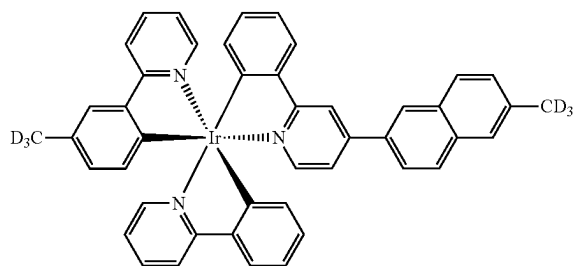
Compound 68



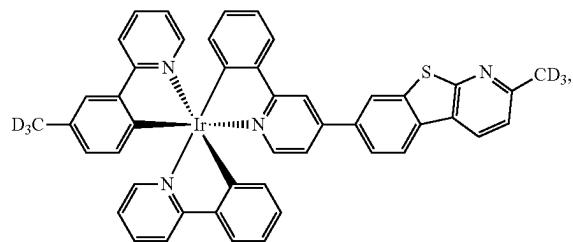
Compound 70



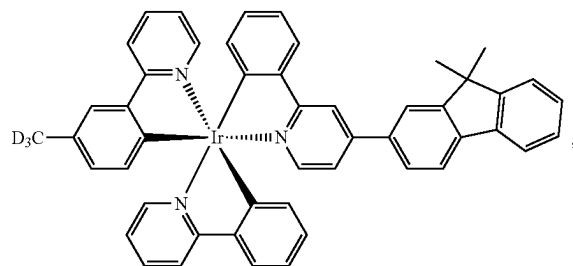
Compound 72



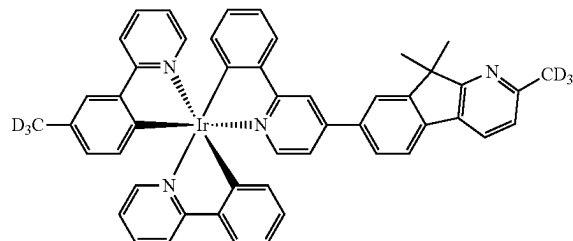
Compound 65



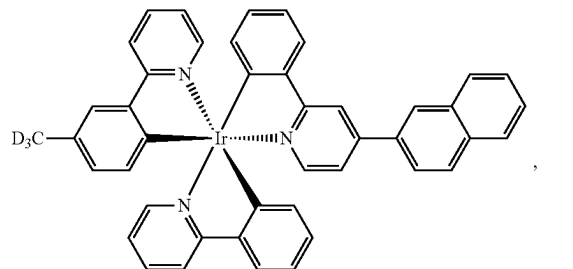
Compound 67



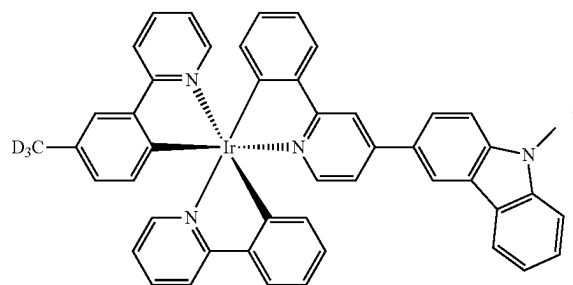
Compound 69



Compound 71



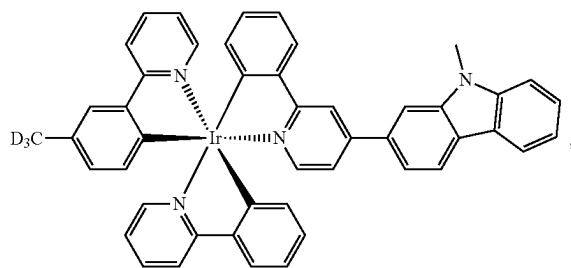
Compound 73



59

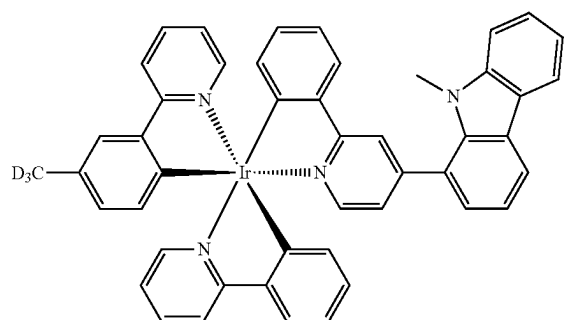
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Compound 74

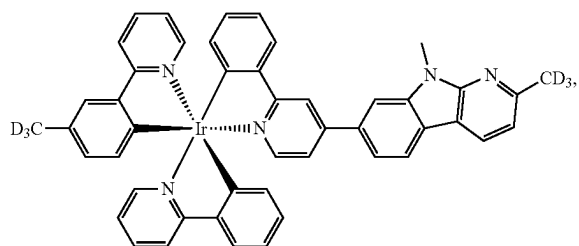


60

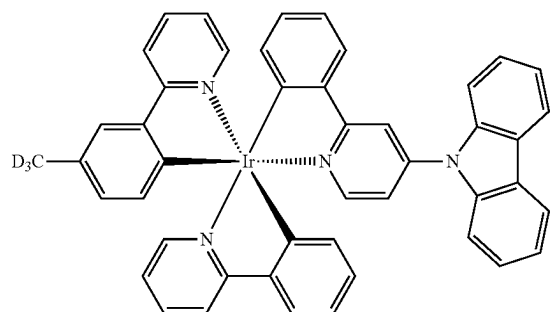
Compound 75



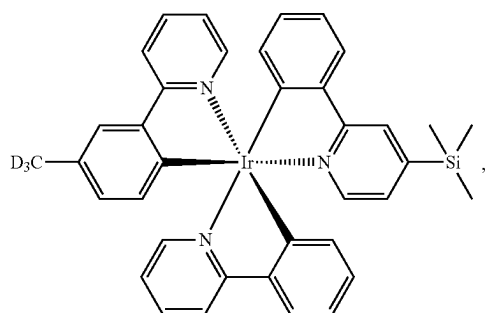
Compound 76



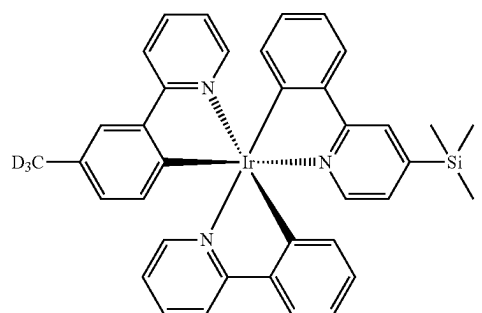
Compound 77



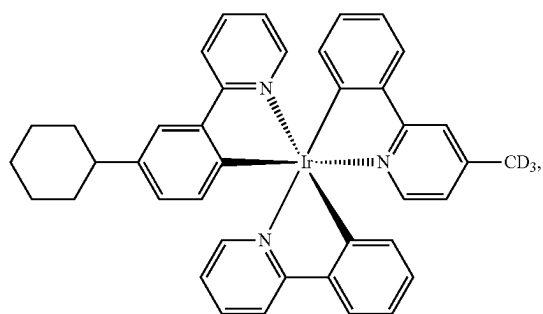
Compound 78



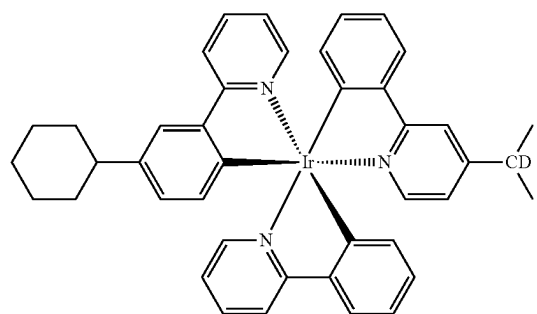
Compound 79



Compound 80



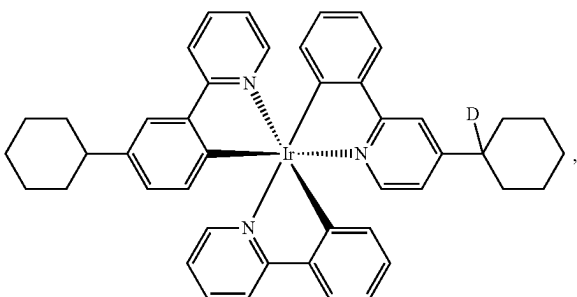
Compound 81



61

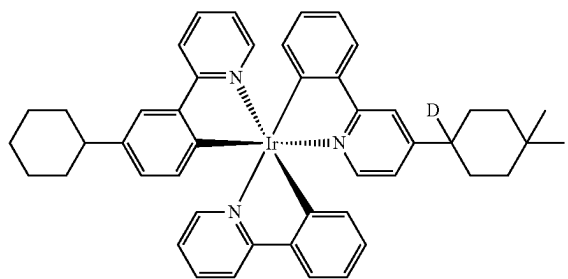
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Compound 82

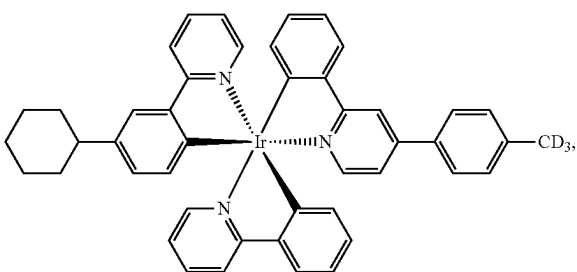


62

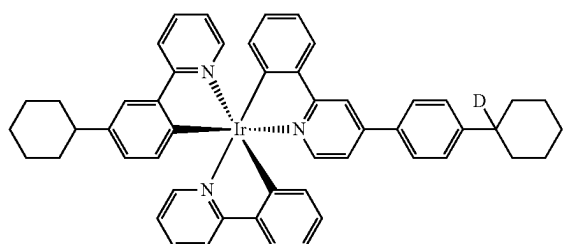
Compound 83



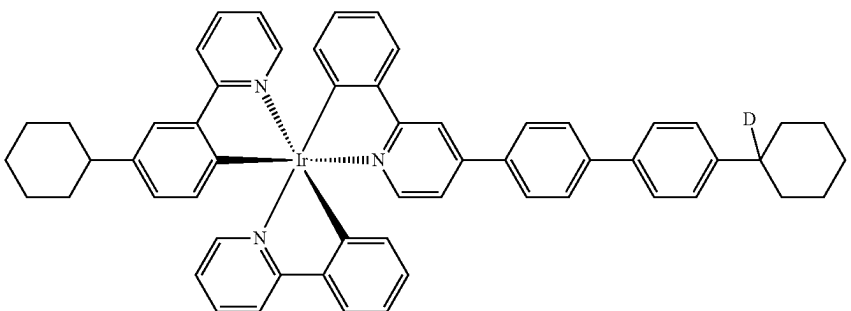
Compound 84



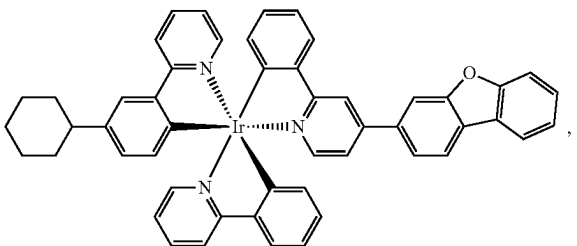
Compound 85



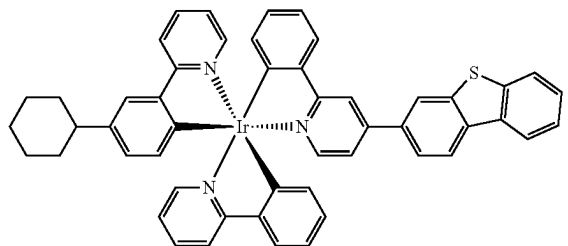
Compound 86



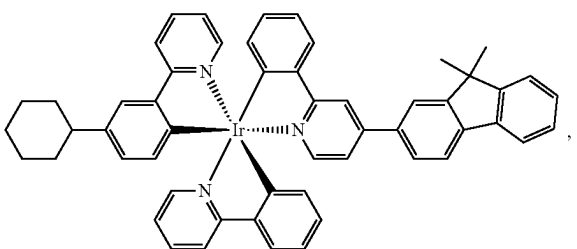
Compound 87



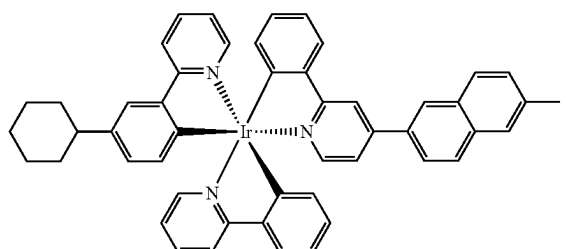
Compound 88



Compound 89



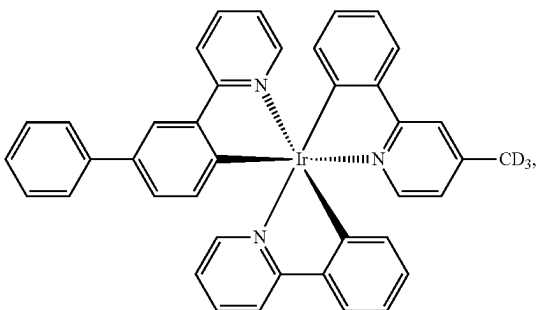
Compound 90



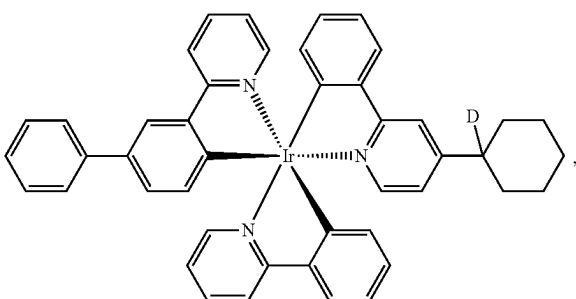
63

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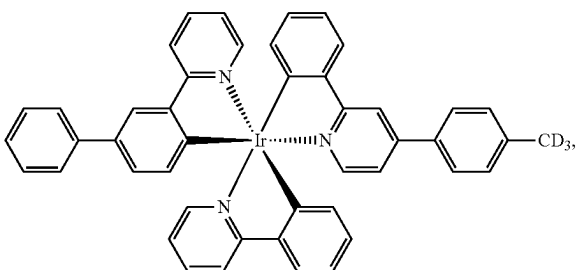
Compound 91



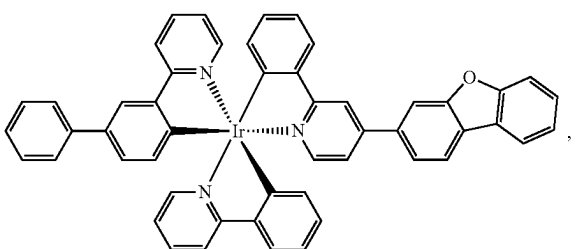
Compound 93



Compound 95

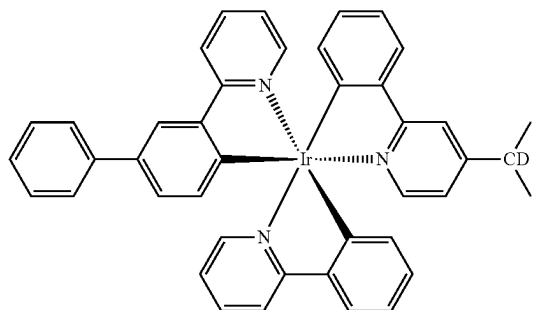


Compound 98

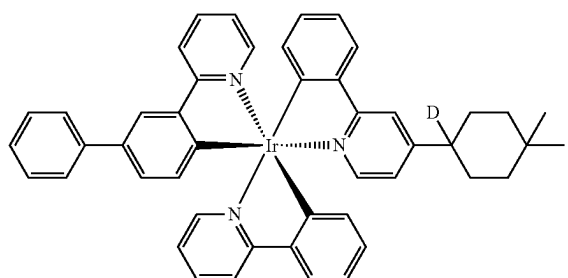


64

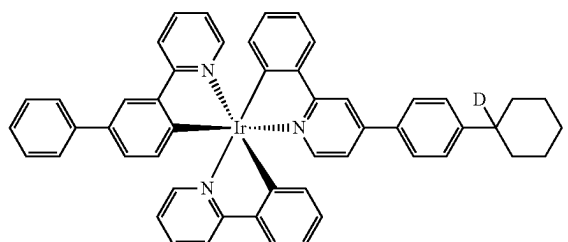
Compound 92



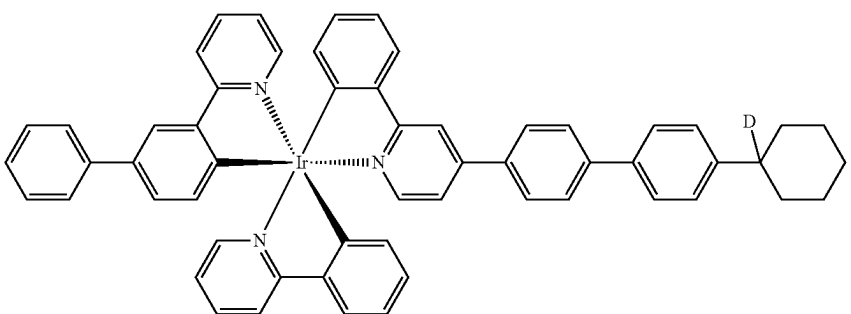
Compound 94



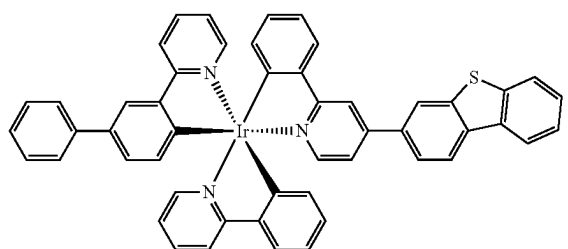
Compound 96



Compound 97



Compound 99

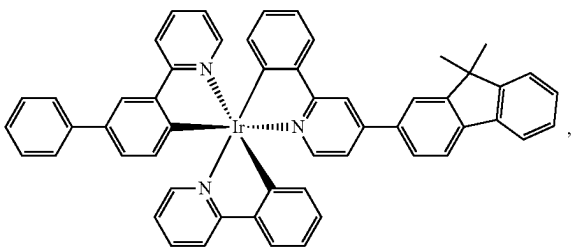


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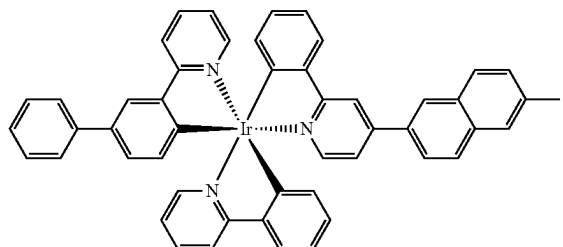
66

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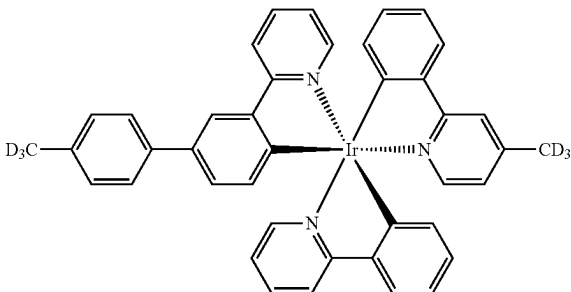
Compound 100



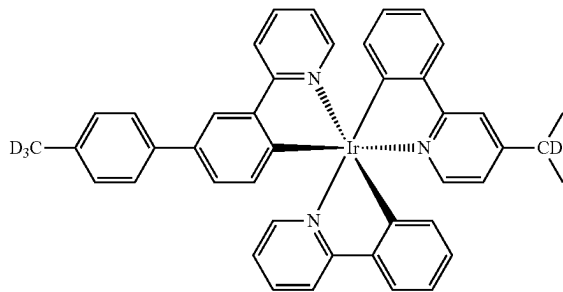
Compound 101



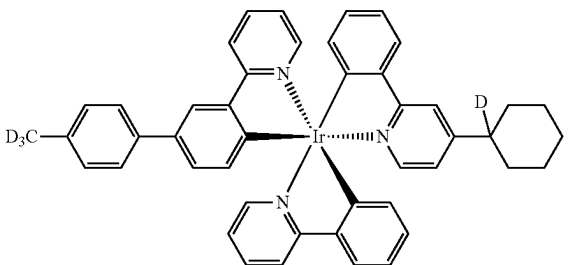
Compound 102



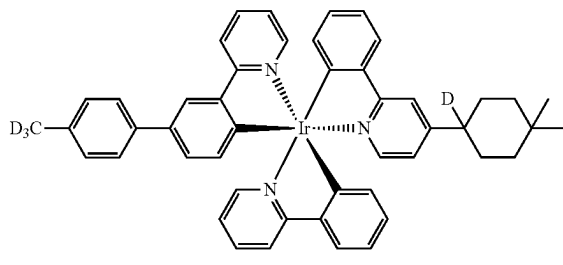
Compound 103



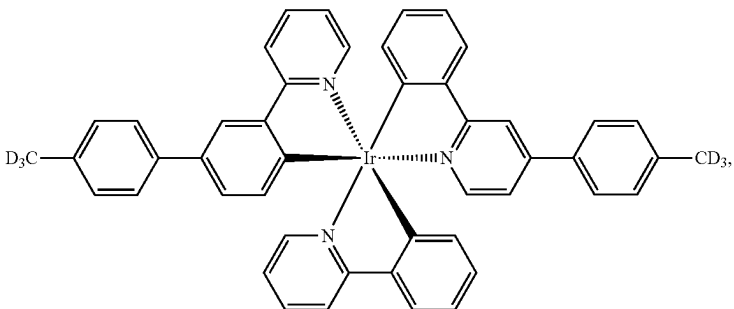
Compound 104



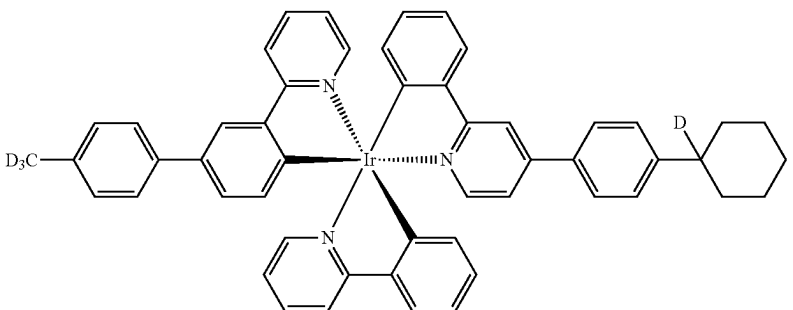
Compound 105



Compound 106



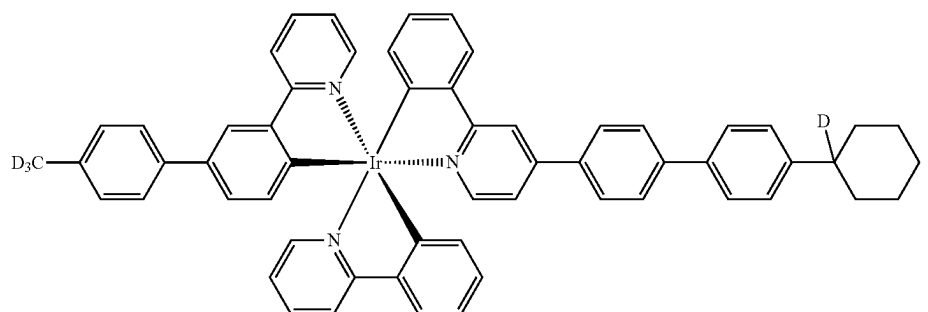
Compound 107



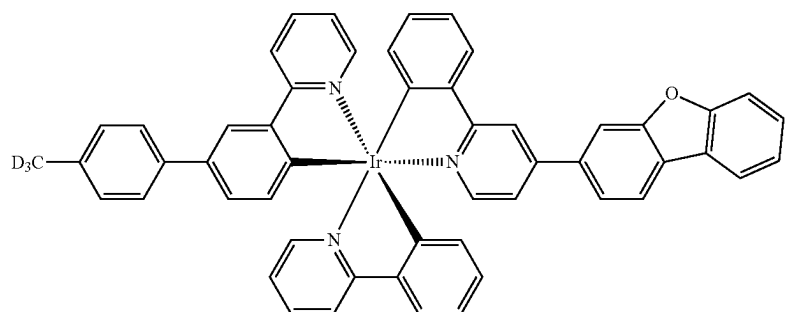
67

68

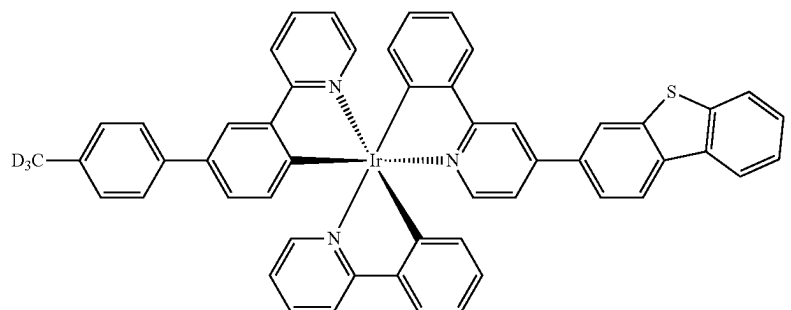
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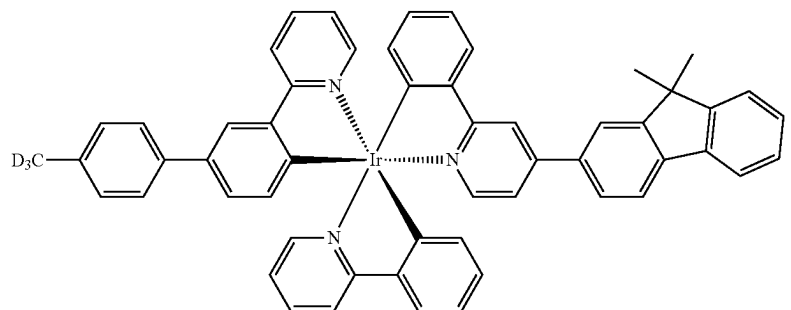
Compound 108



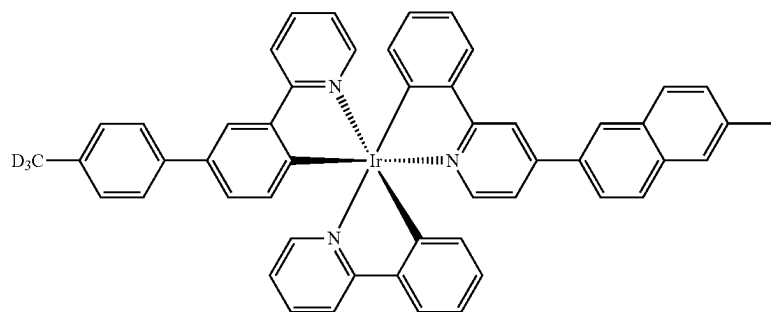
Compound 109



Compound 110

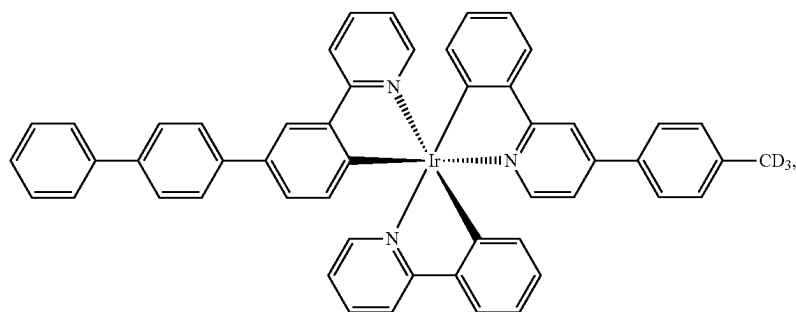


Compound 111

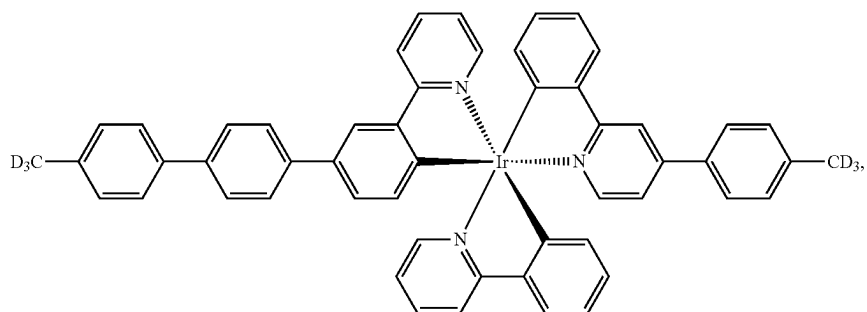


Compound 112

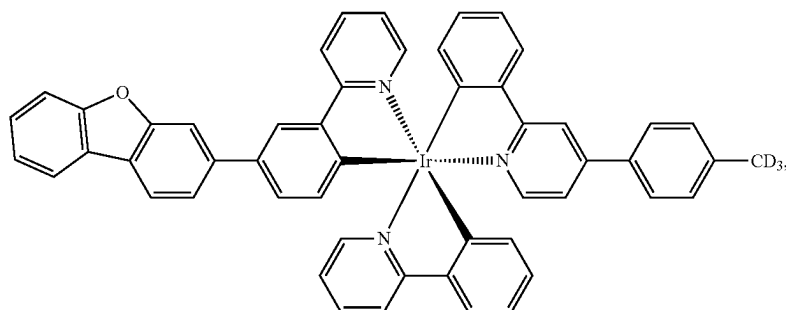
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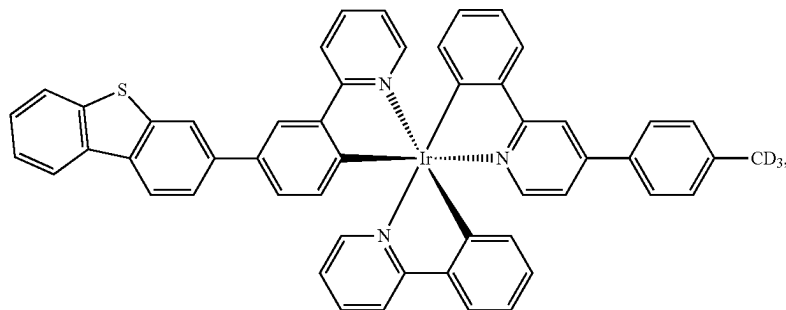
Compound 113



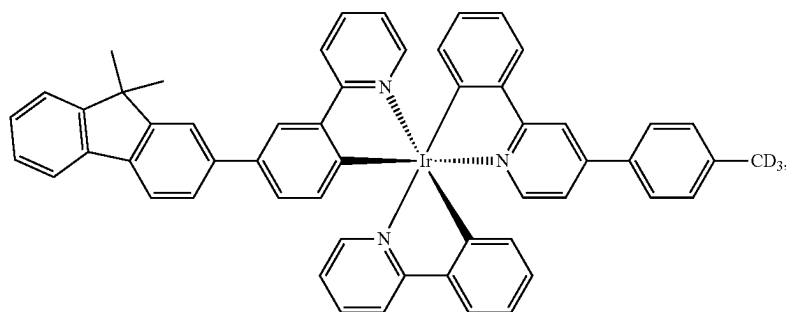
Compound 114



Compound 115



Compound 116



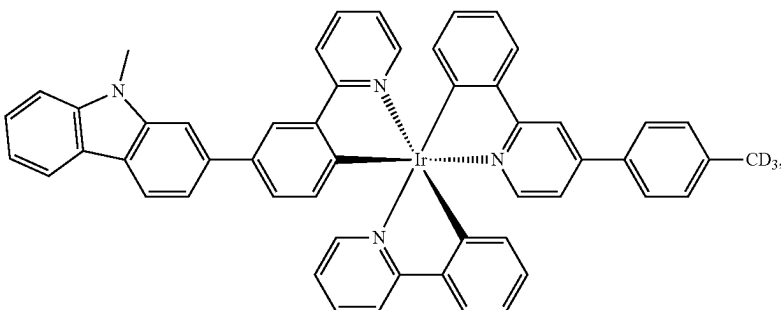
Compound 117

71

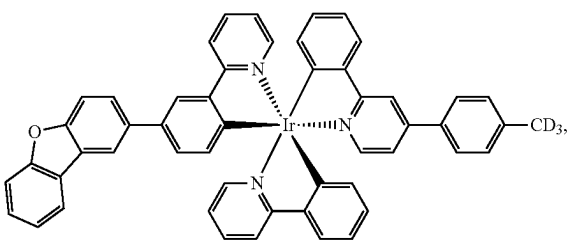
72

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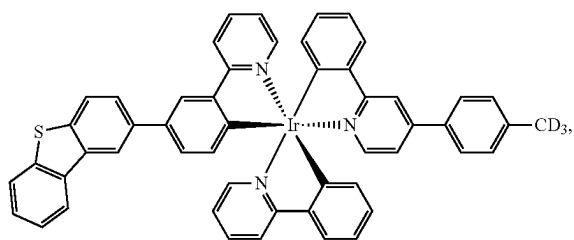
Compound 118



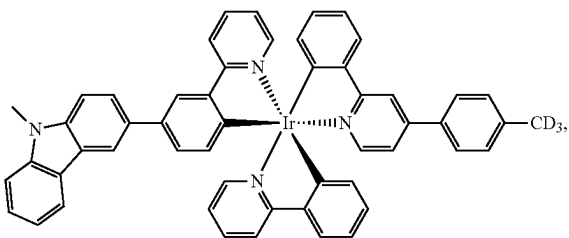
Compound 119



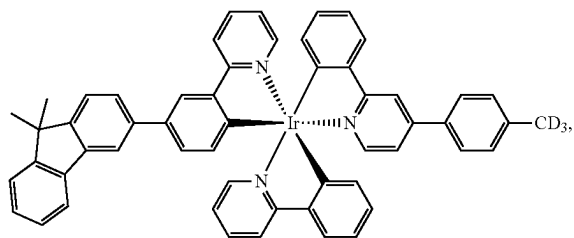
Compound 120



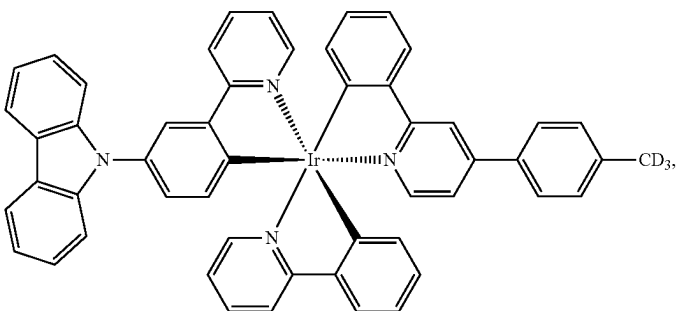
Compound 121



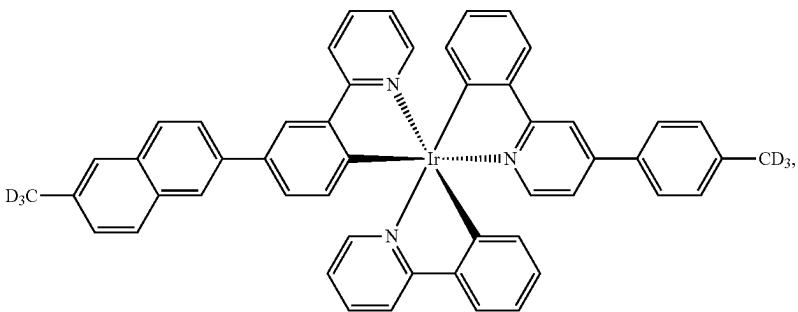
Compound 122



Compound 123



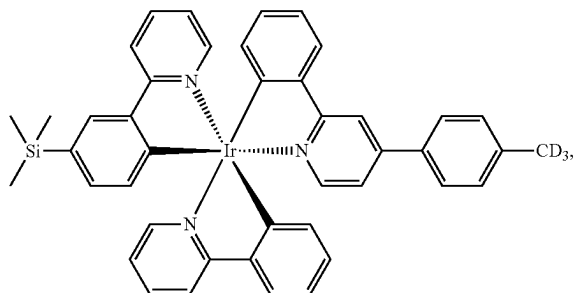
Compound 124



73

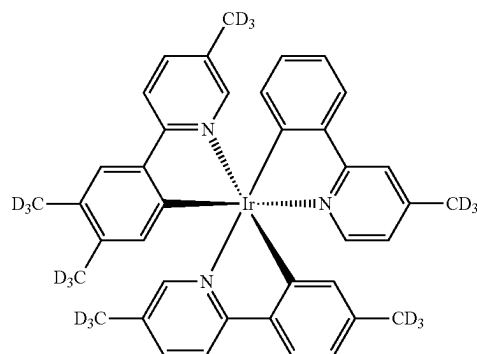
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Compound 125

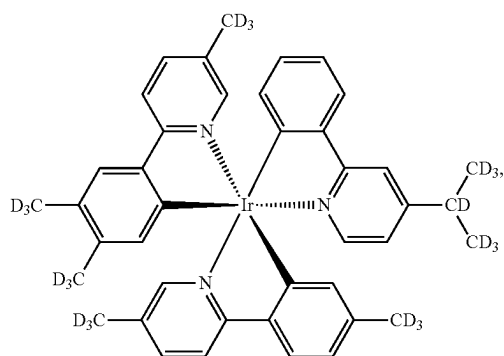


74

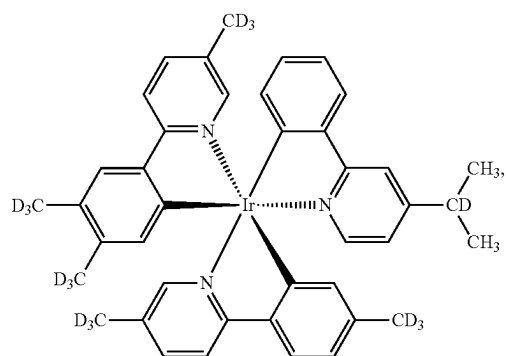
Compound 126



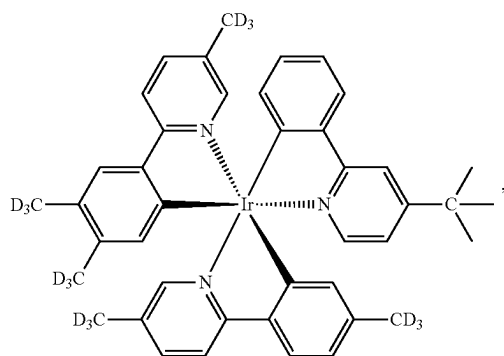
Compound 127



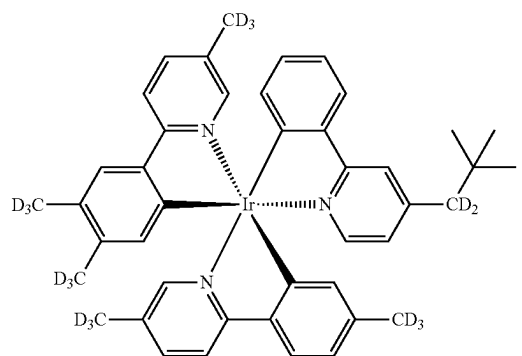
Compound 128



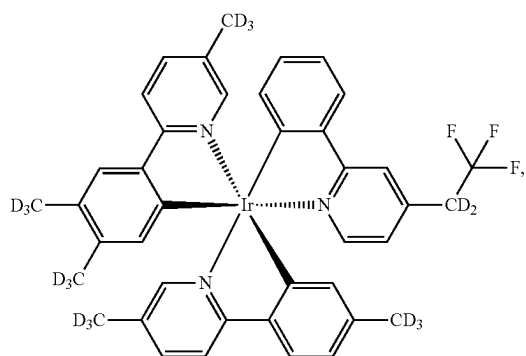
Compound 129



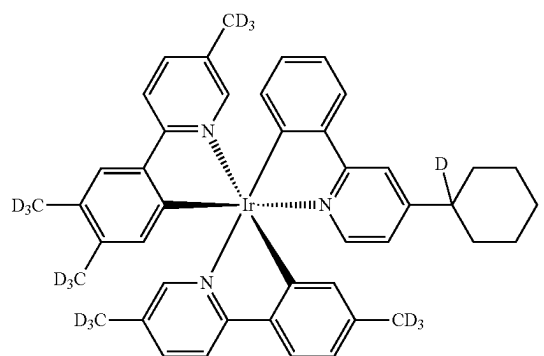
Compound 130



Compound 131



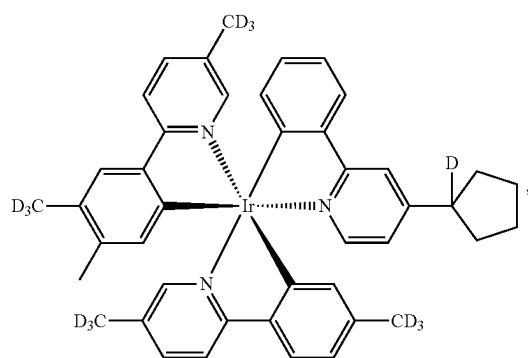
Compound 132



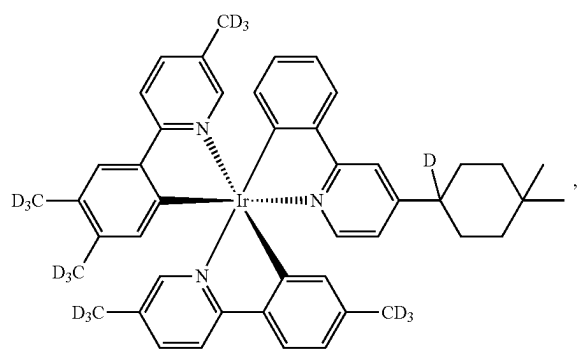
75

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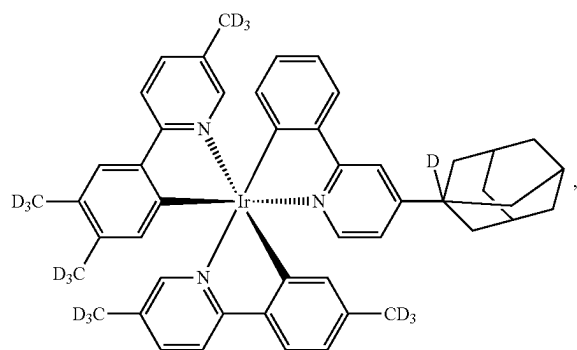
Compound 133



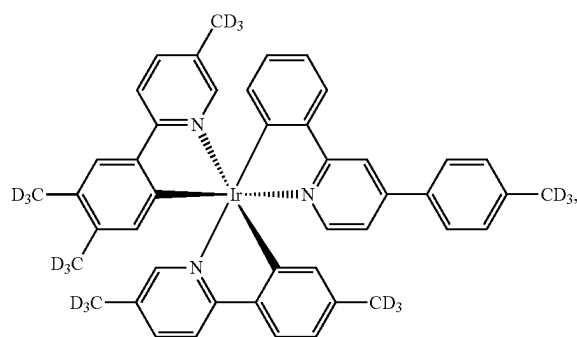
Compound 135



Compound 137

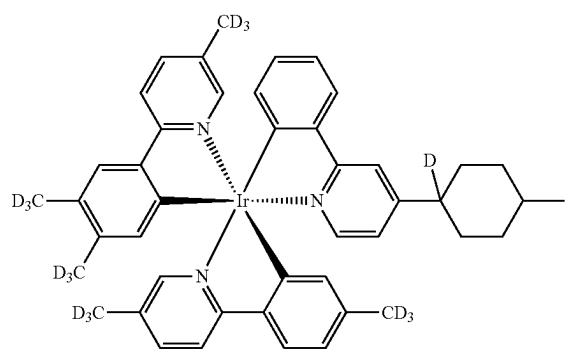


Compound 139

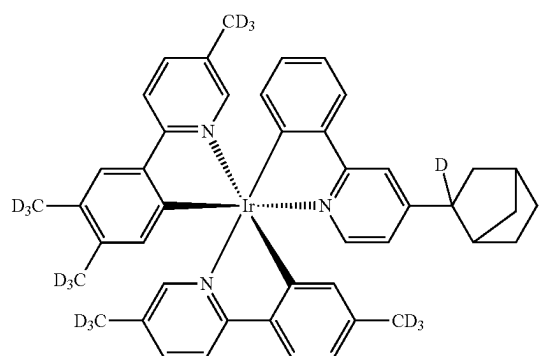


76

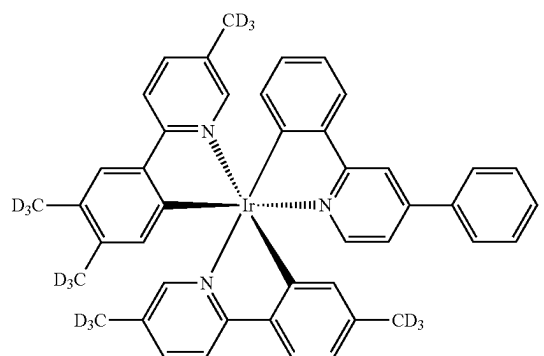
Compound 134



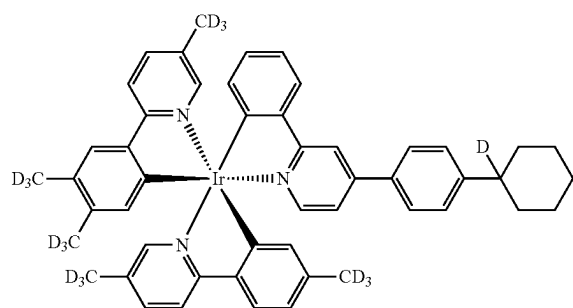
Compound 136



Compound 138



Compound 140



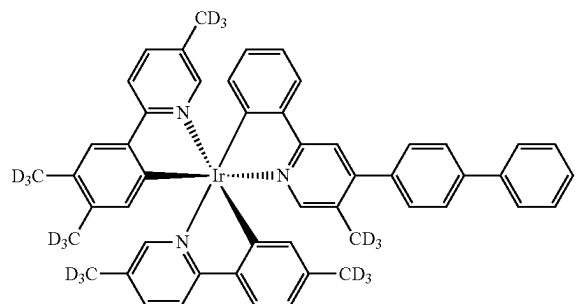
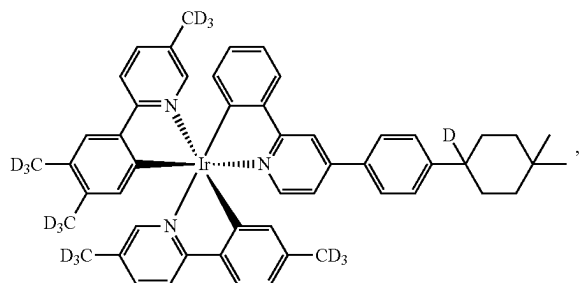
77

78

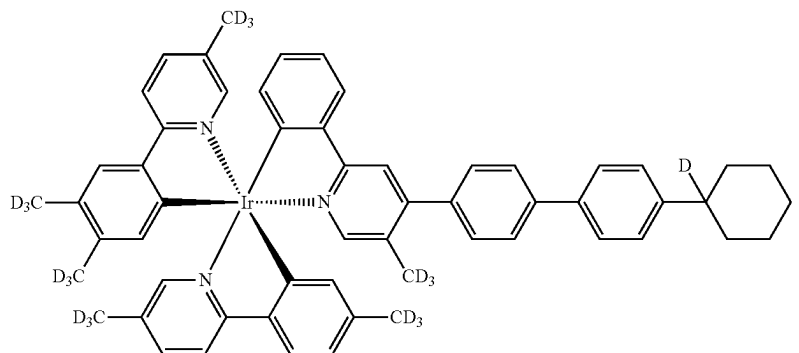
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Compound 141

Compound 142

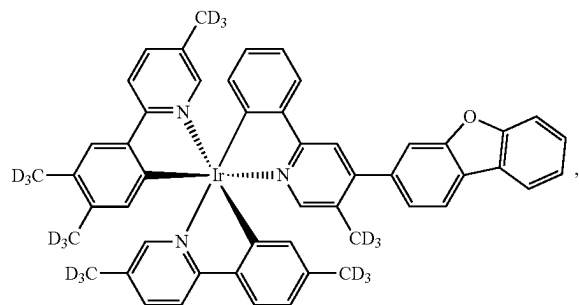
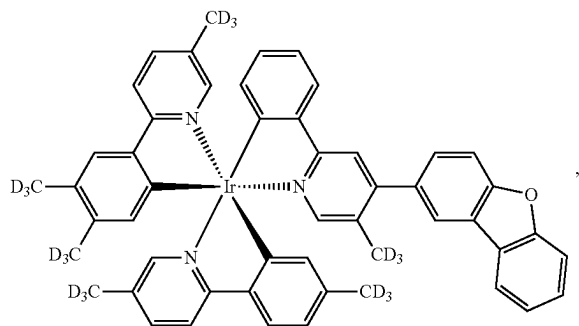


Compound 143



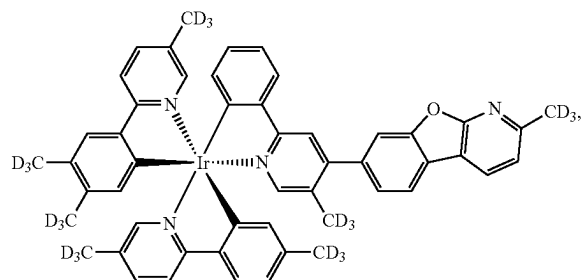
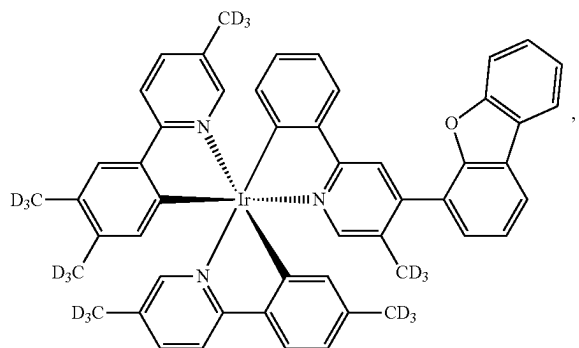
Compound 144

Compound 145



Compound 146

Compound 147



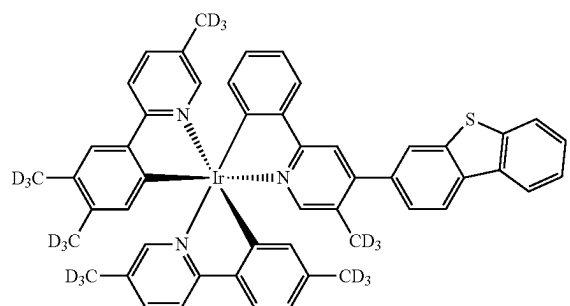
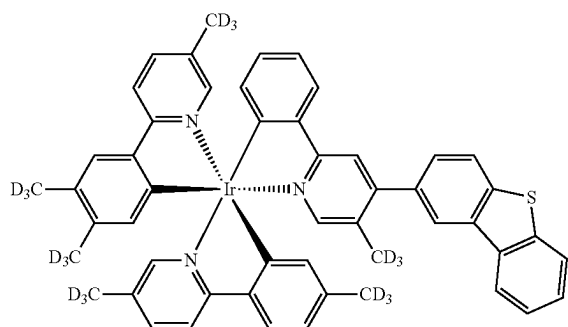
79

80

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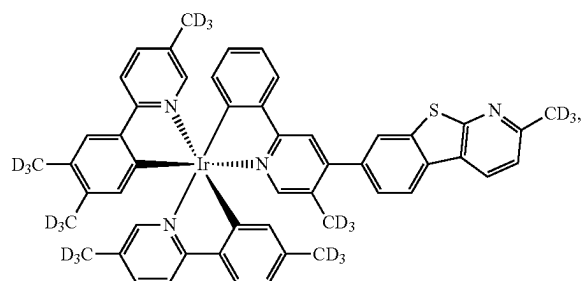
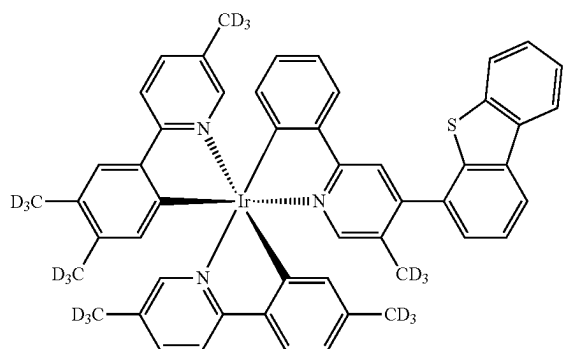
Compound 148

Compound 149



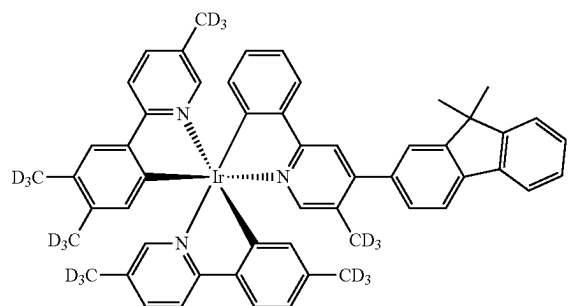
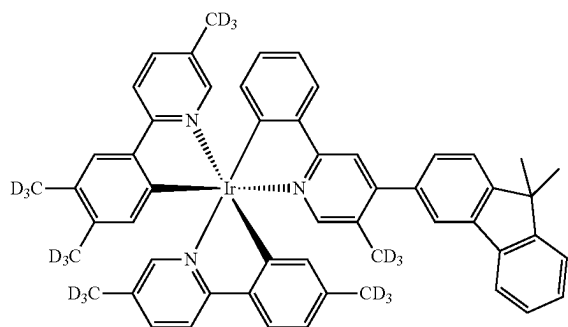
Compound 150

Compound 151



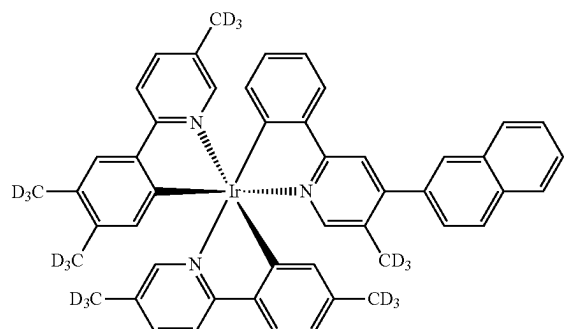
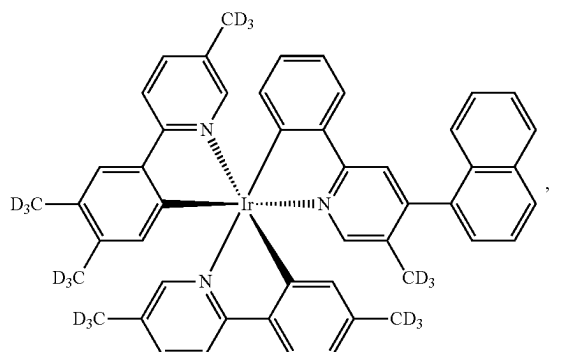
Compound 152

Compound 153



Compound 154

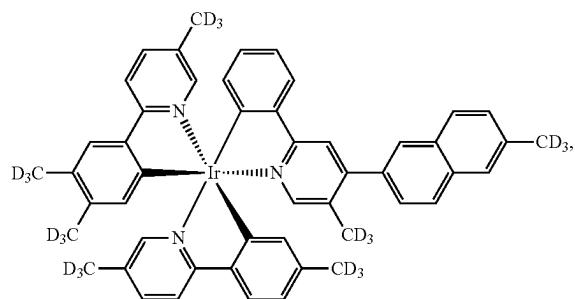
Compound 155



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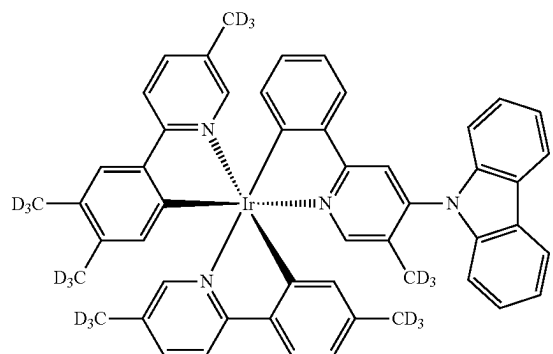
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Compound 156

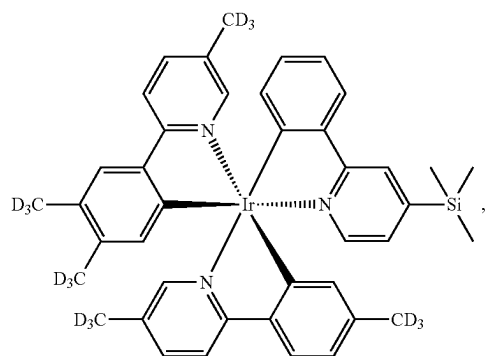


82

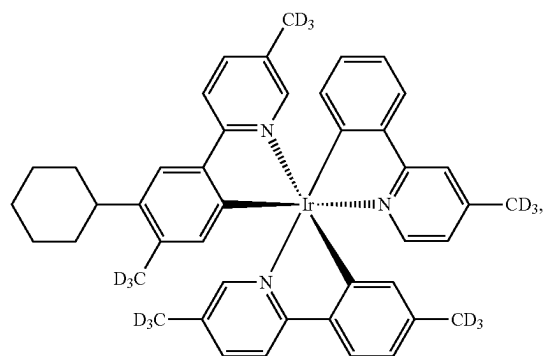
Compound 157



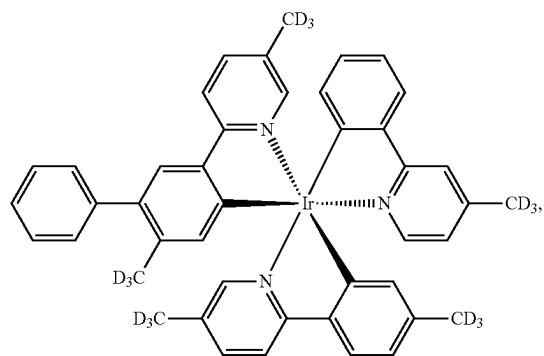
Compound 158



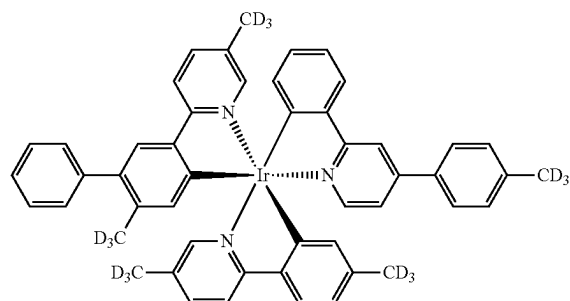
Compound 159



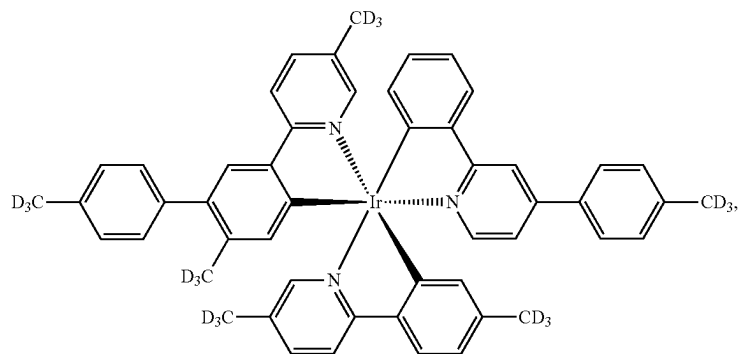
Compound 160



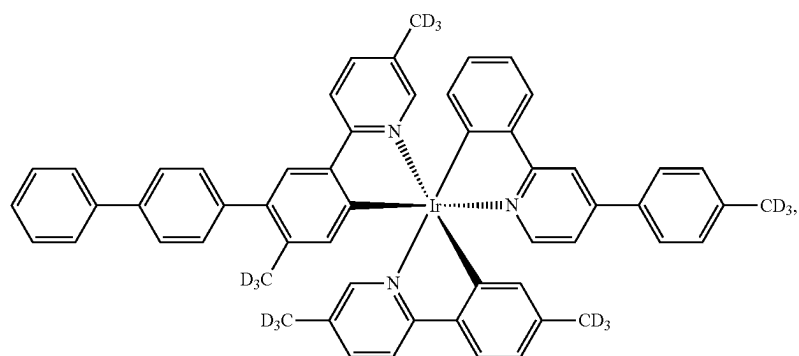
Compound 161



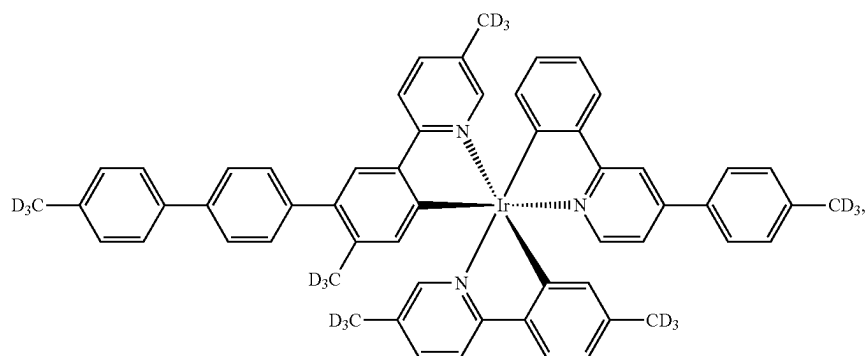
Compound 162



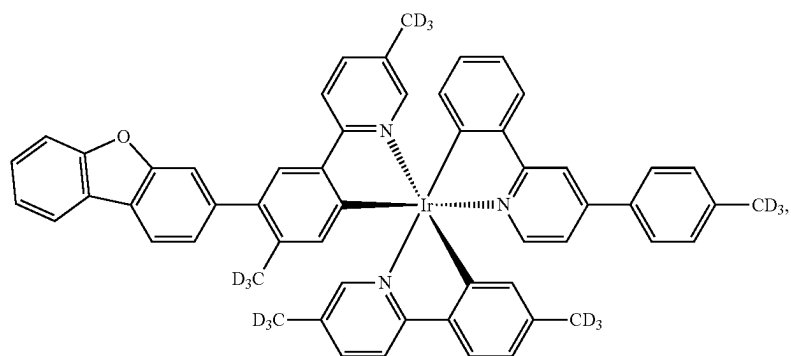
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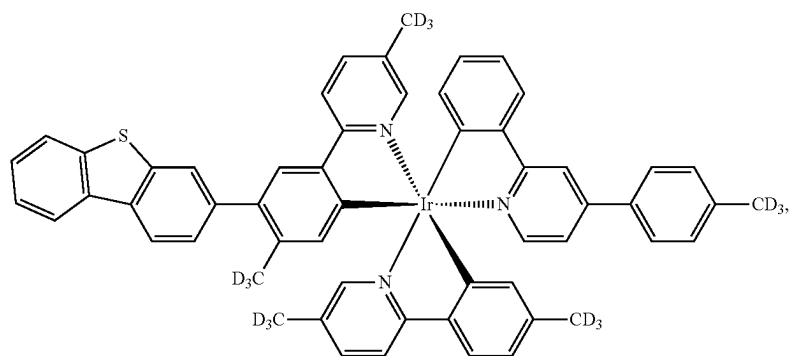
Compound 163



Compound 164



Compound 165



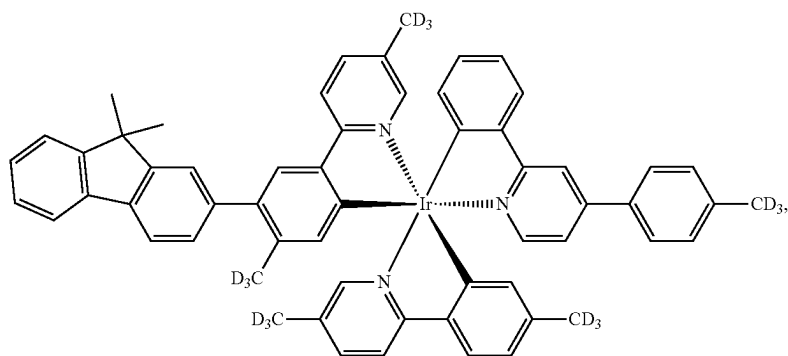
Compound 166

85

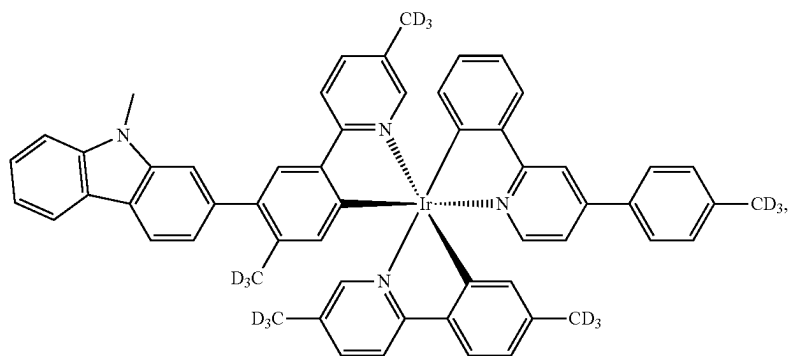
86

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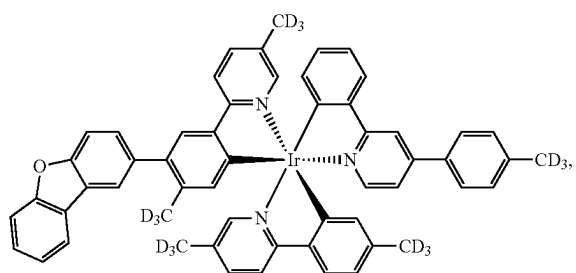
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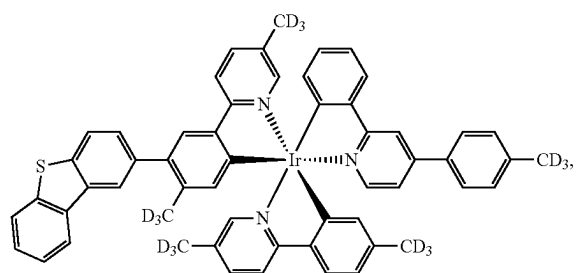
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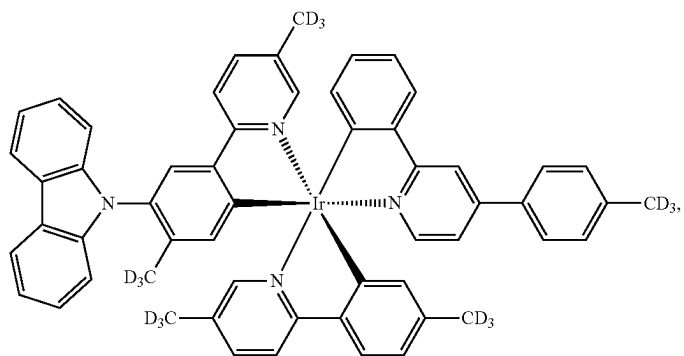
Compound 169



Compound 170



Compound 171

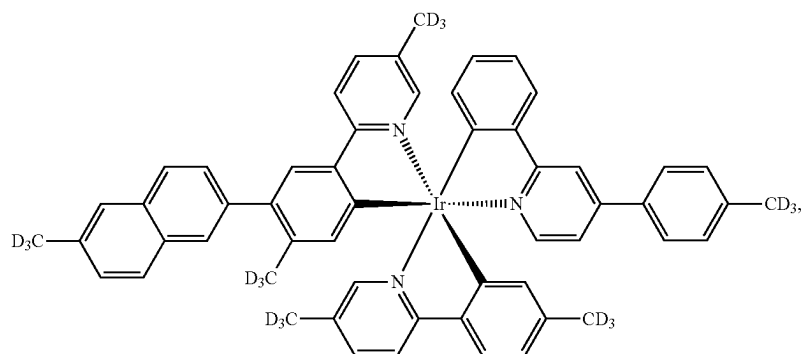


87

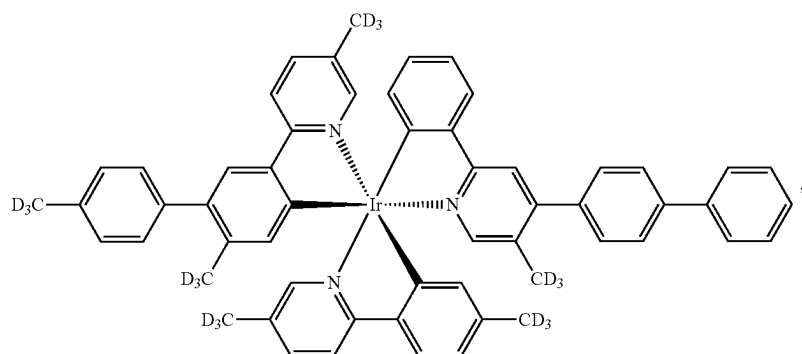
88

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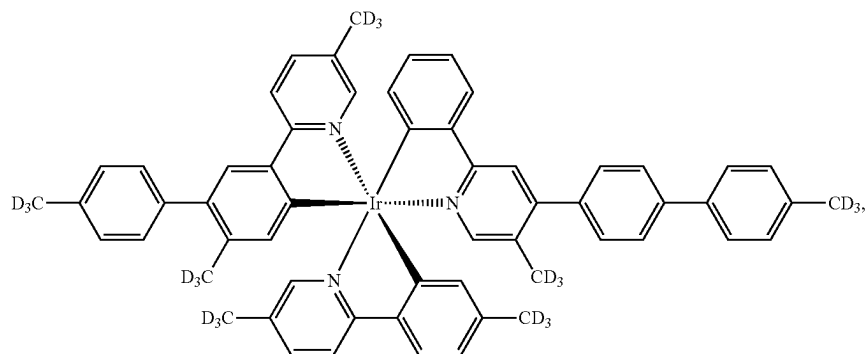
Compound 172



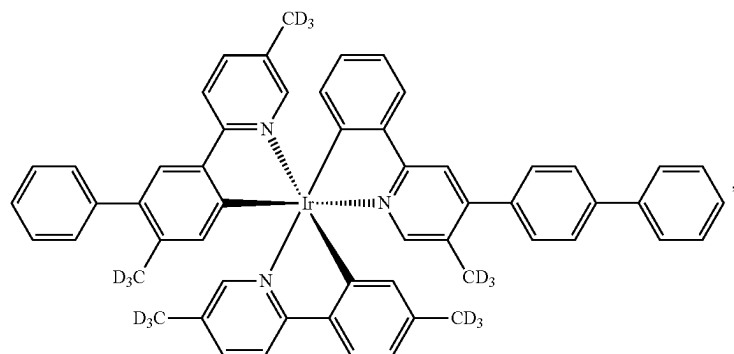
Compound 173



Compound 174

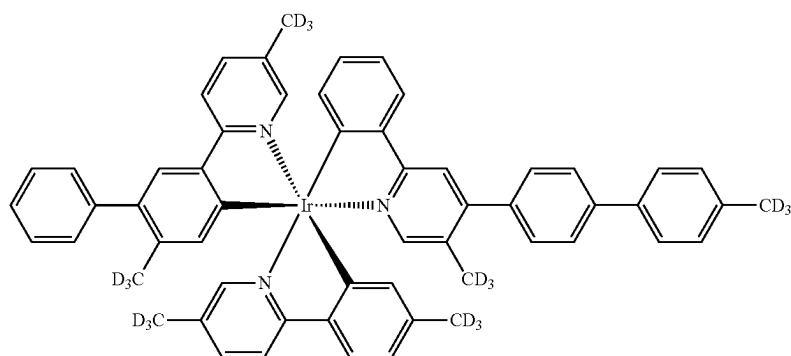


Compound 175

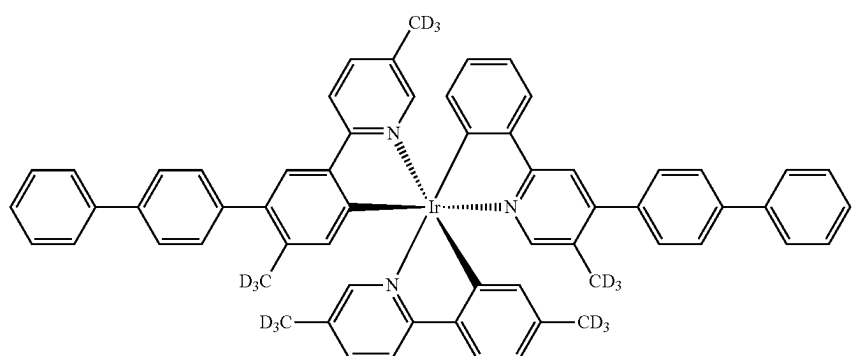


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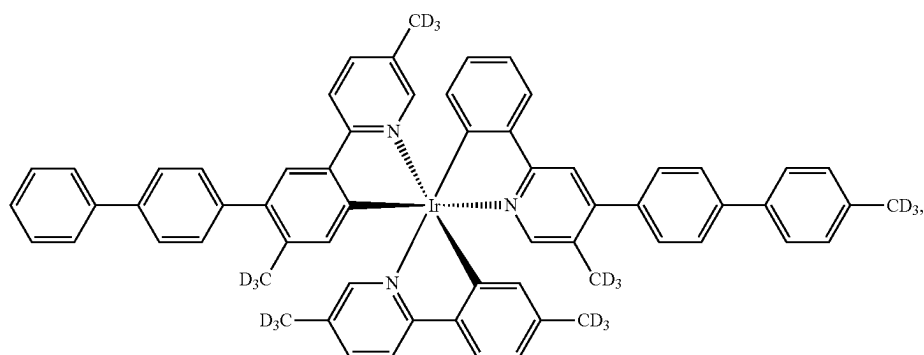
Compound 176



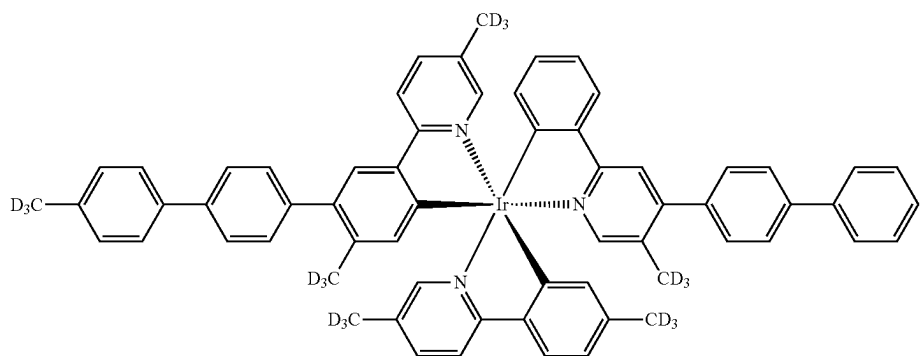
Compound 177



Compound 178



Compound 179

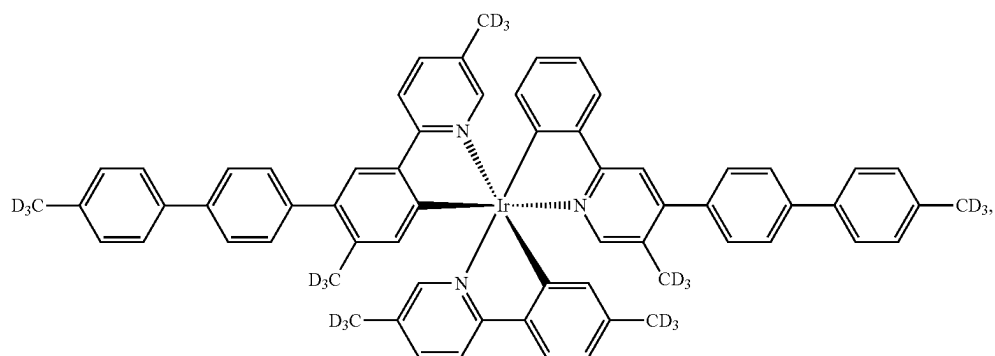


91

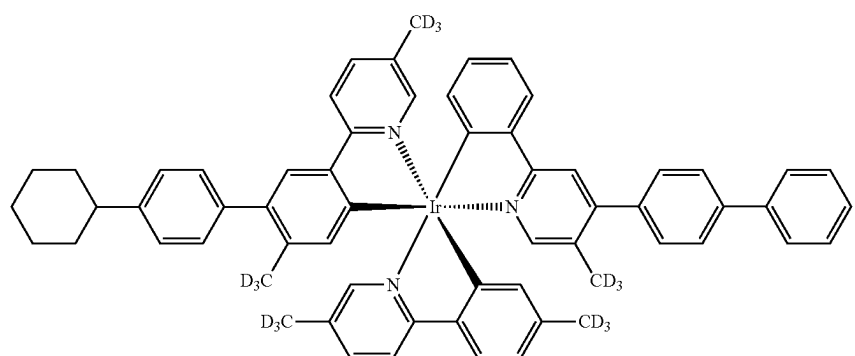
92

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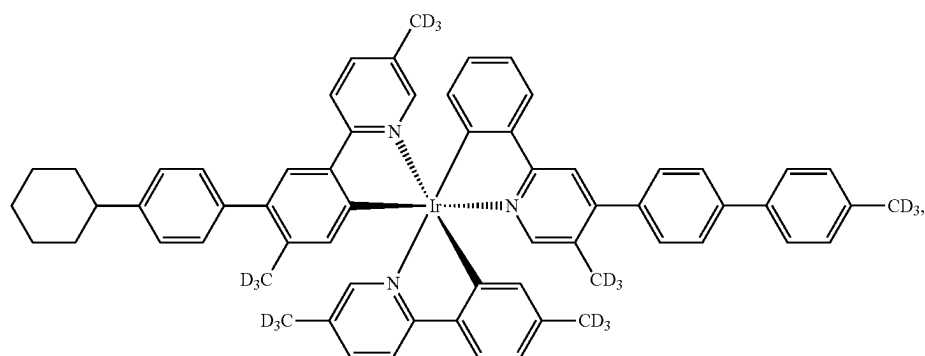
Compound 180



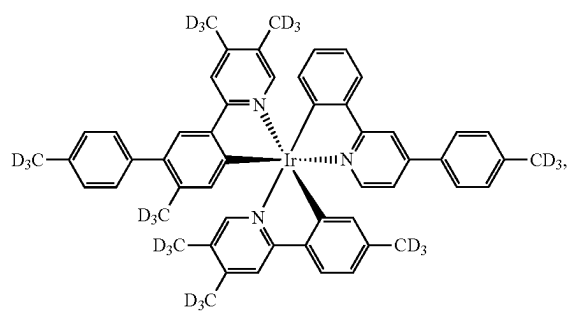
Compound 181



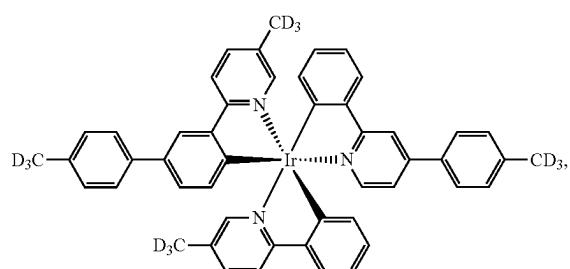
Compound 182



Compound 183



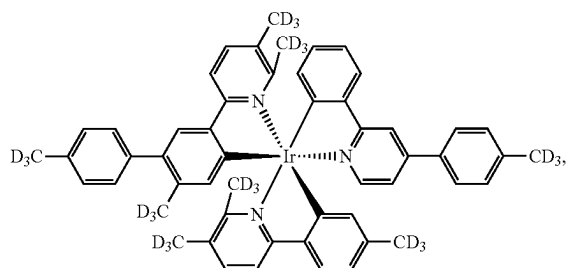
Compound 184



93

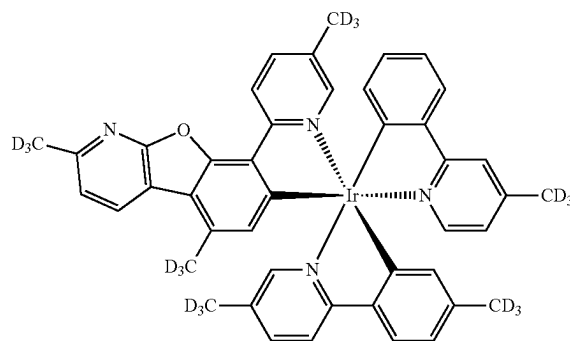
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Compound 185

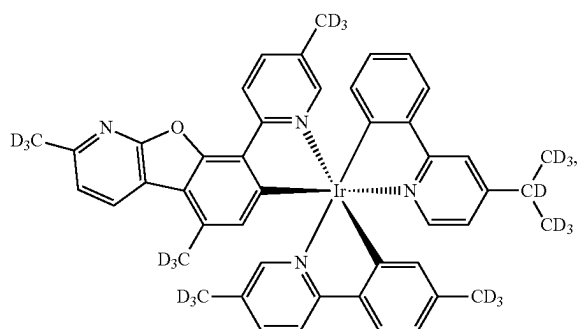


94

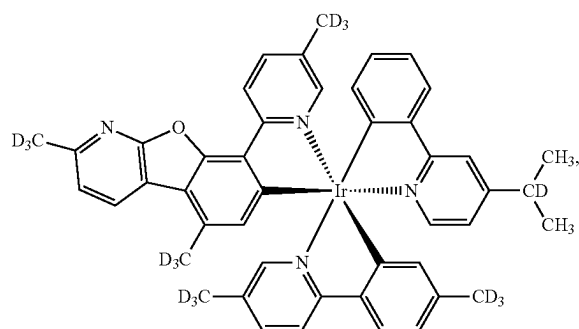
Compound 186



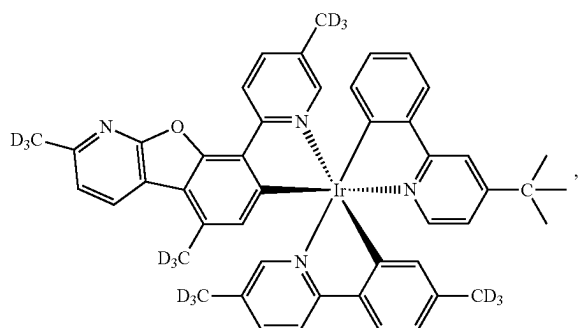
Compound 187



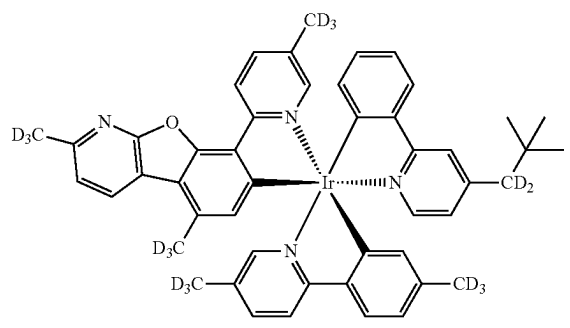
Compound 188



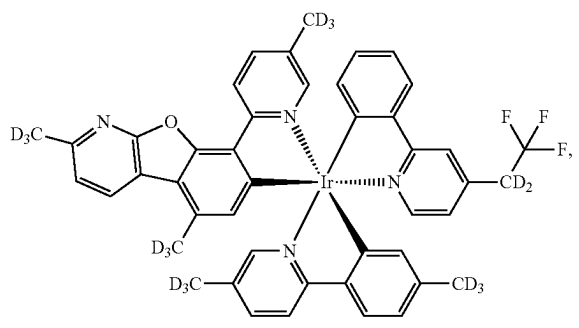
Compound 189



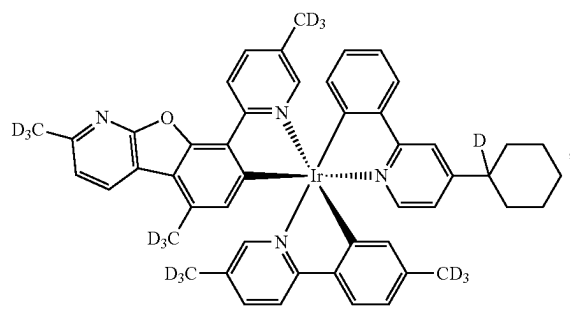
Compound 190



Compound 191

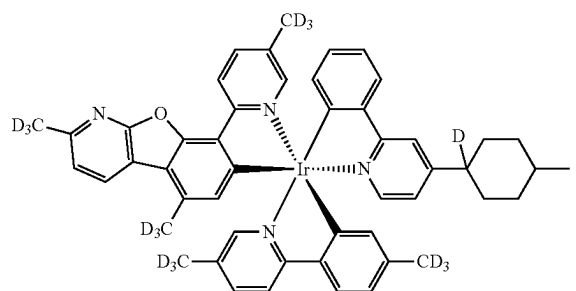


Compound 192



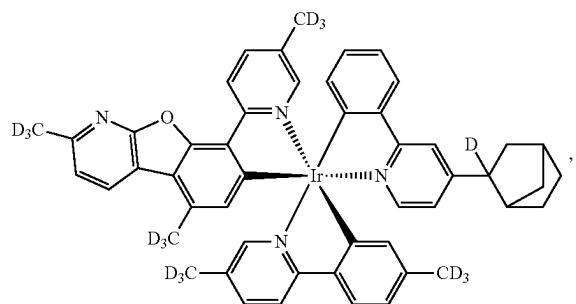
Compound 193

Compound 194



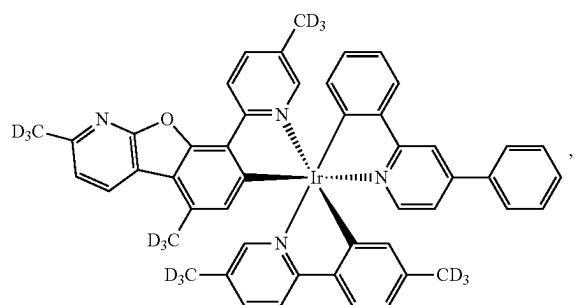
Compound 195

Compound 196

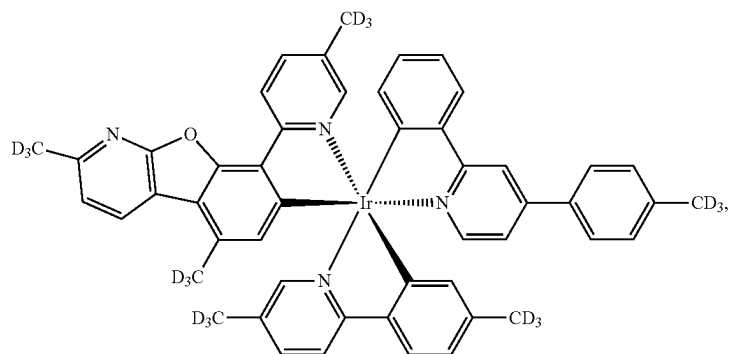


Compound 197

Compound 198

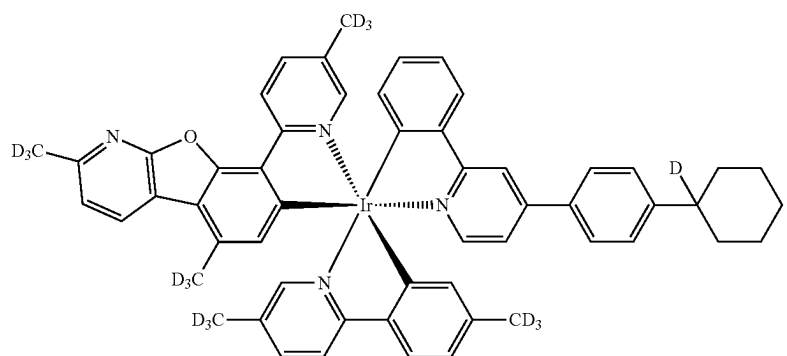


Compound 199

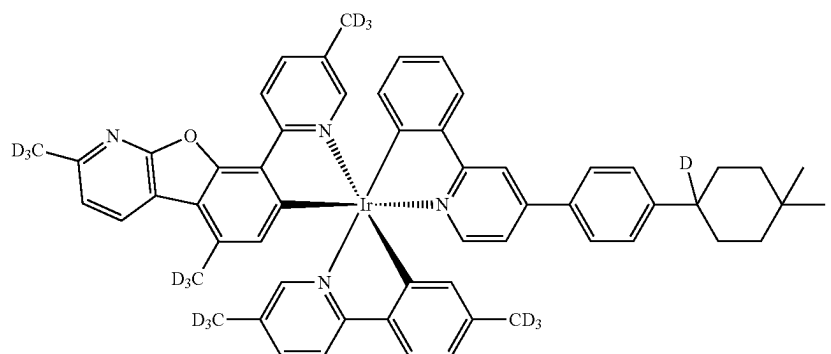


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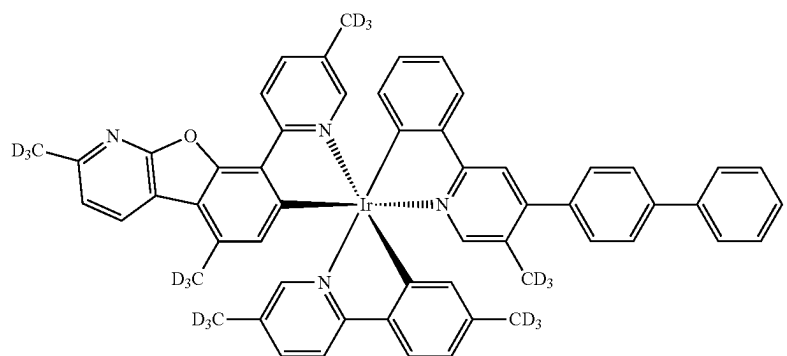
Compound 200



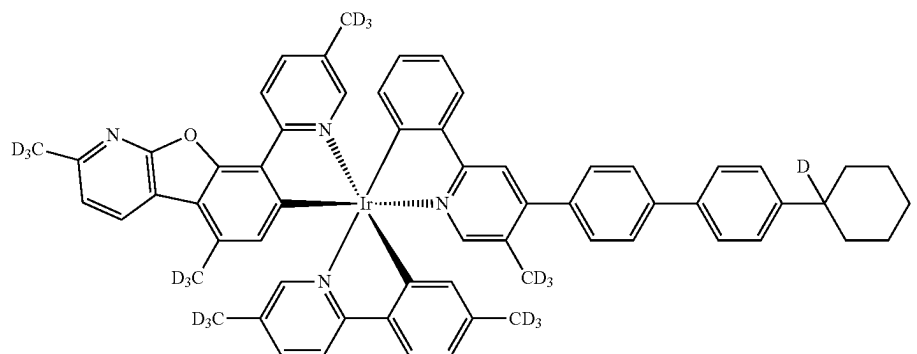
Compound 201



Compound 202

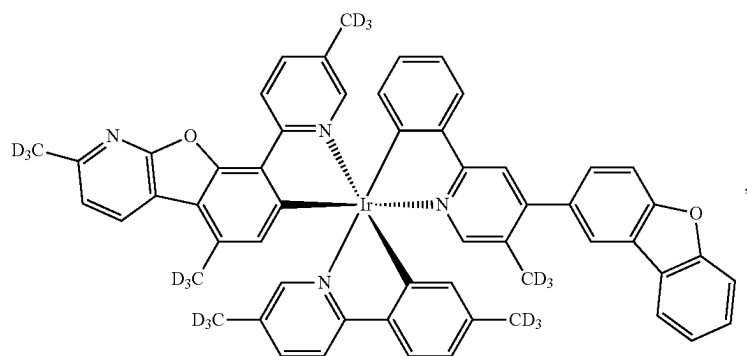


Compound 203

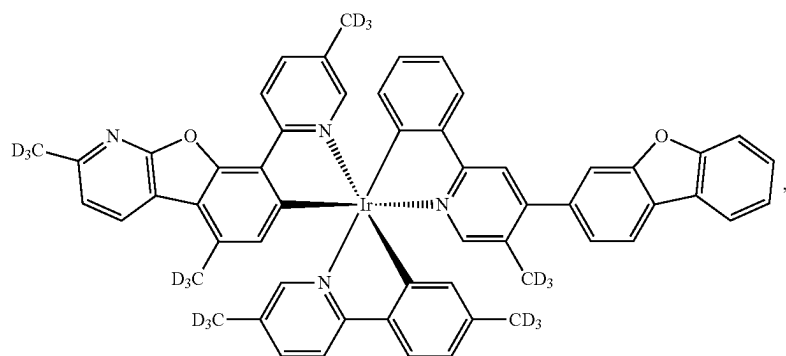


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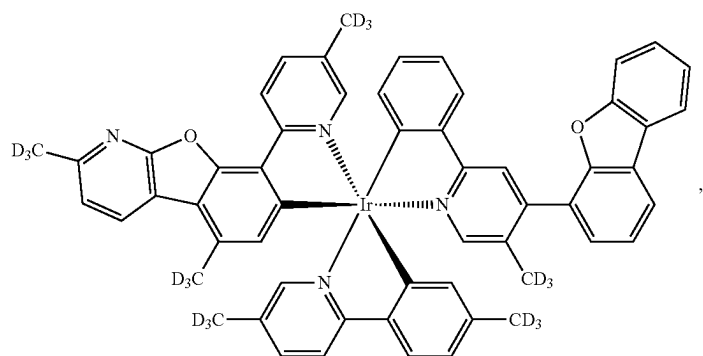
Compound 204



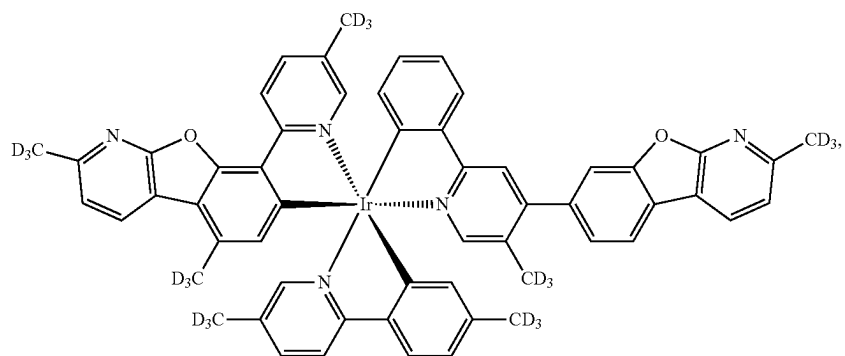
Compound 205



Compound 206

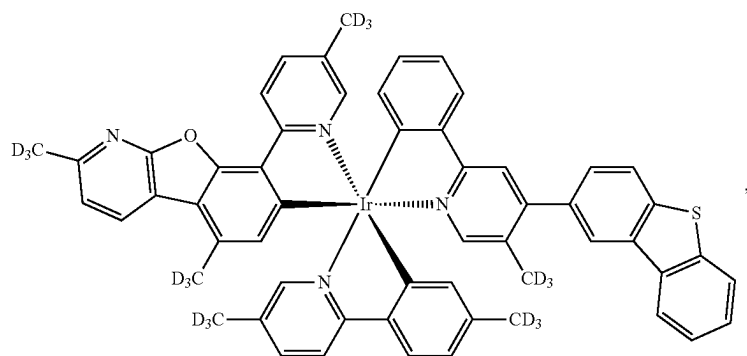


Compound 207

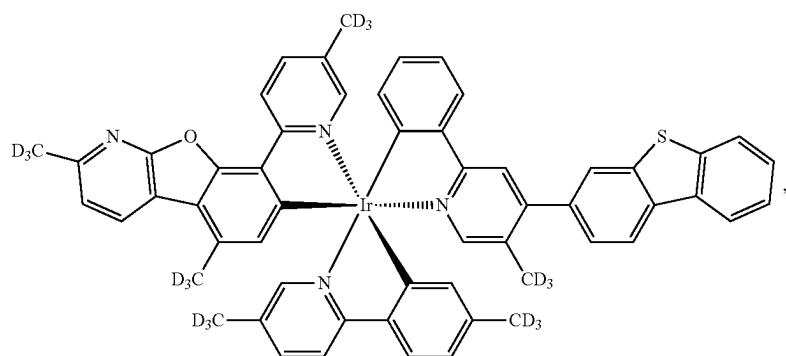


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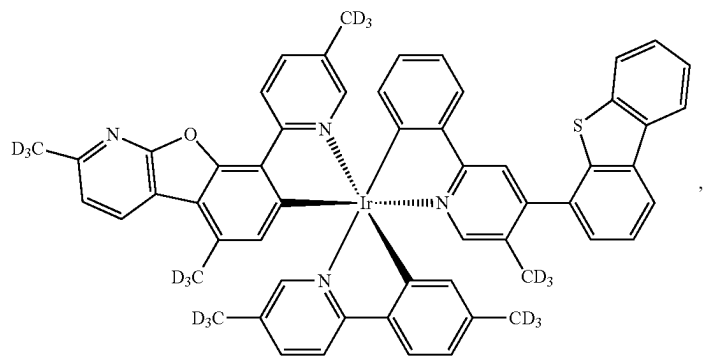
Compound 208



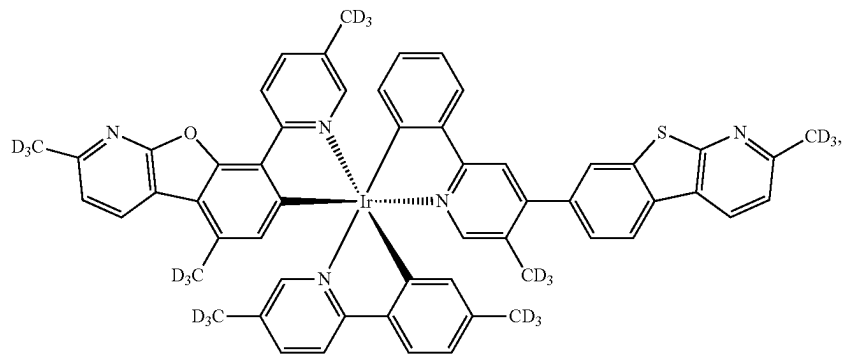
Compound 209



Compound 210



Compound 211

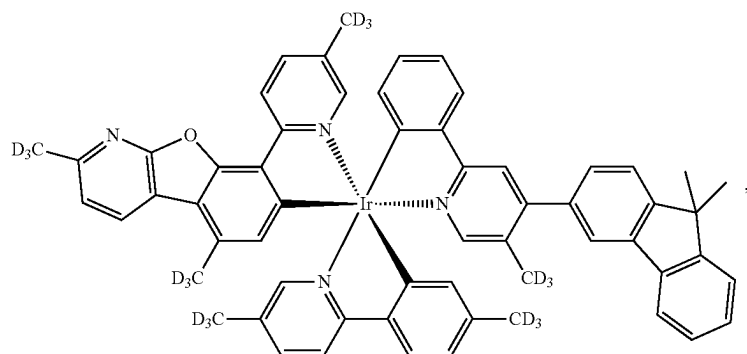


103

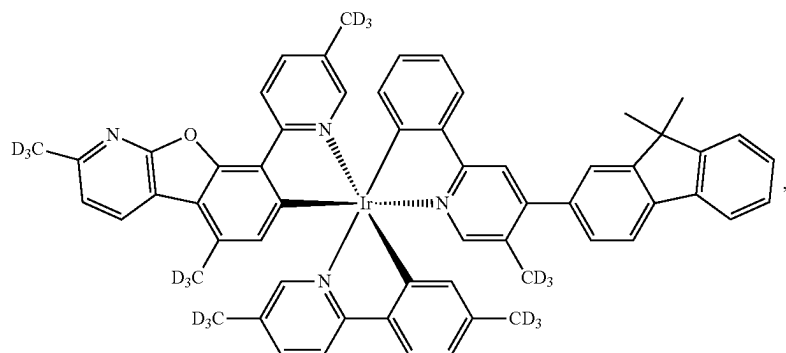
104

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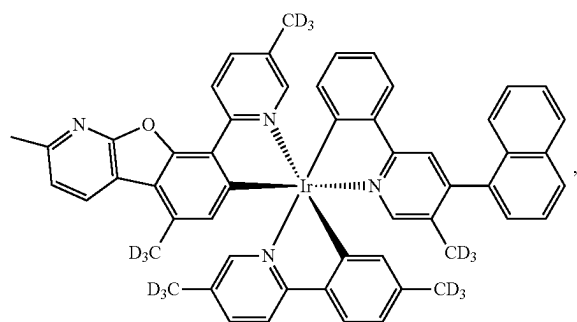
Compound 212



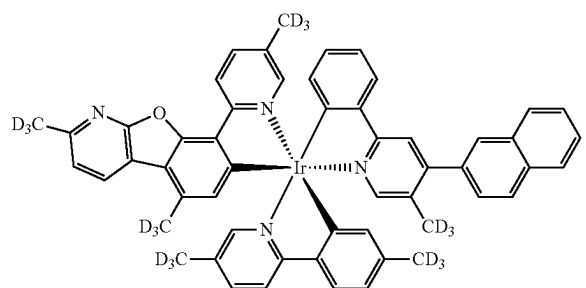
Compound 213



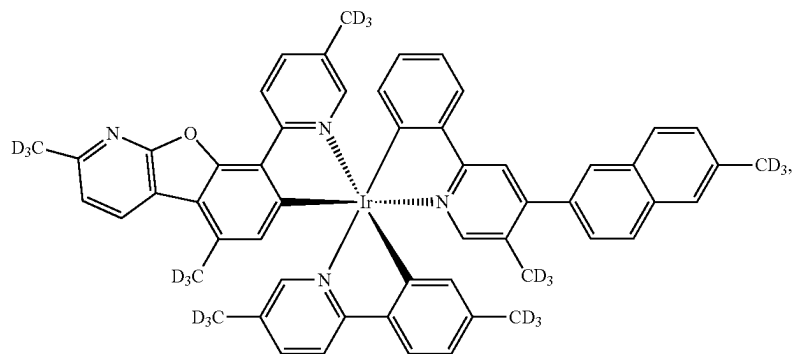
Compound 214



Compound 215



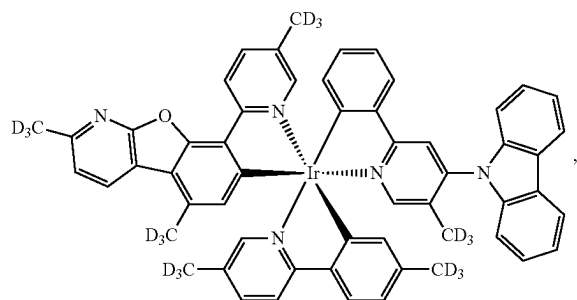
Compound 216



105

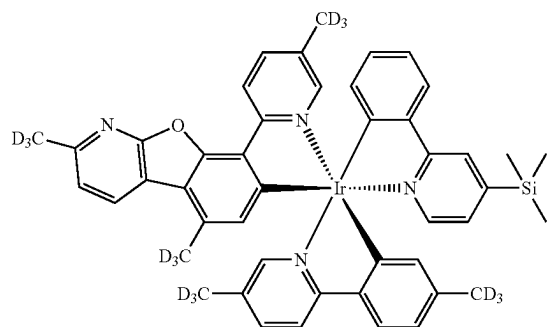
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Compound 217

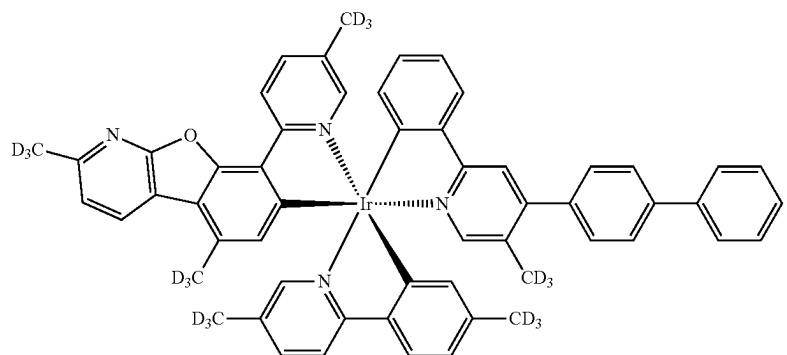


106

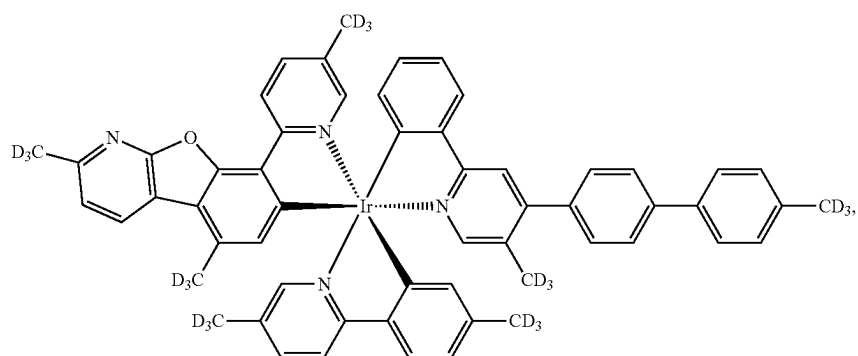
Compound 218



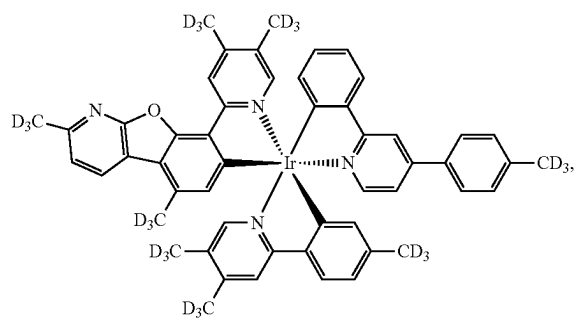
Compound 219



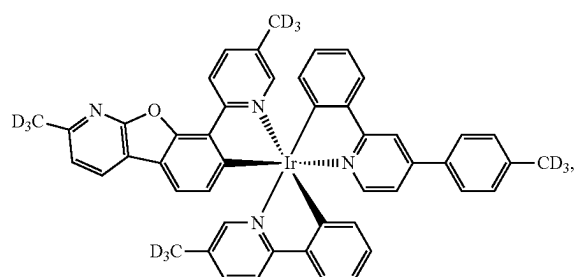
Compound 220



Compound 221



Compound 222



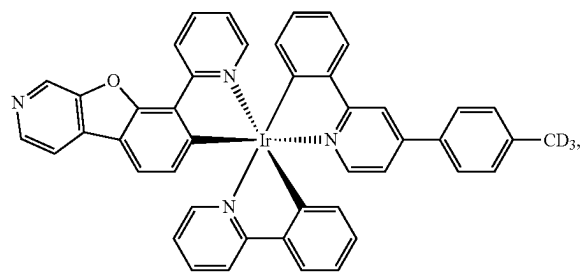
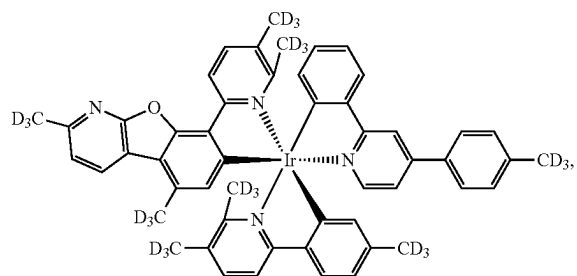
107

108

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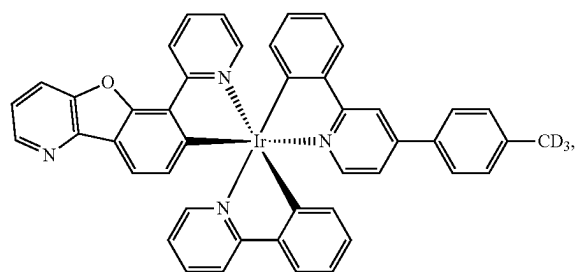
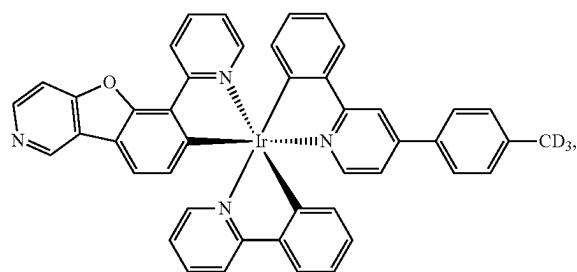
Compound 223

Compound 224



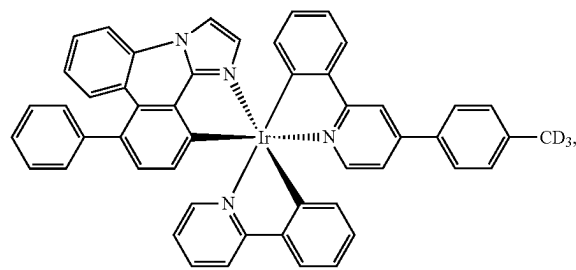
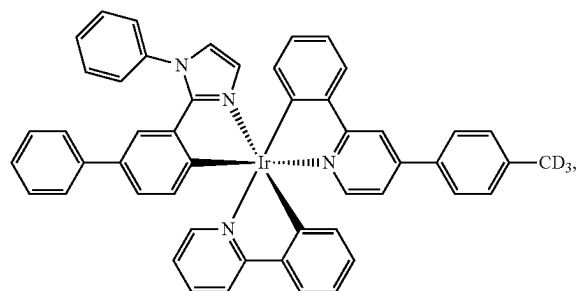
Compound 225

Compound 226



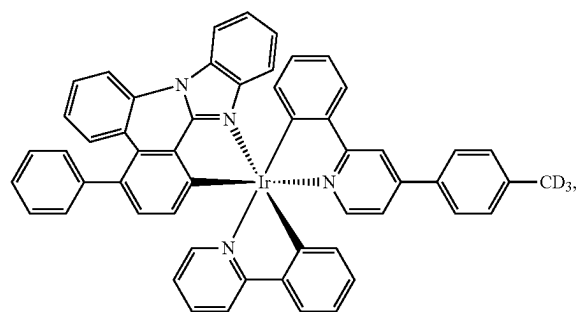
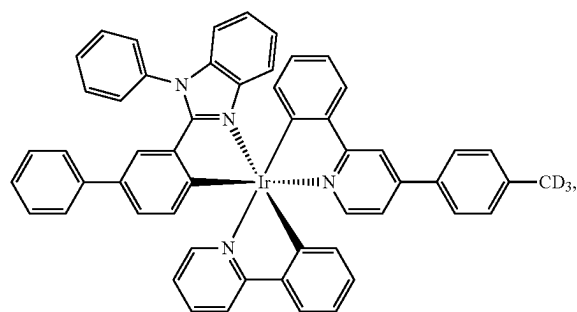
Compound 227

Compound 228



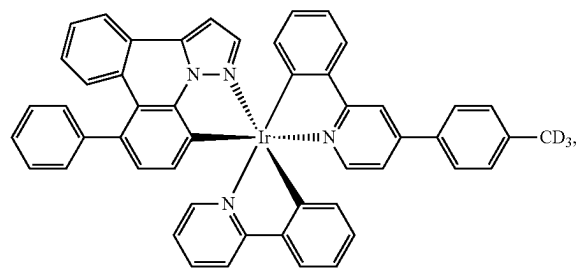
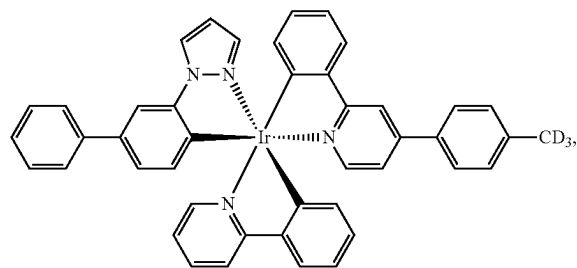
Compound 229

Compound 230



Compound 231

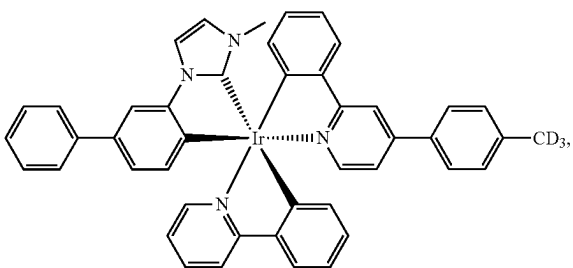
Compound 232



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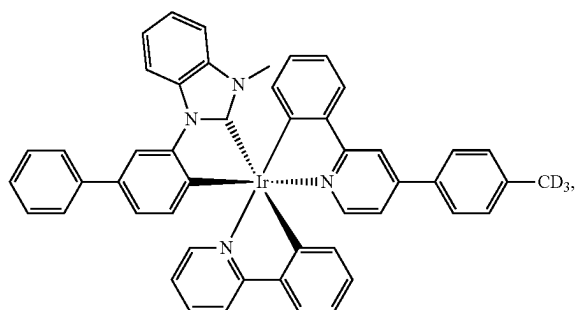
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Compound 233

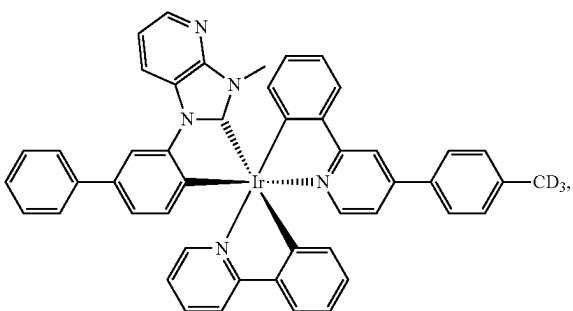


110

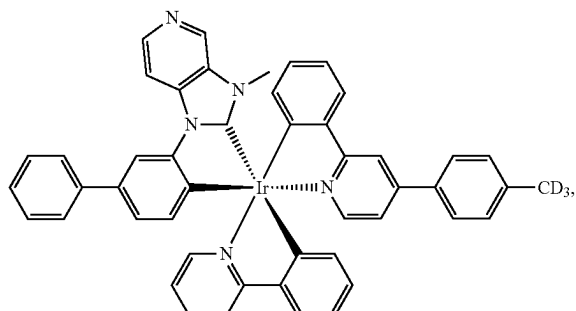
Compound 234



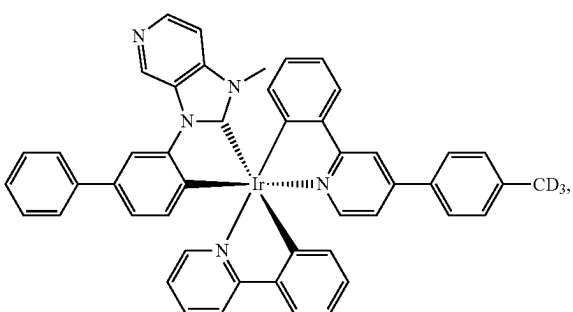
Compound 235



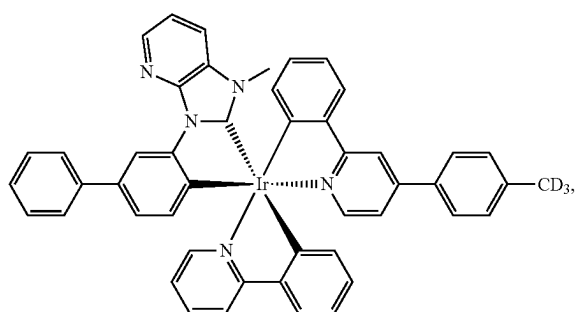
Compound 236



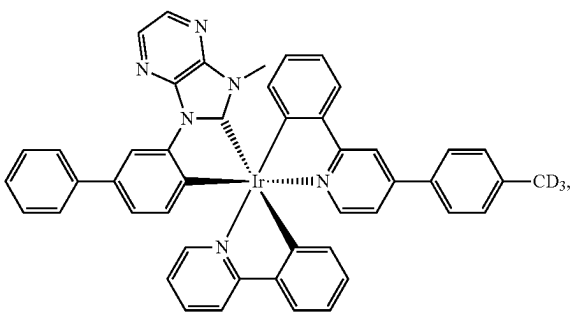
Compound 237



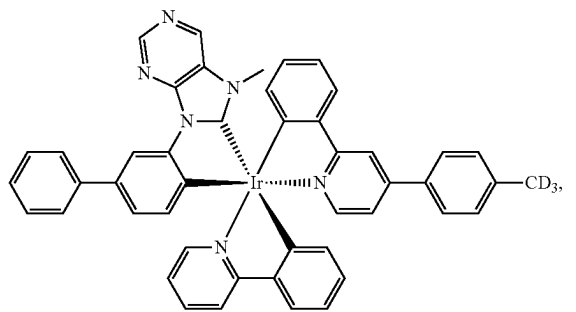
Compound 238



Compound 239



Compound 240



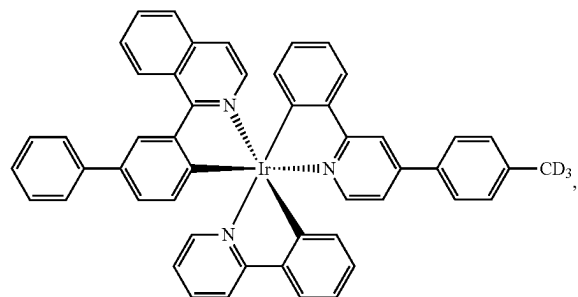
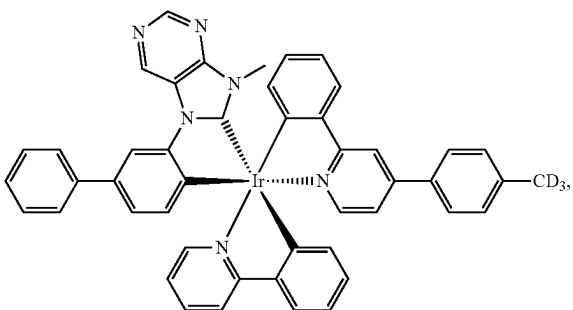
111

112

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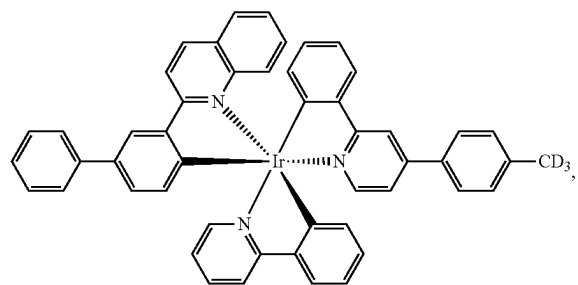
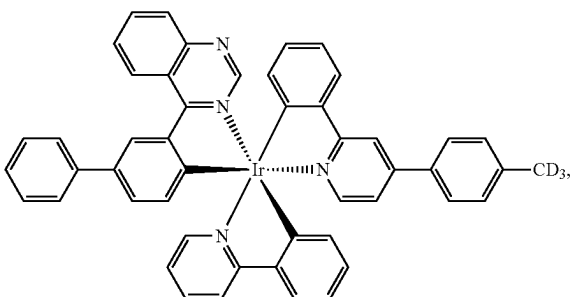
Compound 241

Compound 242

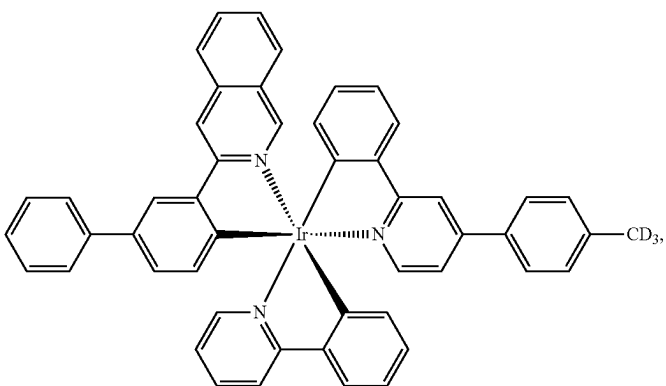


Compound 243

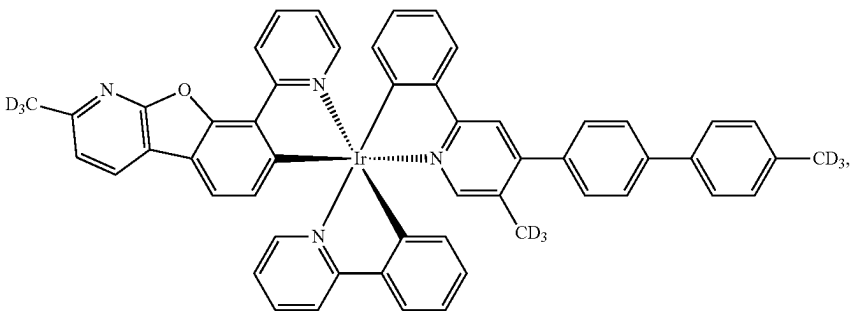
Compound 244



Compound 245

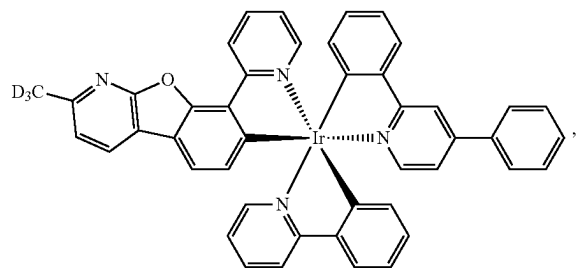
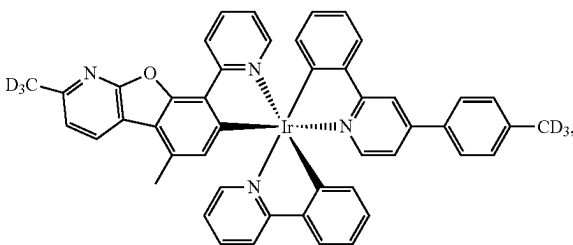


Compound 246



Compound 247

Compound 248

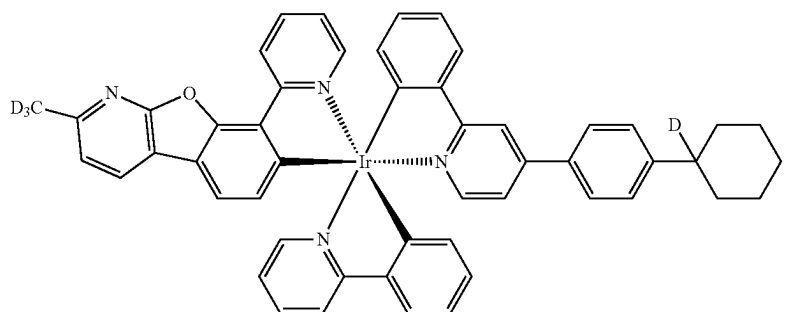


113

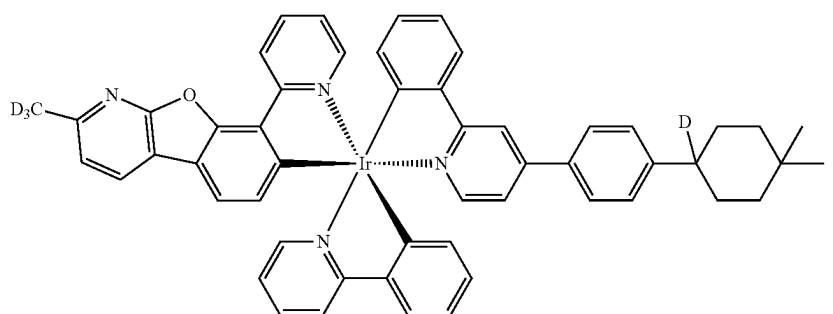
114

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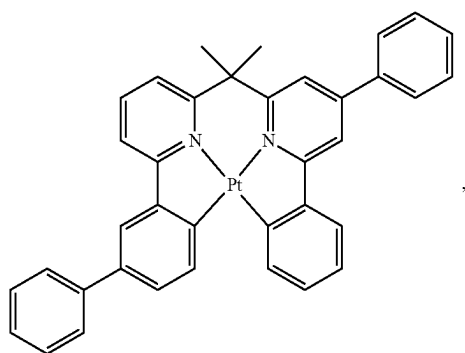
Compound 249



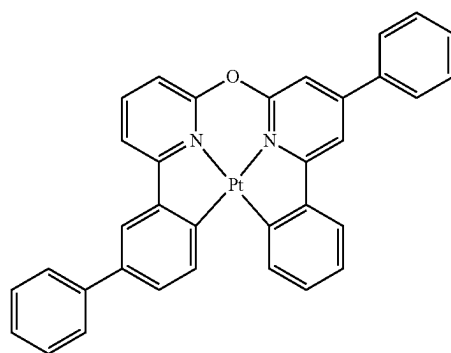
Compound 250



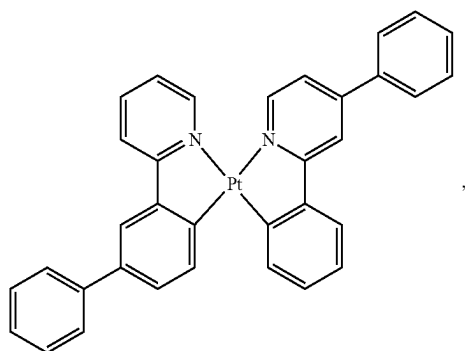
Compound 251



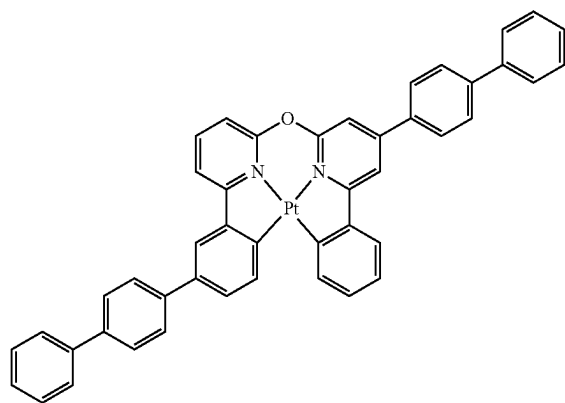
Compound 252



Compound 253



Compound 254

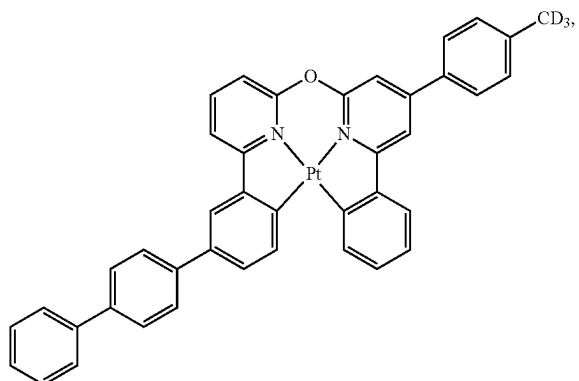


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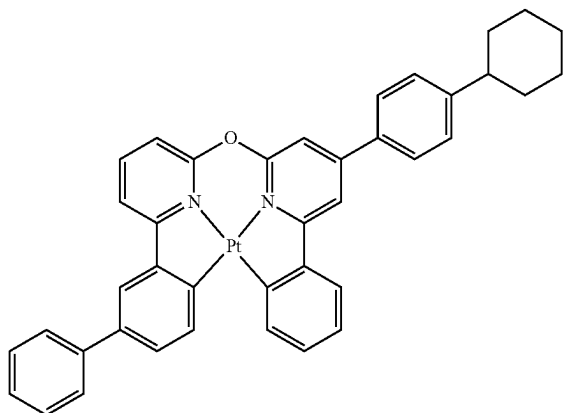
116

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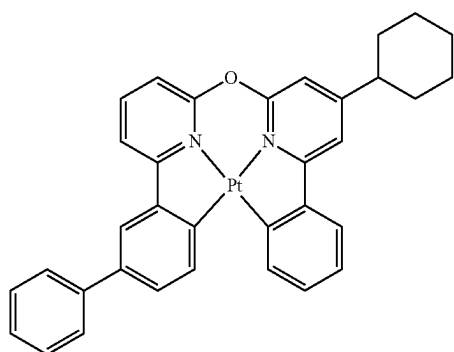
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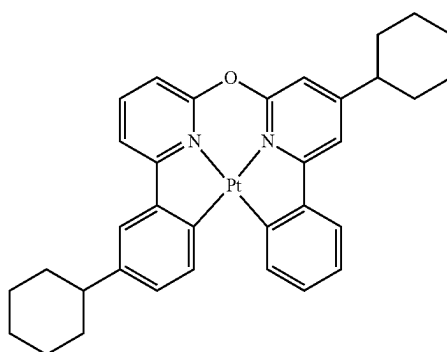
Compound 256



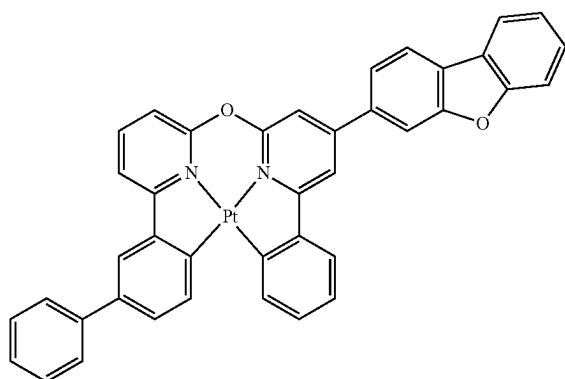
Compound 257



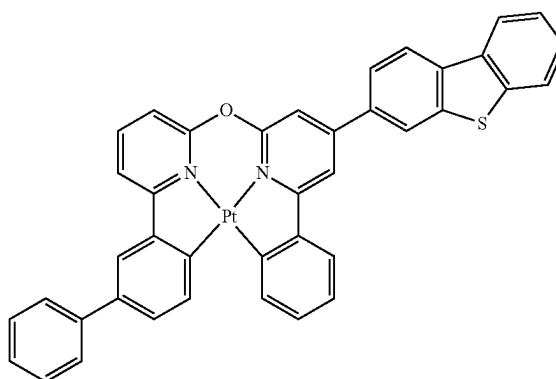
Compound 258



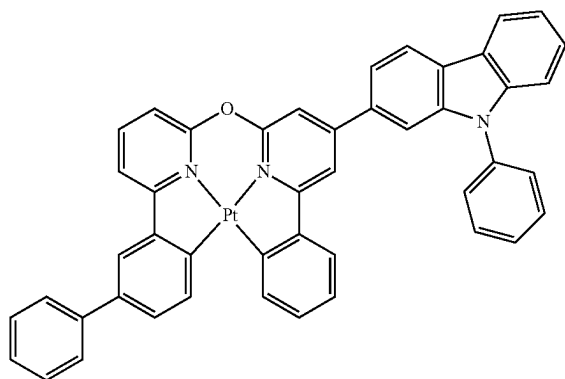
Compound 259



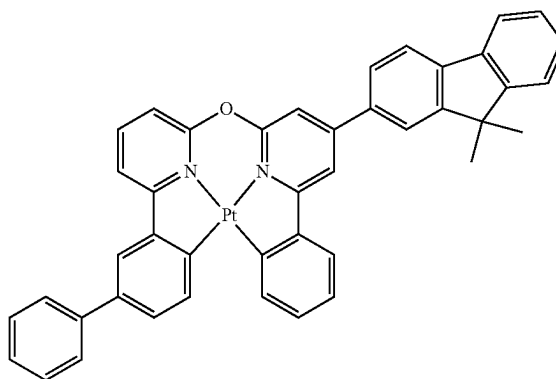
Compound 260



Compound 261



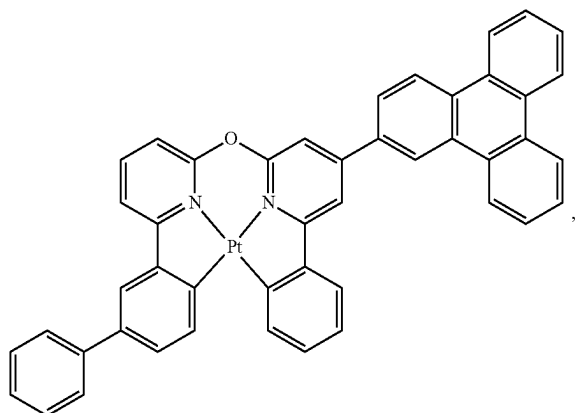
Compound 262



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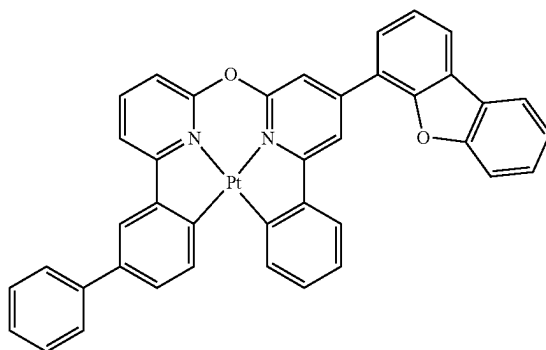
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Compound 263

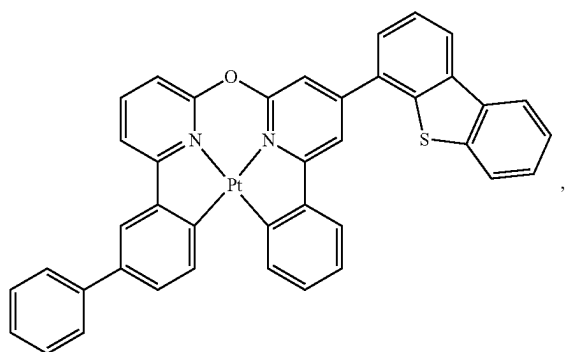


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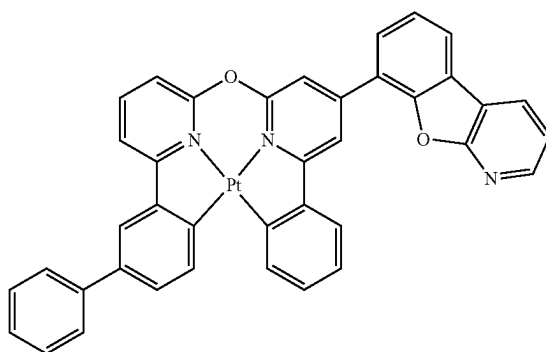
Compound 264



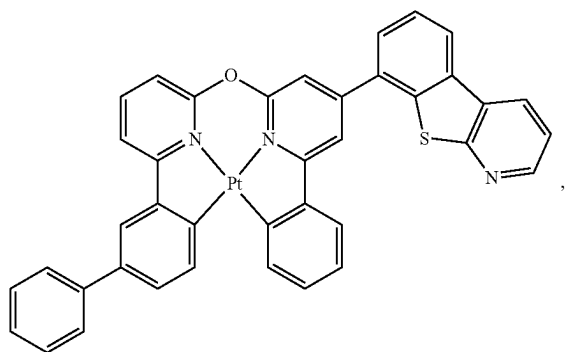
Compound 265



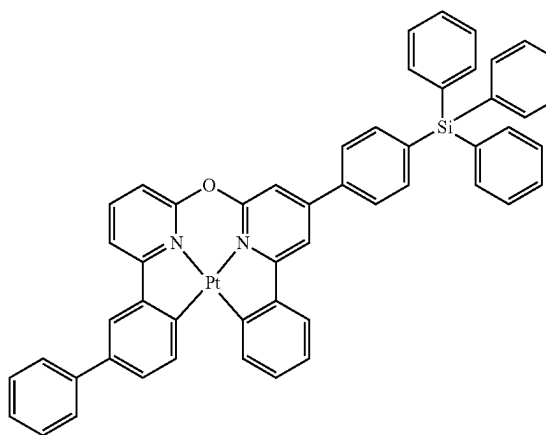
Compound 266



Compound 267



Compound 268



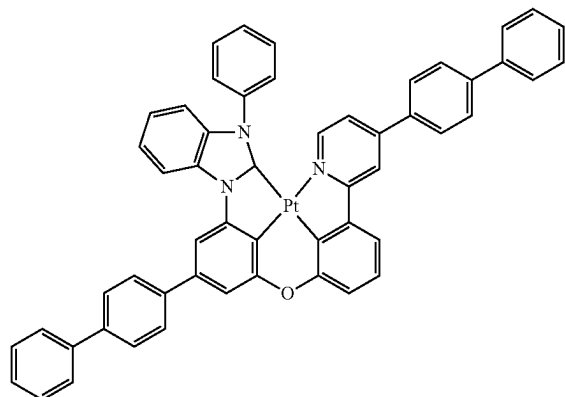
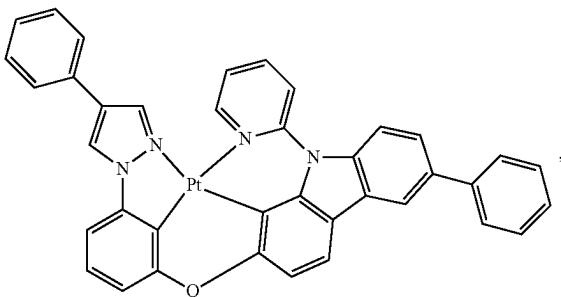
119

120

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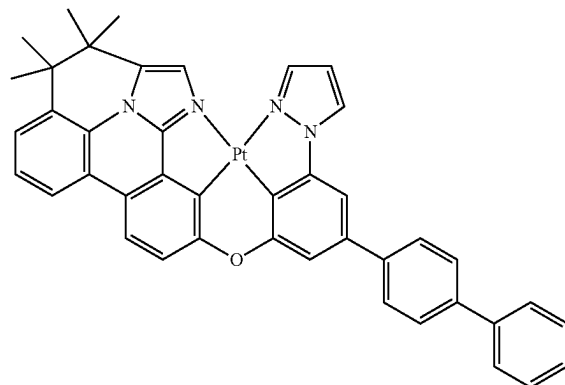
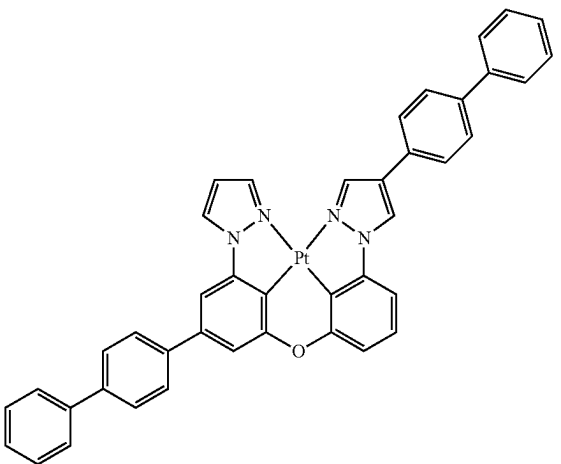
Compound 269

Compound 270



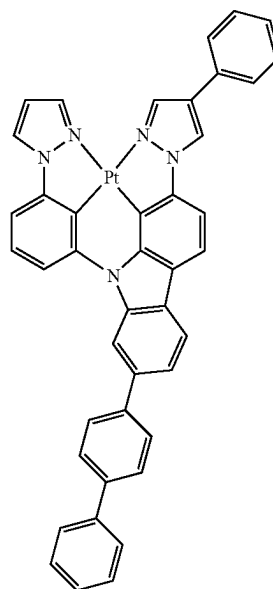
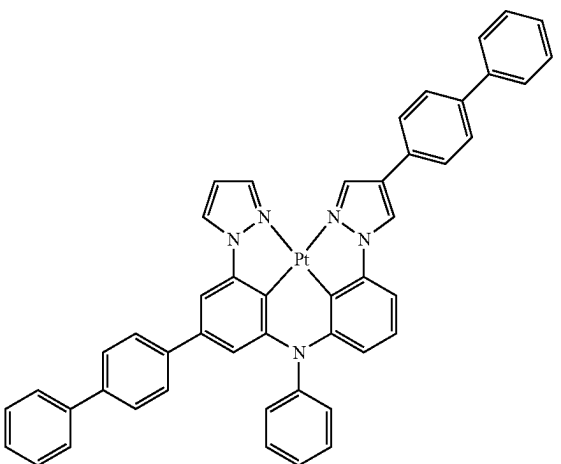
Compound 271

Compound 272

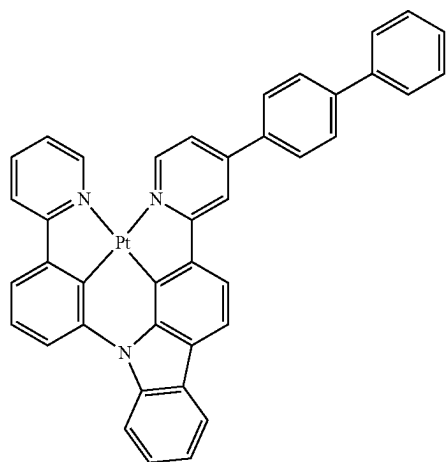


Compound 273

Compound 274

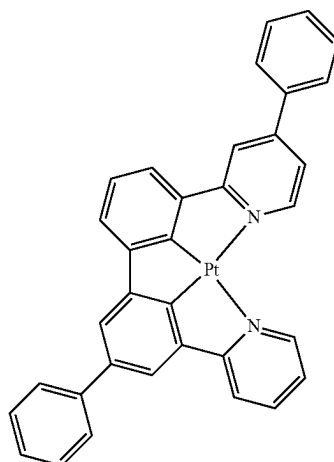


121



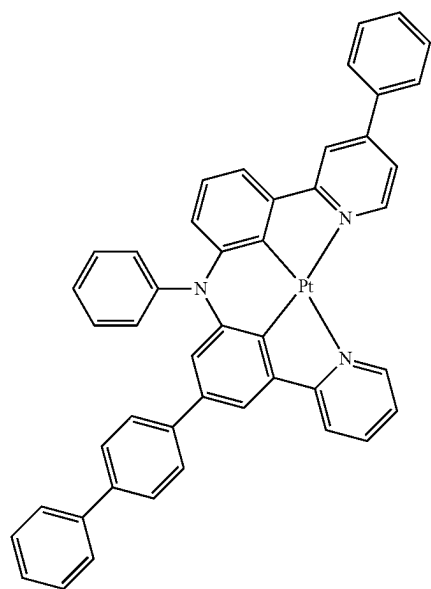
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Compound 275

122

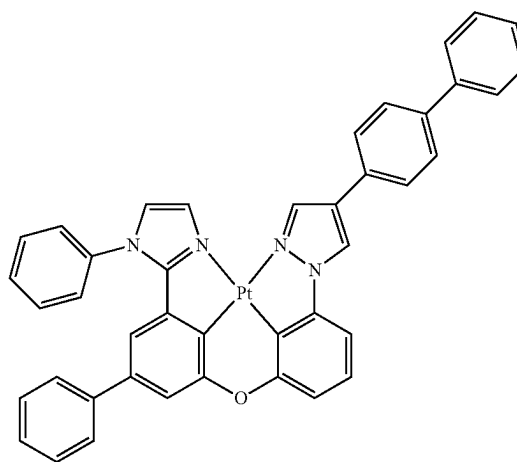


Compound 276

Compound 277



Compound 278



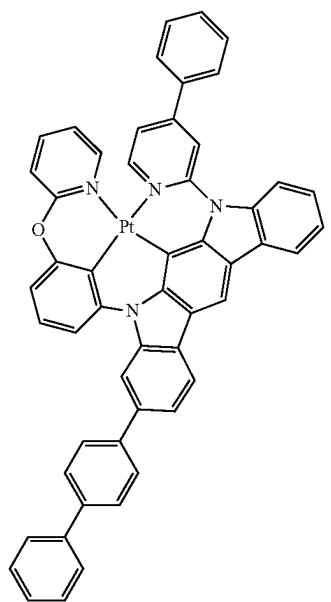
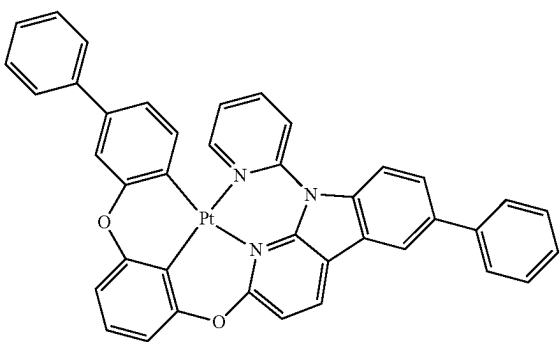
123

124

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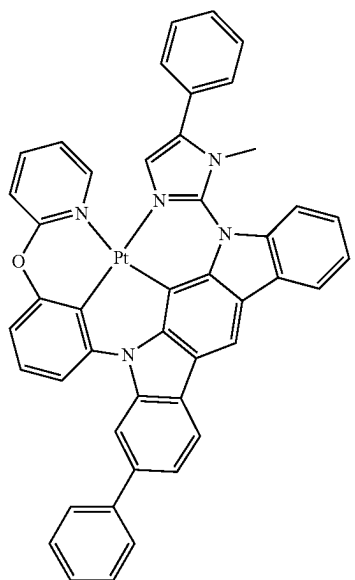
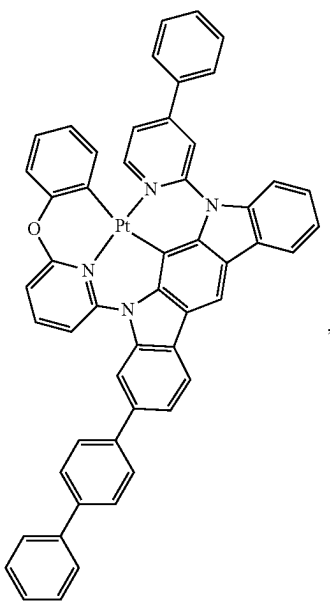
Compound 279

Compound 280



Compound 281

Compound 282



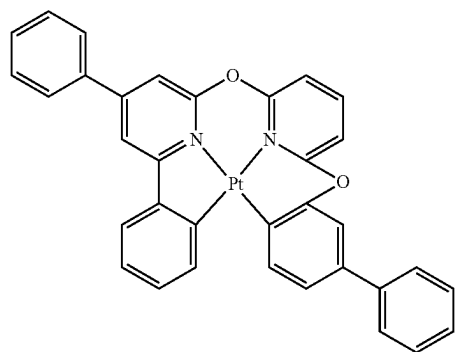
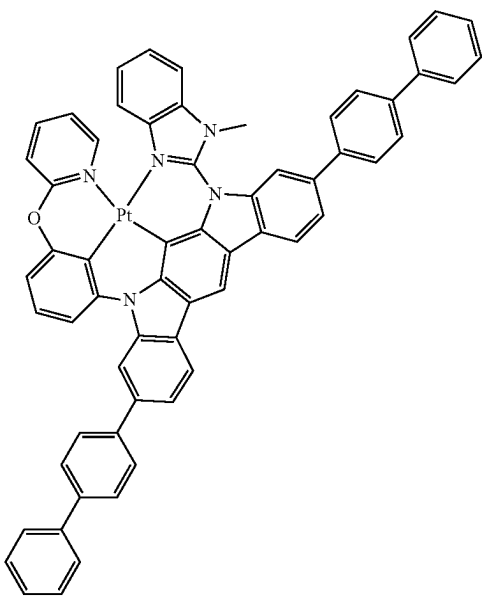
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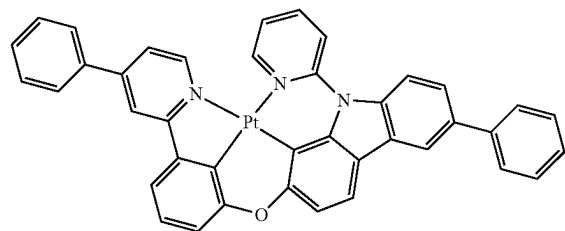
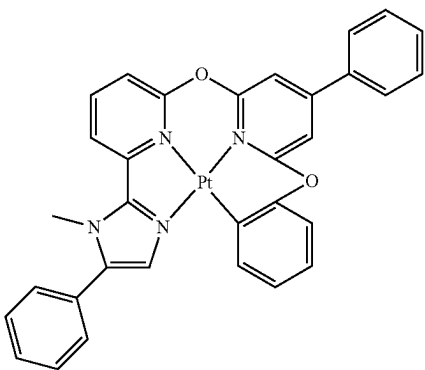
Compound 283

Compound 284



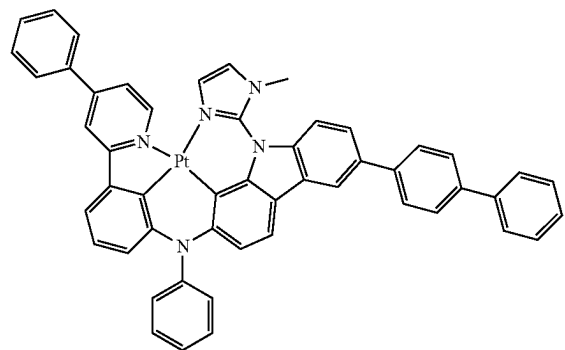
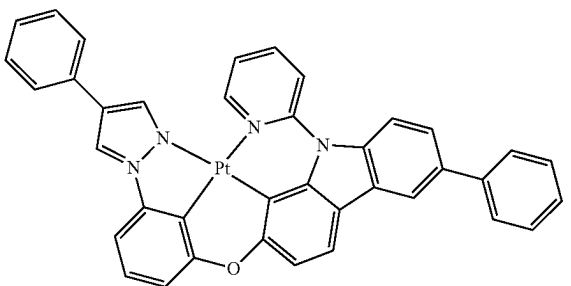
Compound 285

Compound 286



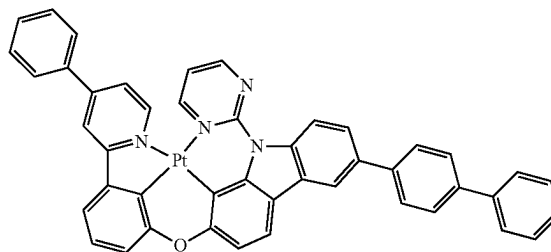
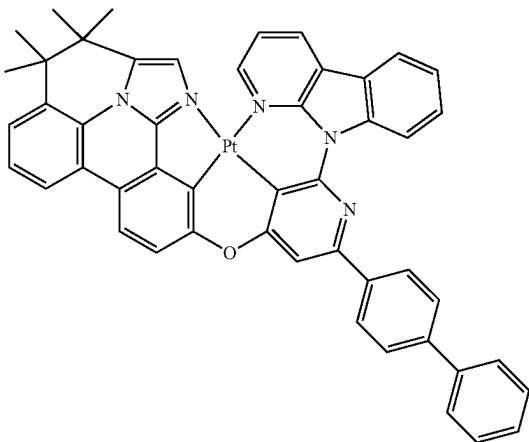
Compound 287

Compound 288



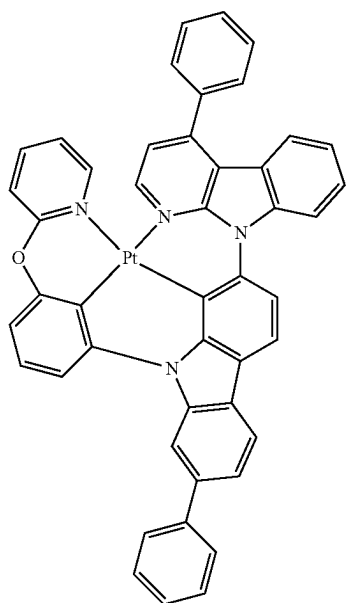
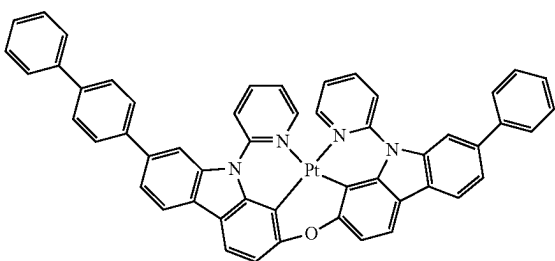
Compound 289

Compound 290

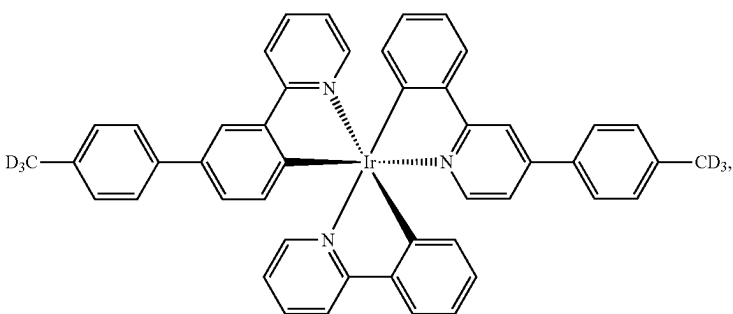


Compound 291

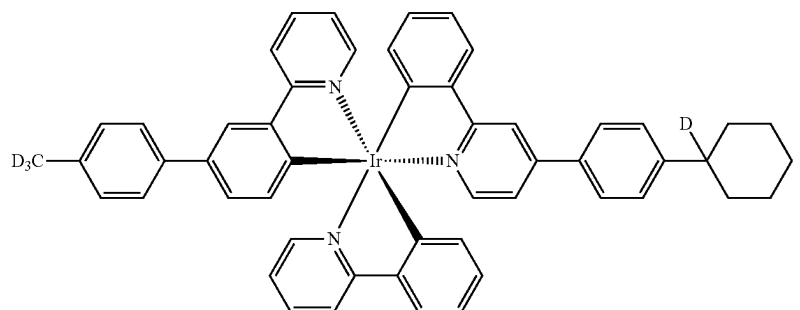
Compound 292



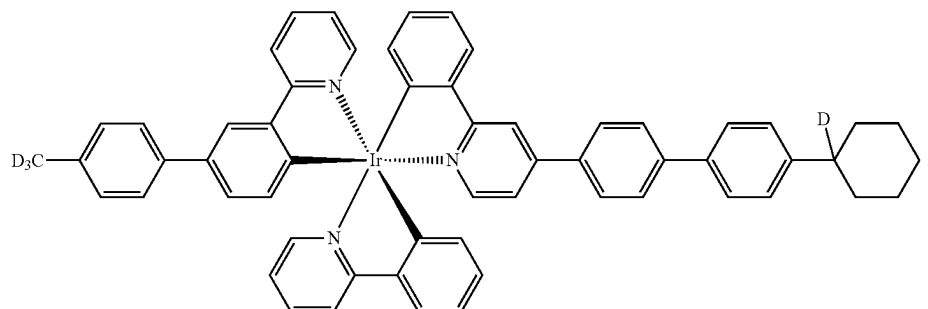
Compound 106



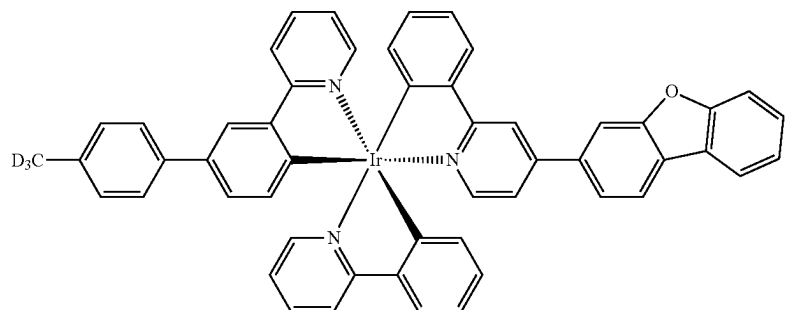
Compound 107



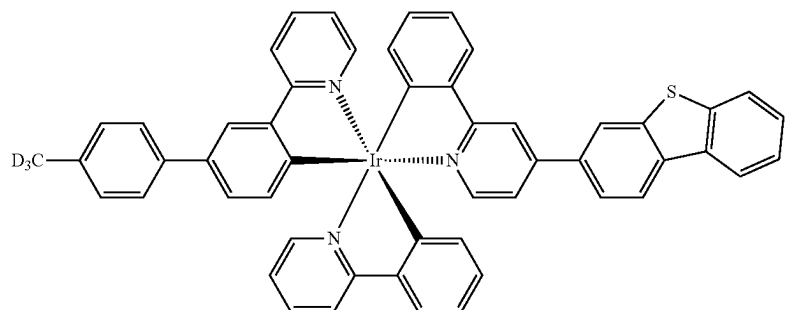
Compound 108



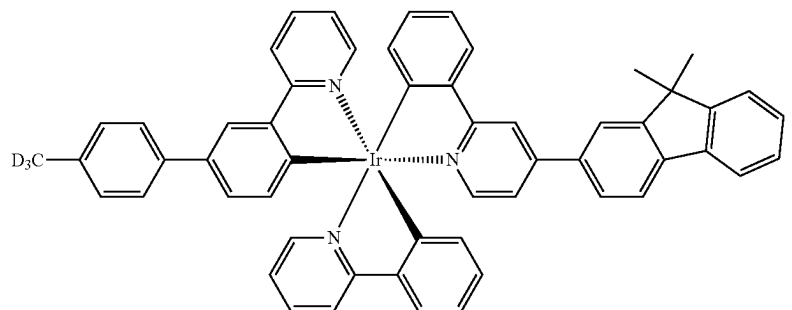
Compound 109



Compound 110



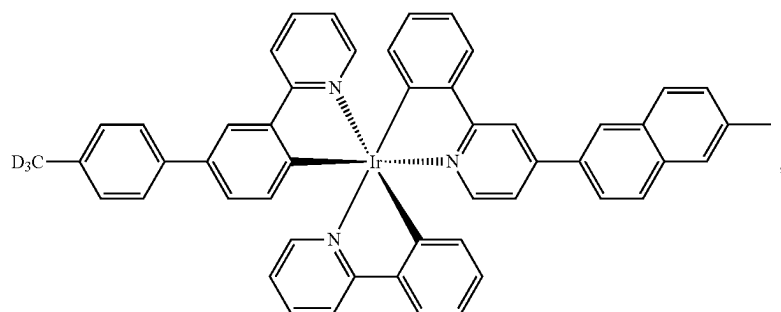
Compound 111



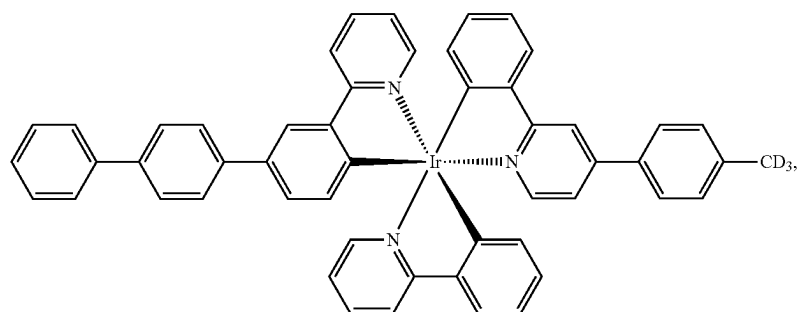
131

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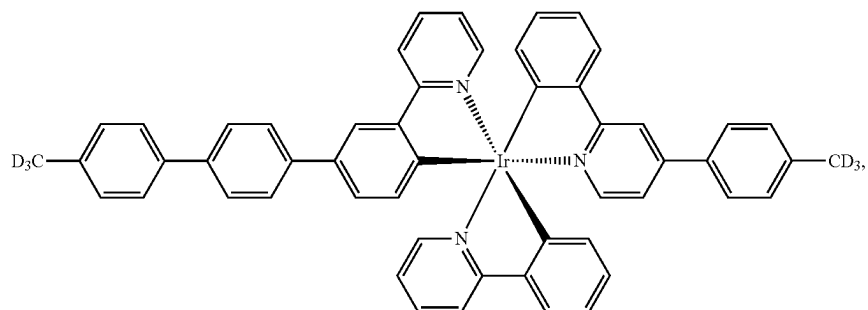
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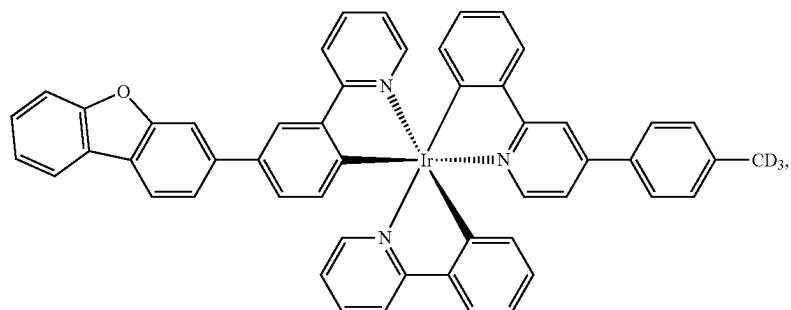
Compound 112



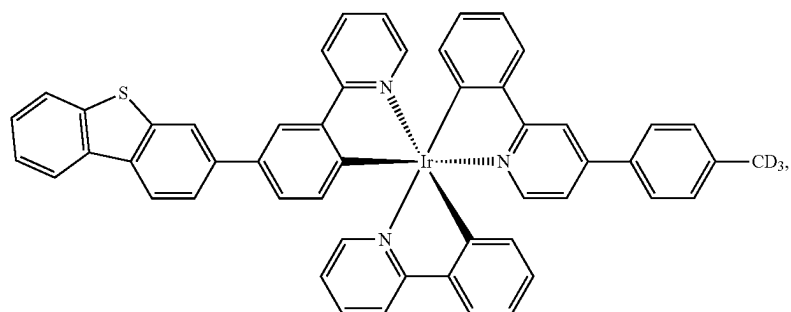
Compound 113



Compound 114



Compound 115



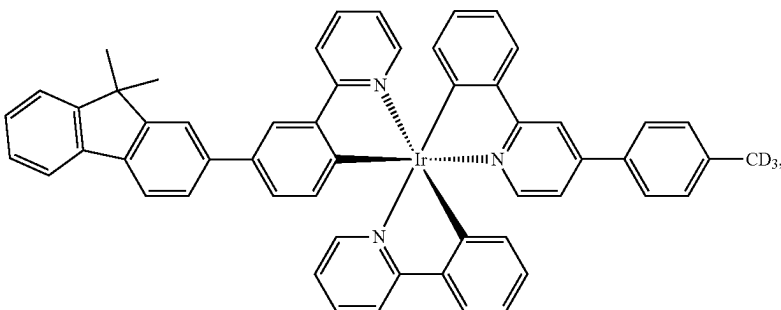
Compound 116

133

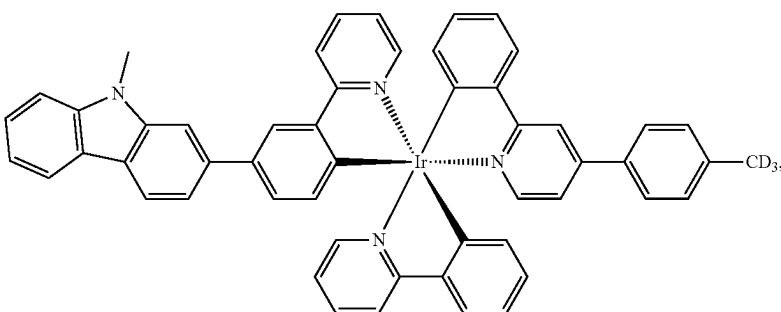
134

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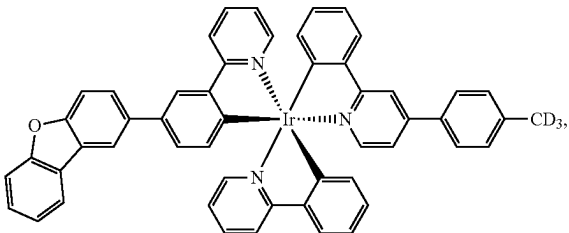
Compound 117



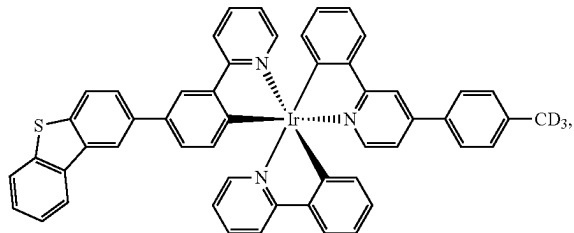
Compound 118



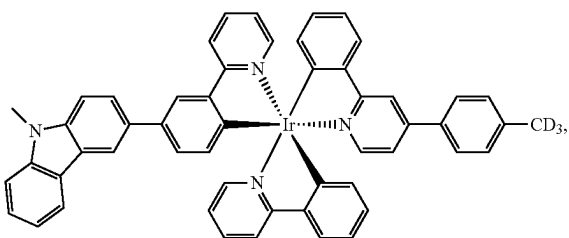
Compound 119



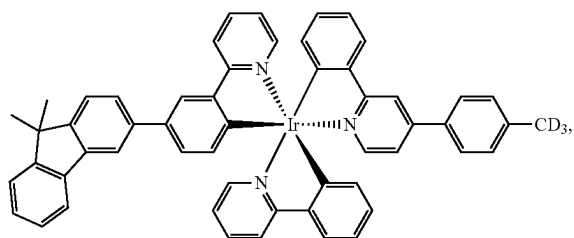
Compound 120



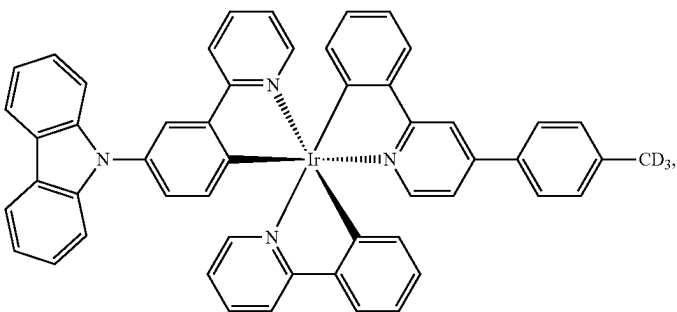
Compound 121



Compound 122



Compound 123

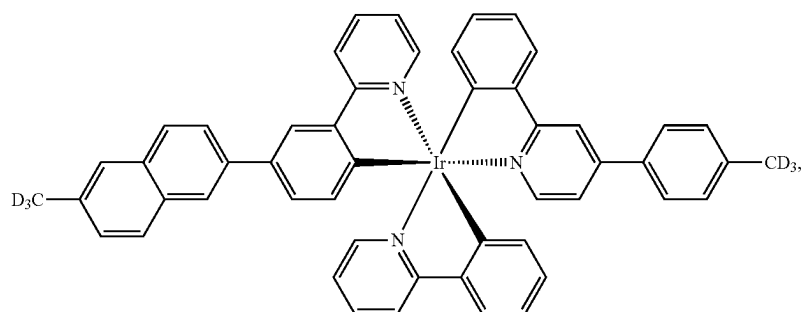


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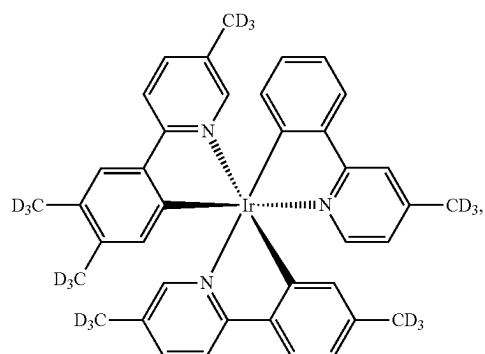
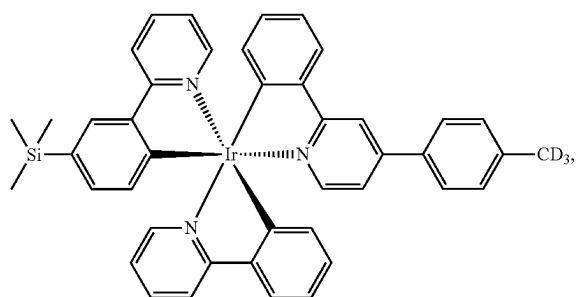
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Compound 124



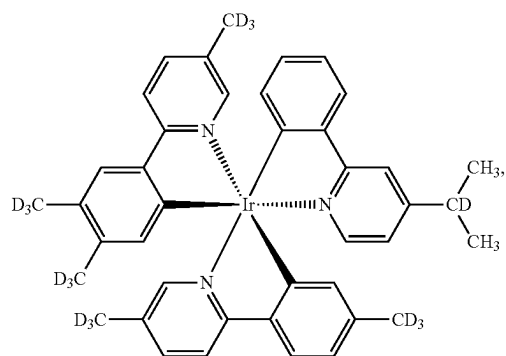
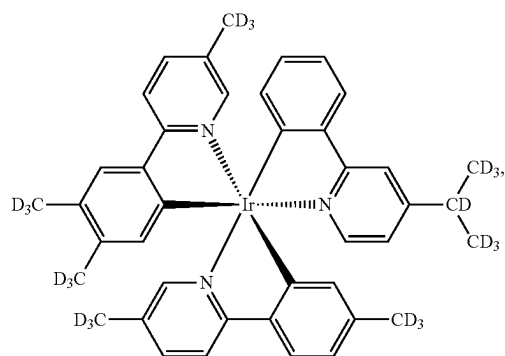
Compound 125

Compound 126



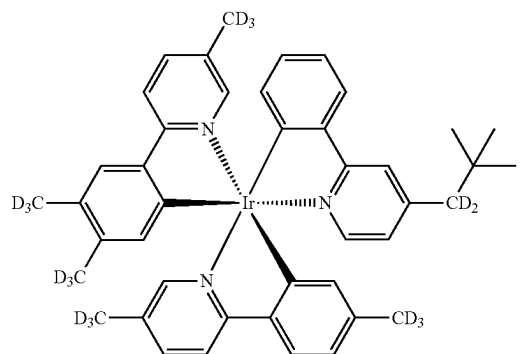
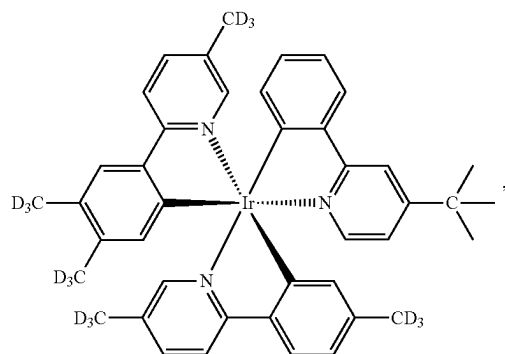
Compound 127

Compound 128



Compound 129

Compound 130



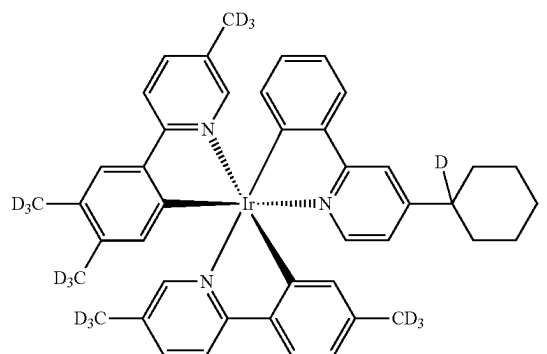
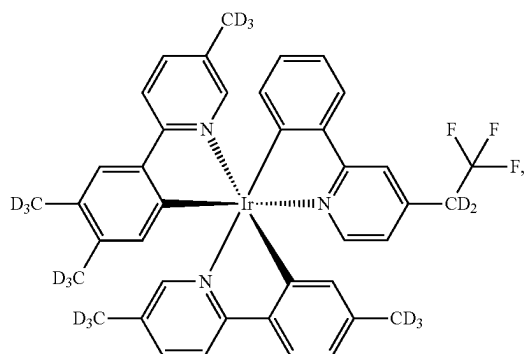
137

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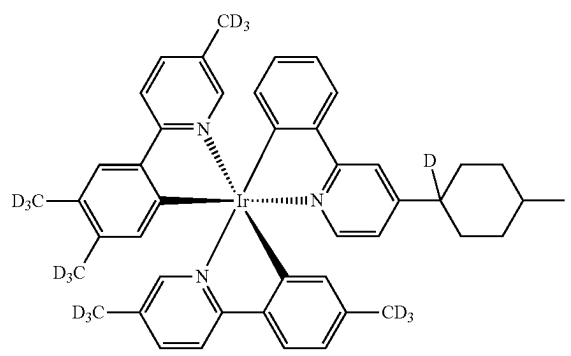
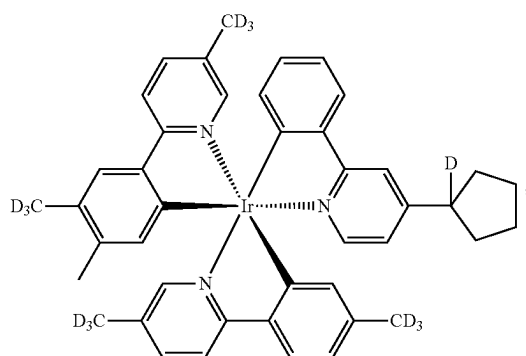
Compound 131

Compound 132



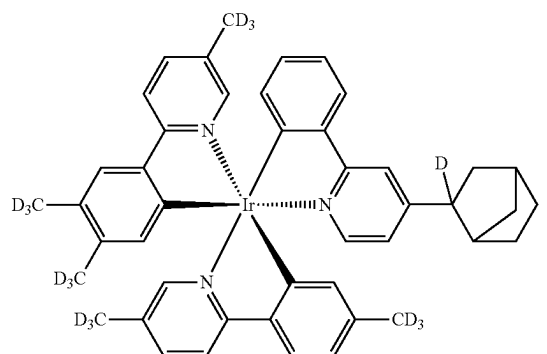
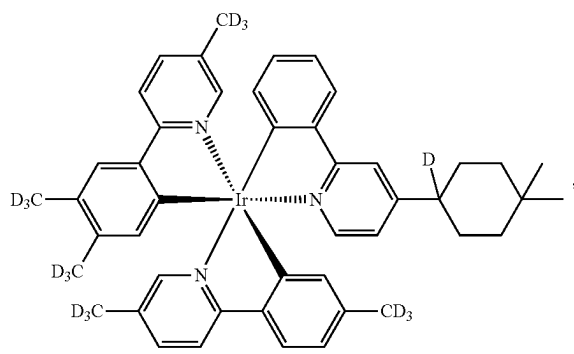
Compound 133

Compound 134



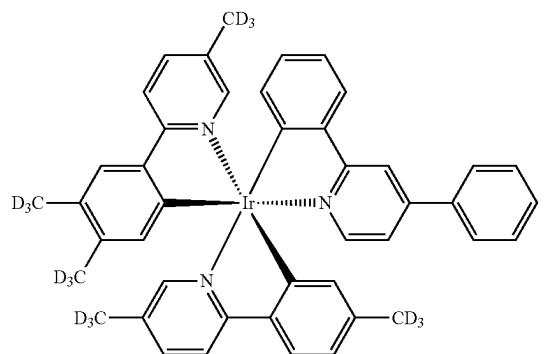
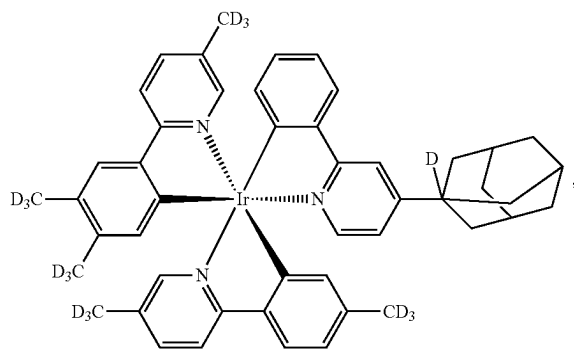
Compound 135

Compound 136



Compound 137

Compound 138



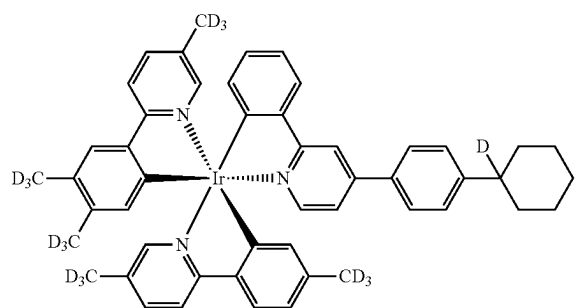
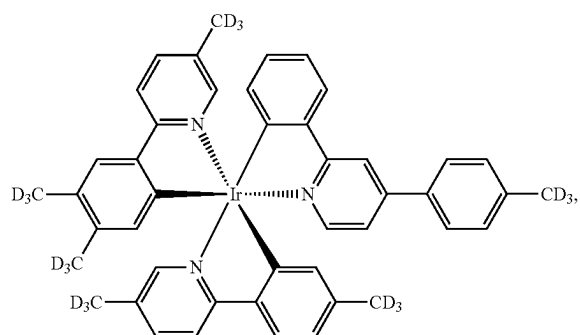
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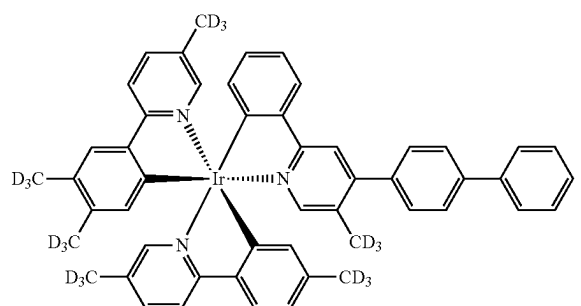
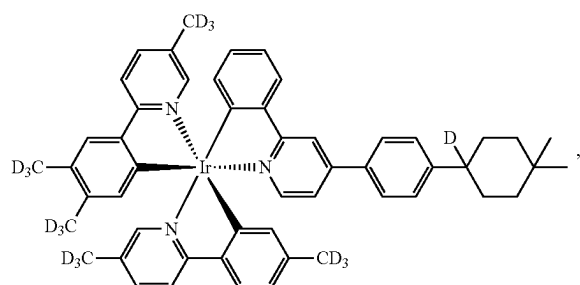
Compound 139

Compound 140

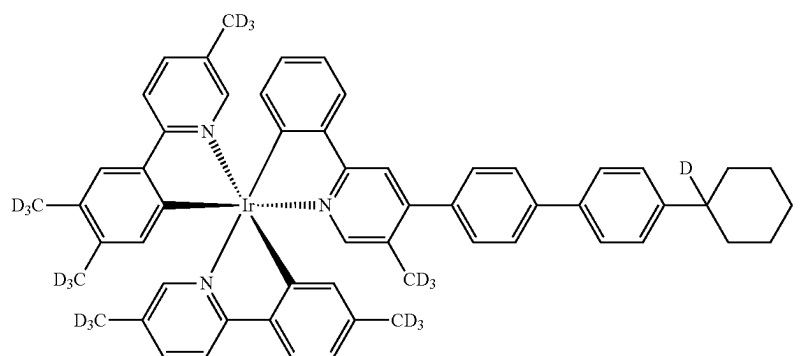


Compound 141

Compound 142

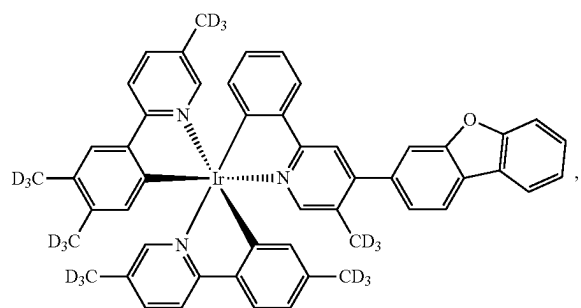
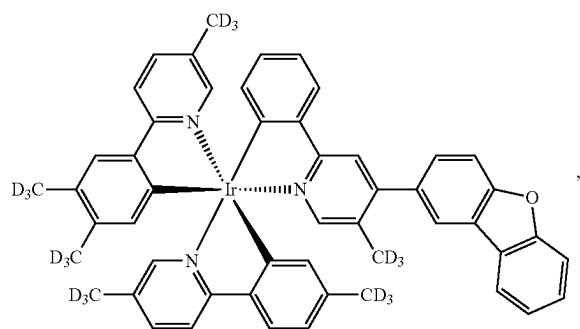


Compound 143



Compound 144

Compound 145

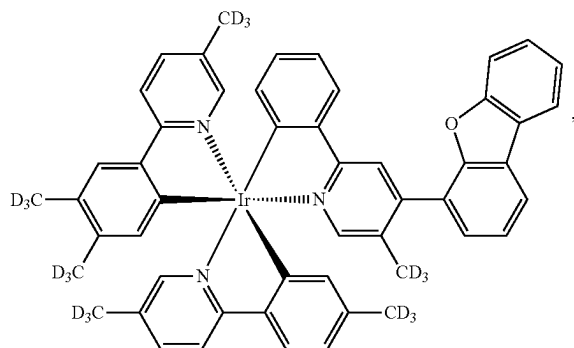


141

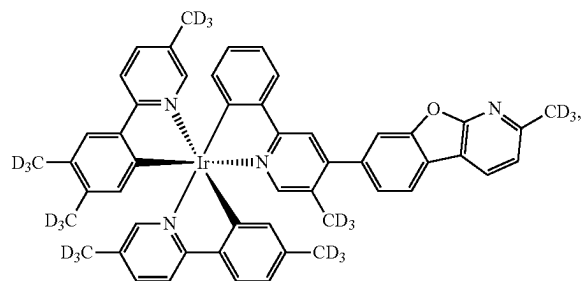
142

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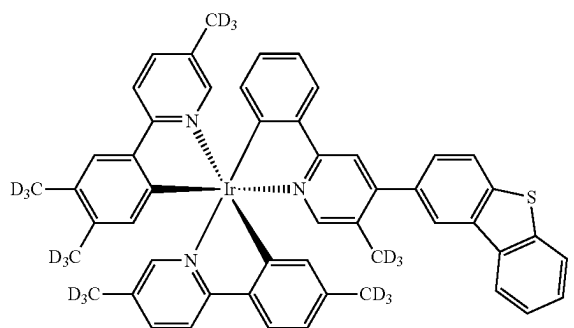
Compound 146



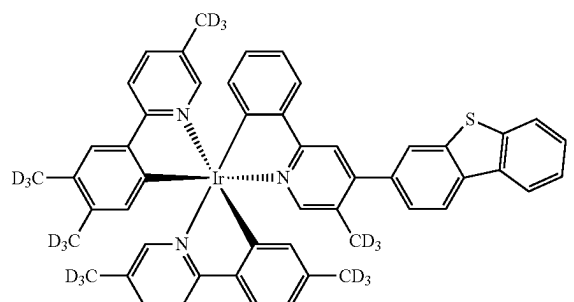
Compound 147



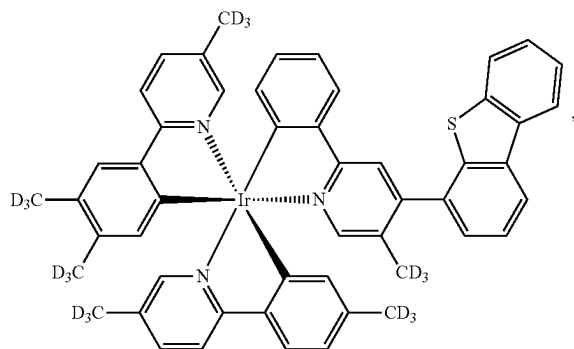
Compound 148



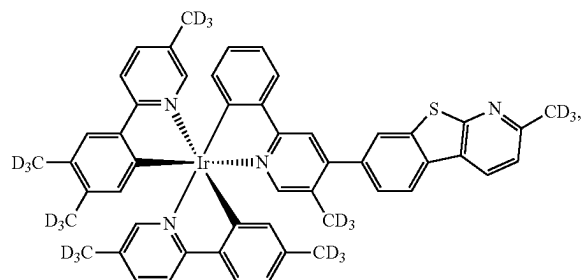
Compound 149



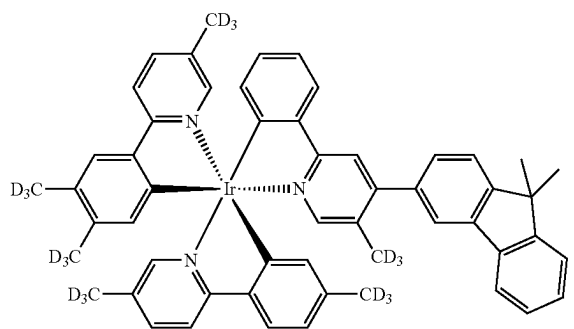
Compound 150



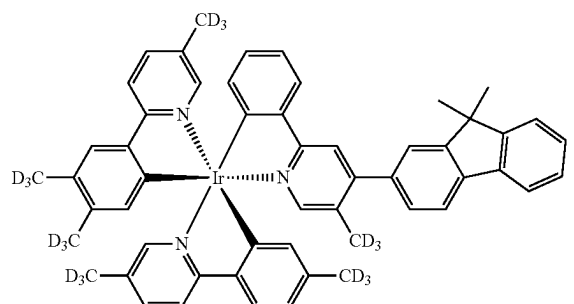
Compound 151



Compound 152



Compound 153



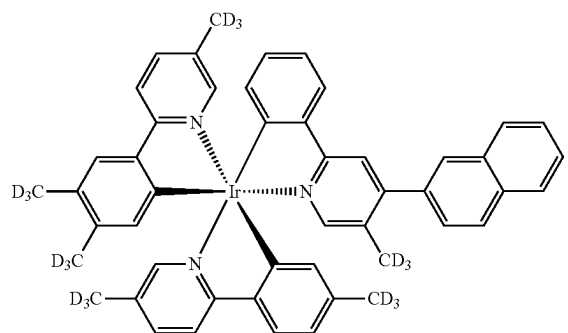
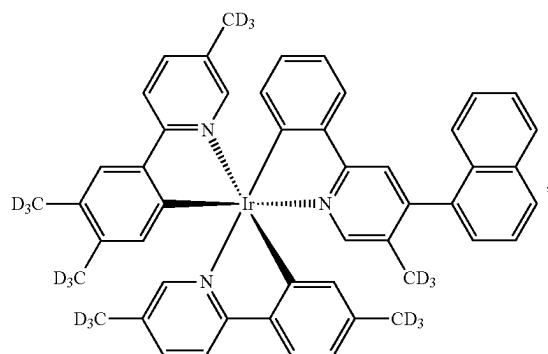
143

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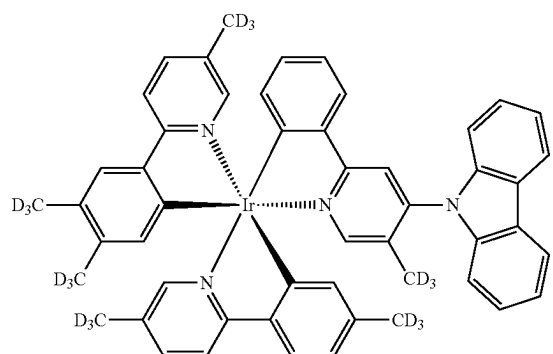
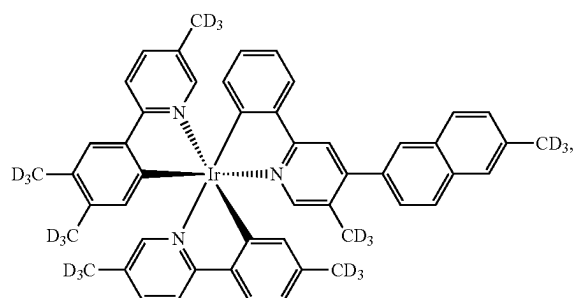
Compound 154

Compound 155



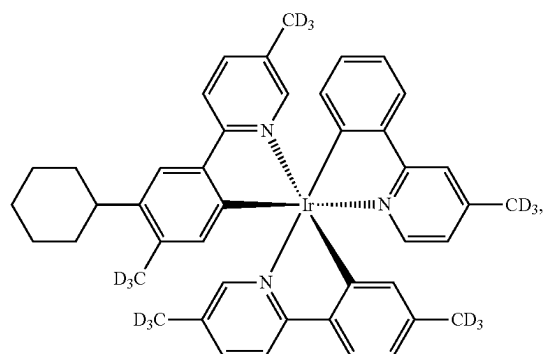
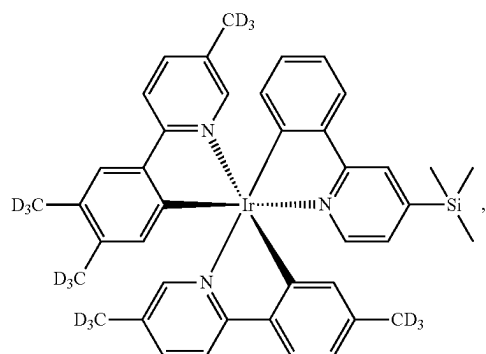
Compound 156

Compound 157



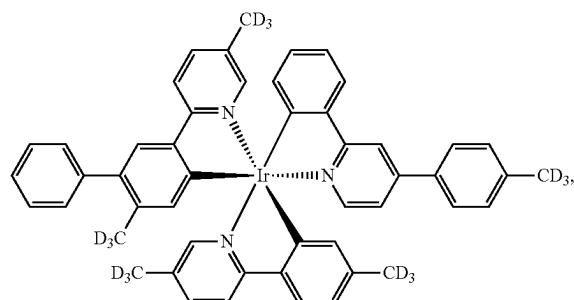
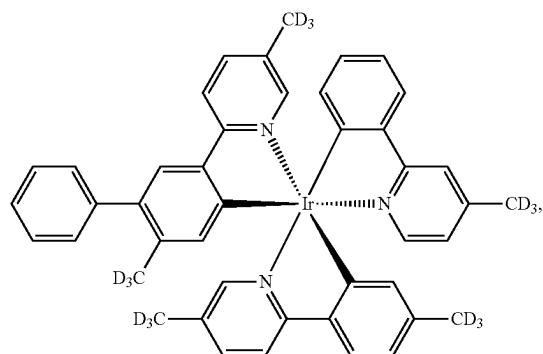
Compound 158

Compound 159



Compound 160

Compound 161

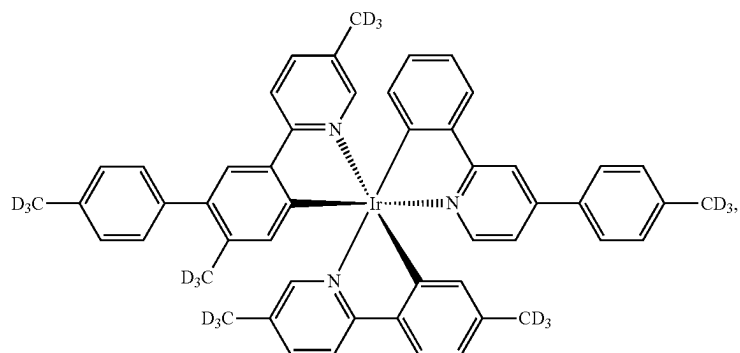


145

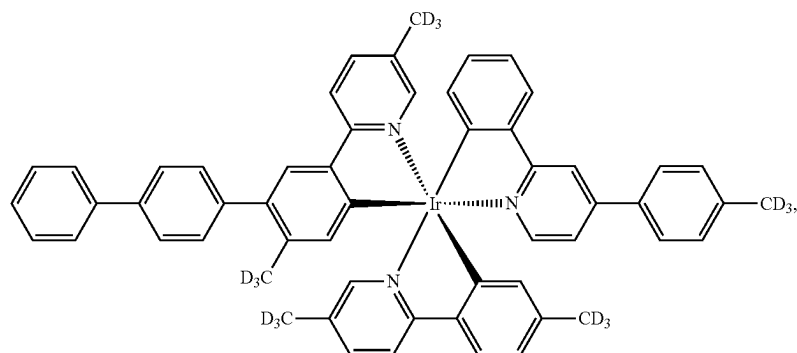
146

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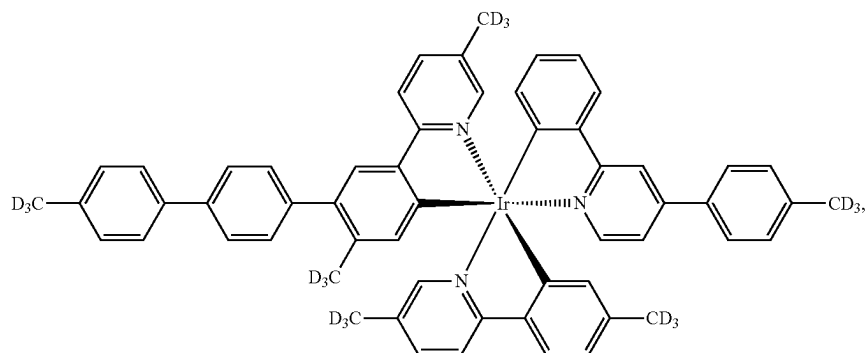
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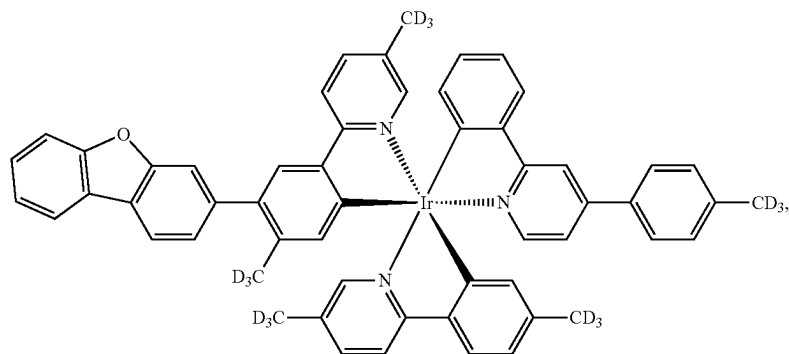
Compound 163



Compound 164



Compound 165

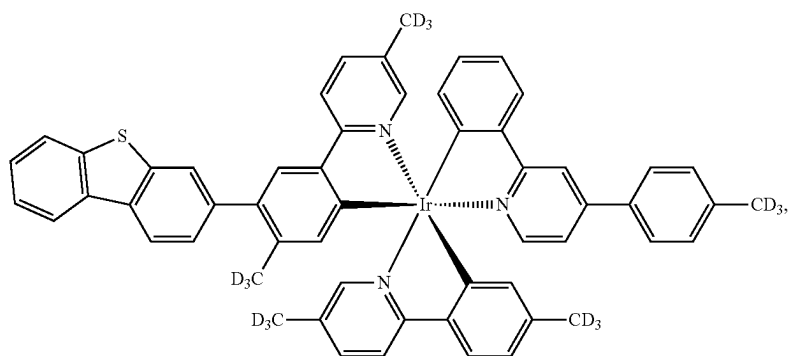


147

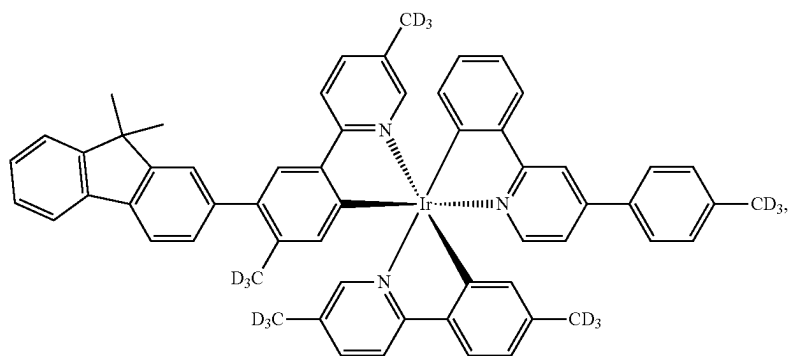
148

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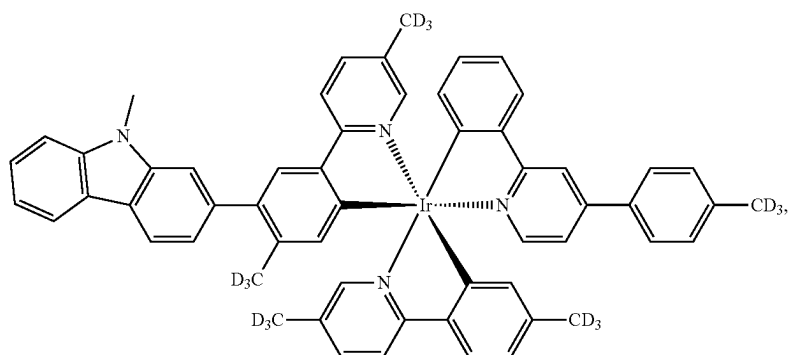
Compound 166



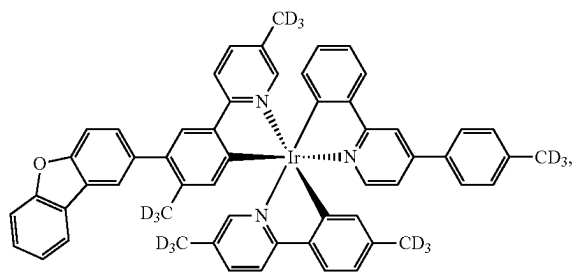
Compound 167



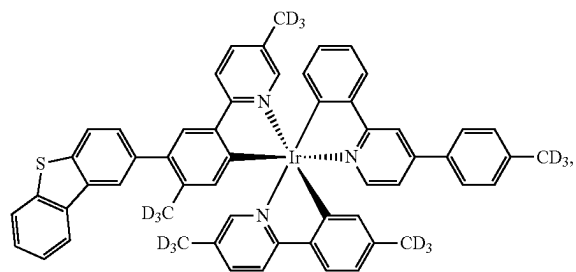
Compound 168



Compound 169



Compound 170

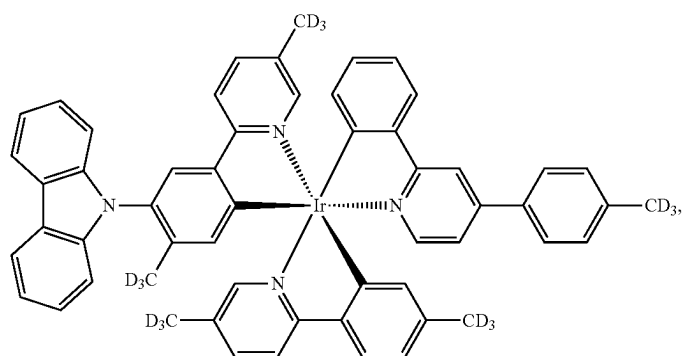


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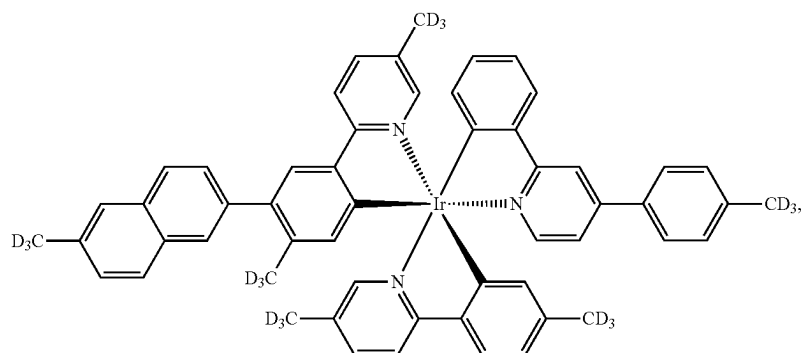
150

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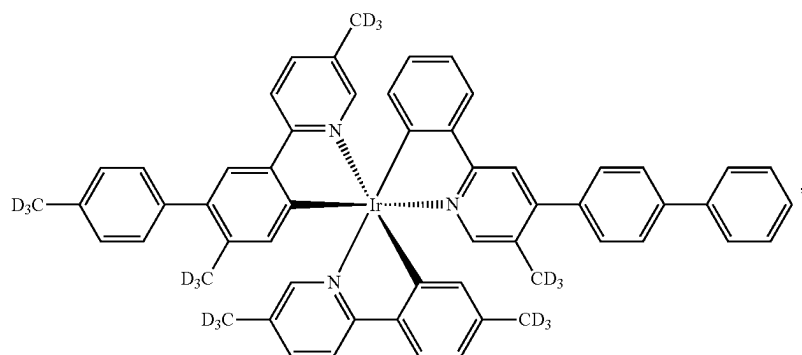
Compound 171



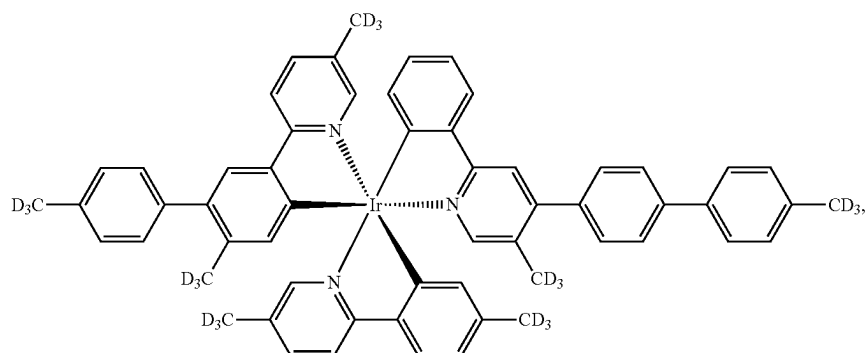
Compound 172



Compound 173



Compound 174

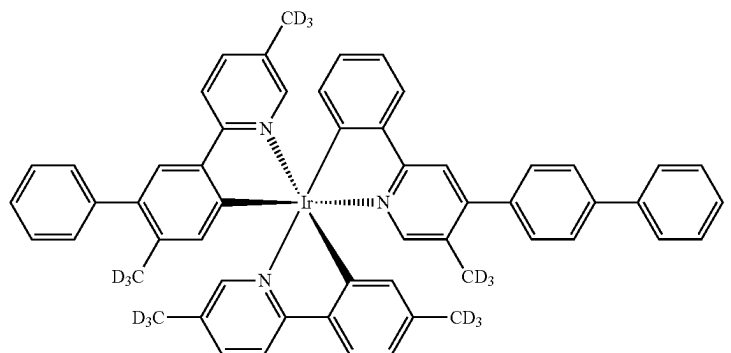


151

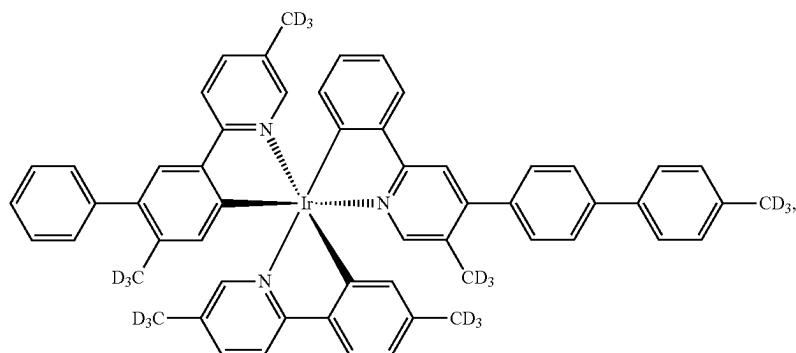
152

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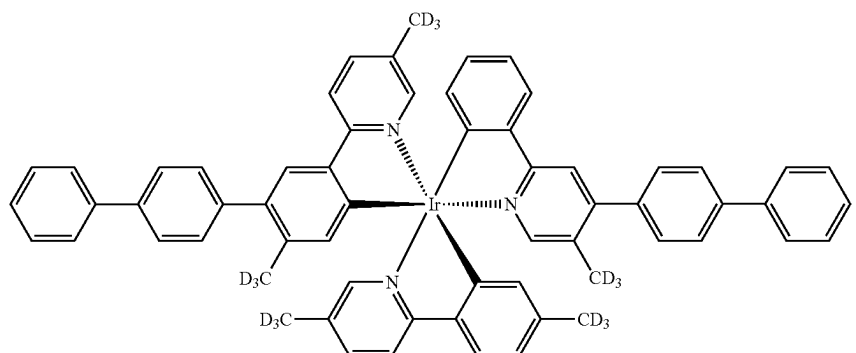
Compound 175



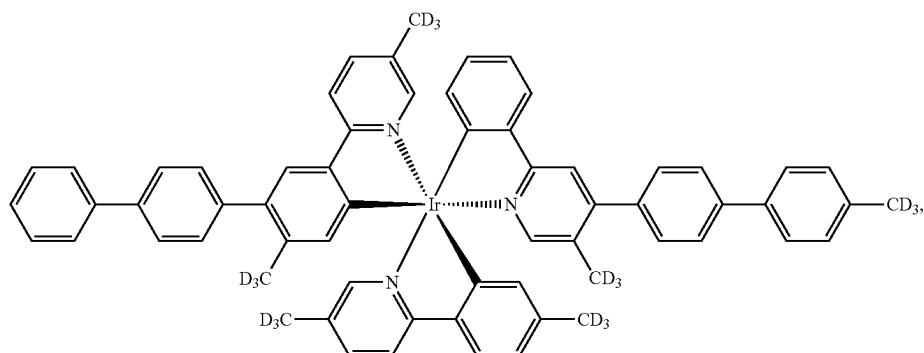
Compound 176



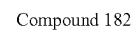
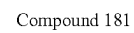
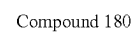
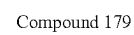
Compound 177



Compound 178



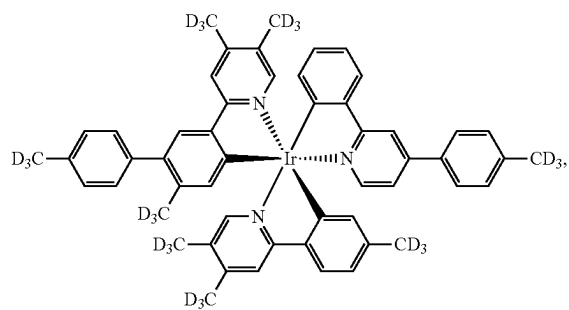
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155

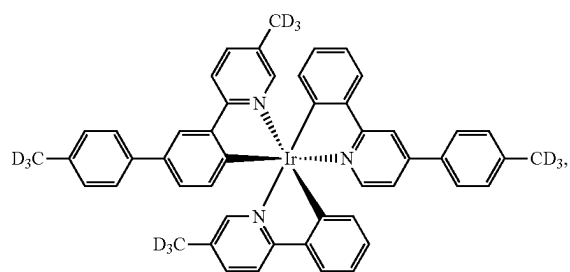
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Compound 183

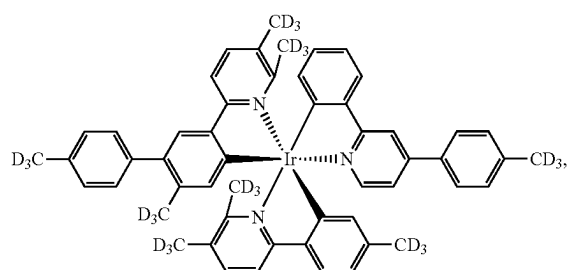


156

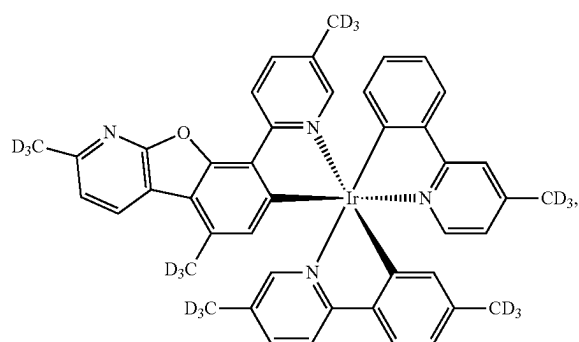
Compound 184



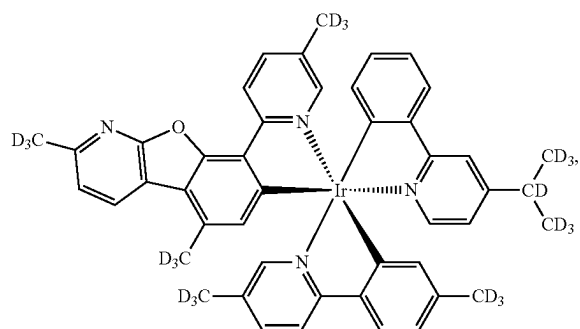
Compound 185



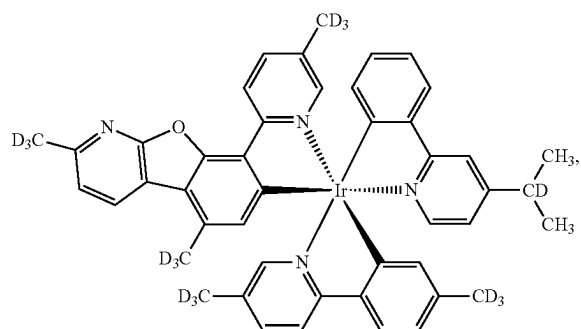
Compound 186



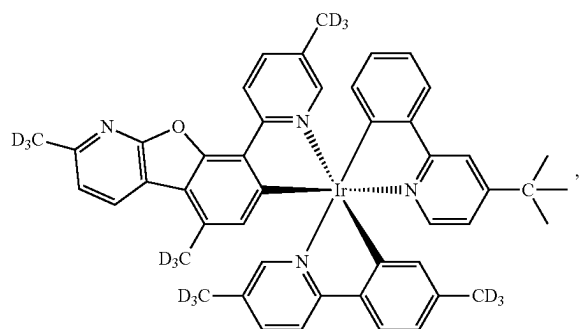
Compound 187



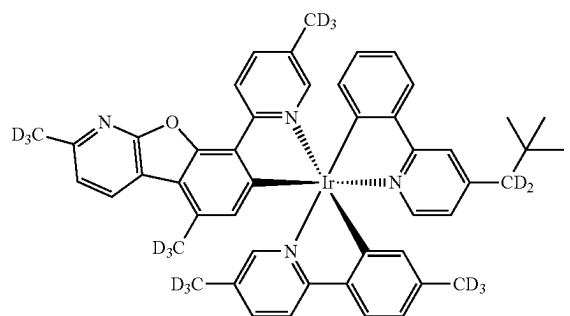
Compound 188



Compound 189



Compound 190



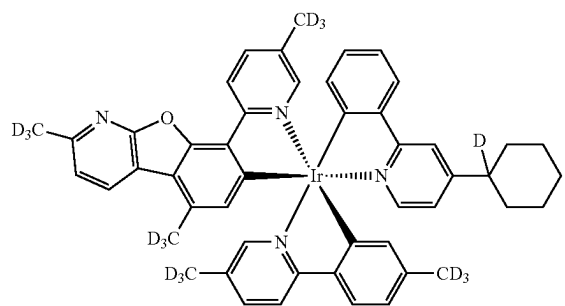
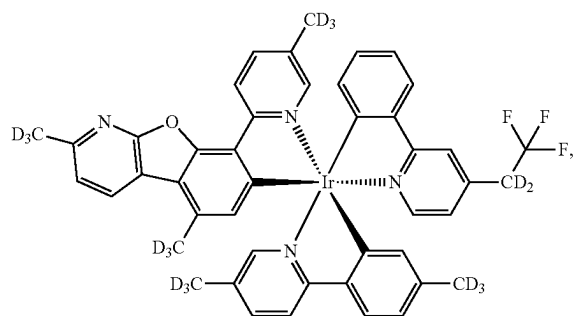
157

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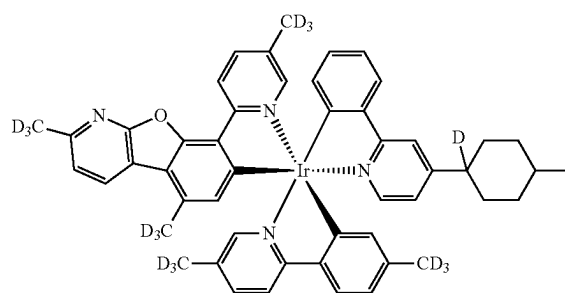
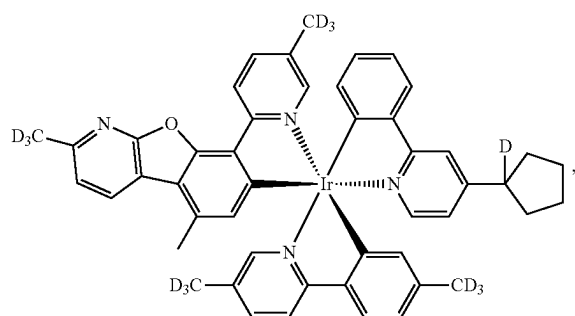
Compound 191

Compound 192



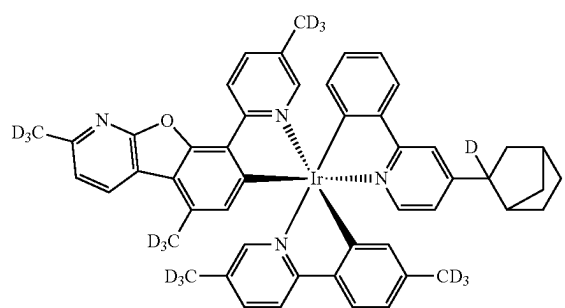
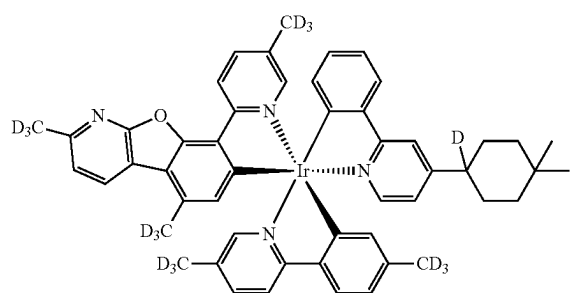
Compound 193

Compound 194



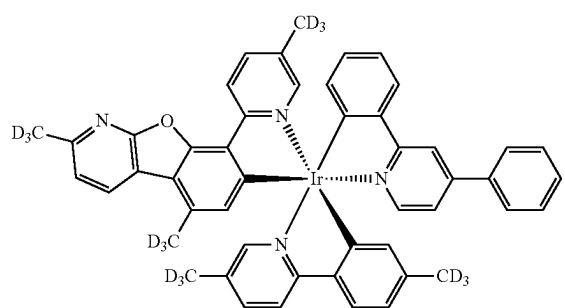
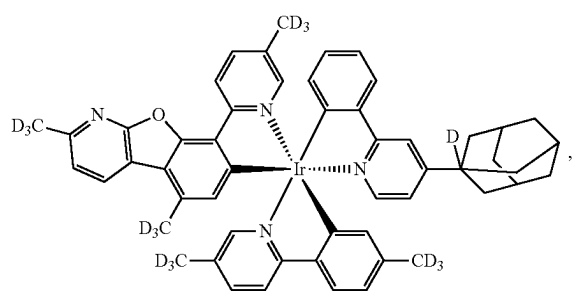
Compound 195

Compound 196



Compound 197

Compound 198

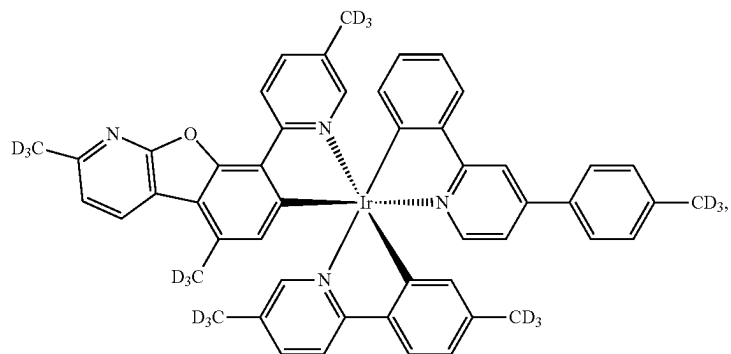


159

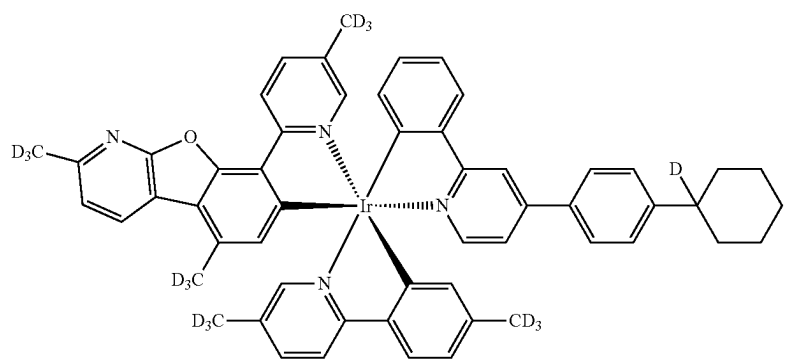
160

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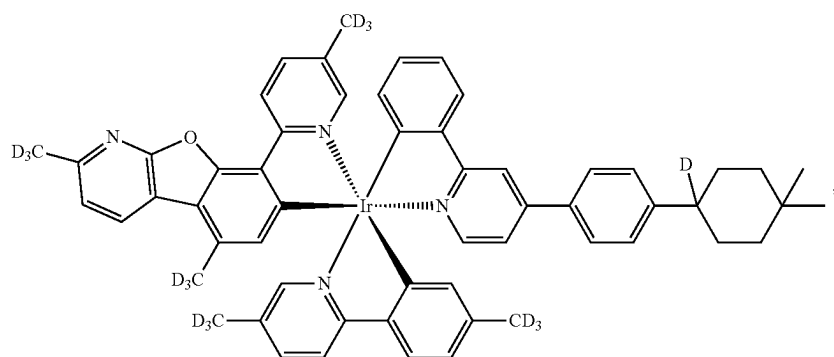
Compound 199



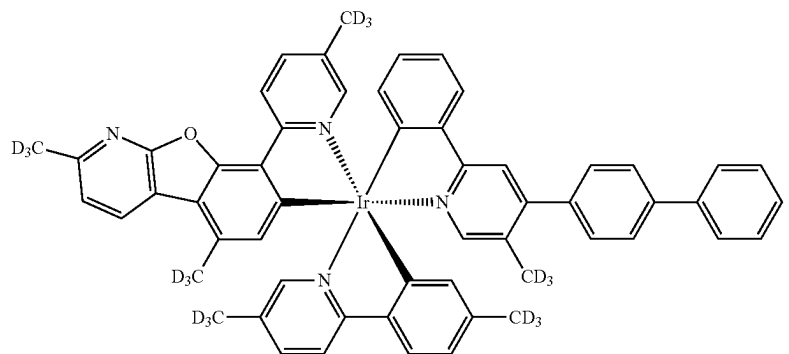
Compound 200



Compound 201

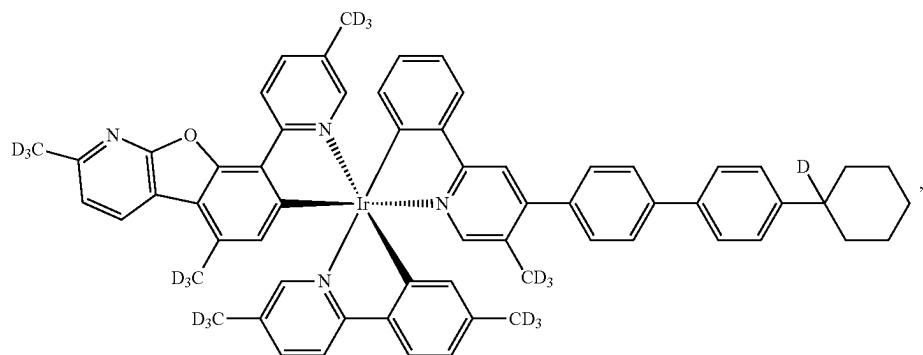


Compound 202

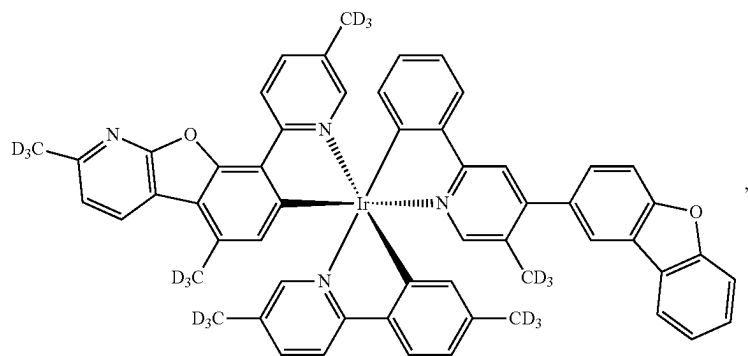


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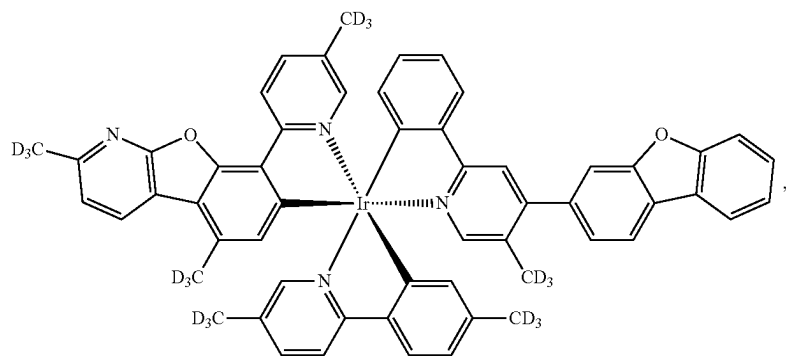
Compound 203



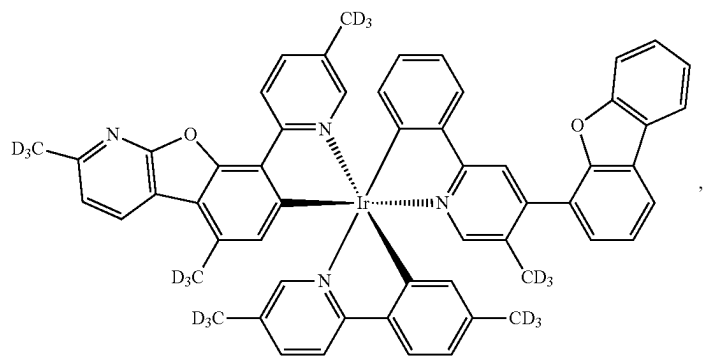
Compound 204



Compound 205

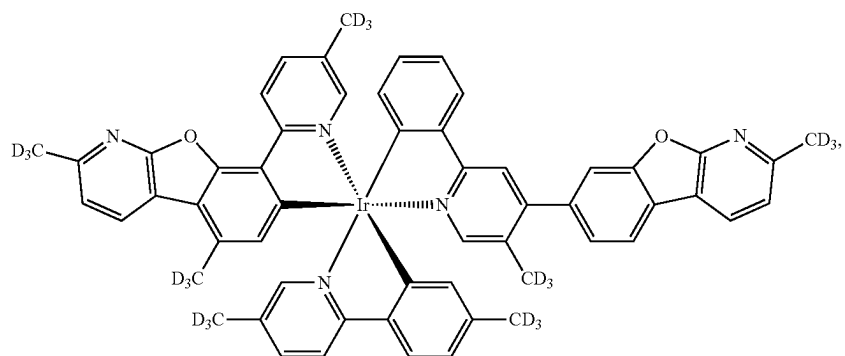


Compound 206

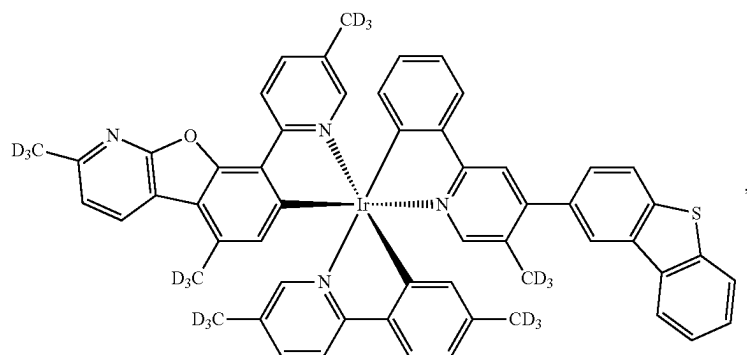


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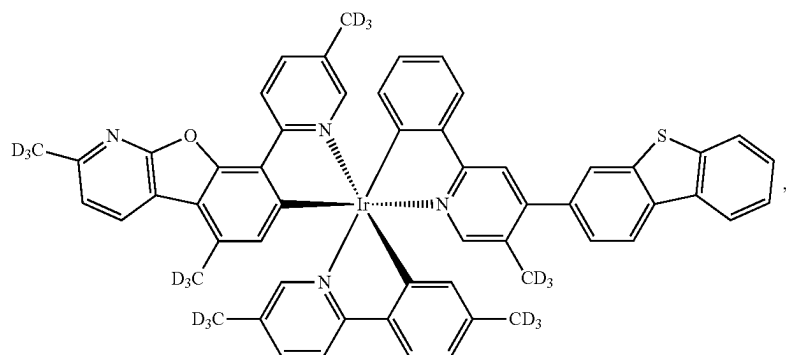
Compound 207



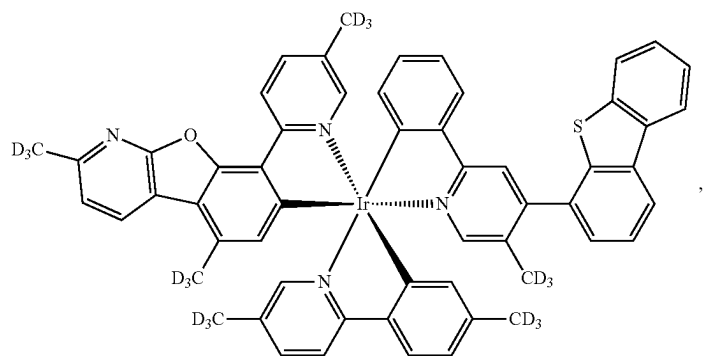
Compound 208



Compound 209



Compound 210

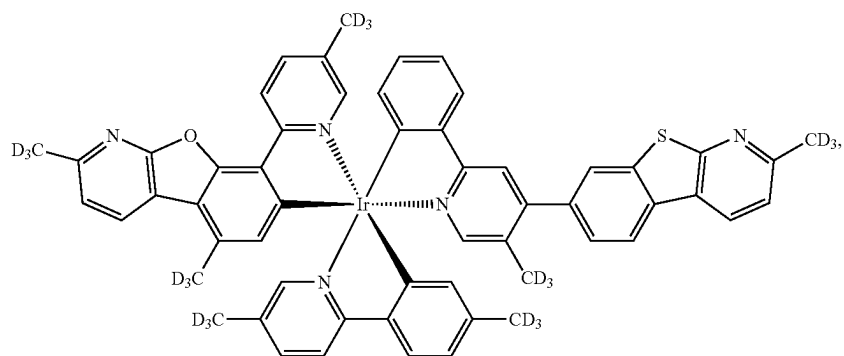


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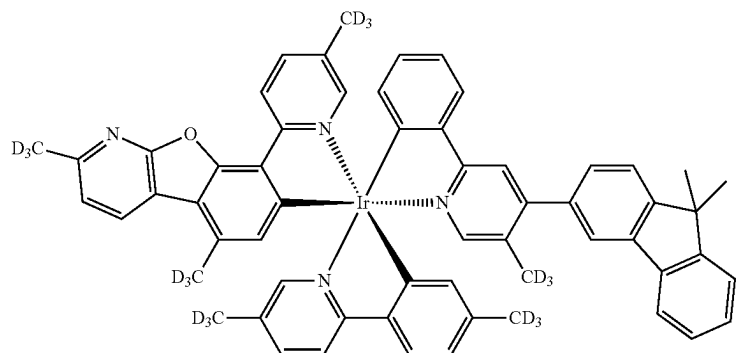
166

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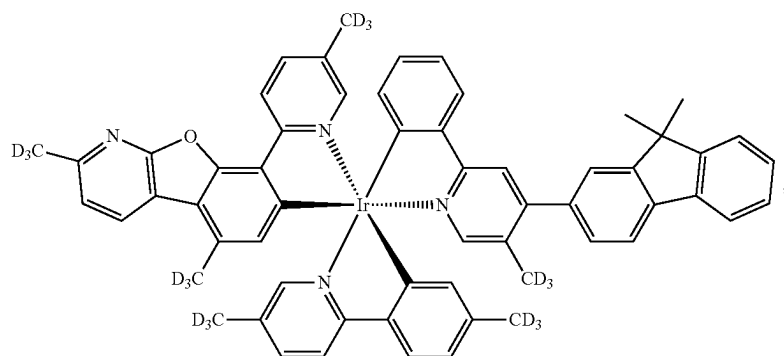
Compound 211



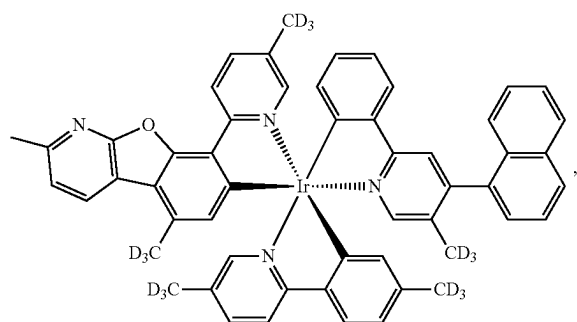
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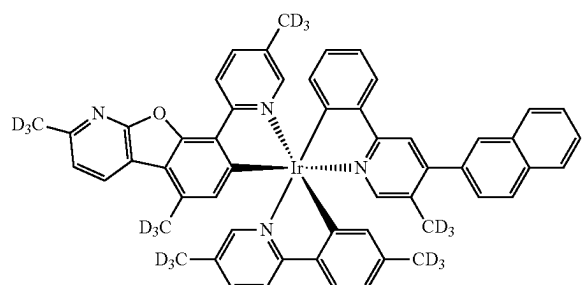
Compound 213



Compound 214



Compound 215

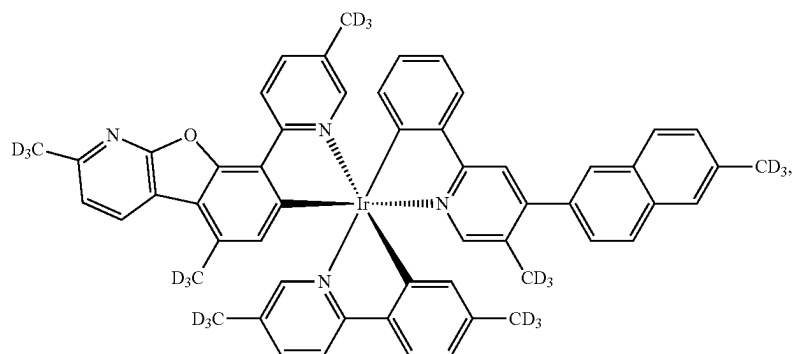


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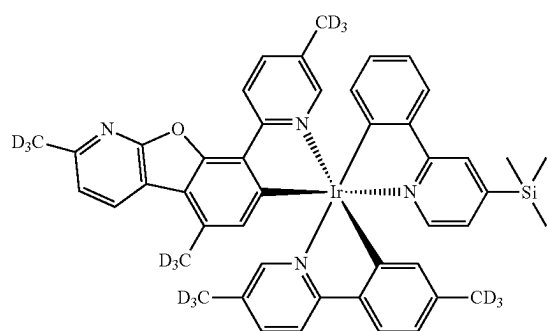
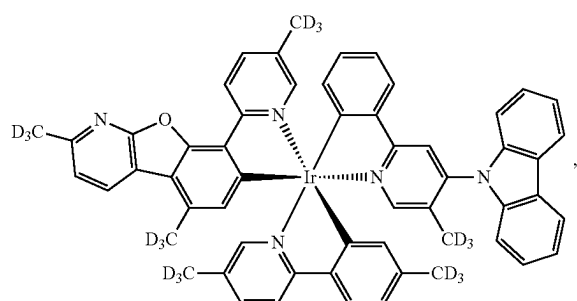
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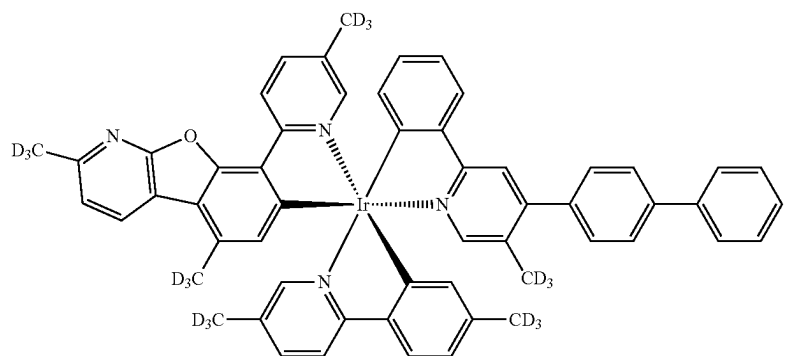


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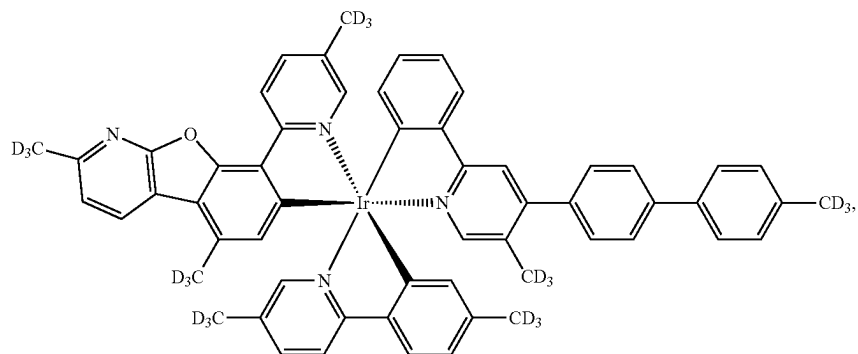
Compound 218



Compound 219



Compound 220



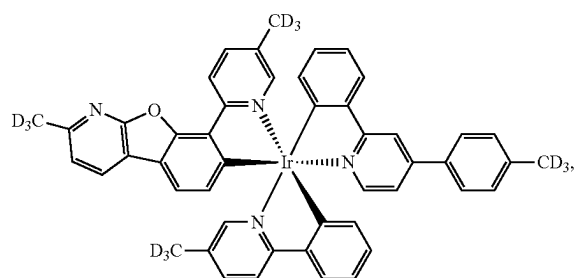
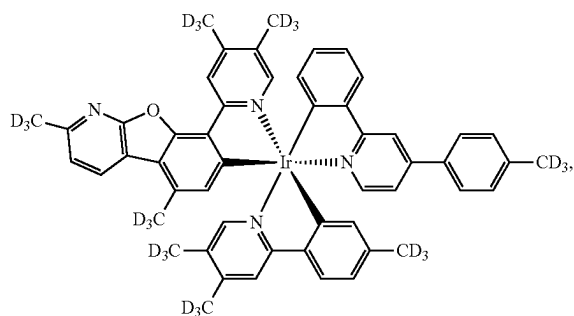
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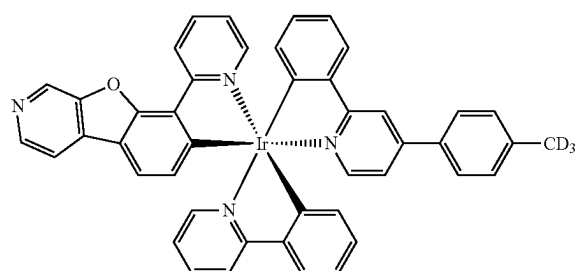
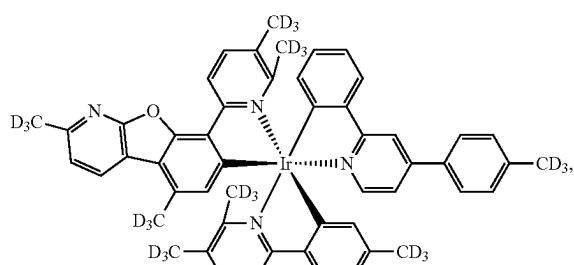
Compound 221

Compound 222



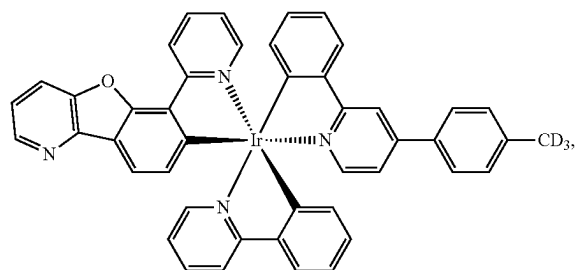
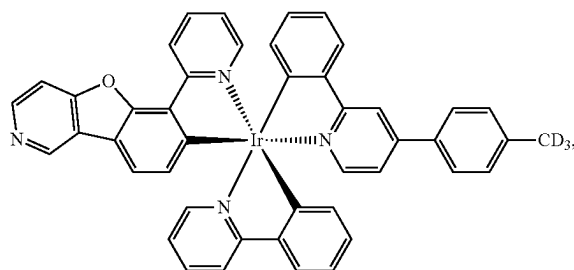
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Compound 224



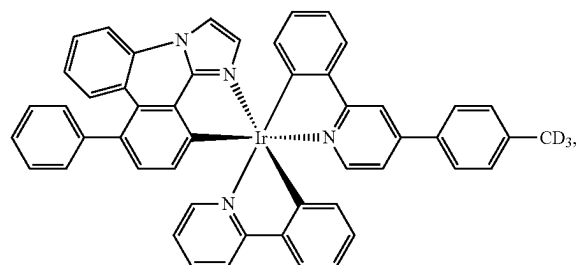
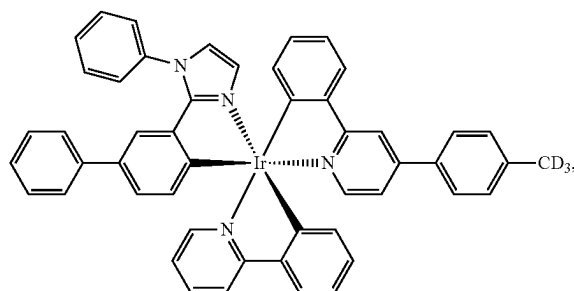
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Compound 226



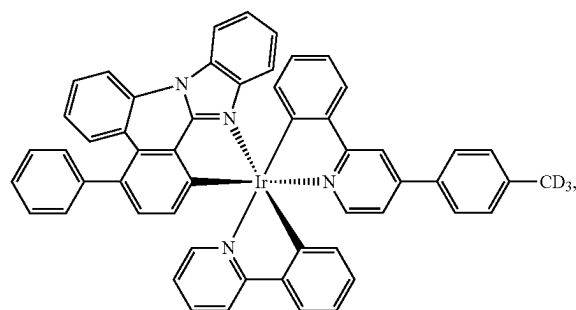
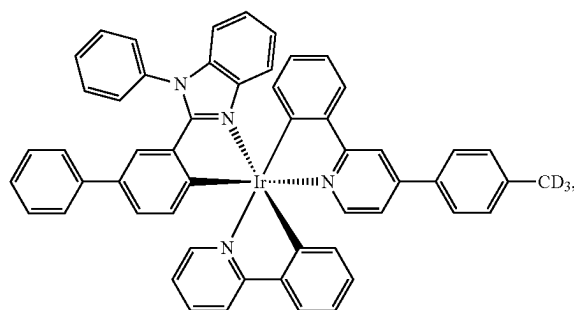
Compound 227

Compound 228



Compound 229

Compound 230



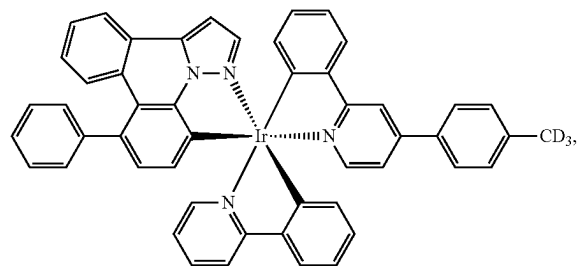
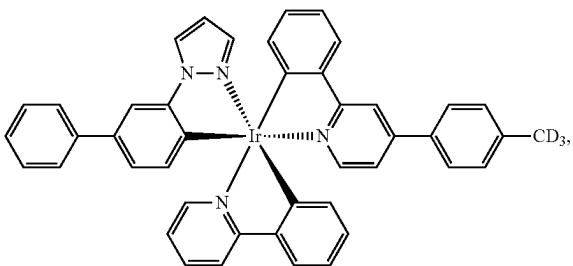
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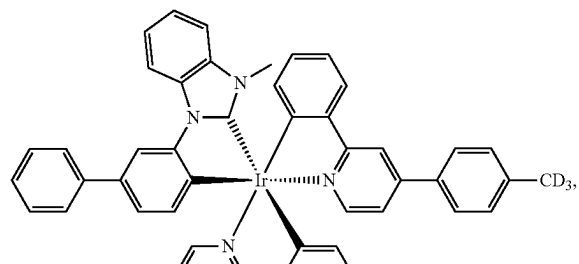
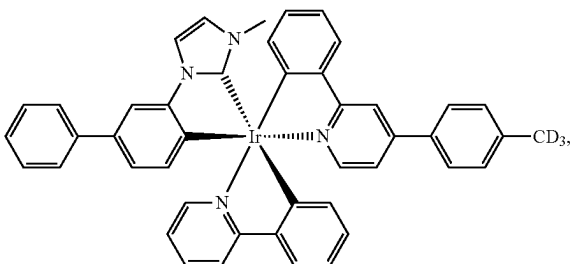
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Compound 232



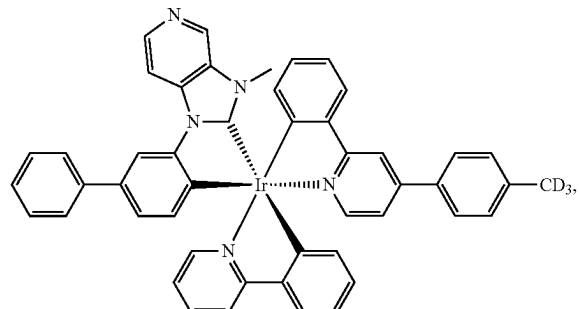
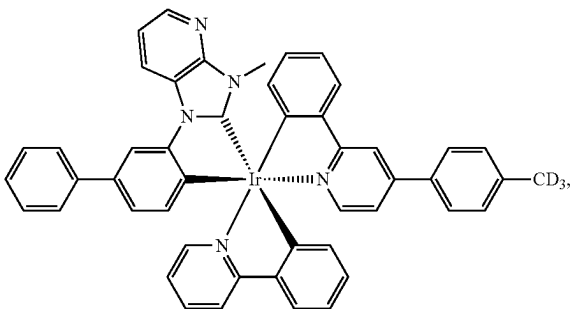
Compound 233

Compound 234



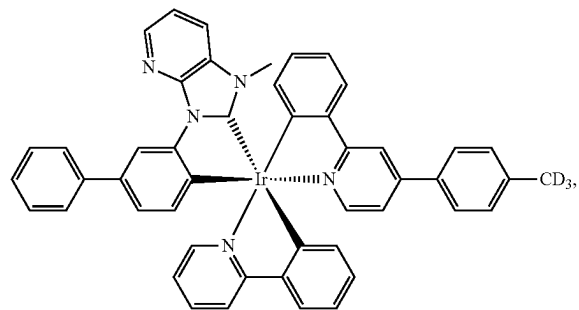
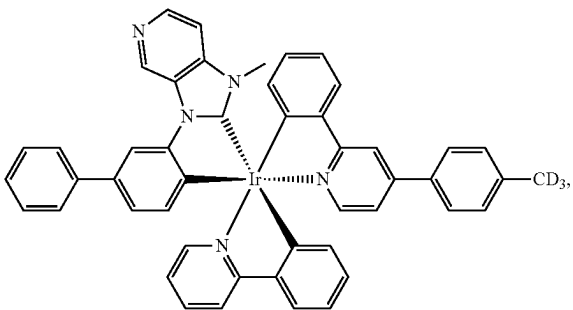
Compound 235

Compound 236



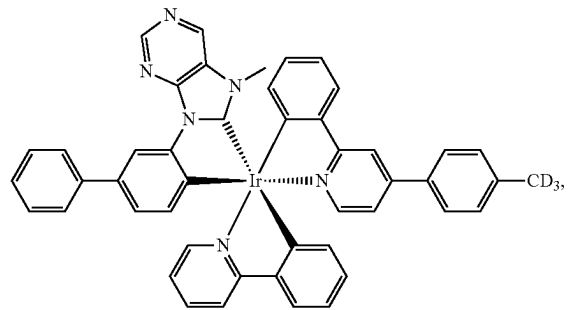
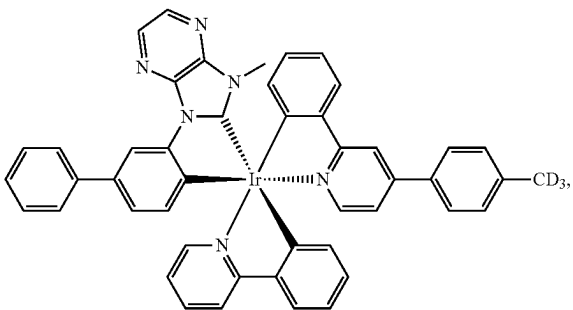
Compound 237

Compound 238



Compound 239

Compound 240



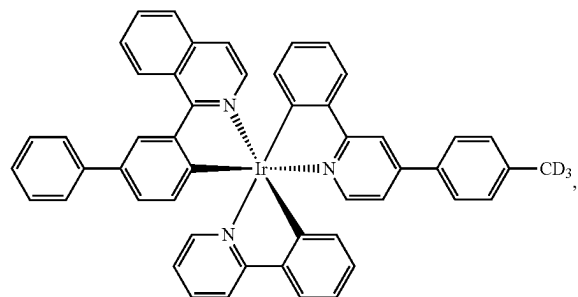
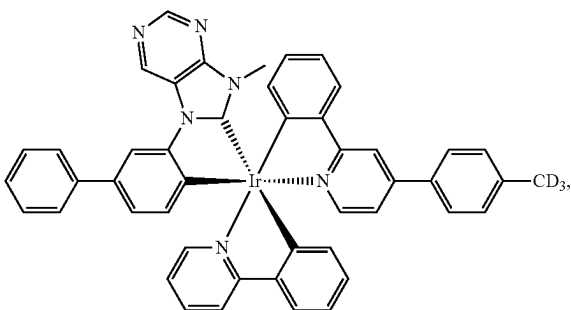
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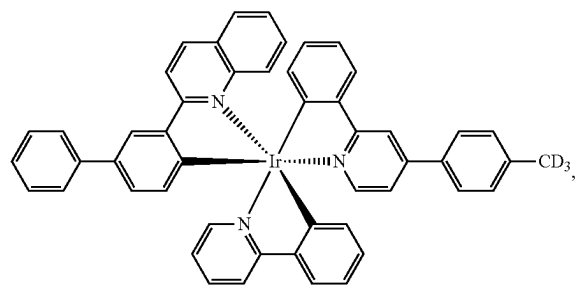
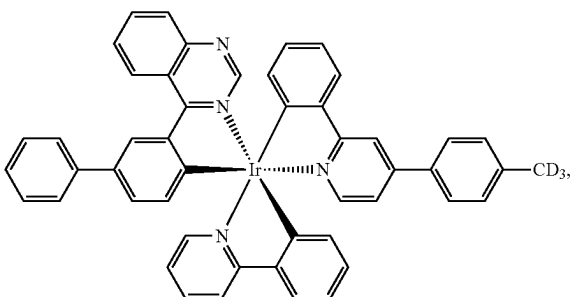
Compound 241

Compound 242

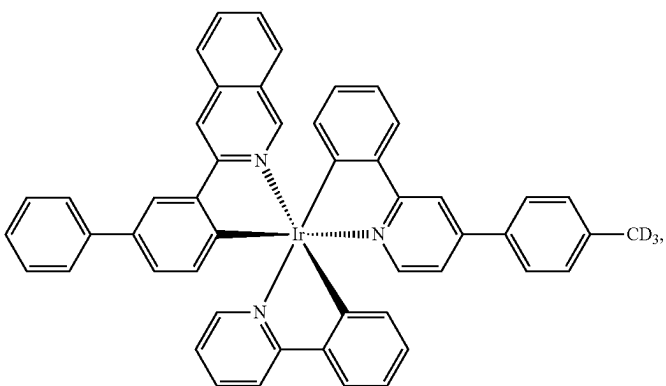


Compound 243

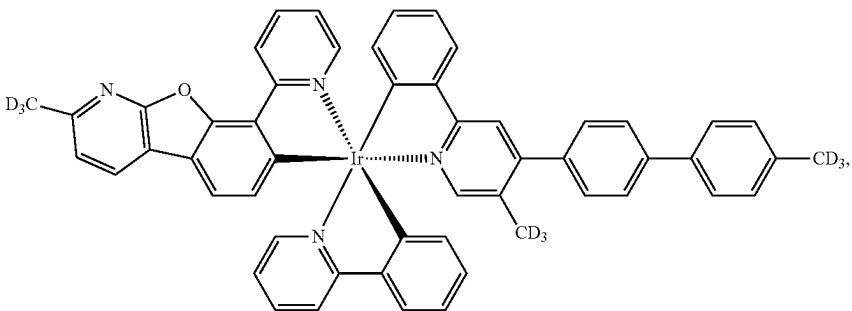
Compound 244



Compound 245

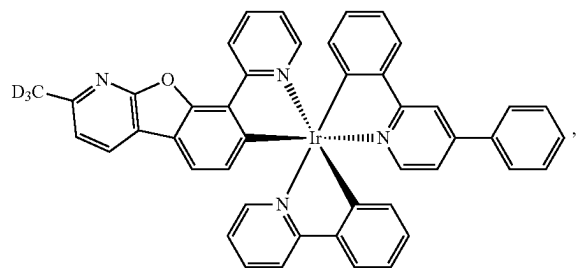
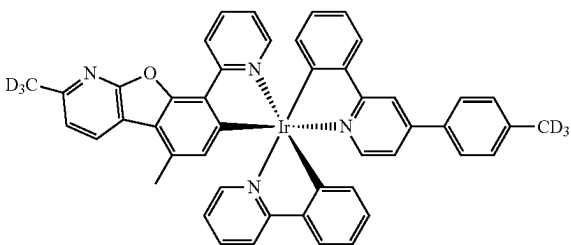


Compound 246



Compound 247

Compound 248

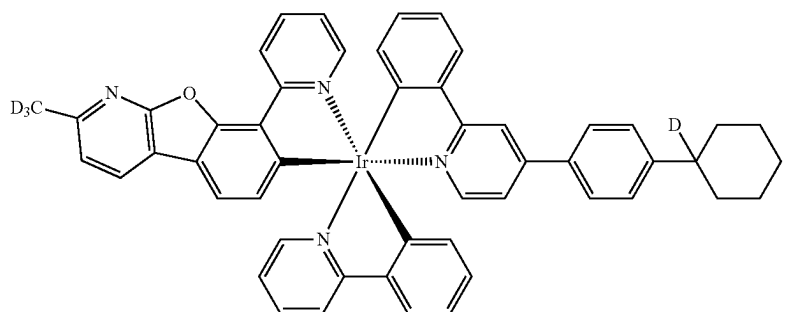


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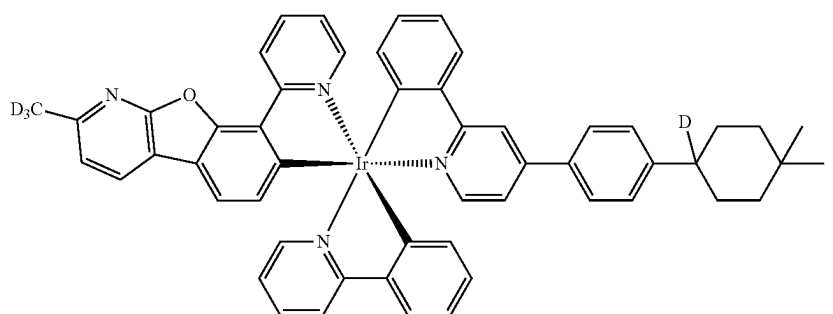
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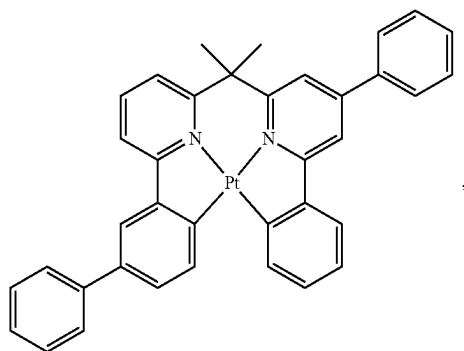
Compound 249



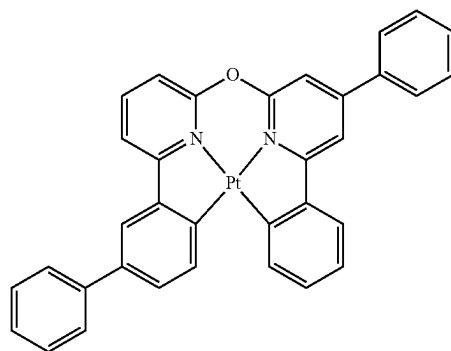
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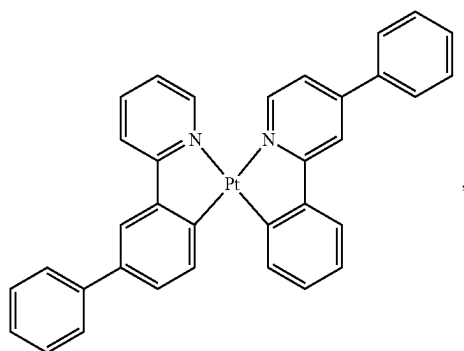
Compound 251



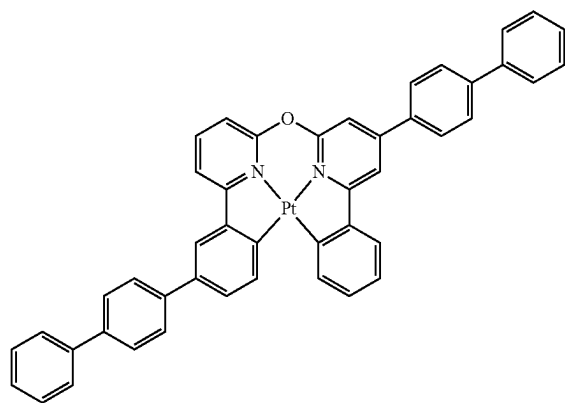
Compound 252



Compound 253



Compound 254

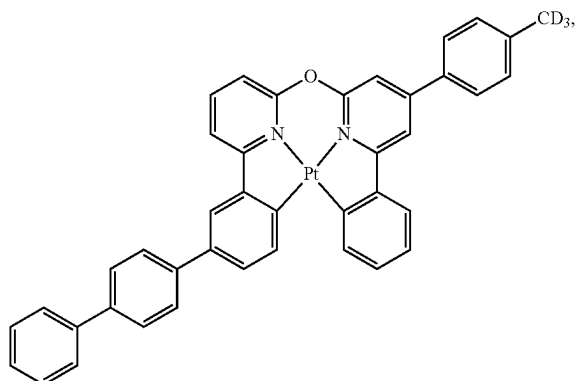


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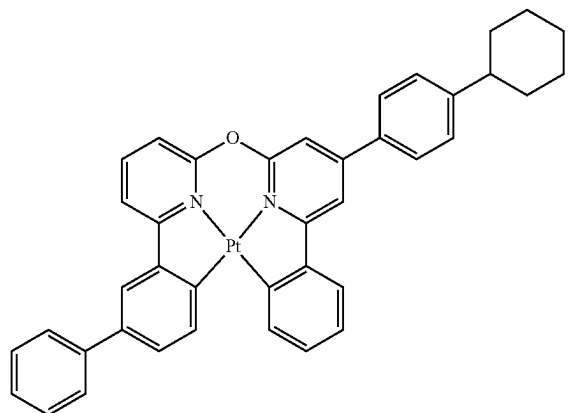
178

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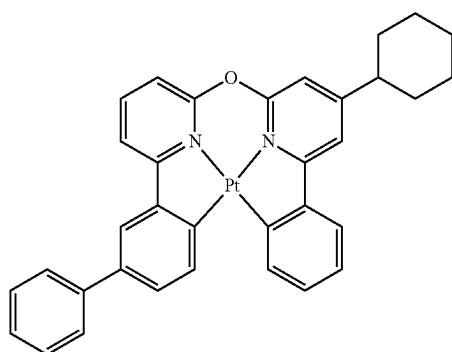
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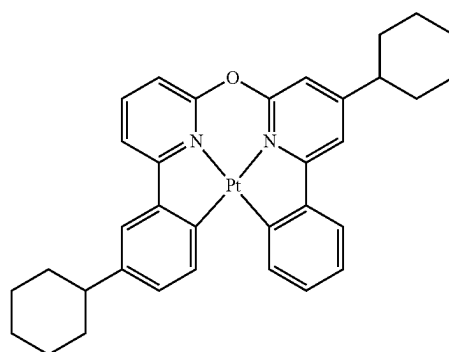
Compound 256



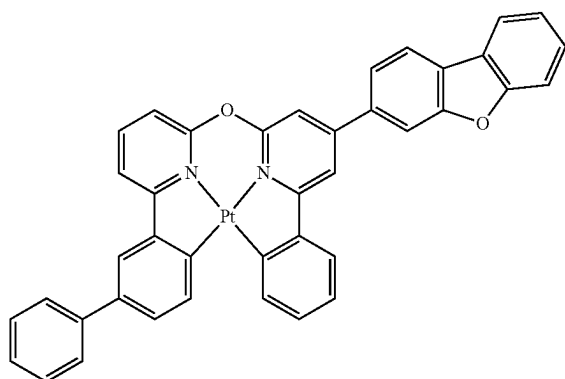
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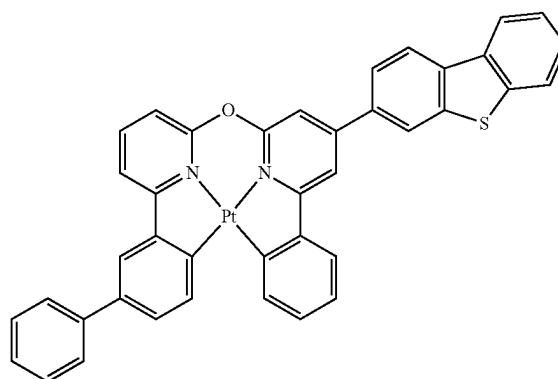
Compound 258



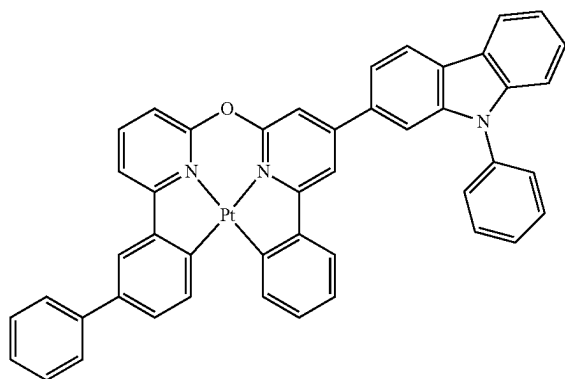
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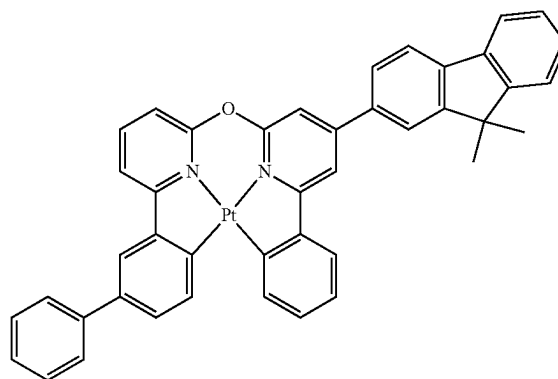
Compound 260



Compound 261



Compound 262



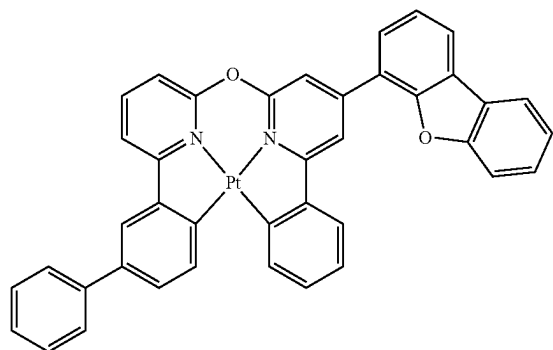
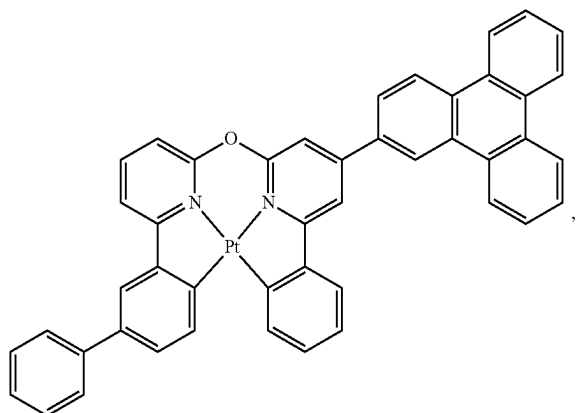
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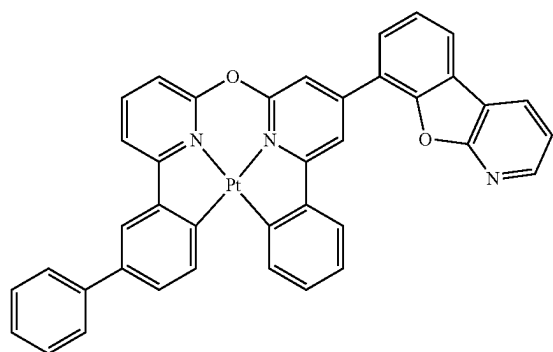
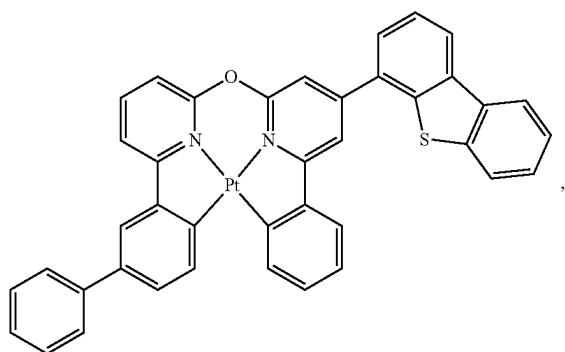
Compound 263

Compound 264



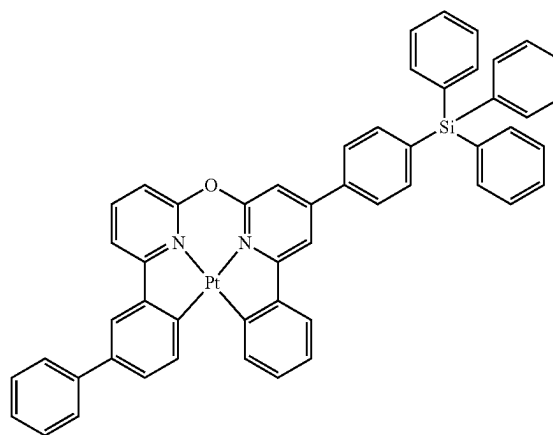
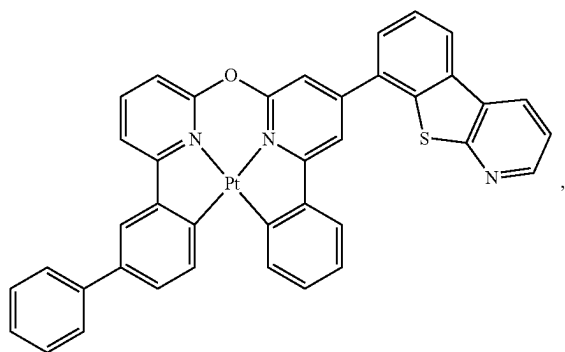
Compound 265

Compound 266



Compound 267

Compound 268



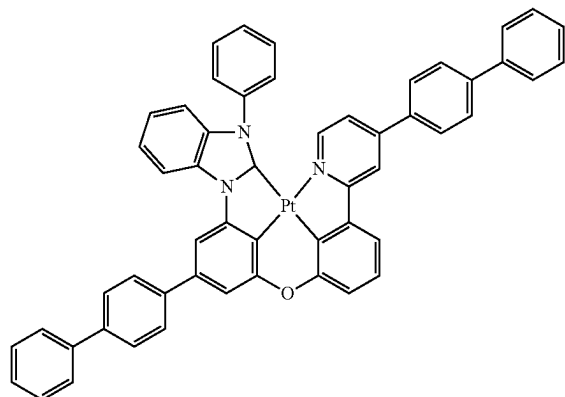
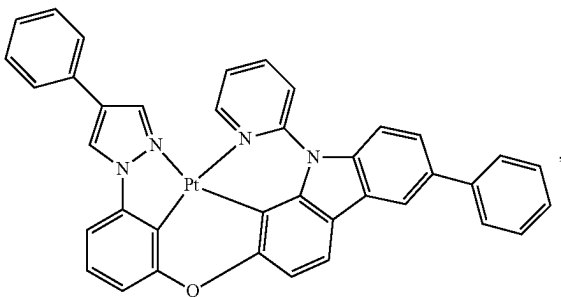
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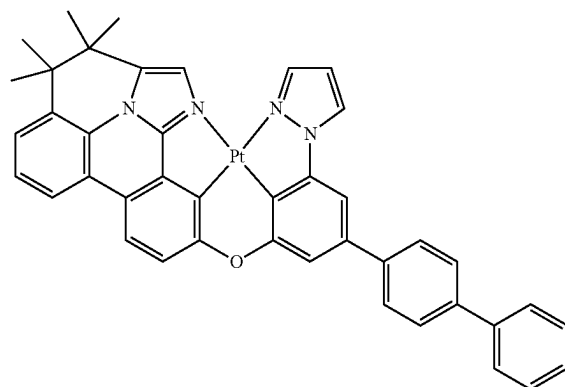
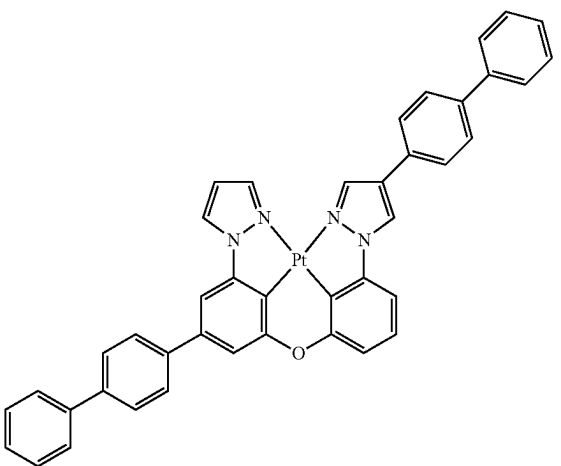
Compound 269

Compound 270



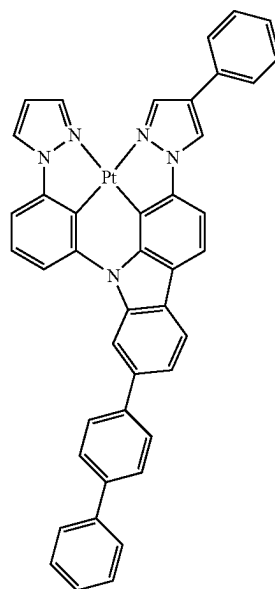
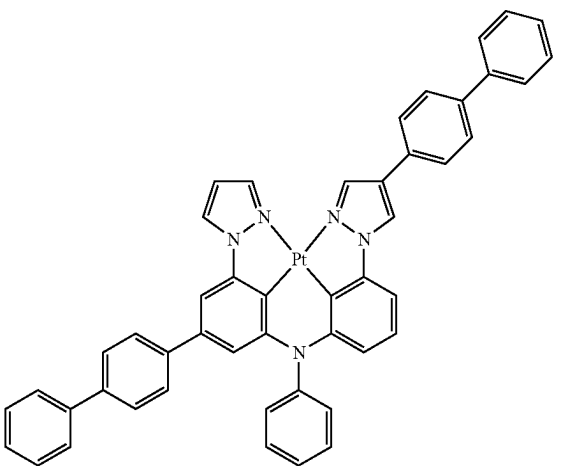
Compound 271

Compound 272

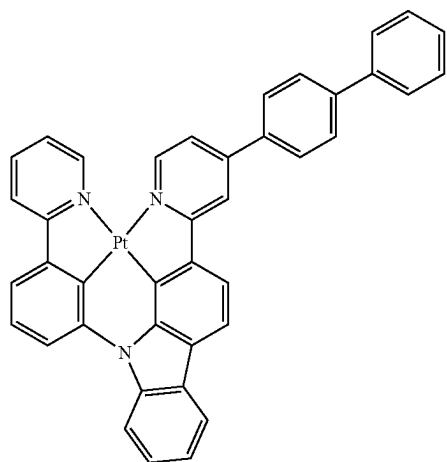


Compound 273

Compound 274



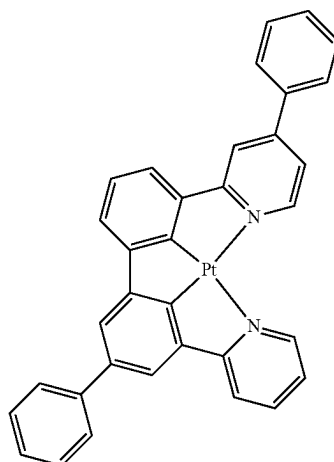
183



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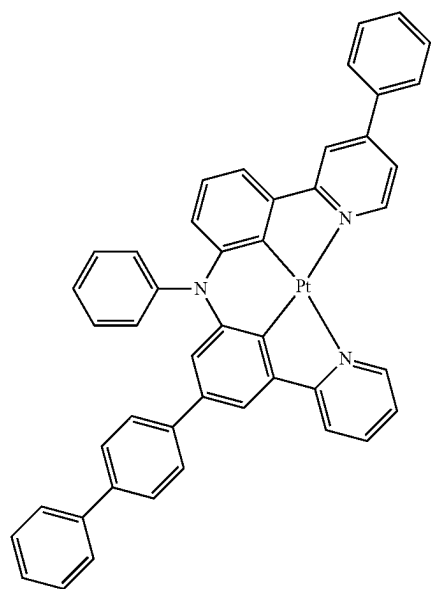
Compound 275

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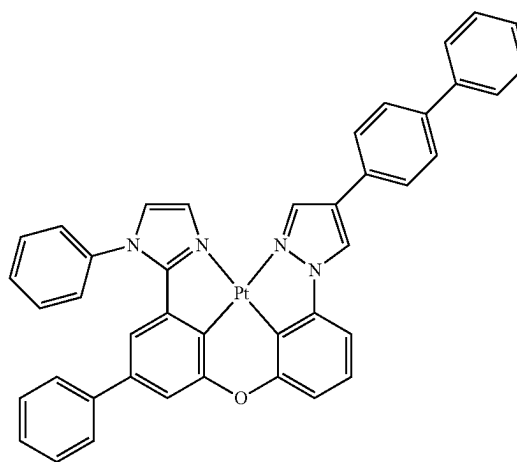


Compound 276

Compound 277



Compound 278



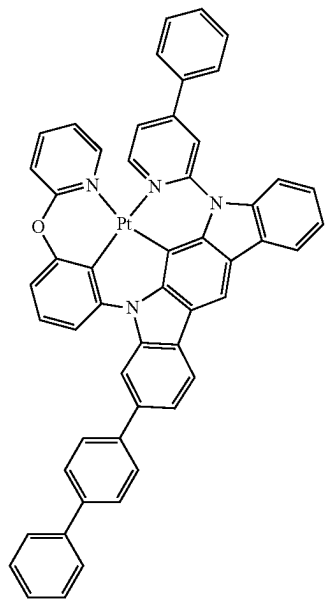
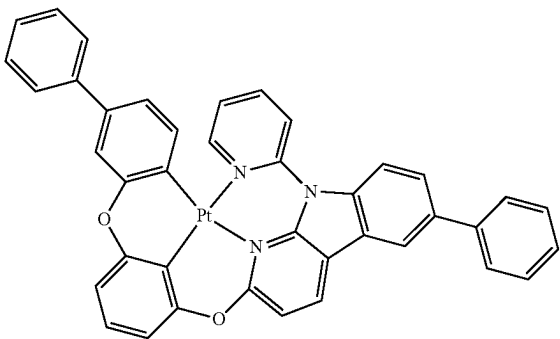
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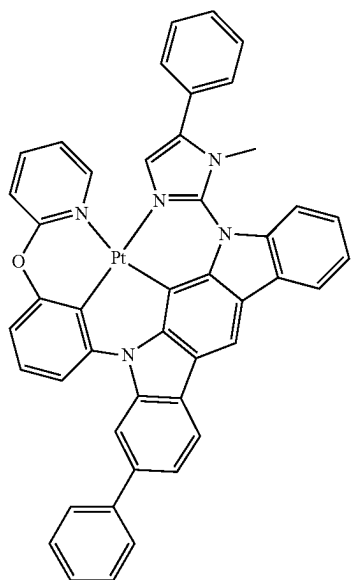
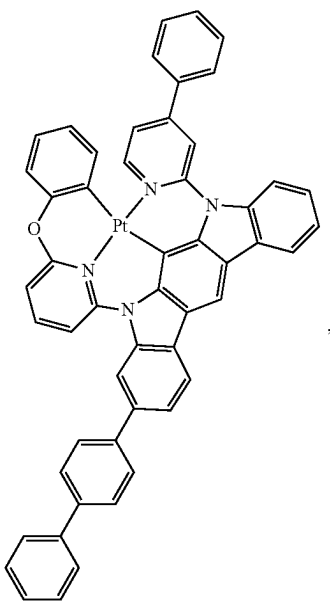
Compound 279

Compound 280



Compound 281

Compound 282



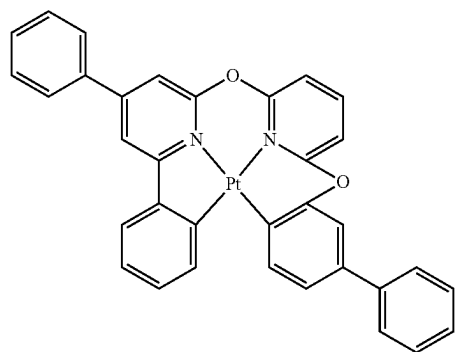
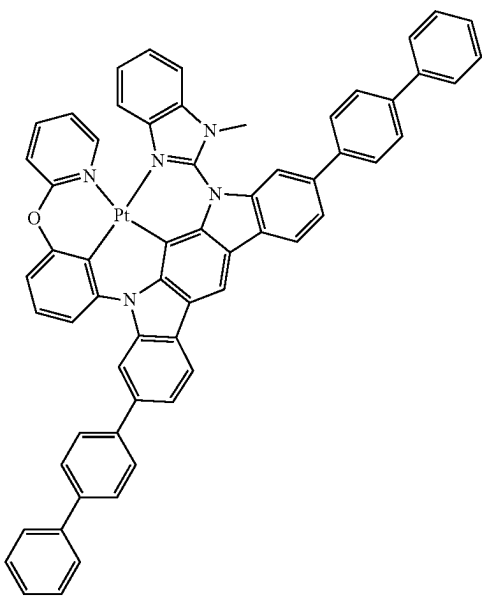
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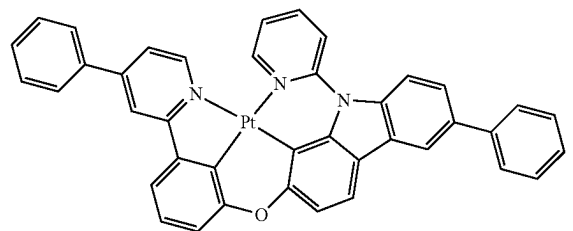
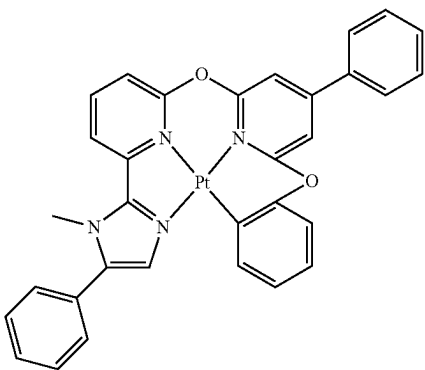
Compound 283

Compound 284



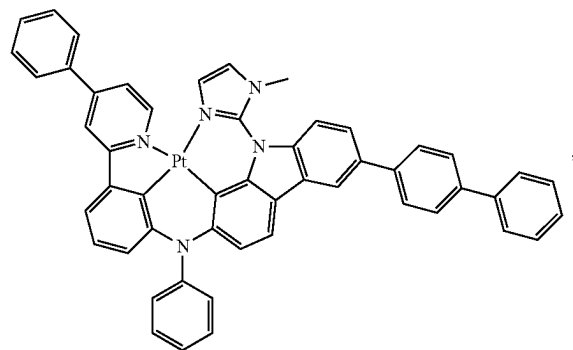
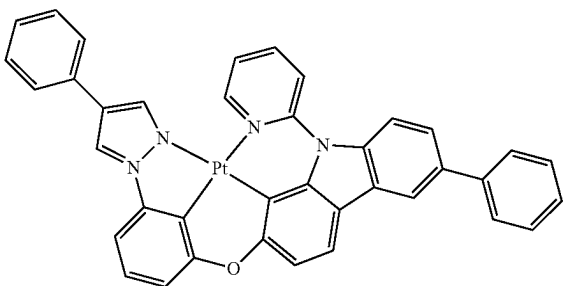
Compound 285

Compound 286



Compound 287

Compound 288



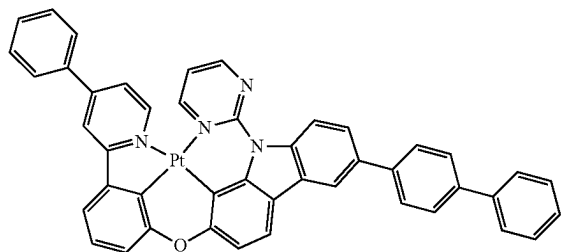
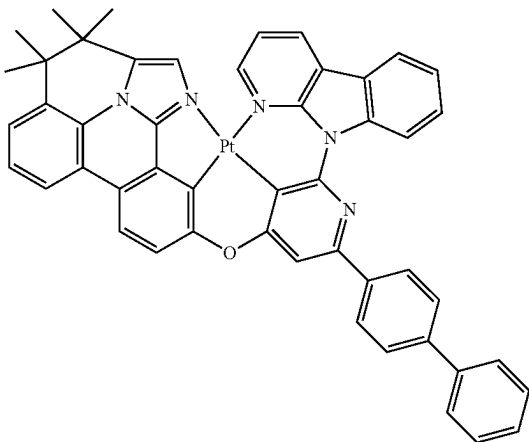
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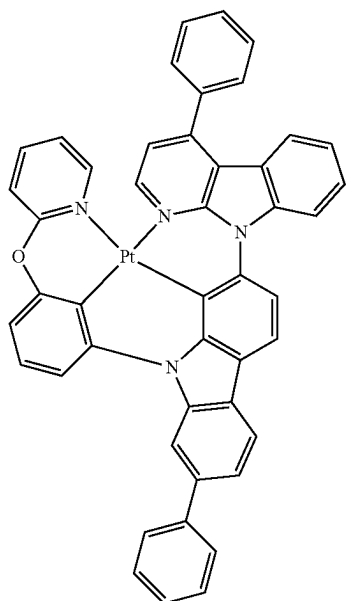
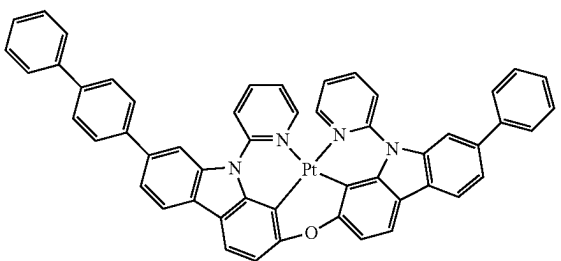
Compound 289

Compound 290

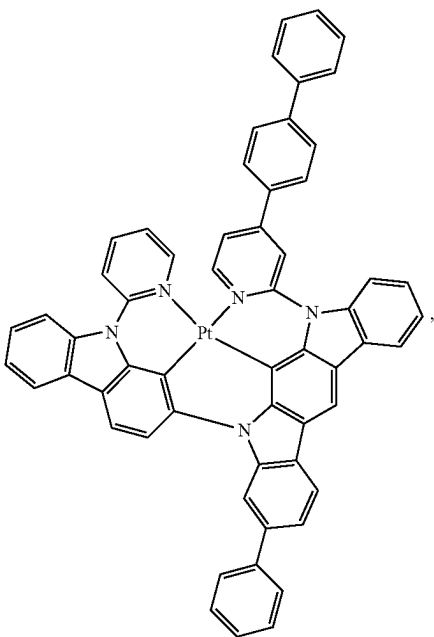


Compound 291

Compound 292



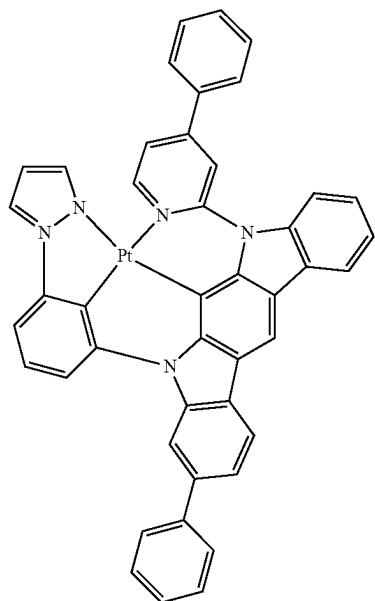
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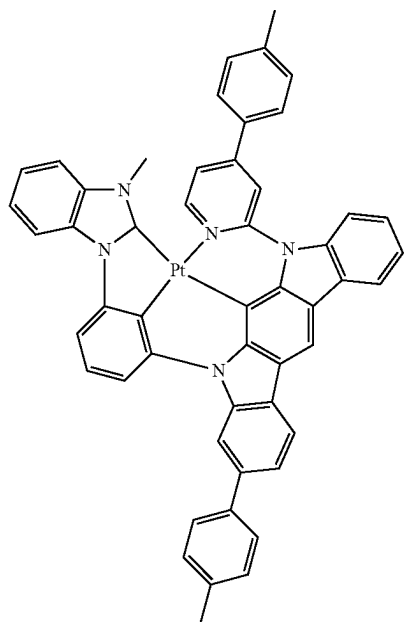
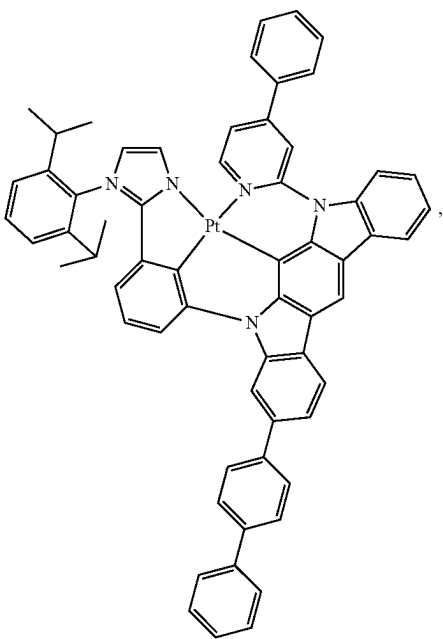
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Compound 293

Compound 294



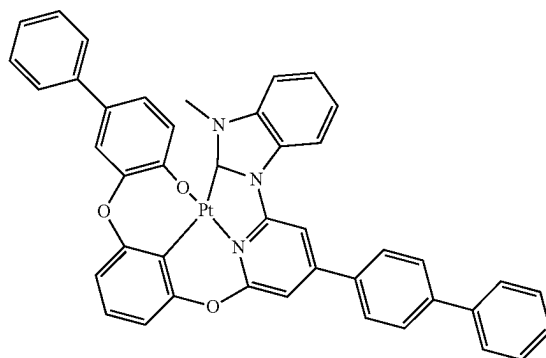
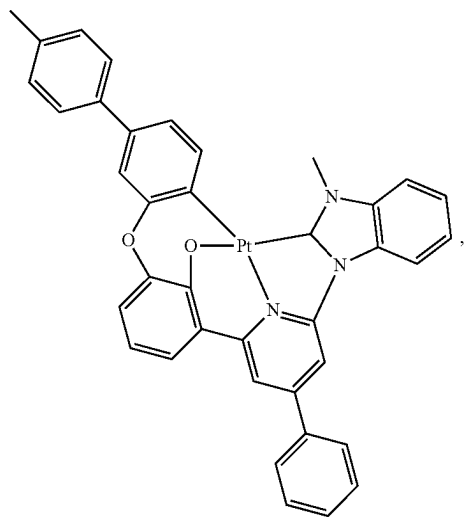
Compound 295

Compound 296

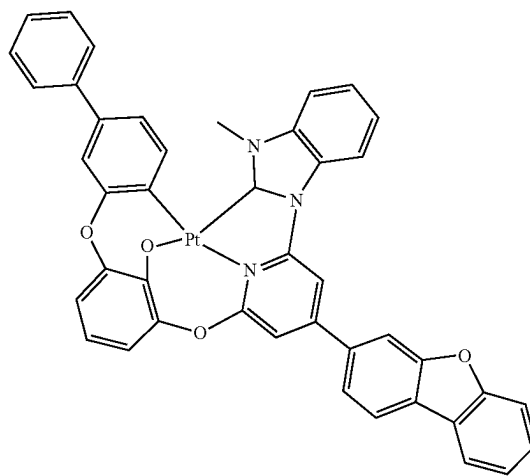
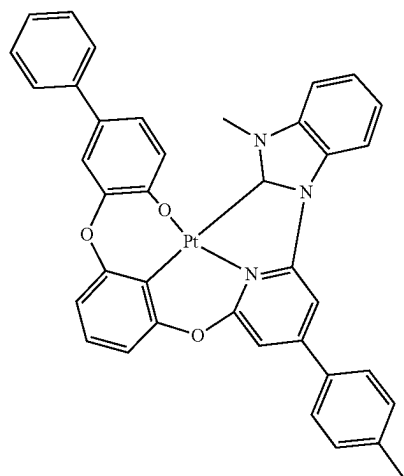


194

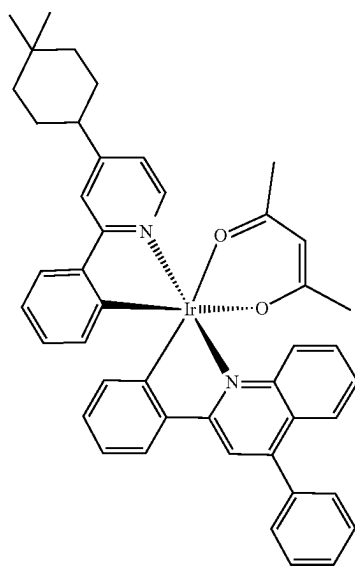
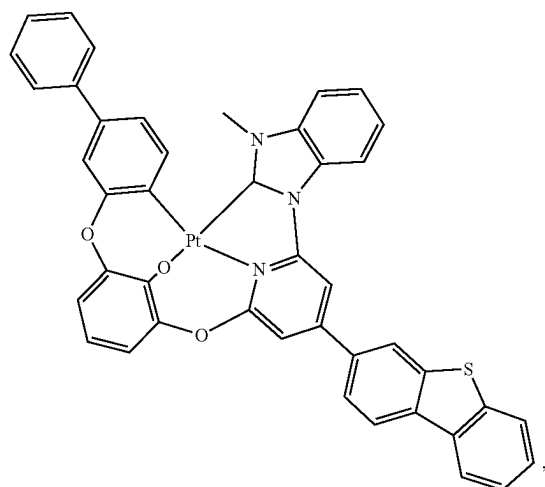
Compound 298



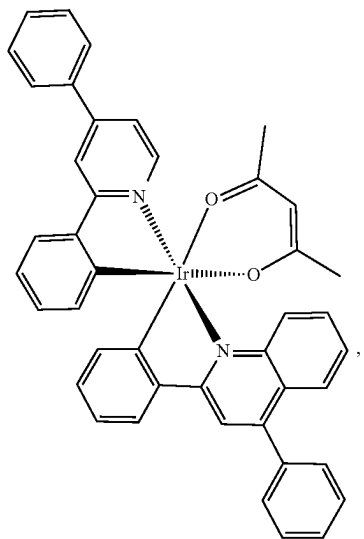
Compound 300



Compound 302



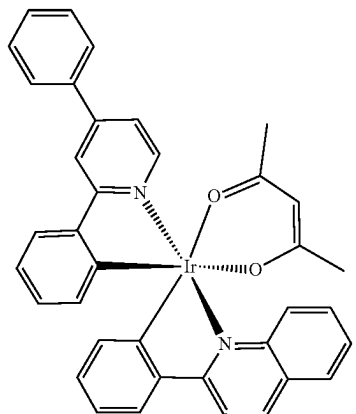
195



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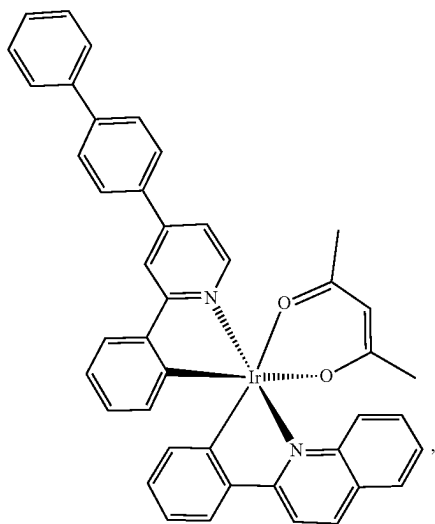
Compound 303

196

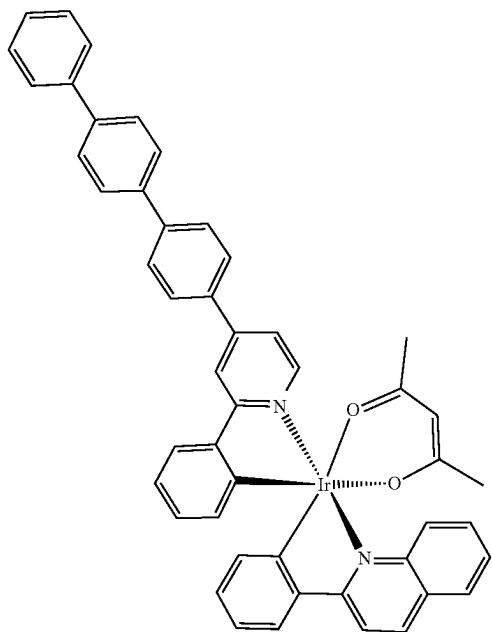


Compound 304

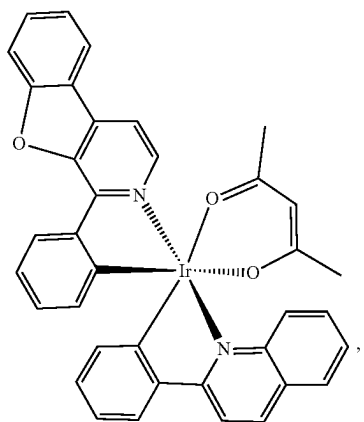
Compound 305



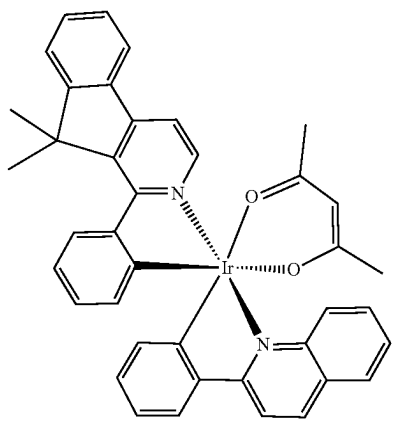
Compound 306



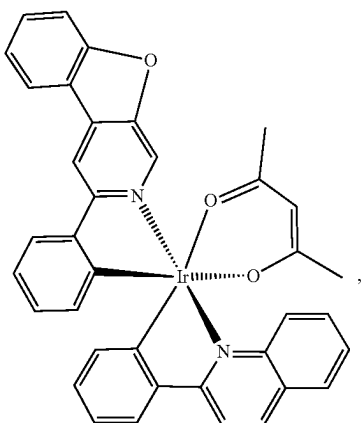
Compound 307



Compound 308



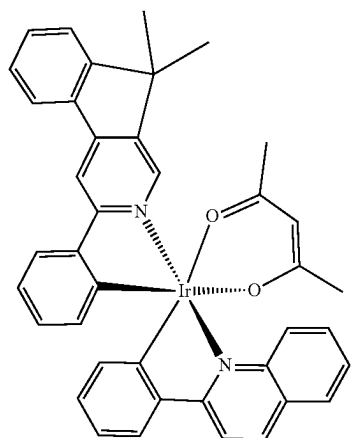
197



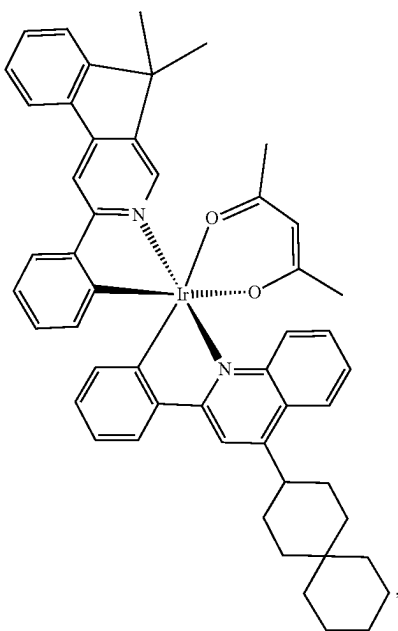
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Compound 309

198

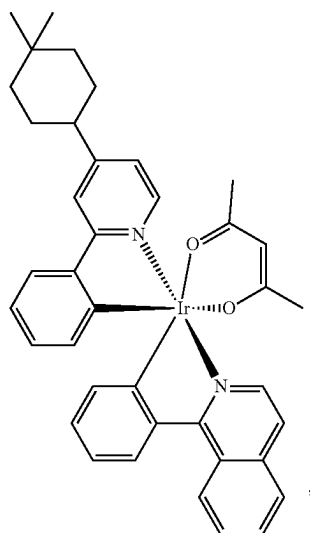


Compound 310



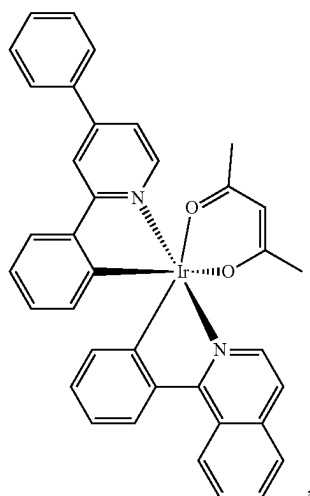
Compound 311

199



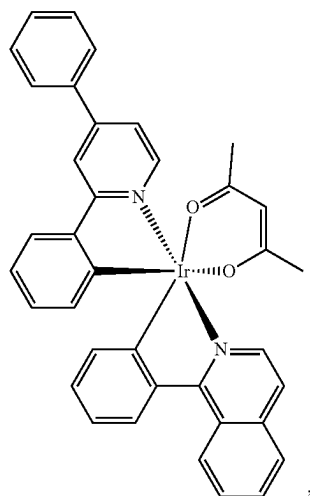
200

Compound 312

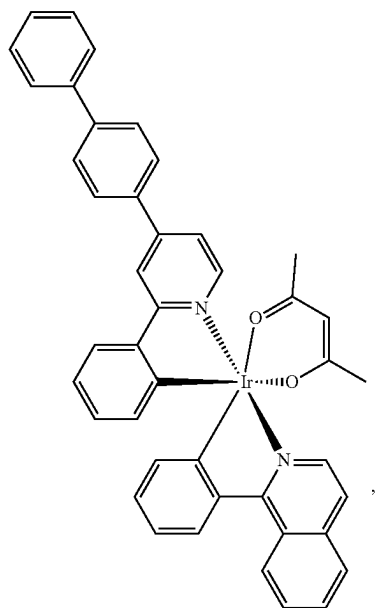


Compound 313

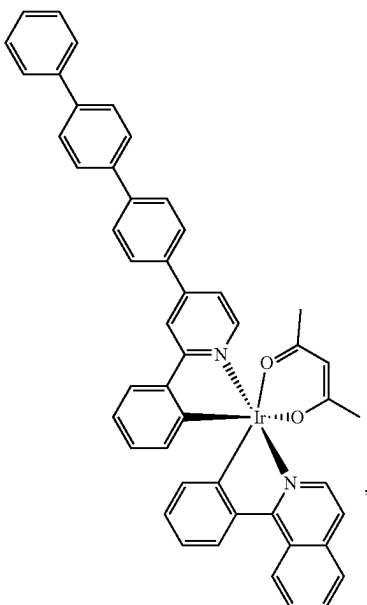
Compound 314



Compound 315



201

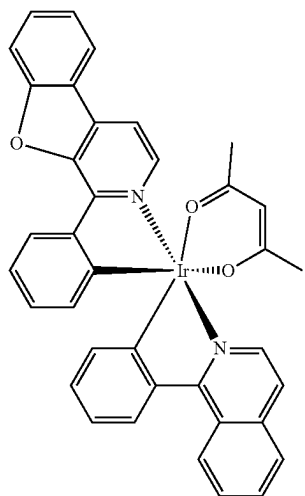


202

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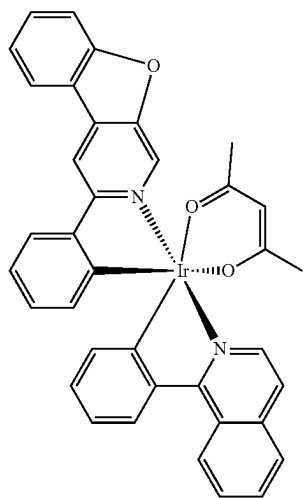
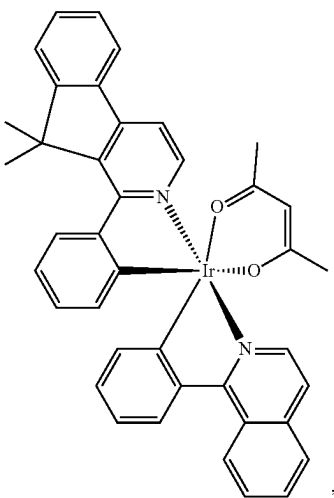
Compound 316

Compound 317

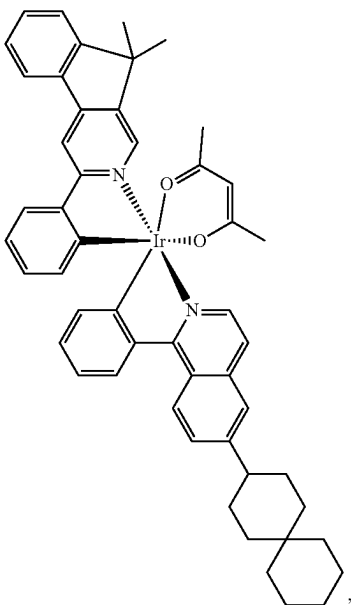


Compound 318

Compound 319

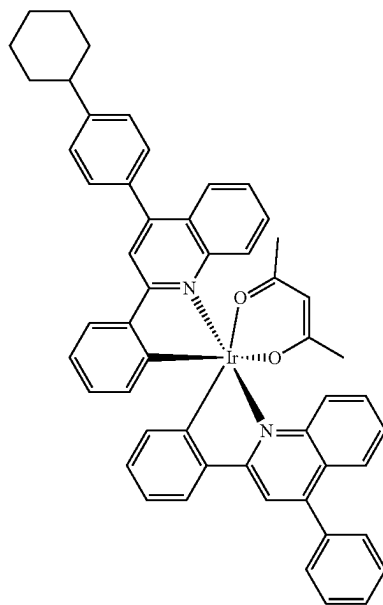


203



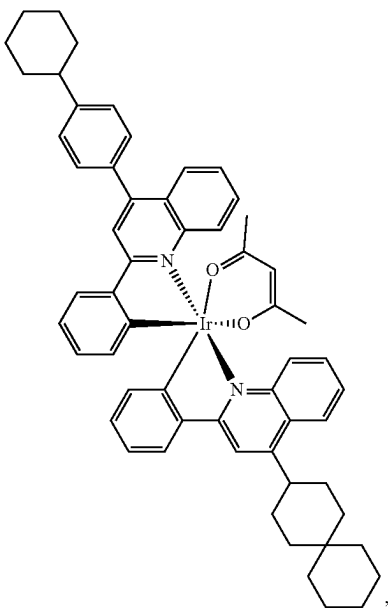
204

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Compound 320

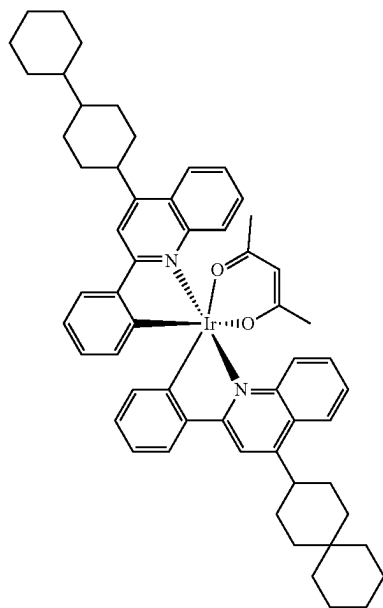


Compound 321

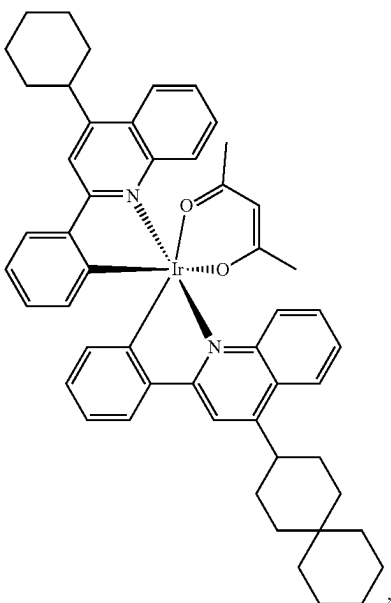
Compound 322



Compound 323

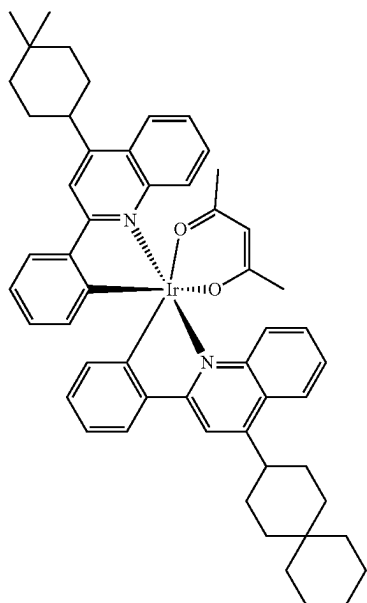


205



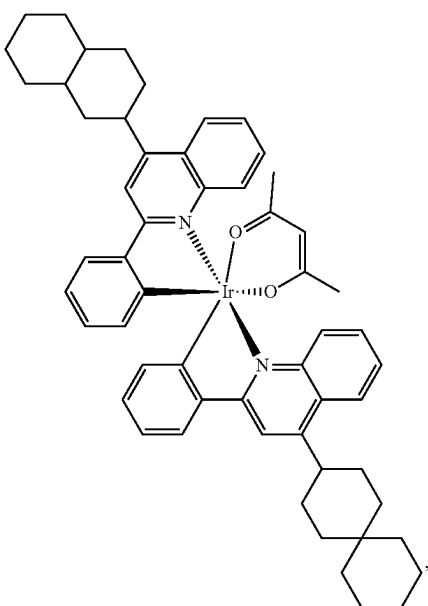
206

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Compound 324

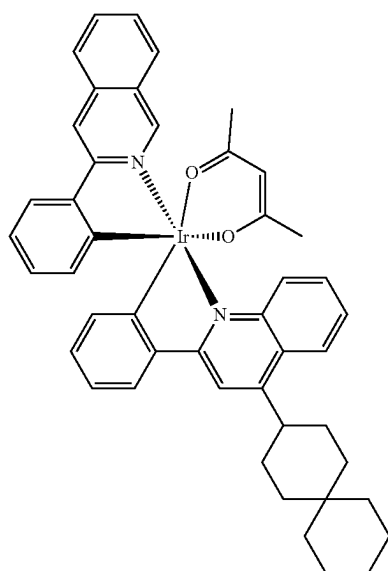


Compound 325

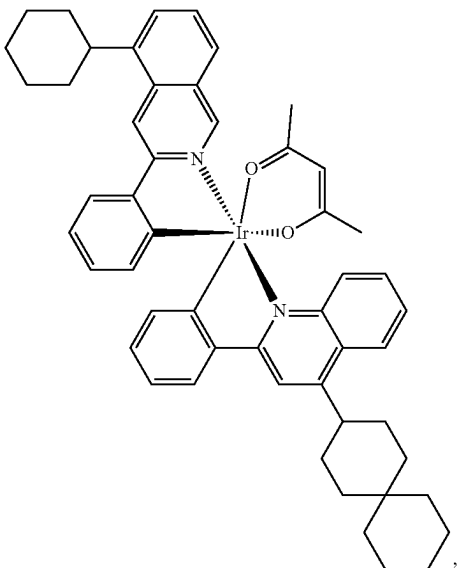
Compound 326



Compound 327



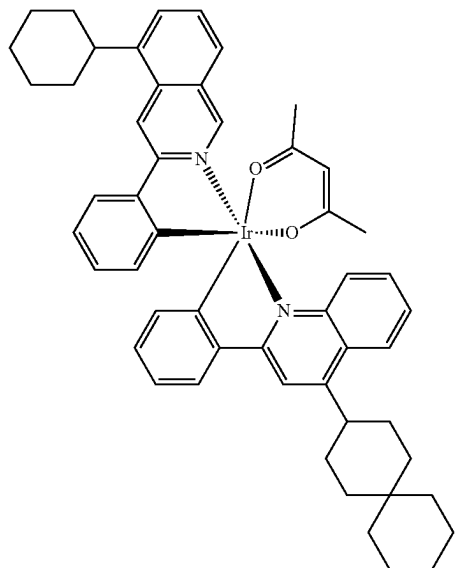
207



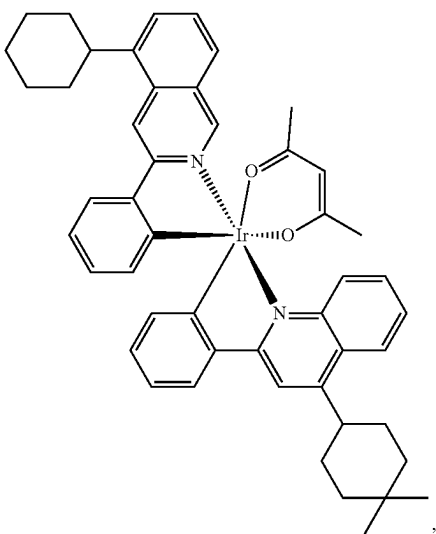
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Compound 328

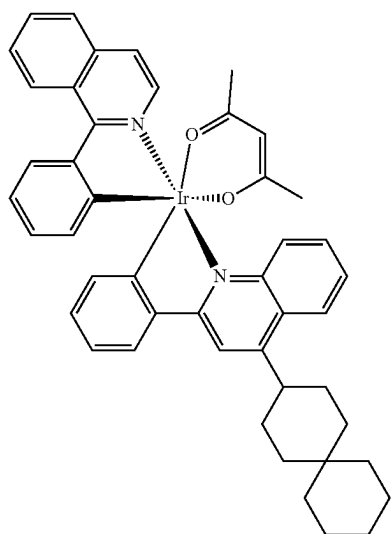
208



Compound 329



Compound 330



Compound 331

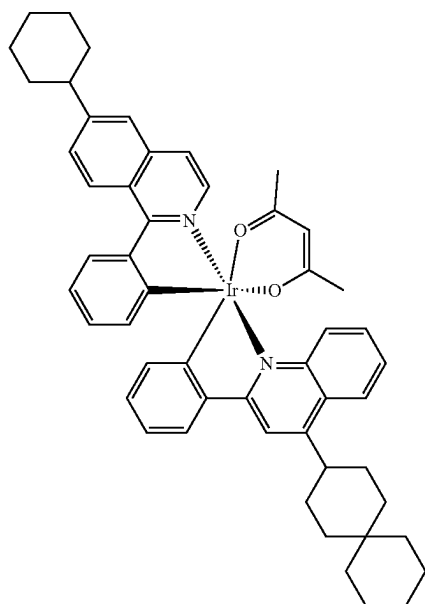
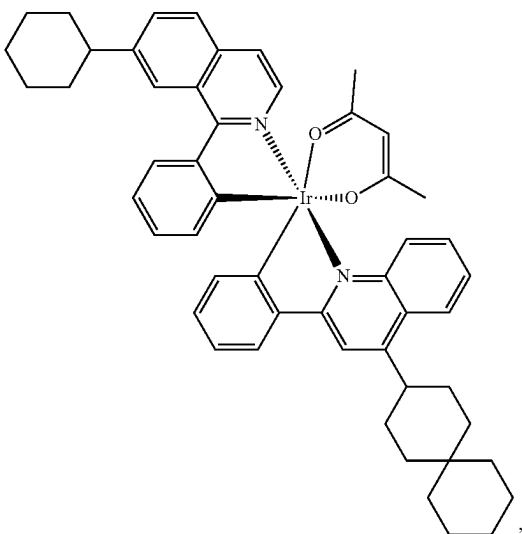
209

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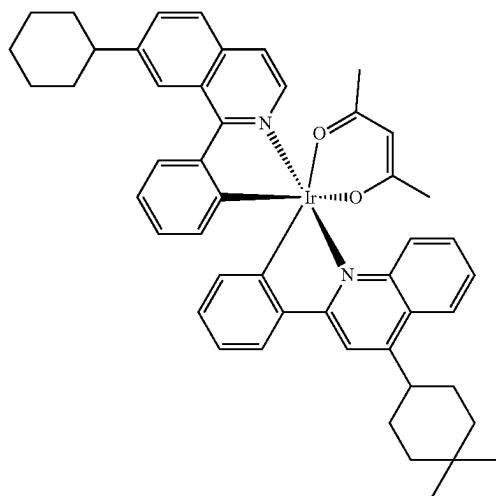
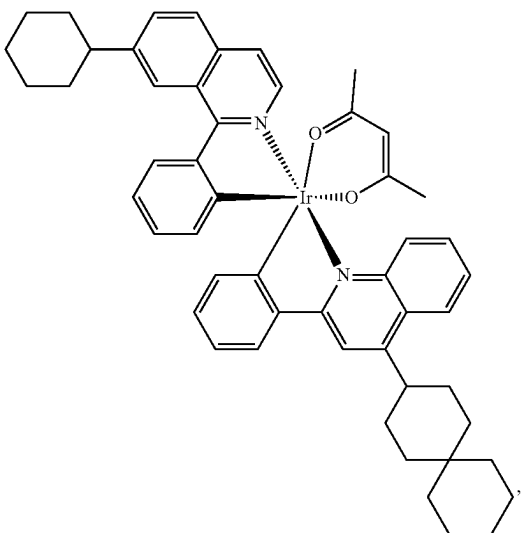
Compound 332

Compound 333



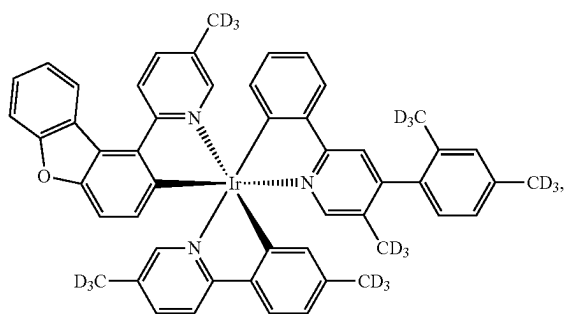
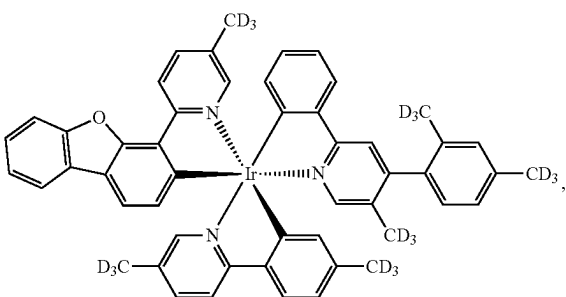
Compound 334

Compound 335



Compound 336

Compound 337



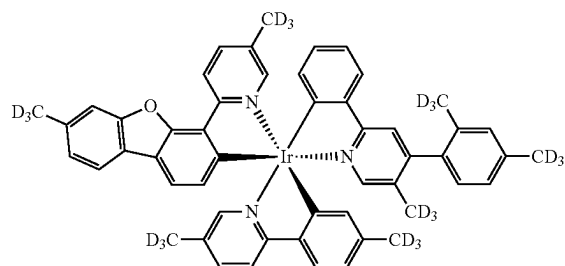
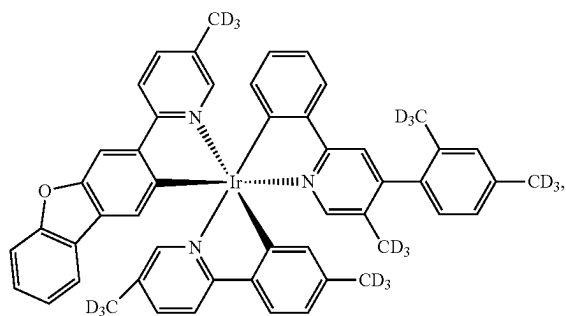
211

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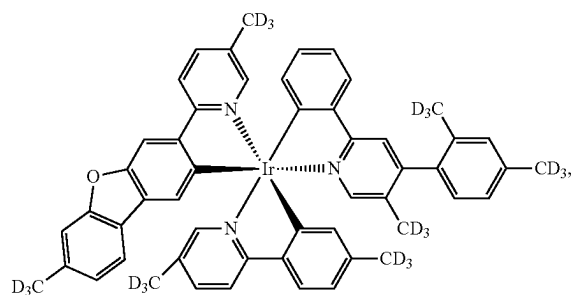
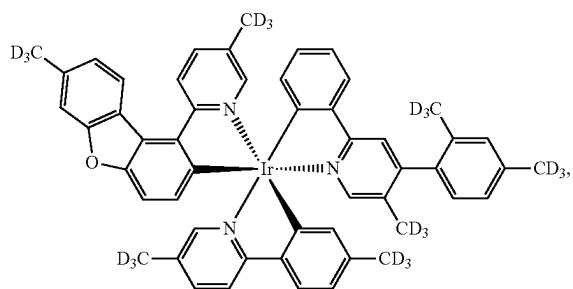
Compound 338

Compound 339



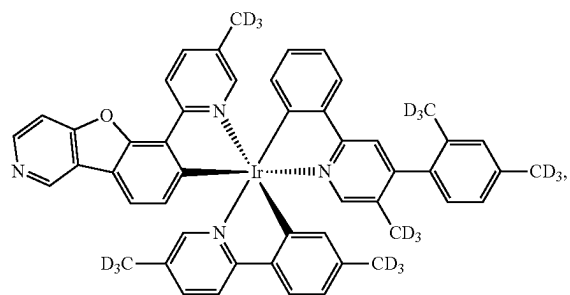
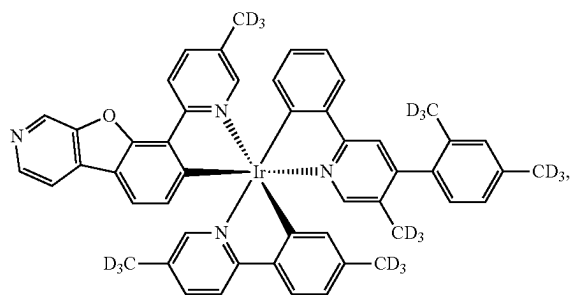
Compound 340

Compound 341



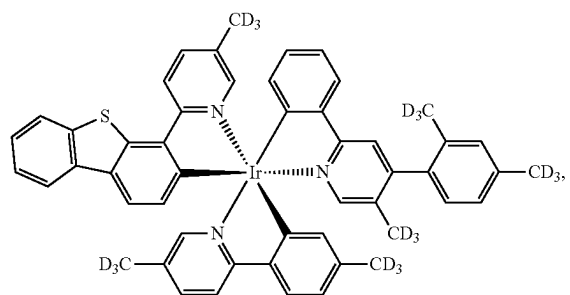
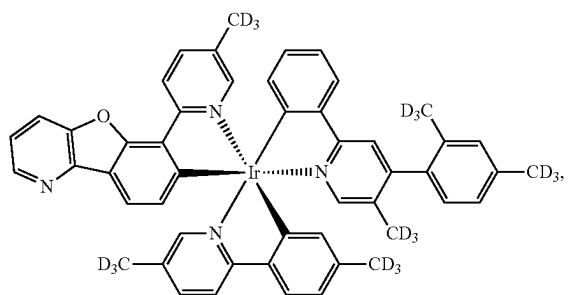
Compound 342

Compound 343



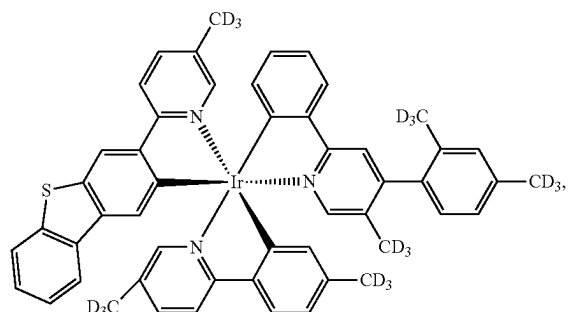
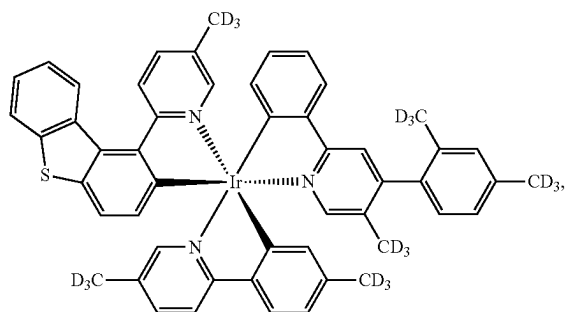
Compound 344

Compound 345



Compound 346

Compound 347



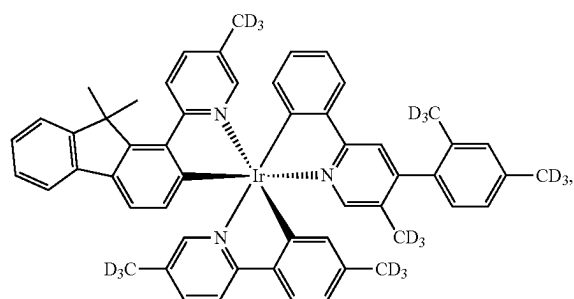
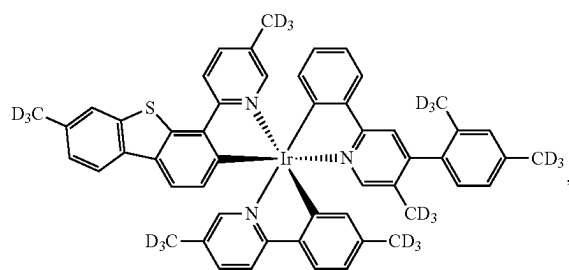
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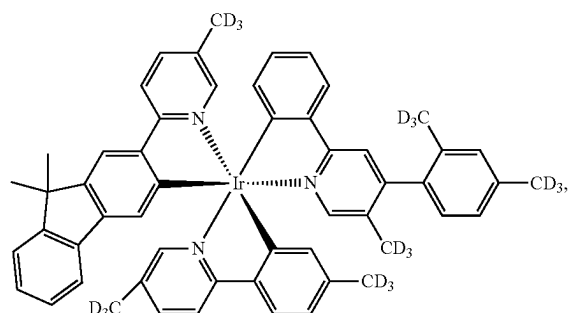
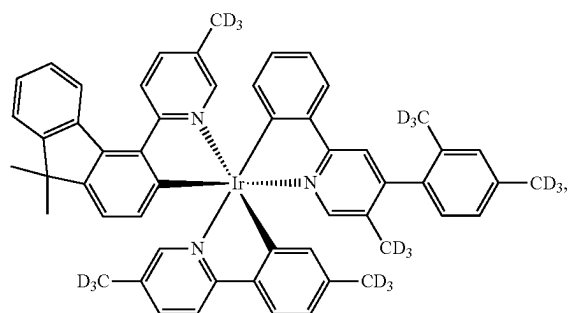
Compound 348

Compound 349



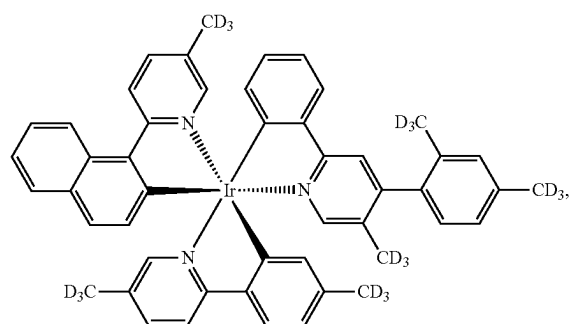
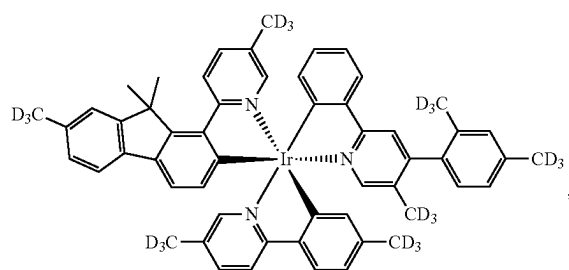
Compound 350

Compound 351



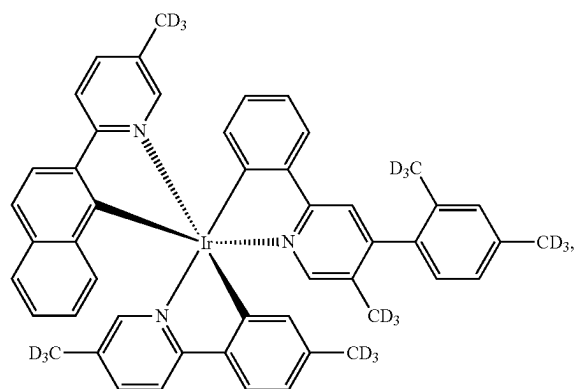
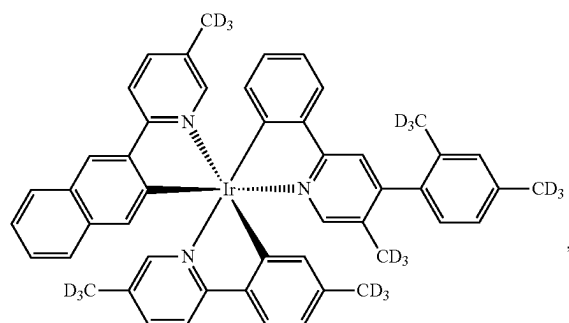
Compound 352

Compound 353



Compound 354

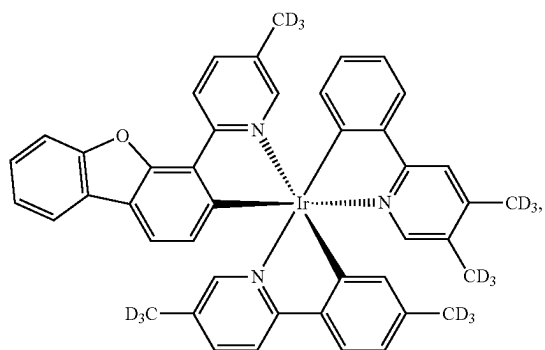
Compound 355



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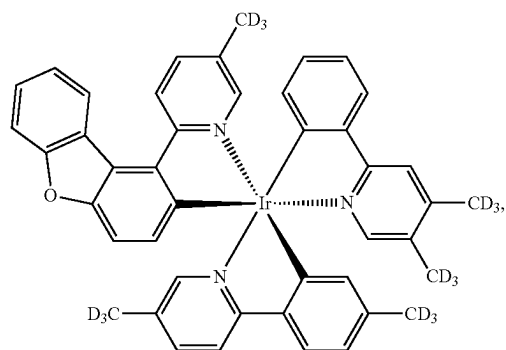
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Compound 356



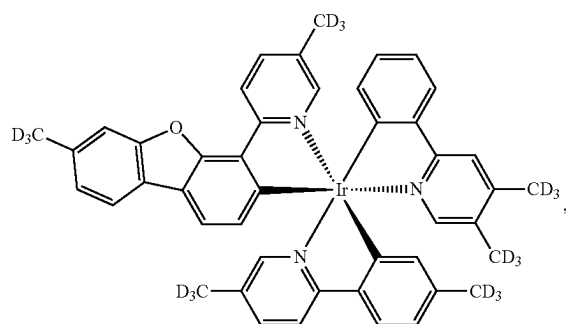
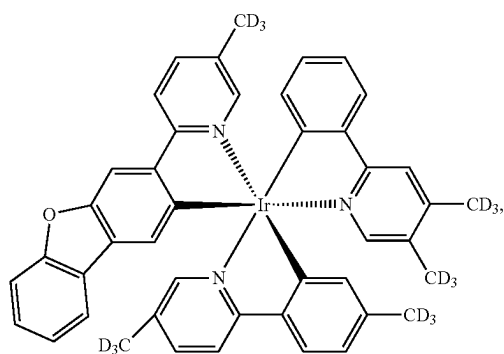
216

Compound 357



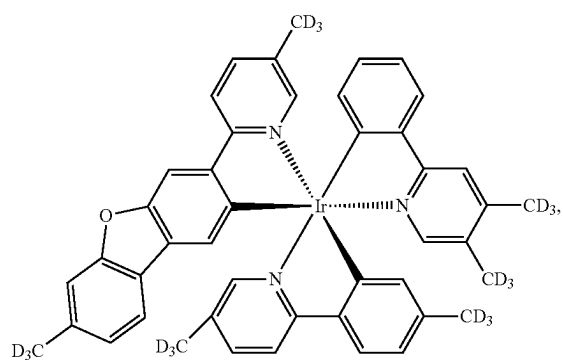
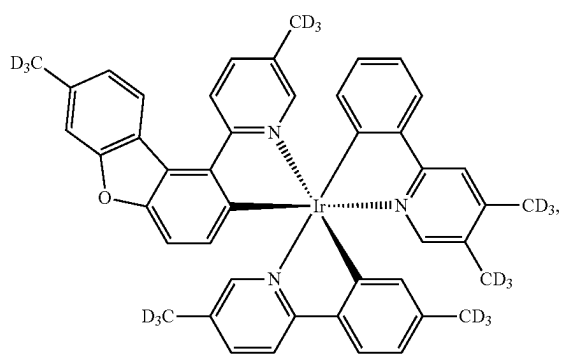
Compound 358

Compound 359



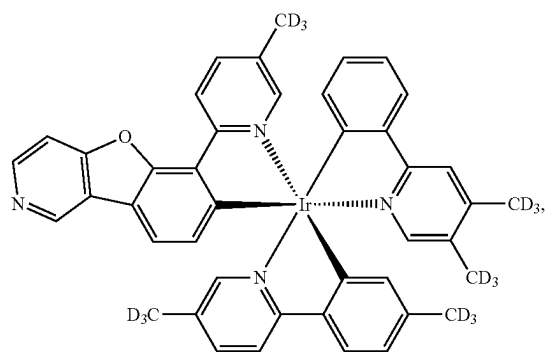
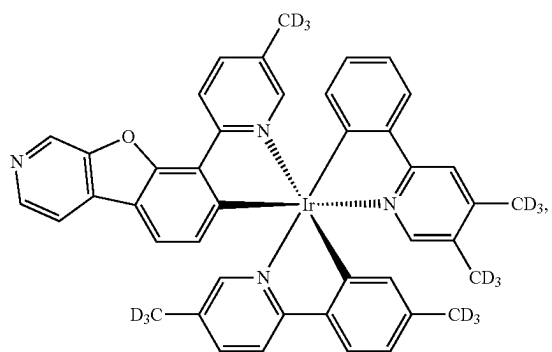
Compound 360

Compound 361



Compound 362

Compound 363



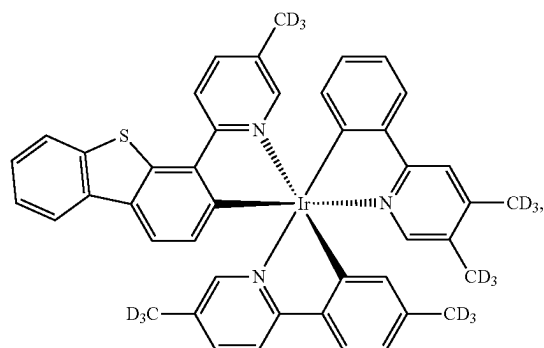
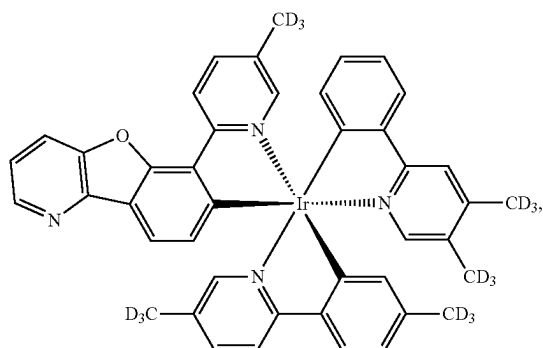
217

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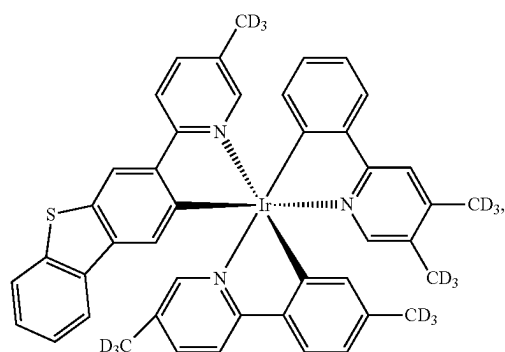
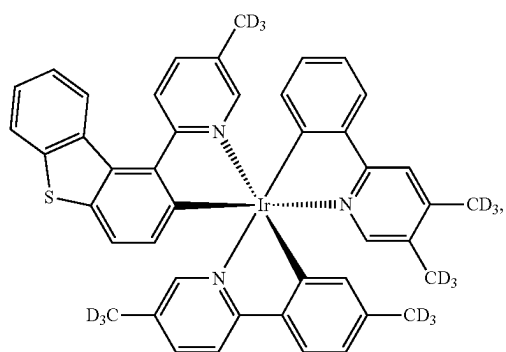
Compound 364

Compound 365



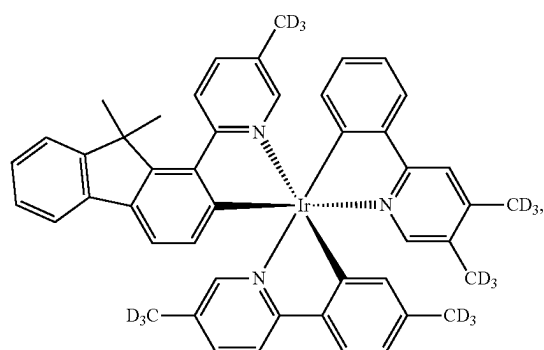
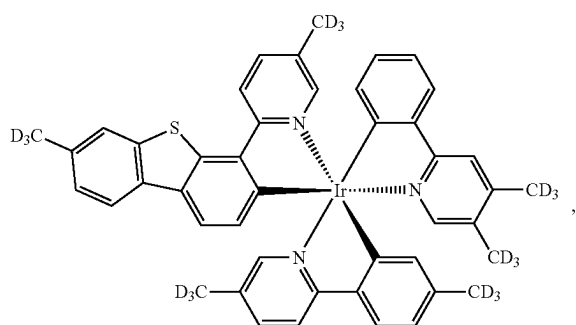
Compound 366

Compound 367



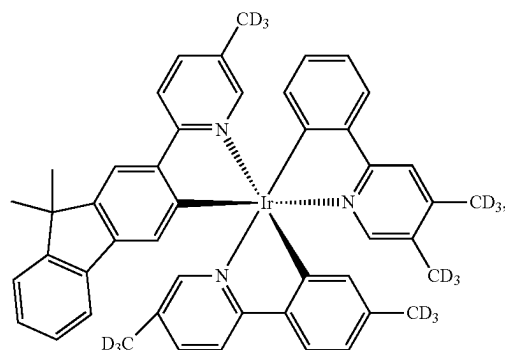
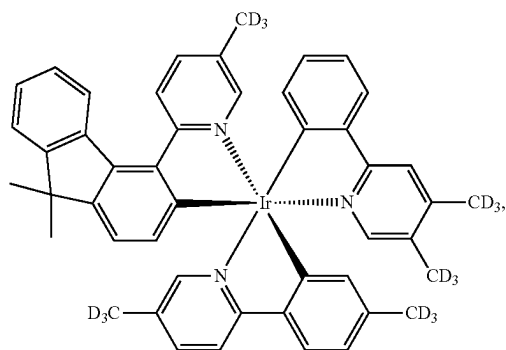
Compound 368

Compound 369



Compound 370

Compound 371



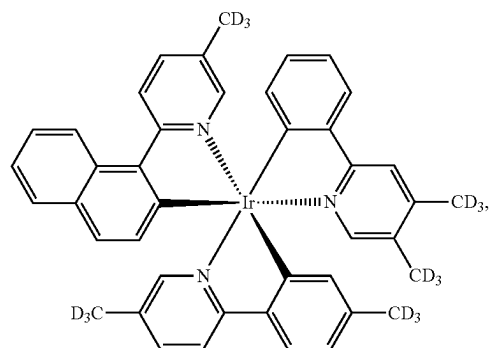
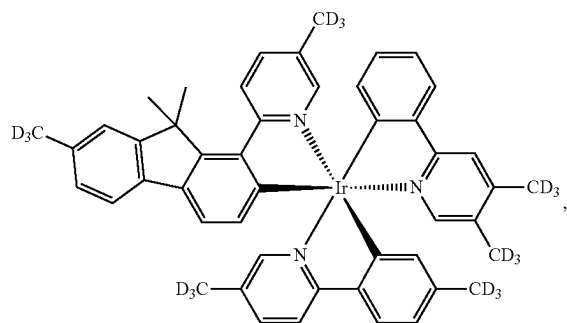
219

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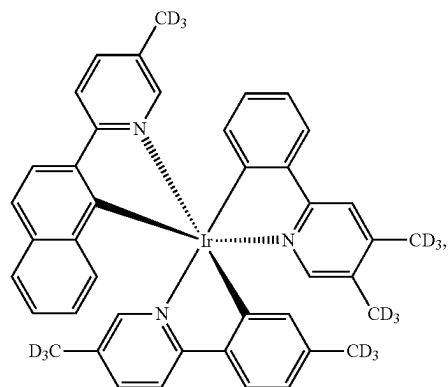
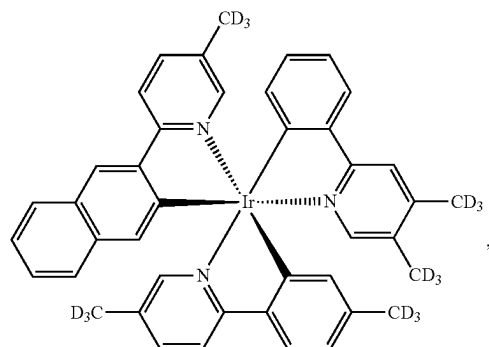
Compound 372

Compound 373



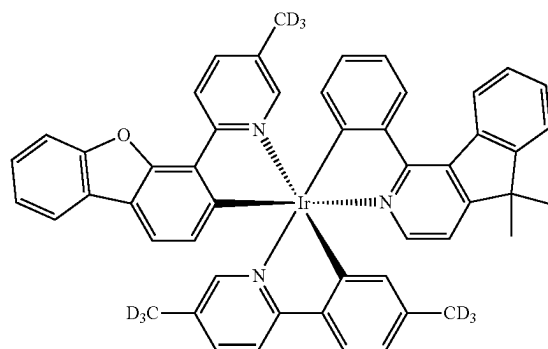
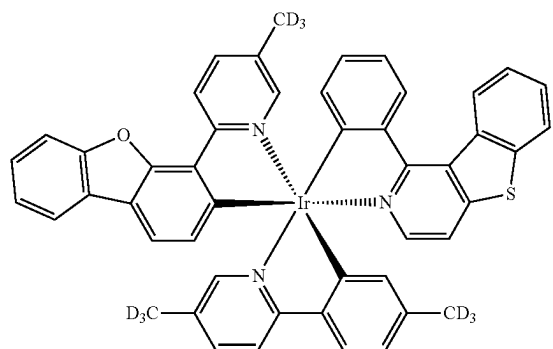
Compound 374

Compound 375



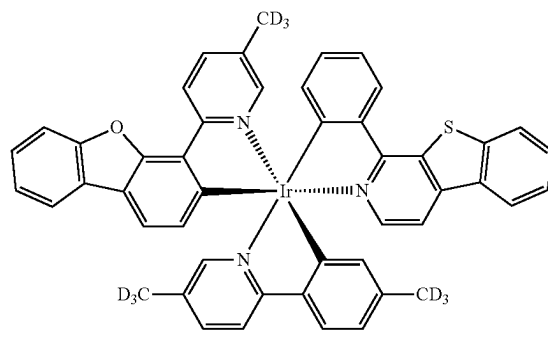
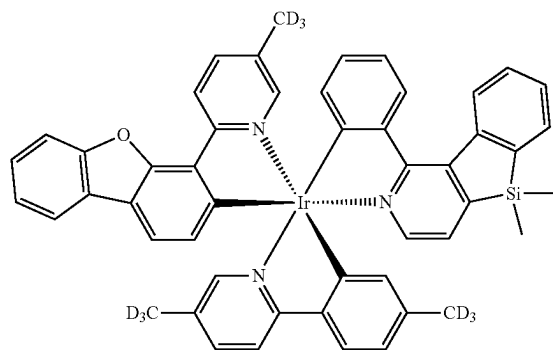
Compound 376

Compound 377



Compound 378

Compound 379



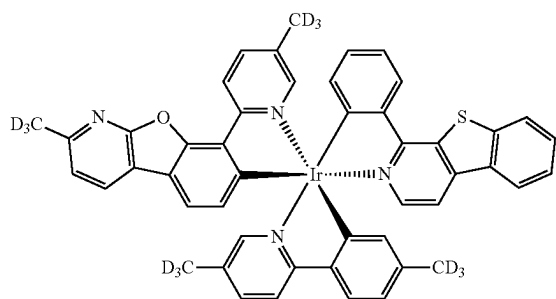
221

222

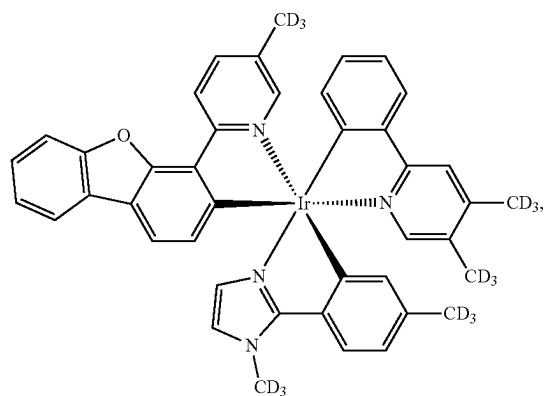
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Compound 380

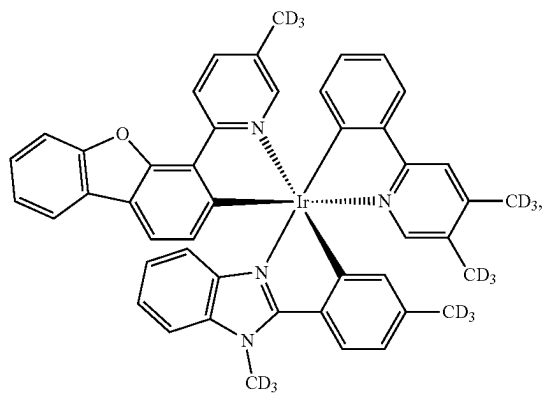
Compound 381



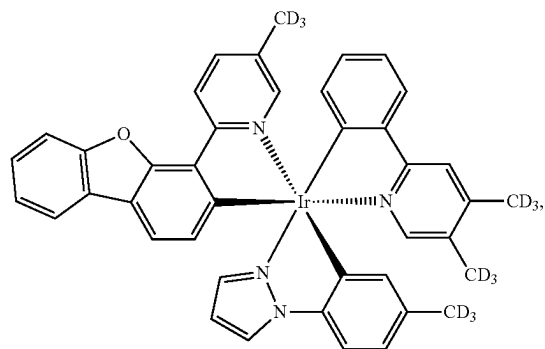
Compound 382



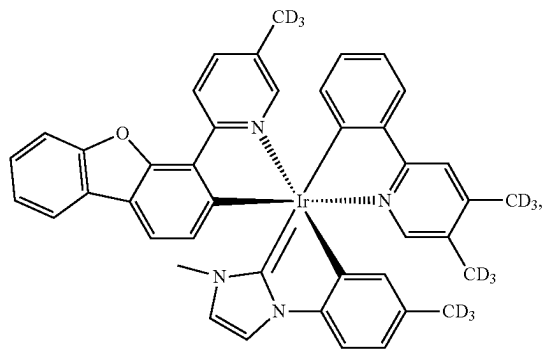
Compound 383



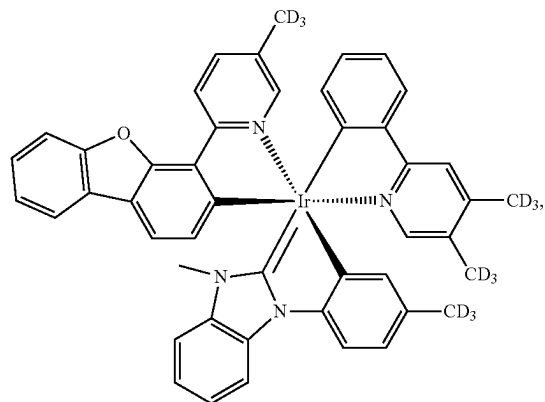
Compound 384



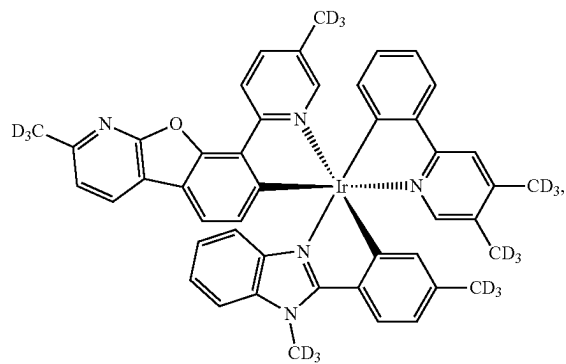
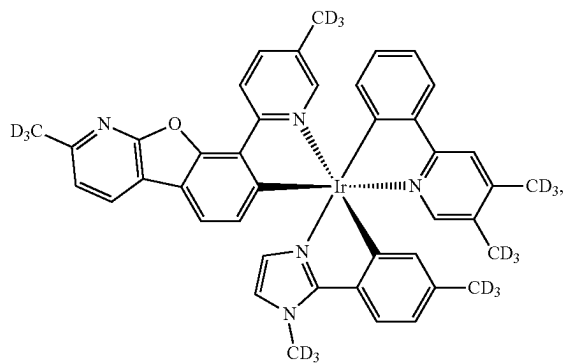
Compound 385



Compound 386



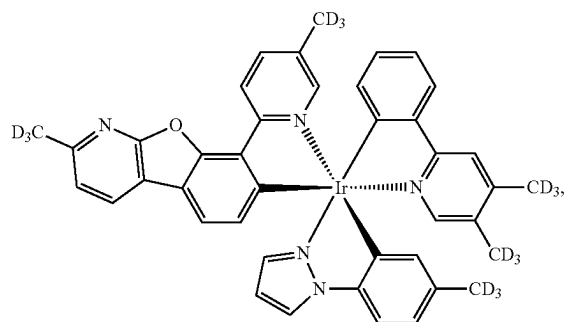
Compound 387



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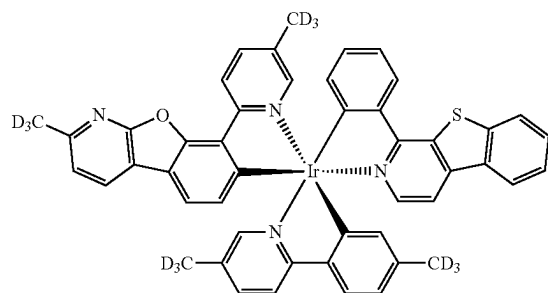
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Compound 388

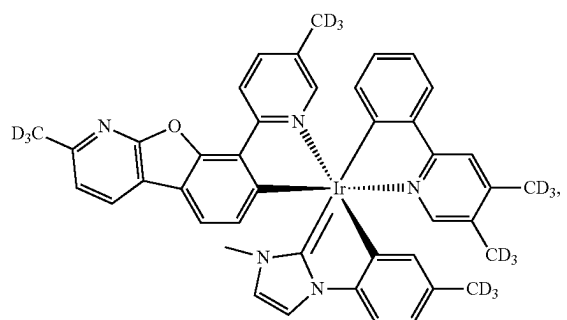


224

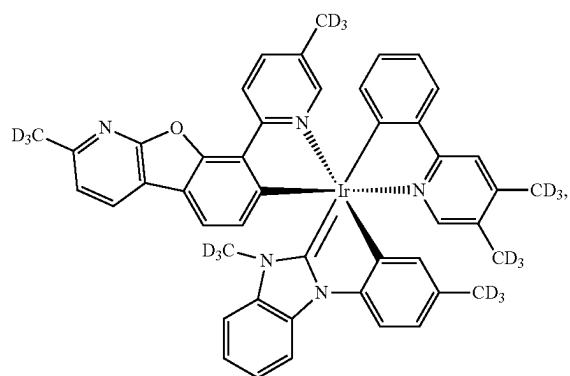
Compound 389



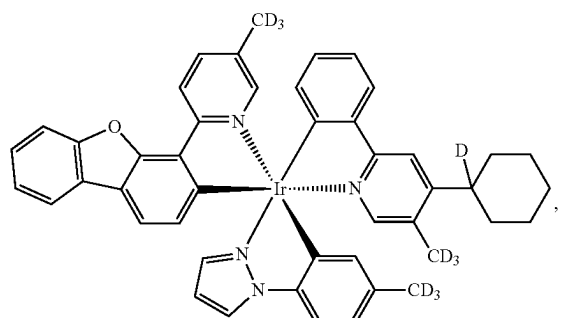
Compound 390



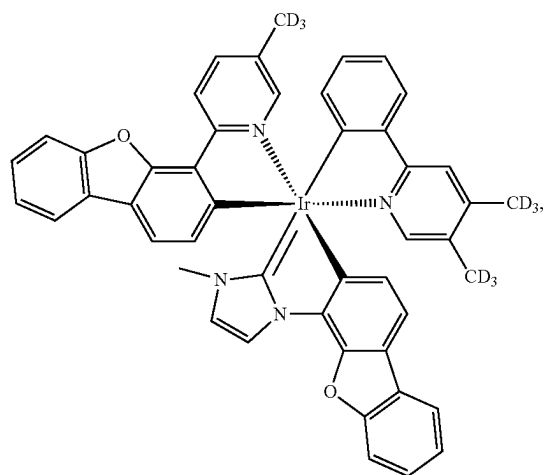
Compound 391



Compound 392

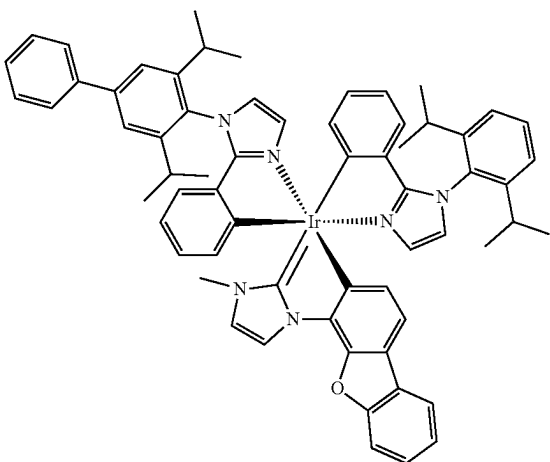


Compound 393



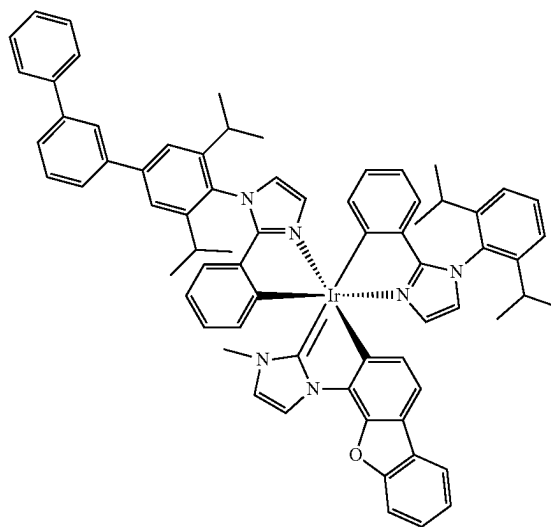
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Compound 394

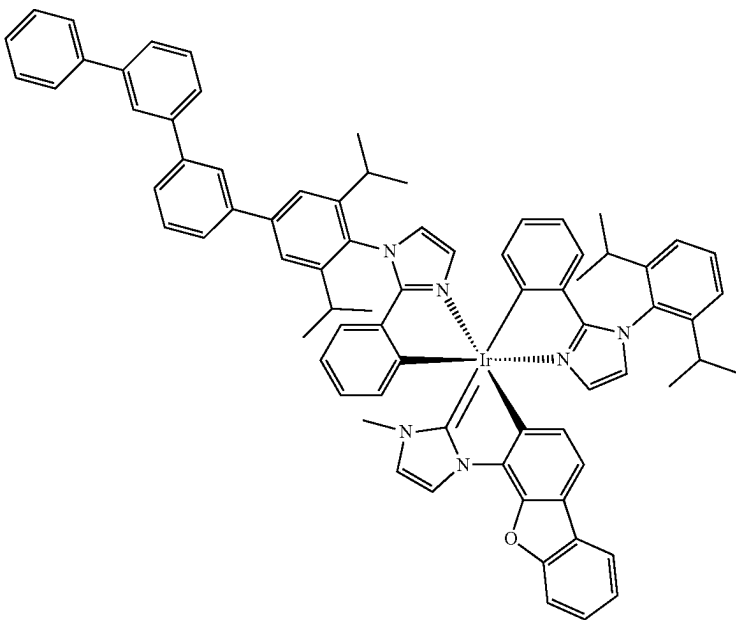


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Compound 395



Compound 396

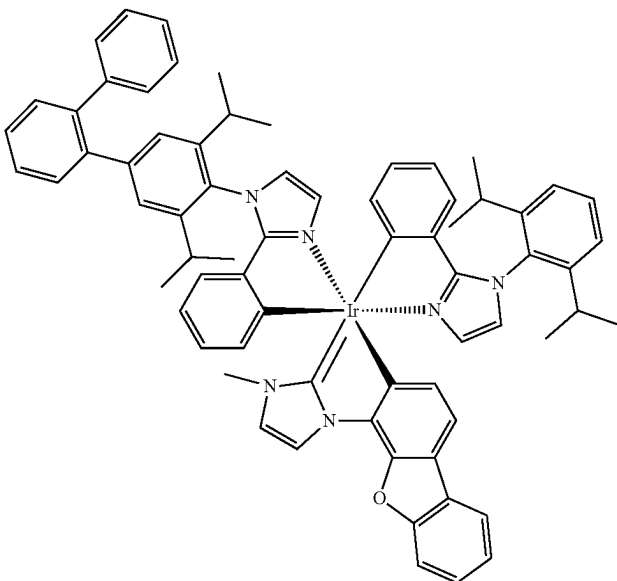


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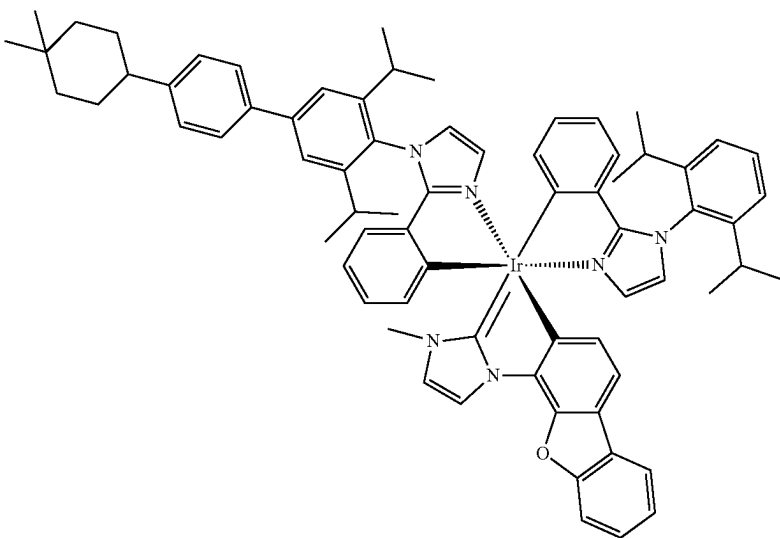
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Compound 397



Compound 398

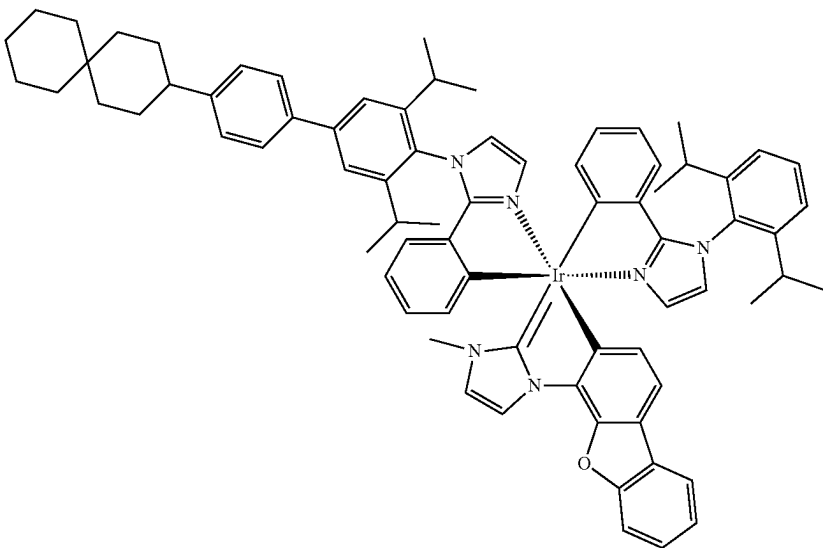


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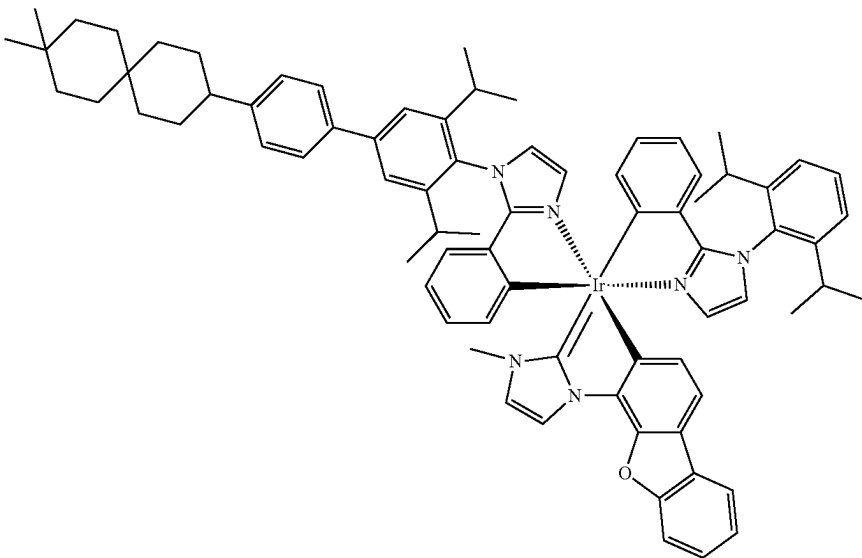
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Compound 399



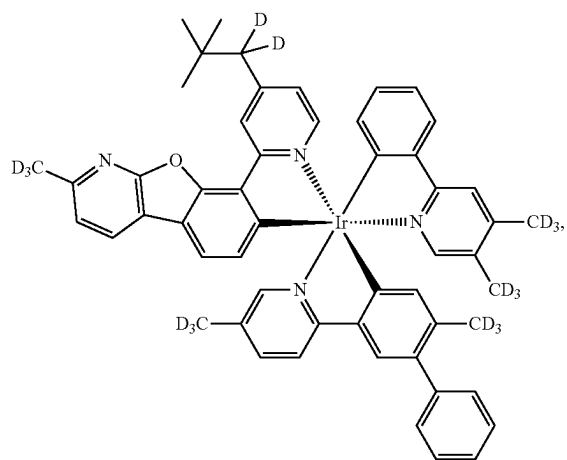
Compound 400



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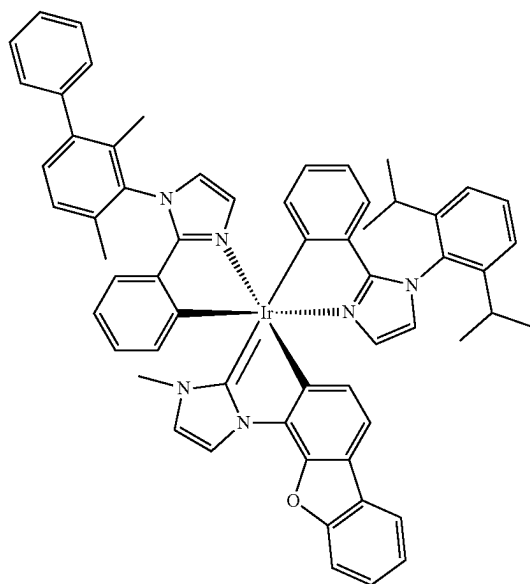
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Compound 401

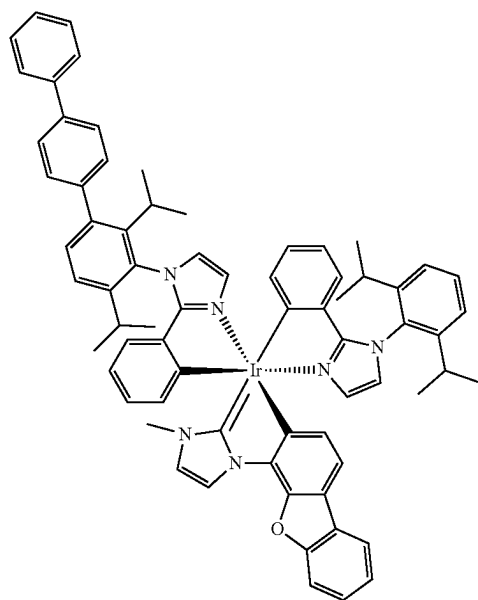


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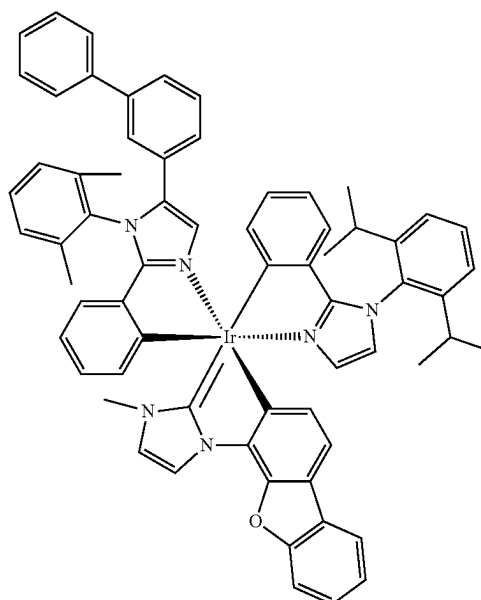
Compound 402



Compound 403



Compound 404

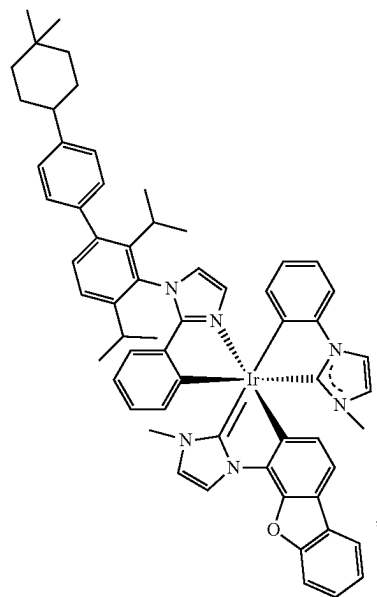
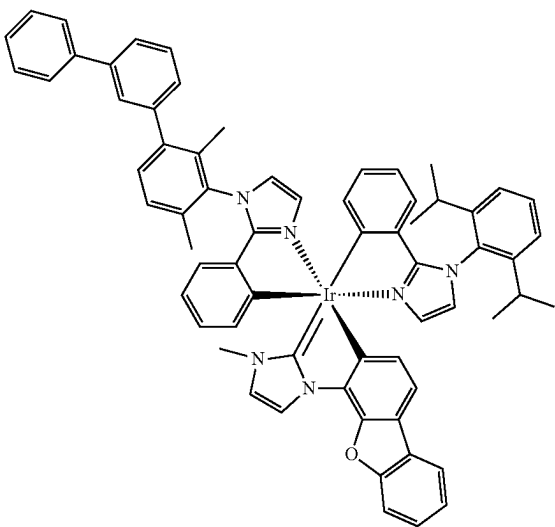


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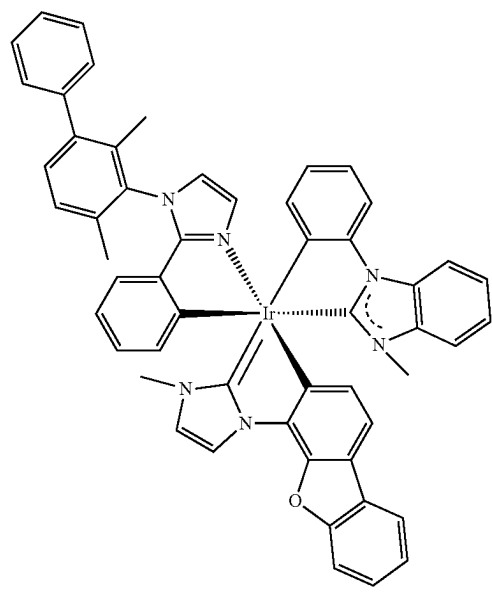
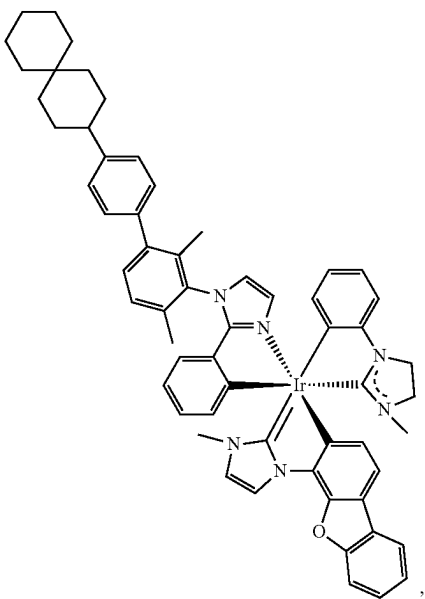
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Compound 405

Compound 407

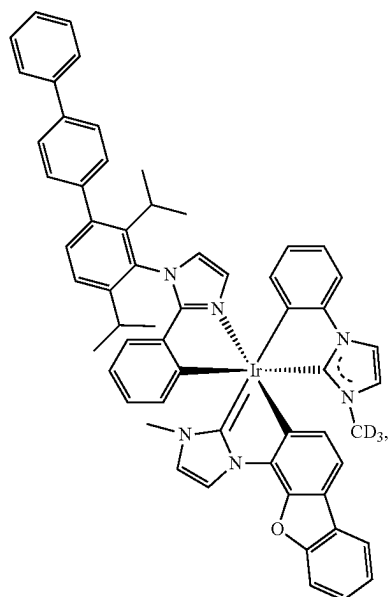


Compound 408

Compound 409



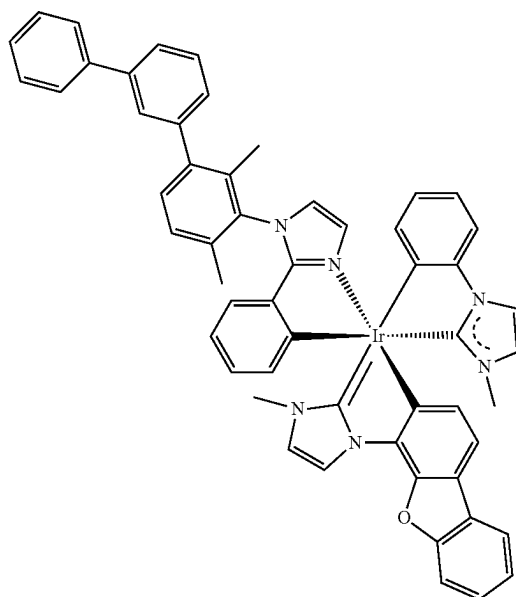
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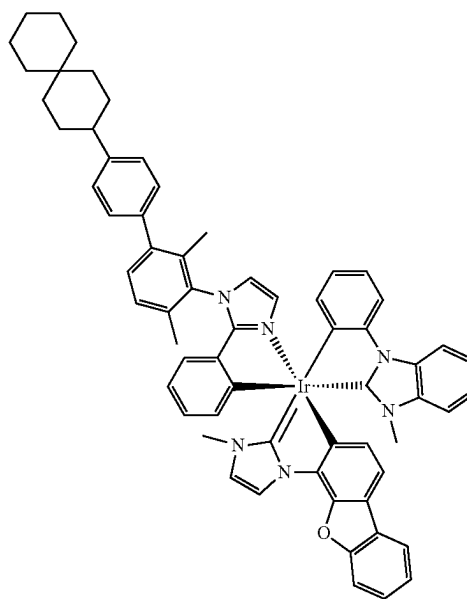
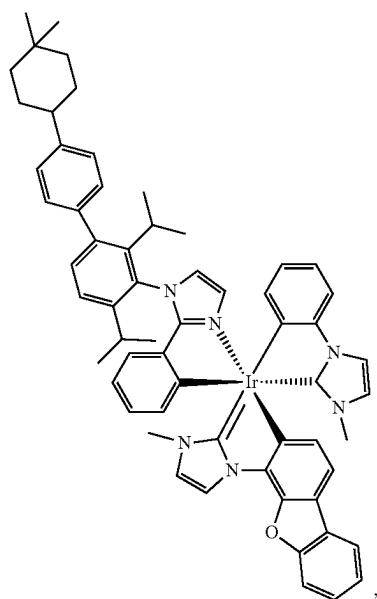
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Compound 410

Compound 411

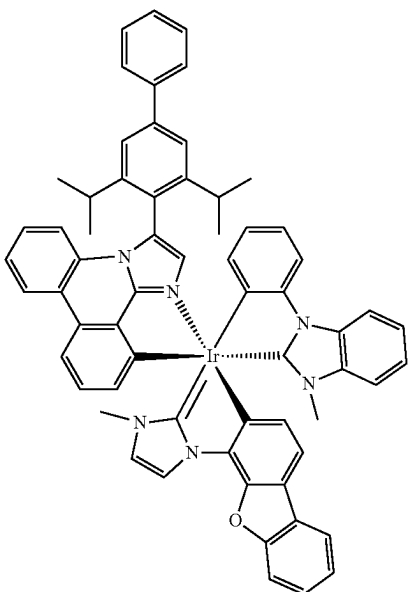


Compound 412

Compound 413



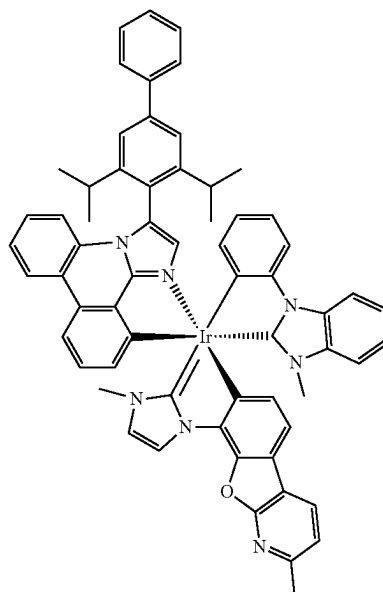
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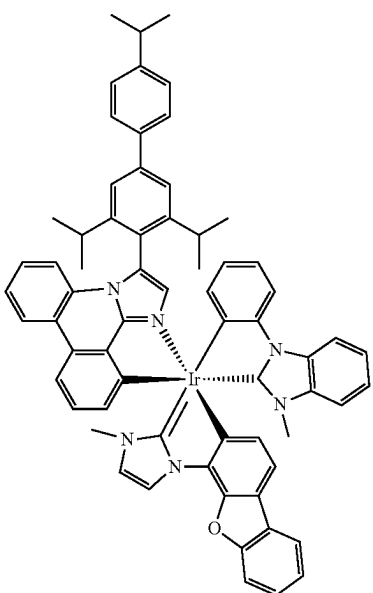
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Compound 414

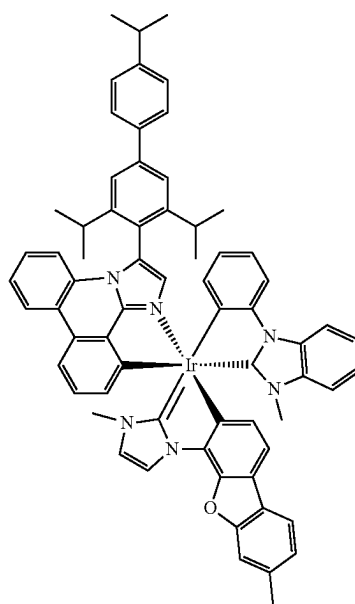
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Compound 415



Compound 416



Compound 417

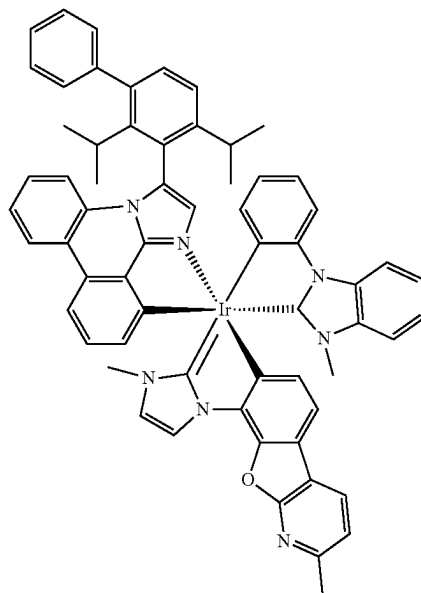
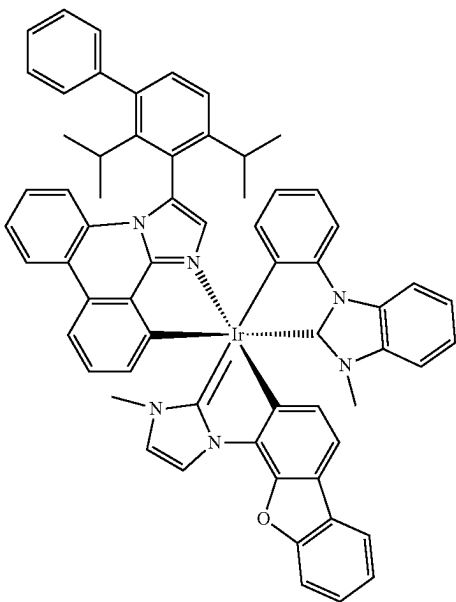
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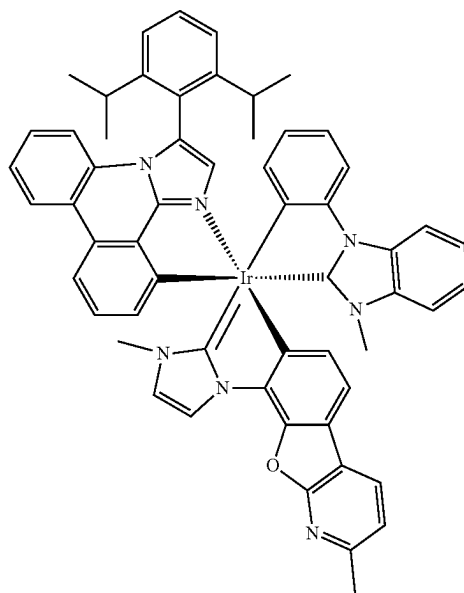
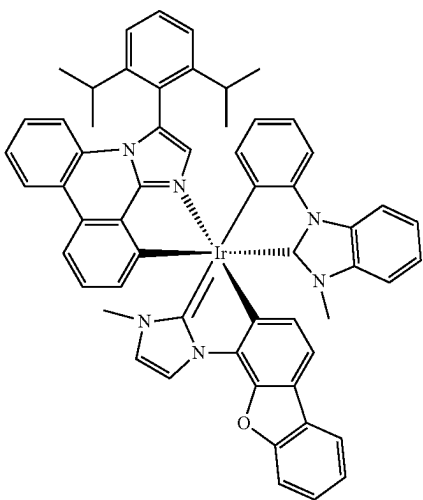
Compound 418

Compound 419



Compound 420

Compound 421

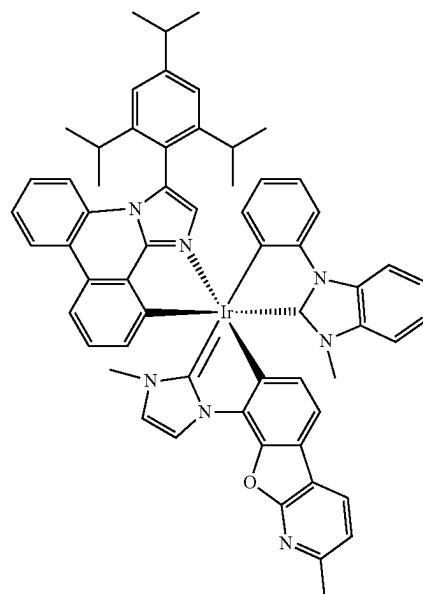
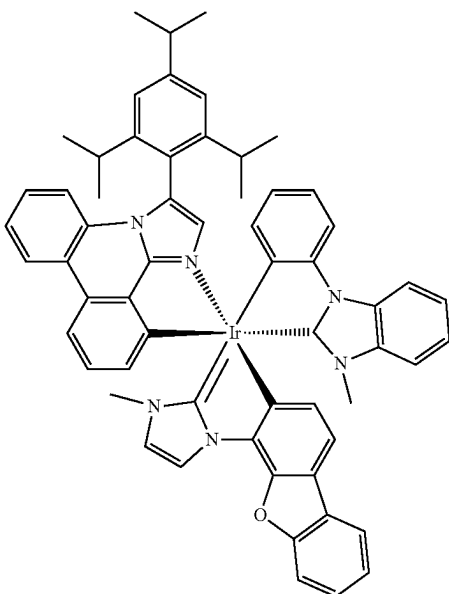


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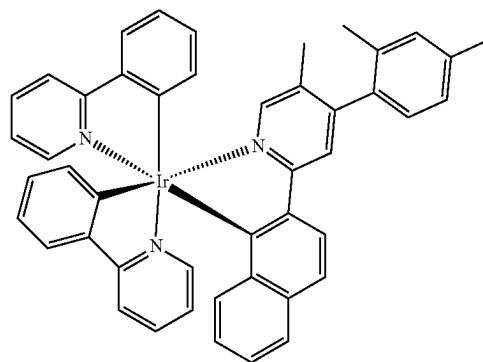
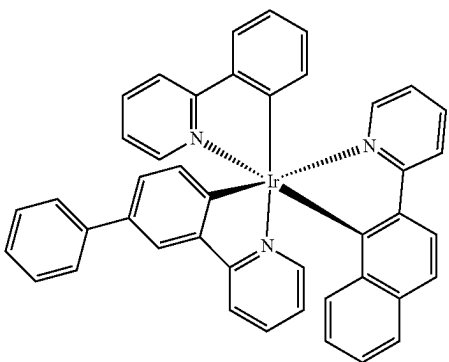
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Compound 422

Compound 423



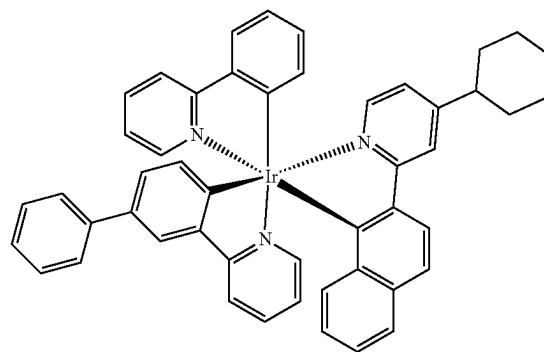
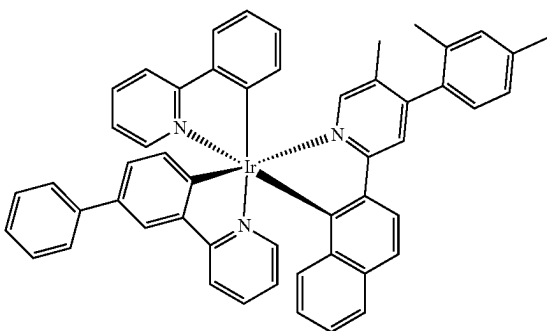
Compound 424

Compound 425



Compound 426

Compound 427

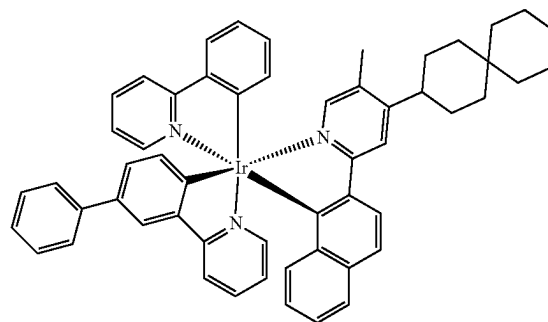
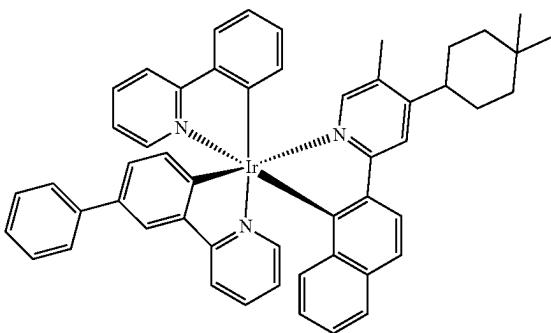


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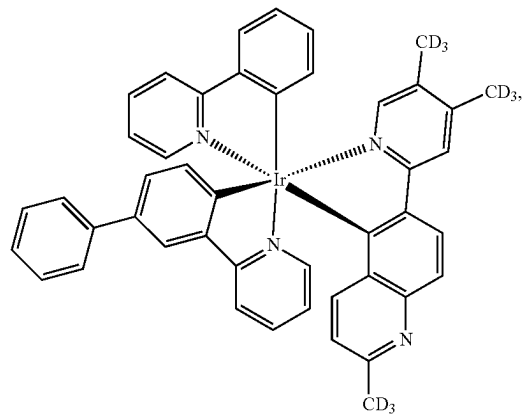
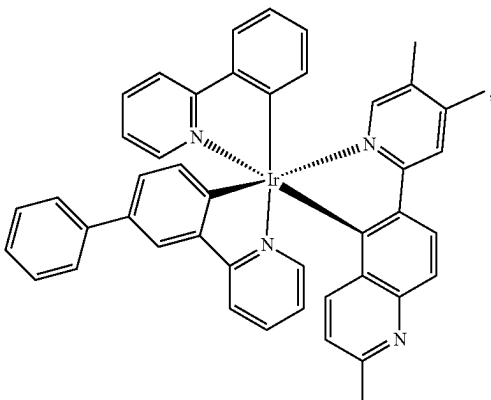
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Compound 428

Compound 429



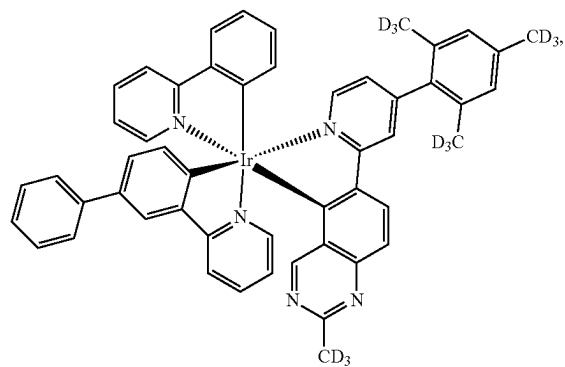
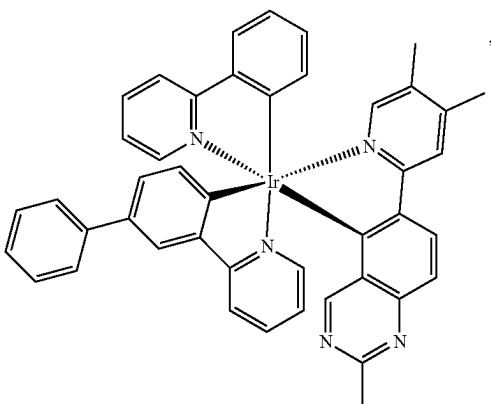
Compound 430

Compound 431



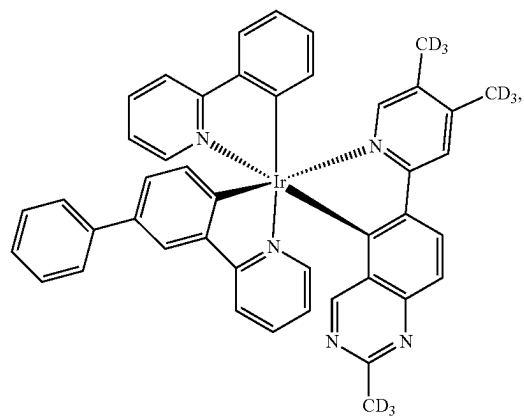
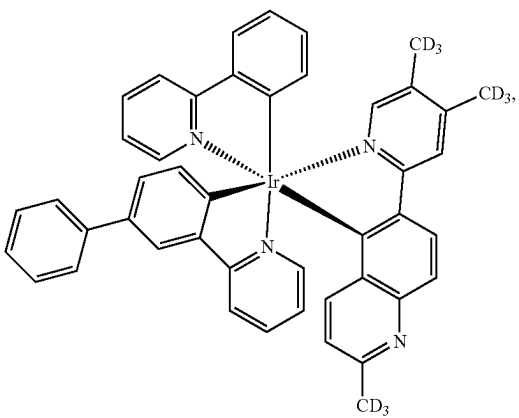
Compound 432

Compound 433

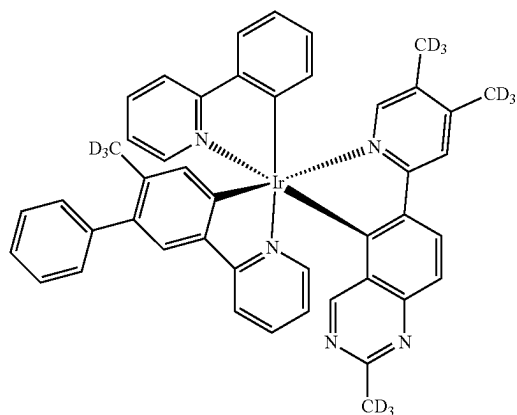


Compound 434

Compound 435

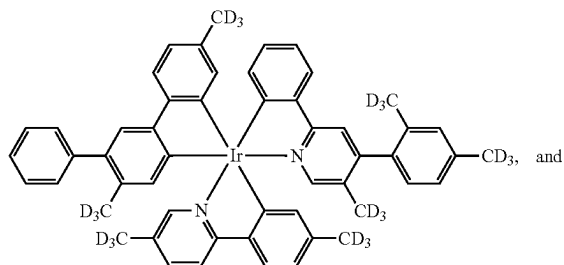


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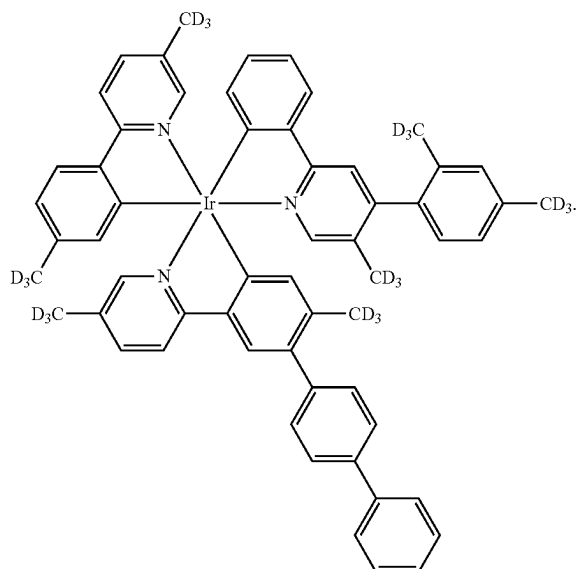
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Compound 436

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Compound 437



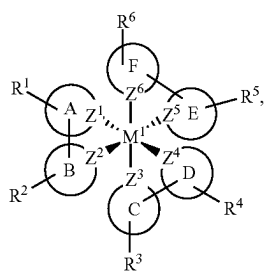
Compound 438



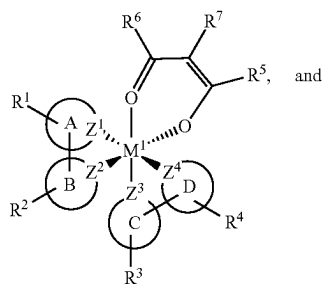
According to another aspect of the present disclosure, an organic light-emitting device (OLED) is disclosed where the OLED comprises: an anode; a cathode; and an organic layer, disposed between the anode and the cathode. The organic layer comprises a compound having the Formula selected from the group consisting of:

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Formula I



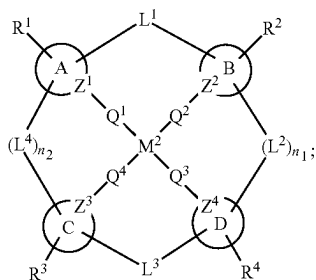
Formula II



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Formula III

wherein rings A, B, C, D, E, and F are each a 5 or 6-membered carbocyclic or heterocyclic ring;

wherein in Formula I: A-B, C-D, and E-F form three bidentate ligands coordinated to metal M^1 ; wherein A-B, C-D, and E-F are different from each other; wherein ring A is trans to ring D, ring B is trans to ring E, and ring C is trans to ring F in a octahedral coordination configuration;

wherein in Formula II: A-B, C-D, and one acetylacetonate ligand form three bidentate ligands coordinated to metal M^1 ; wherein A-B, and C-D are different from each other; wherein ring A is trans to ring D, ring B is trans to oxygen atom, and ring C is trans to oxygen atom in a octahedral coordination configuration;

wherein in Formula III: L^1 and L^3 each independently selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof; n_1 , n_2 each independently is 0 or 1; when n_1 or n_2 is 1, L^2 or L^4 is selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof; when n_1 or n_2 is 0, L^2 or L^4 is not present; Q^1 , Q^2 , Q^3 , and Q^4 each independently selected from the group consisting of direct bond and oxygen; and when any of Z^1 , Z^2 , Z^3 , and Z^4 is nitrogen, the Q^1 , Q^2 , Q^3 , and Q^4 attached thereto is a direct bond;

ring A is trans to ring D, and ring B is trans to ring C in a square-planar coordination configuration;

wherein R^1 , R^2 , R^3 , R^4 , R^5 , R^6 , and R^7 each represents mono to the maximum possible number of substitution, or no substitution;

wherein Z^1 , Z^2 , Z^3 , Z^4 , Z^5 , and Z^6 are each independently selected from the group consisting of carbon and nitrogen;

wherein M^1 is a metal selected from the group consisting of Ir, Os, Rh, Ru, and Re; M^2 is a metal selected from the group consisting of Pt and Pd;

wherein a first distance is the distance between the farthest atom away from M^1 in R^1 to the farthest atom away from M^1 in R^4 ;

wherein a second distance is the distance between the farthest atom away from M^1 in R^2 to the farthest atom away from M^1 in R^5 ;

wherein a third distance is the distance between the farthest atom away from M^1 in R^3 to the farthest atom away from M^1 in R^6 ;

wherein a fourth distance is the distance between the farthest atom away from M^2 in R^1 to the farthest atom away from M^2 in R^4 ;

wherein a fifth distance is the distance between the farthest atom away from M^2 in R^2 to the farthest atom away from M^2 in R^3 ;

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wherein the first distance is longer than the second distance and the third distance each by at least 1.5 Å;

wherein the fourth distance is longer than the fifth distance by at least 1.5 Å;

wherein R, R', R^1 , R^2 , R^3 , R^4 , R^5 , R^6 , and R^7 are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and

wherein any two substituents among R, R', R^1 , R^2 , R^3 , R^4 , R^5 , R^6 , and R^7 are optionally joined or fused into a ring.

In some embodiments of the OLED, where the compound in the organic layer is of Formula I, at least one of (i), (ii), and (iii) is true, wherein (i) one R^1 connects to one R^2 , (ii) one R^3 connects to one R^4 , (iii) one R^5 connects to one R^6 . In some embodiments where the compound is of Formula II, at least one of (i) and (ii) is true, wherein (i) one R^1 connects to one R^2 , (ii) one R^3 connects to one R^4 .

In some embodiments of the OLED, the organic layer is an emissive layer and the compound is an emissive dopant or a non-emissive dopant.

In some embodiments of the OLED, the organic layer further comprises a host, wherein the host comprises a triphenylene containing benzo-fused thiophene or benzo-fused furan;

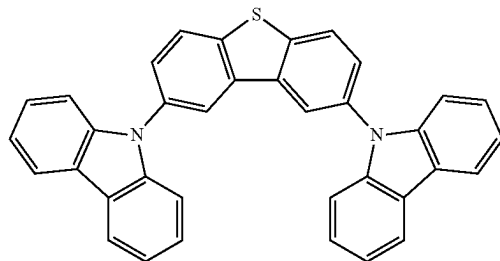
wherein any substituent in the host is an unfused substituent independently selected from the group consisting of C_nH_{2n+1} , OC_nH_{2n+1} , OAr_1 , $N(C_nH_{2n+1})_2$, $N(Ar_1)(Ar_2)$, $CH=CH-C_nH_{2n+1}$, $C\equiv CC_nH_{2n+1}$, Ar_1 , Ar_1-Ar_2 , and $C_nH_{2n}-Ar_1$, or the host has no substitutions;

wherein n is from 1 to 10; and

wherein Ar_1 and Ar_2 are independently selected from the group consisting of benzene, biphenyl, naphthalene, triphenylene, carbazole, and heteroaromatic analogs thereof.

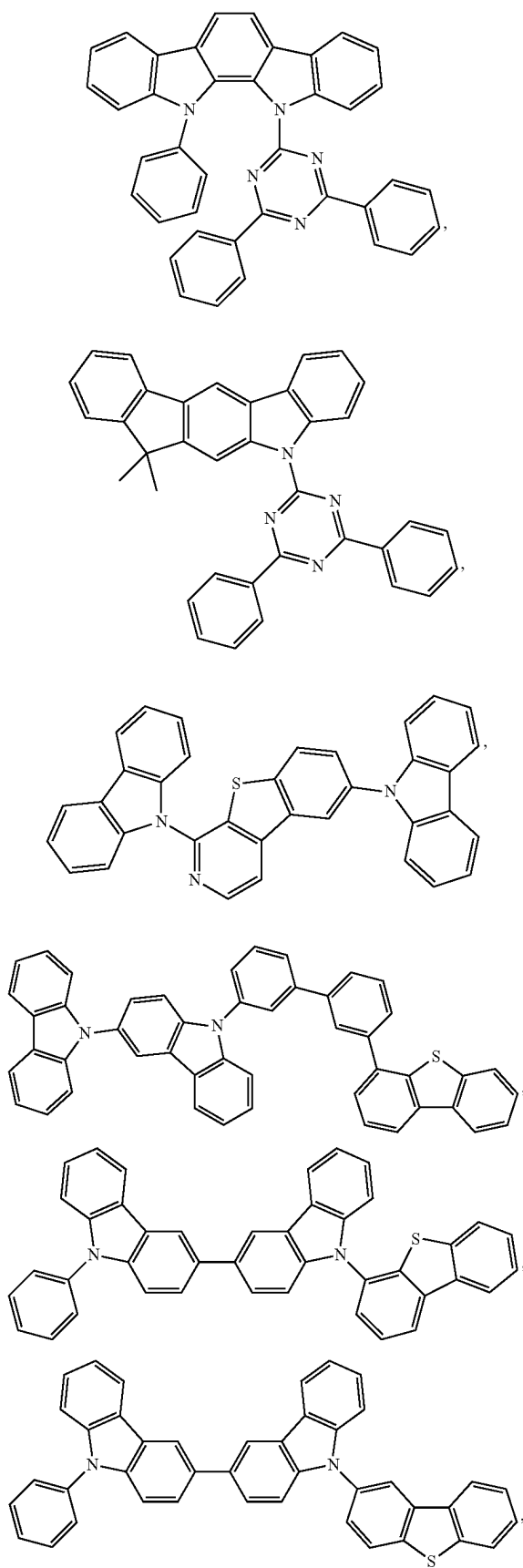
In some embodiments of the OLED, the organic layer further comprises a host, wherein host comprises at least one chemical group selected from the group consisting of triphenylene, carbazole, dibenzothiophene, dibenzofuran, dibenzoselenophene, azatriphenylene, azacarbazole, aza-dibenzothiophene, aza-dibenzofuran, and aza-dibenzoselenophene.

In some embodiments of the OLED, the organic layer further comprises a host, wherein the host is selected from the group consisting of:

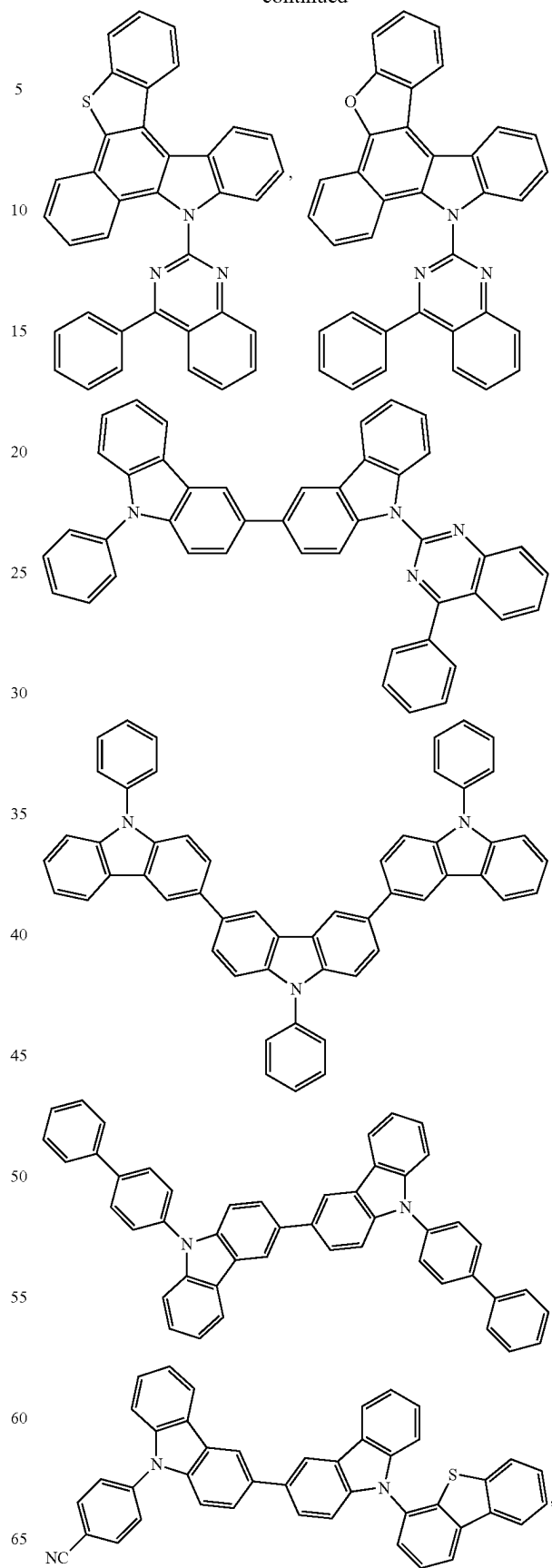


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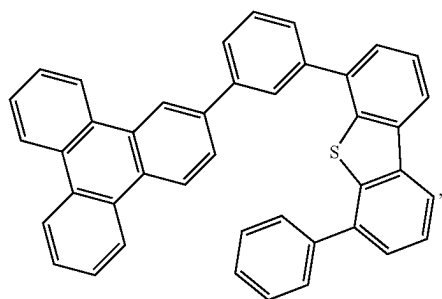
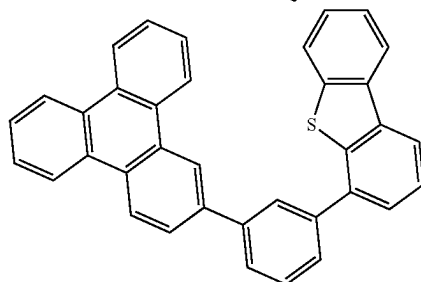
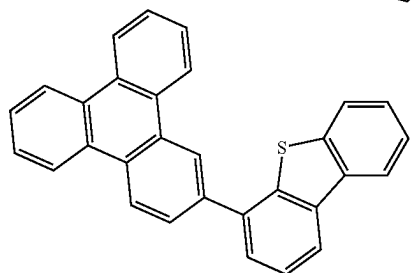
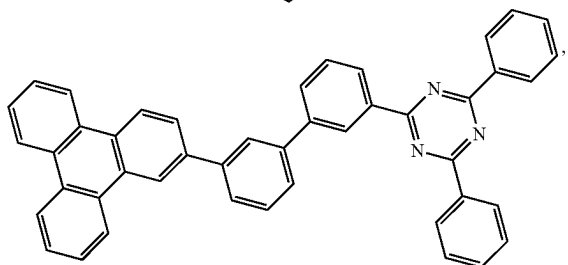
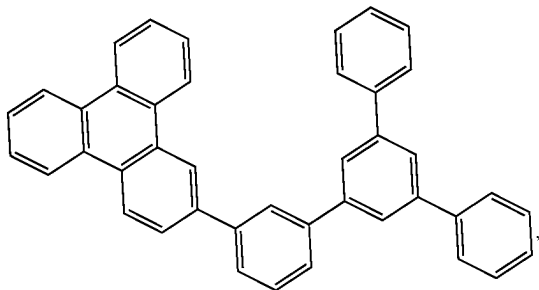
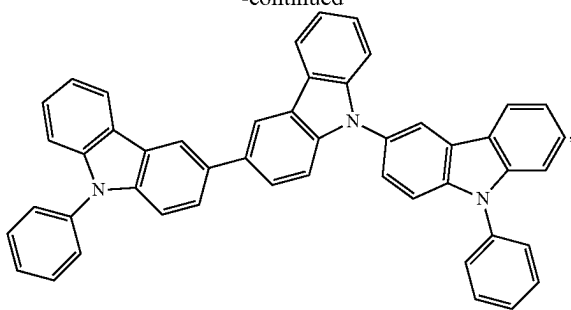
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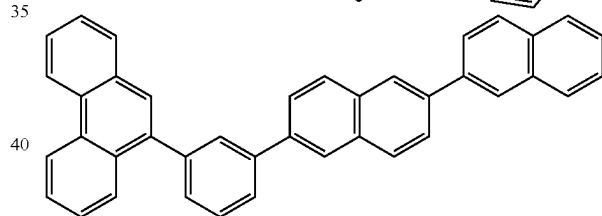
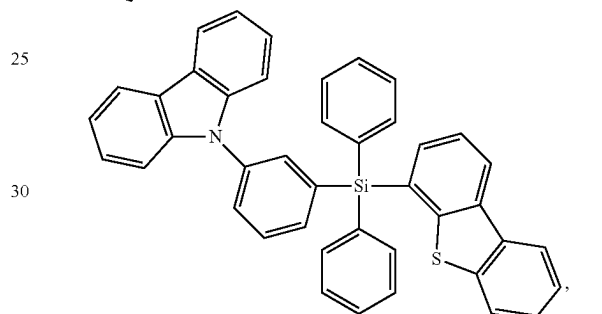
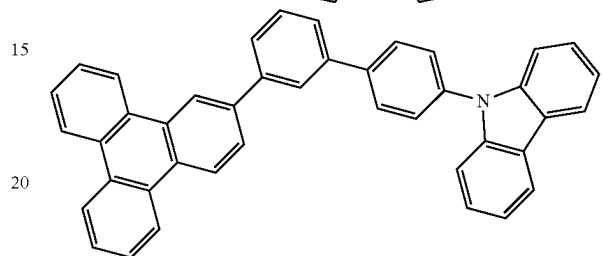
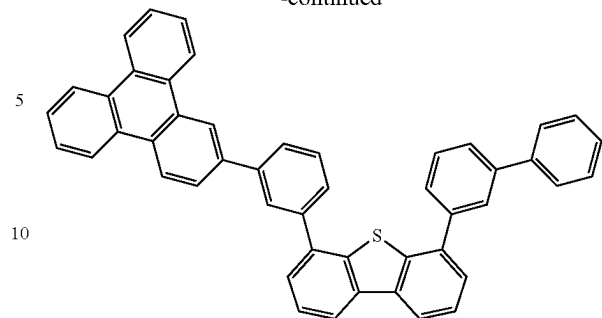


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**252**

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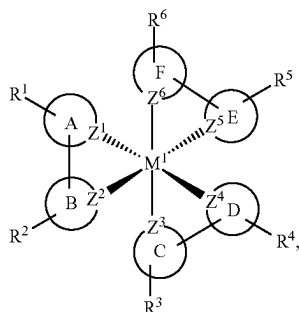
45 and combinations thereof.

In some embodiments of the OLED, the organic layer further comprises a host, wherein the host comprises a metal complex.

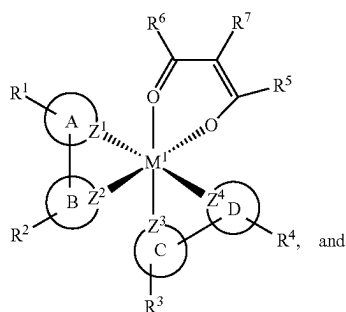
50 According to another aspect, a consumer product comprising the OLED described above is disclosed. In some embodiments of the consumer product, the consumer product is selected from the group consisting of flat panel displays, computer monitors, medical monitors, televisions, billboards, lights for interior or exterior illumination and/or signaling, heads-up displays, fully or partially transparent displays, flexible displays, laser printers, telephones, mobile phones, tablets, phablets, personal digital assistants (PDAs), wearable devices, laptop computers, digital cameras, camcorders, viewfinders, micro-displays, 3-D displays, virtual reality or augmented reality displays, vehicles, video walls comprising multiple displays tiled together, theater or stadium screen, and a sign.

65 According to another aspect, a formulation comprising the compound having a formula selected from the group consisting of:

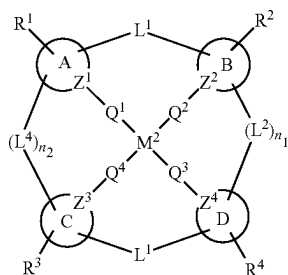
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Formula I



Formula II



Formula III

is disclosed, wherein rings A, B, C, D, E, and F are each a 5 or 6-membered carbocyclic or heterocyclic ring;

wherein in Formula I: A-B, C-D, and E-F form three bidentate ligands coordinated to metal M¹; wherein A-B, C-D, and E-F are different from each other; wherein ring A is trans to ring D, ring B is trans to ring E, and ring C is trans to ring F in an octahedral coordination configuration;

wherein in Formula II: A-B, C-D, and one acetylacetonate ligand form three bidentate ligands coordinated to metal M¹; wherein A-B, and C-D are different from each other; wherein ring A is trans to ring D, ring B is trans to oxygen atom, and ring C is trans to oxygen atom in an octahedral coordination configuration;

wherein in Formula III: L¹ and L³ each independently selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof; n₁, n₂ each independently is 0 or 1; when n₁ or n₂ is 1, L² or L⁴ is selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR',

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GeRR', alkyl, and combinations thereof; when n₁ or n₂ is 0, L² or L⁴ is not present; Q¹, Q², Q³, and Q⁴ each independently selected from the group consisting of direct bond and oxygen; when any of Z¹, Z², Z³, and Z⁴ is nitrogen, the Q¹, Q², Q³, and Q⁴ attached thereto is a direct bond;

ring A is trans to ring D, ring B is trans to ring C in a square-planar coordination configuration;

wherein R¹, R², R³, R⁴, R⁵, R⁶, and R⁷ each represents mono to the maximum possible number of substitution, or no substitution;

wherein Z¹, Z², Z³, Z⁴, Z⁵, and Z⁶ are each independently selected from the group consisting of carbon and nitrogen;

wherein M¹ is a metal selected from the group consisting of Ir, Os, Rh, Ru, and Re; M² is a metal selected from the group consisting of Pt and Pd;

wherein a first distance is the distance between the farthest atom away from M¹ in R¹ to the farthest atom away from M¹ in R⁴;

wherein a second distance is the distance between the farthest atom away from M¹ in R² to the farthest atom away from M¹ in R⁵;

wherein a third distance is the distance between the farthest atom away from M¹ in R³ to the farthest atom away from M¹ in R⁶;

wherein a fourth distance is the distance between the farthest atom away from M² in R¹ to the farthest atom away from M² in R⁴;

wherein a fifth distance is the distance between the farthest atom away from M² in R² to the farthest atom away from M² in R³;

wherein the first distance is longer than the second distance and the third distance each by at least 1.5 Å;

wherein the fourth distance is longer than the fifth distance by at least 1.5 Å;

wherein R, R¹, R², R³, R⁴, R⁵, R⁶, and R⁷ are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alk-enyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and

wherein any two substituents among R, R¹, R², R³, R⁴, R⁵, R⁶, and R⁷ are optionally joined or fused into a ring.

In some embodiments, the compound can be an emissive dopant. In some embodiments, the compound can produce emissions via phosphorescence, fluorescence, thermally activated delayed fluorescence, i.e., TADF (also referred to as E-type delayed fluorescence), triplet-triplet annihilation, or combinations of these processes.

The OLED disclosed herein can be incorporated into one or more of a consumer product, an electronic component module, and a lighting panel. The organic layer can be an emissive layer and the compound can be an emissive dopant in some embodiments, while the compound can be a non-emissive dopant in other embodiments.

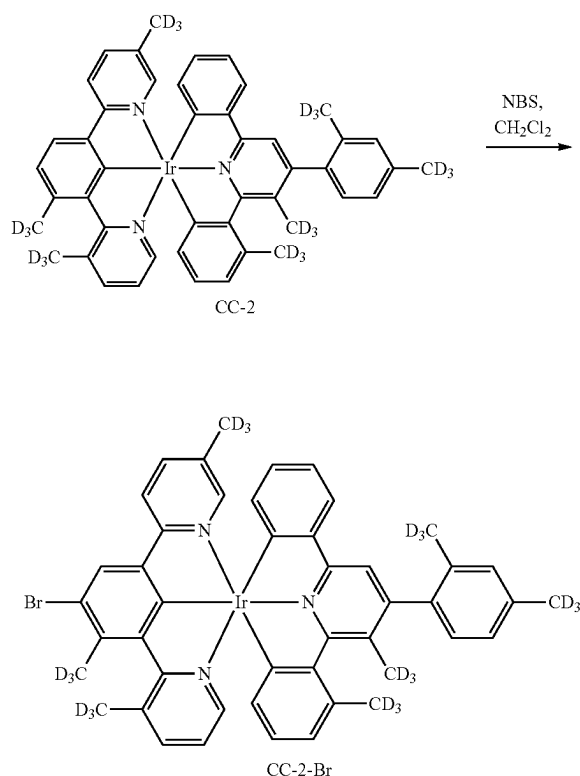
The formulation can include one or more components selected from the group consisting of a solvent, a host, a hole injection material, hole transport material, and an electron transport layer material, disclosed herein.

255

EXPERIMENTAL

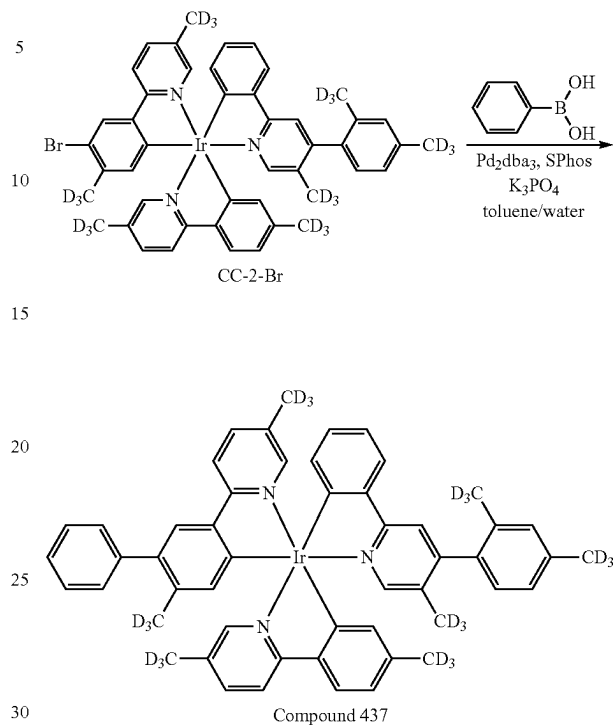
Synthesis of Compound 437

Step 1



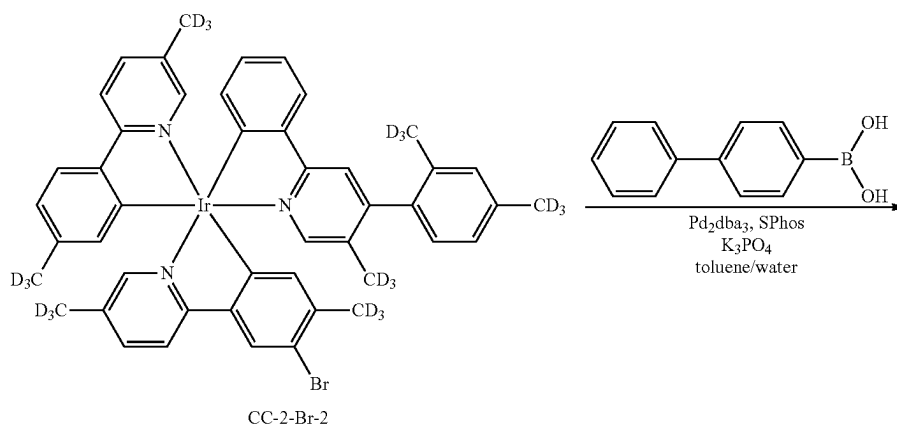
CC-2 (2.3 g, 2.71 mmol) was dissolved in dry dichloromethane (400 ml). The mixture was degassed with N_2 and cooled to $0^\circ C$. 1-Bromopyrrolidine-2,5-dione (0.81 g, 2.71 mmol) was dissolved in DCM (300 mL) and added dropwise. After addition, the temperature was gradually raised to room temperature and stirred for 12 hrs. Saturated $NaHCO_3$ (20 mL) solution was added. The organic phase was separated and collected. The solvent was removed and the residue was coated on Celite and purified on silica gel column eluted with toluene/heptane 70/30 (v/v) to give the product CC-2-Br (0.6 g, 24%).

Step 2

256

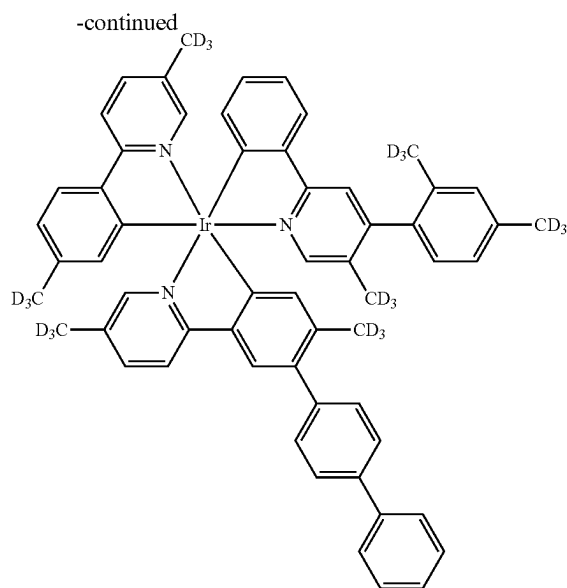
CC-2-Br (0.72 g, 0.775 mmol) was dissolved in a mixture of toluene (40 ml) and water (4 ml). The mixture was purged with N_2 for 10 mins. K_3PO_4 (0.411 g, 1.937 mmol), SPhos (0.095 g, 0.232 mmol), Pd_2dba_3 (0.043 g, 0.046 mmol), and phenylboronic acid (0.189 g, 1.55 mmol) were added. The mixture was heated under N_2 at $110^\circ C$ for 12 hrs. The reaction then was cooled down to room temperature, the product was extracted with DCM. The organic phase was separated and collected. The solvent was removed and the residue was coated on Celite and purified on silica gel column eluted with toluene/heptane 70/30 (v/v). The product was purified by crystallization from toluene/MeOH to give compound 437 (0.7 g).

Synthesis of Compound 438



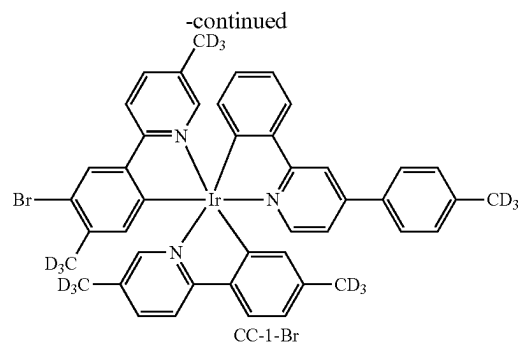
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Compound 438

CC-2-Br-2 (0.6 g, 0.646 mmol) was dissolved in a mixture of toluene (100 ml) and water (10 ml). The mixture was
 30 purged with N₂ for 10 mins. K₃PO₄ (0.343 g 1.61 mmol), SPhos (0.080 g, 0.19 mmol), Pd₂dba₃ (0.035 g, 0.039 mmol), and [1,1-biphenyl]4-ylboronic acid (0.256 g, 1.29
 35 mmol) were added. The mixture was heated under N₂ at 110° C. for 12 hrs. Then the reaction was cooled down to room temperature, the product was extracted with DCM and
 40 organic phase was separated. The solvent was removed and the residue was coated on Celite and purified on silica gel column eluted with toluene/heptane 70/30 (v/v). The product
 45 was purified by crystallization from toluene/MeOH to give compound 438 (0.64 g).

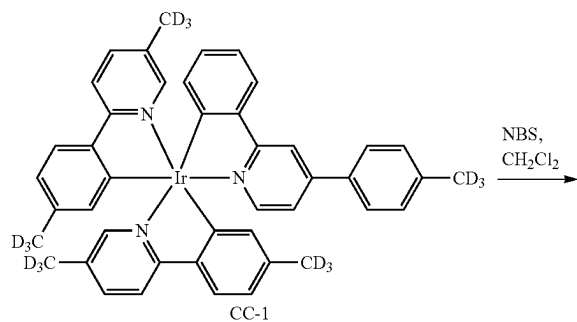


CC-1-Br

Synthesis of Compound 161

50

Step 1

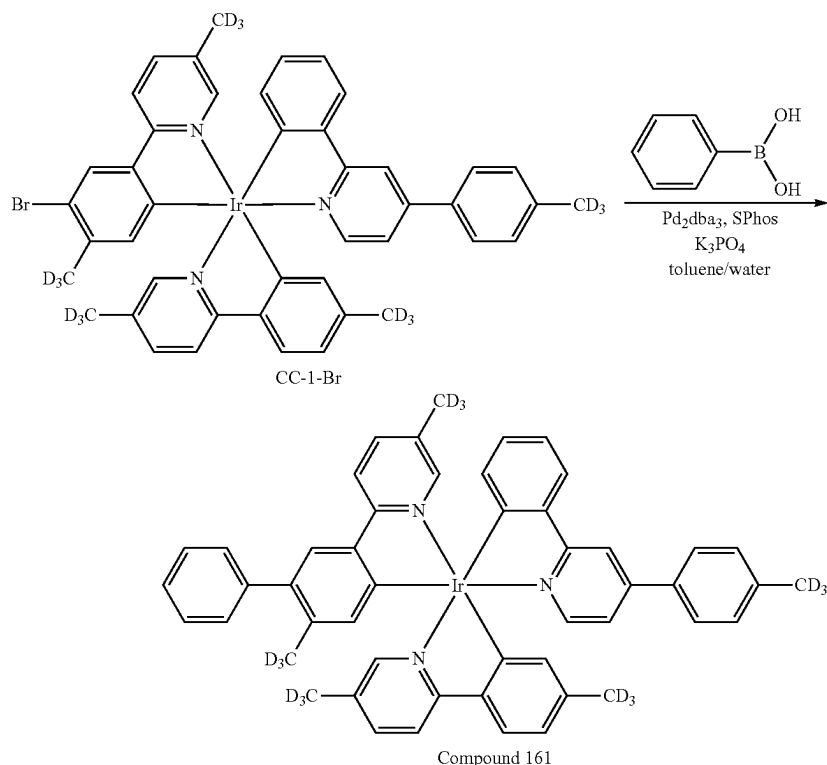


CC-1 (2.04 g, 2.500 mmol) was dissolved in dry dichloromethane (400 ml). The mixture was degassed with N₂ and
 55 cooled to 0° C. 1-bromopyrrolidine-2,5-dione (0.445 g, 2.500 mmol) was dissolved in DCM (200 mL) and dropwise added. After addition, the temperature was gradually raised
 60 to room temperature and stirred for 16 hrs. Sat. NaHCO₃ (20 mL) solution was added. The organic phase was separated and collected. The solvent was removed and the residue was
 65 coated on Celite and purified on silica gel column eluted by using 70/30 toluene/heptane to give the product CC-1-Br (0.6 g).

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Step 2



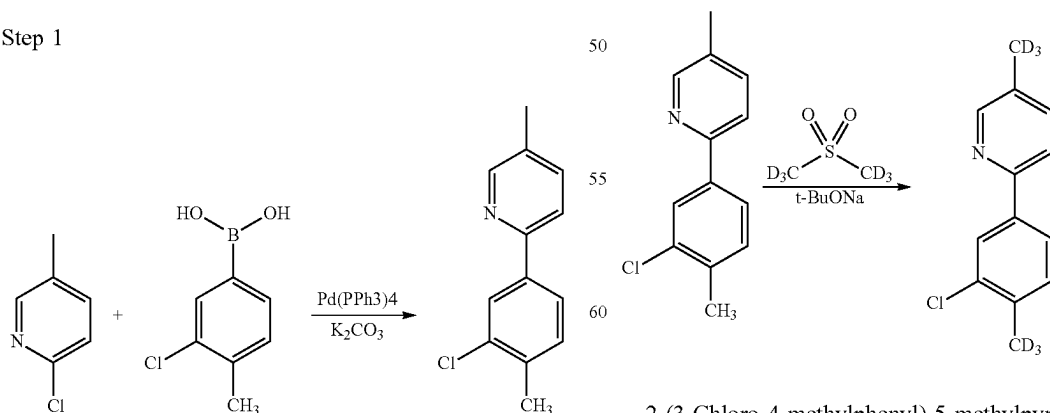
CC-1-Br (1.16 g, 1.296 mmol) was dissolved in a mixture of toluene (120 ml) and water (12.00 ml). The mixture was purged with N₂ for 10 mins. K₃PO₄ (0.688 g, 3.24 mmol), SPhos (0.160 g, 0.389 mmol), Pd₂dba₃ (0.071 g, 0.078 mmol), and phenylboronic acid (0.316 g, 2.59 mmol) were added. The mixture was heated under N₂ at 110° C. for 16 hrs. After the reaction was complete it was cooled down to room temperature, the product was extracted with DCM. The organic phase was separated and collected. The solvent was removed and the residue was coated on Celite and purified on silica gel column eluted by using 70/30 toluene/heptane. The product was purified by recrystallization in toluene/MeOH to give Compound 161 (1.0 g).

and potassium carbonate (21.74 g, 157 mmol) were dissolved in the mixture of DME (150 ml) and water (20 ml) under nitrogen to give a colorless suspension. Pd(PPh₃)₄ (0.909 g, 0.786 mmol) was added to the reaction mixture, then the reaction mixture was degassed and heated to 95° C. for 12 hrs. It was then cooled down to room temperature, organic phase was separated and evaporated. The residue was subjected to column chromatography on silica gel column, eluted with heptanes/THF 9/1 (v/v), providing after crystallization from heptanes 10 g (58% yield) of white solid.

Step 2

Synthesis of Compound 401

Step 1



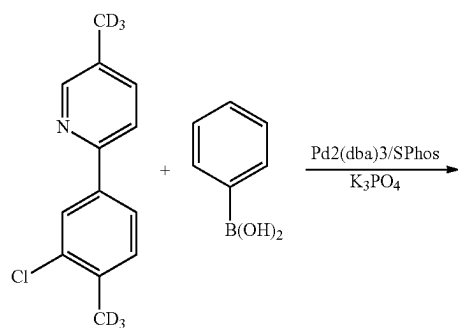
2-Chloro-5-methylpyridine (10.03 g, 79 mmol), (3-chloro-4-methylphenyl)boronic acid (13.4 g, 79 mmol),

2-(3-Chloro-4-methylphenyl)-5-methylpyridine (10 g, 45.9 mmol), ((methyl-d₃)sulfonyl)methane-d₃ (92 g, 919 mmol), and sodium 2-methylpropan-2-olate (2.65 g, 27.6 mmol) were dissolved together under nitrogen to give a dark

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solution. The reaction mixture was heated to 80° C. under nitrogen for 12 hrs, cooled down, diluted with ethyl acetate, washed with water, dried over sodium sulfate, filtered and evaporated. Purified by column chromatography on silica gel, eluted with heptanes/THF 9/1 (v/v), providing white solid, then crystallized from heptanes, providing colorless crystalline material (9.1 g, 81% yield).

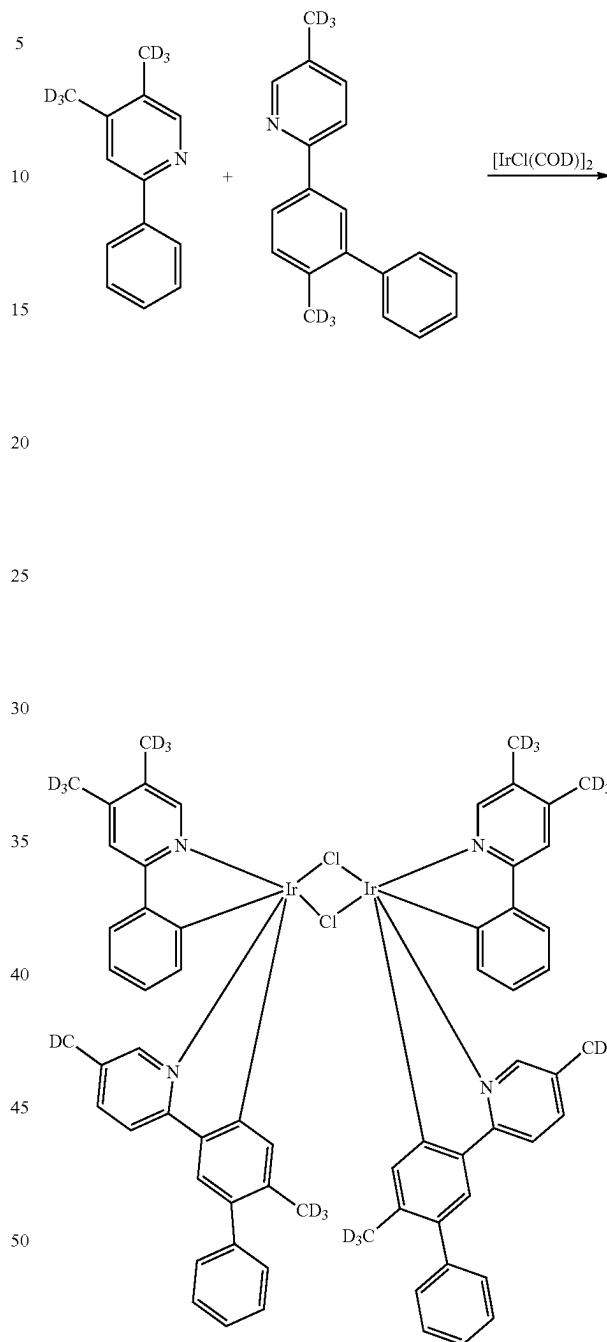
Step 3



2-(3-Chloro-4-(methyl-d3)phenyl)-5-(methyl-d3)pyridine (7.45 g, 33.3 mmol), phenylboronic acid (6.09 g, 49.9 mmol), potassium phosphate (15.34 g, 66.6 mmol), $\text{Pd}_2(\text{dba})_3$ (0.305 g, 0.333 mmol) and dicyclohexyl(2',6'-dimethoxy-[1,1'-biphenyl]-2-yl)phosphane (Sphos, 0.273 g, 0.666 mmol) were dissolved in the mixture of DME (150 ml) and water (25 ml) under nitrogen to give a red suspension. The reaction mixture was degassed and heated to reflux under nitrogen. After overnight heating about 80% conversion was achieved. Addition of more Ph boronic acid and catalyst didn't improve conversion. Separated organic phase, evaporated and purified the residue by column chromatography on silica gel, eluted with heptanes/THF 9/1, then crystallized from heptanes. White solid (6.2 g, 70% yield).

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Step 4

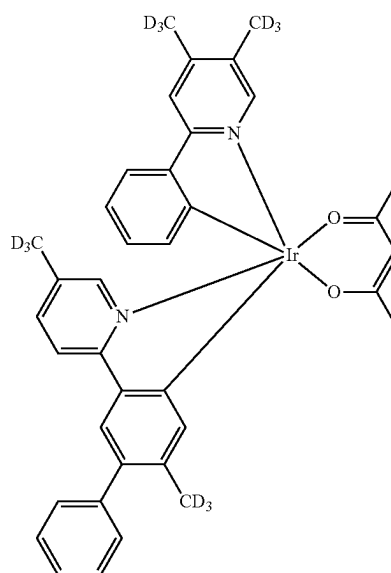
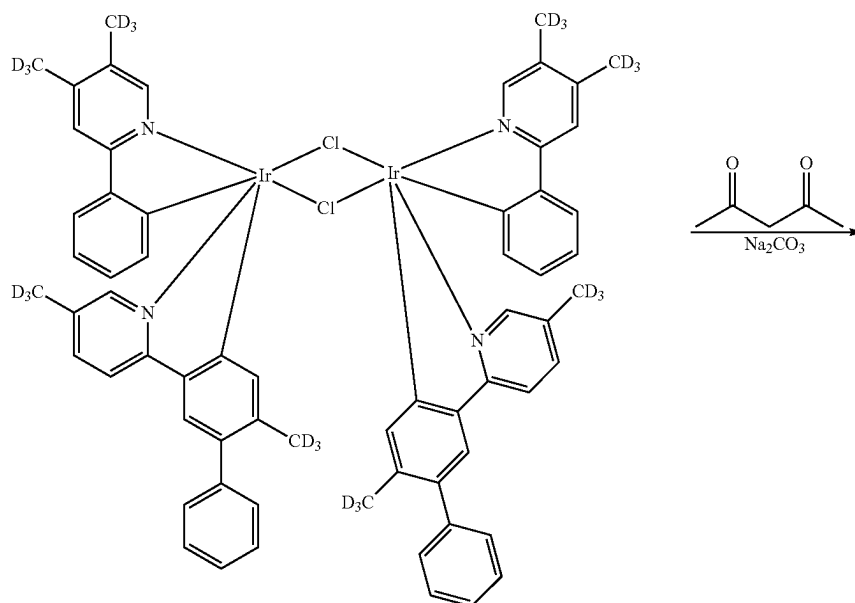


Under nitrogen atmosphere 4,5-bis(methyl-d3)-2-phenylpyridine (1.427 g, 7.54 mmol), 5-(methyl-d3)-2-(6-(methyl-d3)-[1,1'-biphenyl]-3-yl)pyridine (2 g, 7.54 mmol), and $[\text{IrCl}(\text{COD})]_2$ (2.53 g, 3.77 mmol) were dissolved in ethoxyethanol (50 ml) under nitrogen to give a red solution. The reaction mixture was heated to reflux for 1 hr, then precipitate was formed. Added 30 mL more of ethoxyethanol and continued to reflux for 48 hrs, then the reaction mixture was cooled down to room temperature. The crude material was used without additional purification on the next step.

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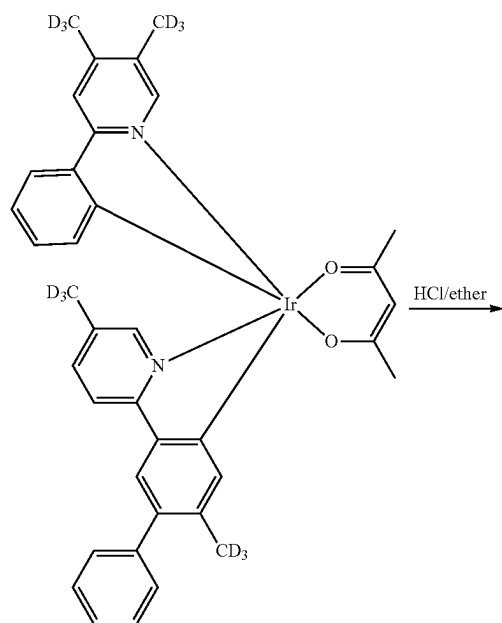
Step 5



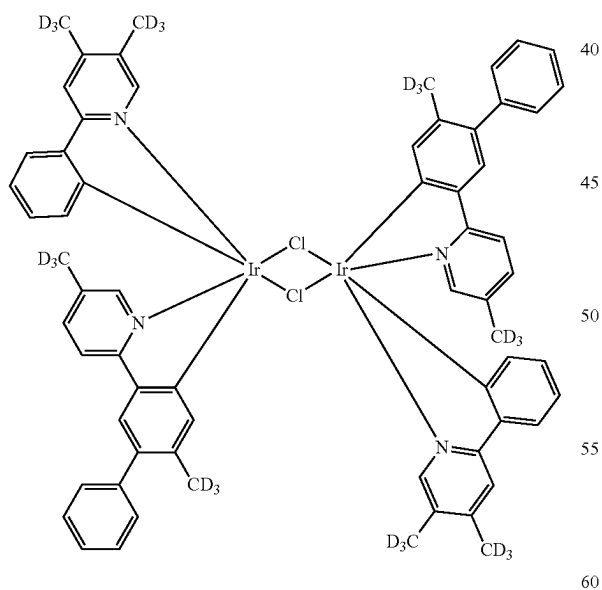
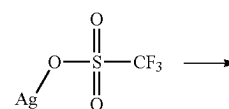
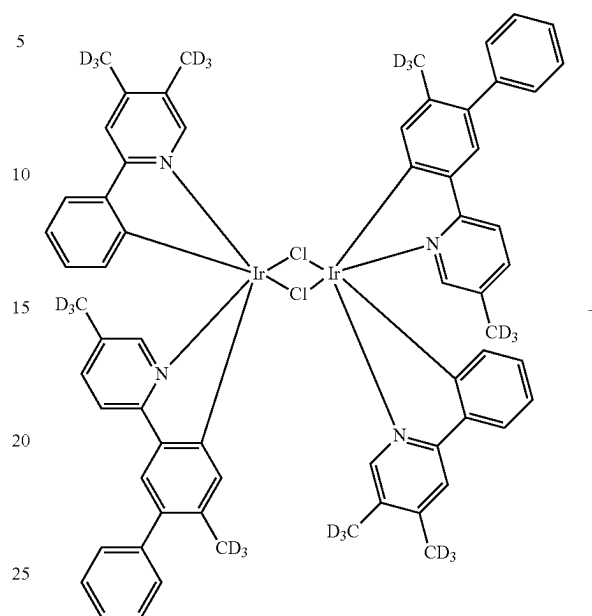
Iridium dimer suspended in ethoxyethanol (from Step 4) was mixed under nitrogen atmosphere with pentane-2,4-⁶⁰
dione (2.59 g, 25.9 mmol) and sodium carbonate (3.43 g, 32.3 mmol) in 50 ml of methanol, stirred 24 hrs under nitrogen at 55° C. and evaporated. The yellow residue was subjected to column chromatography on silica gel column,⁶⁵
eluted with gradient mixture heptanes/toluene, providing 5 g (36% yield) of the target acac complex.

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Step 6

**266**

Step 7

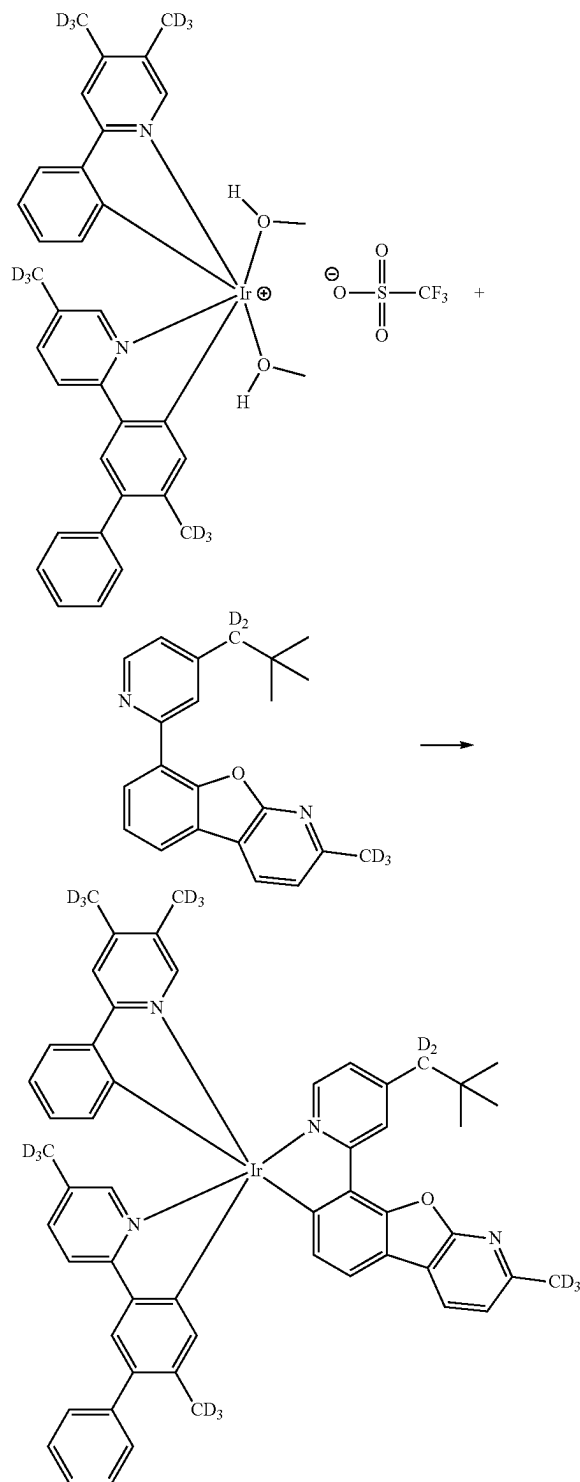


The acac complex (5 g, 6.72 mmol) was dissolved in DCM (20 mL), then HCl in ether (16.80 ml, 33.6 mmol) was added as one portion, stirred for 10 min, evaporated. The residue was triturated in methanol. The solid was filtered and washed with methanol and heptanes to obtain yellow solid (4.55 g, 100% yield).

The Ir dimer (4.55 g, 3.34 mmol) and (((trifluoromethyl) sulfonyl)oxy)silver (2.062 g, 8.03 mmol) were suspended in 50 ml of DCM/methanol 1/1 (v/v) mixture and stirred over 72 hrs at room temperature, filtered through celite and evaporated, providing yellow solid (4.75 g, 83% yield).

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Step 8



The mixture of triflic salt (3 g, 3.5 mmol) and 2-(13-methyl-d2)-8-(4-(2,2-dimethylpropyl-1,1-d2)pyridin-2-yl)benzofuro[2,3-b]pyridine (2.56 g, 7.7 mmol) in 30 mL of methanol were stirred under nitrogen at 65° C. for 5 days. Then material was cooled down, and methanol was evaporated. The residue was subjected to column chromatography on the silica gel column, eluted with 2% of ethyl acetate in

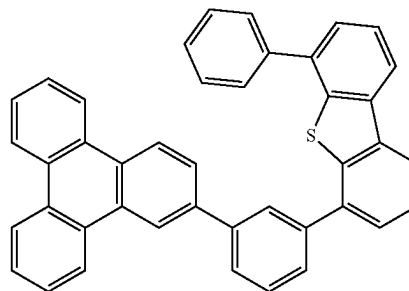
268

toluene, providing two isomers of the product (1.7 g with high R_r and 0.7 g of complex with low R_r). Complex with low R_r is the target compound 401.

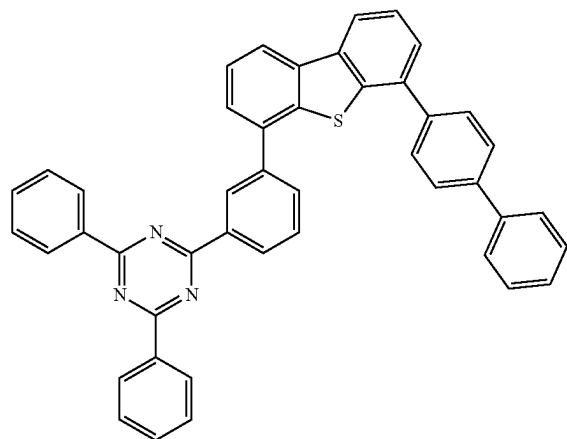
Device Examples

All example devices were fabricated by high vacuum (<10⁻⁷ Torr) thermal evaporation. The anode electrode was 750 Å of indium tin oxide (ITO). The cathode consisted of 10 Å of Liq (8-hydroxyquinoline lithium) followed by 1,000 Å of Al. All devices were encapsulated with a glass lid sealed with an epoxy resin in a nitrogen glove box (<1 ppm of H₂O and O₂) immediately after fabrication with a moisture getter incorporated inside the package. The organic stack of the device examples consisted of sequentially, from the ITO Surface: 100 Å of HAT-CN as the hole injection layer (HIL); 450 Å of HTM as a hole transporting layer (HTL); emissive layer (EML) with thickness 400 Å. Emissive layer containing H-host (H1): E-host (H2) in 6:4 ratio and 12 weight % of green emitter. 350 Å of Liq (8-hydroxyquinoline lithium) doped with 40% of ETM as the ETL. Device structure is shown in the table 1. Table 1 shows the schematic device structure. The chemical structures of the device materials are shown below.

H1



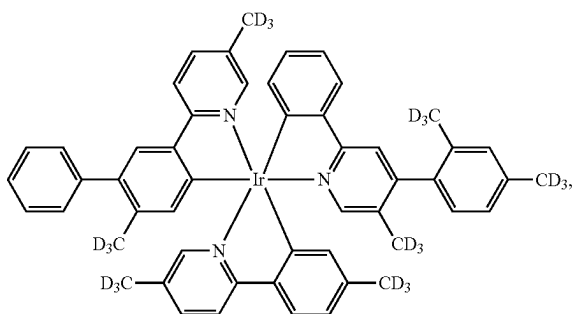
H2



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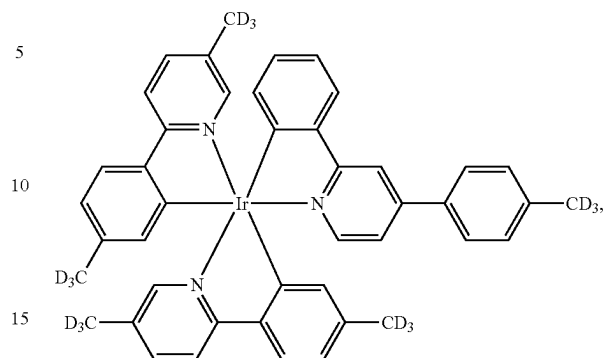
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Compound 437

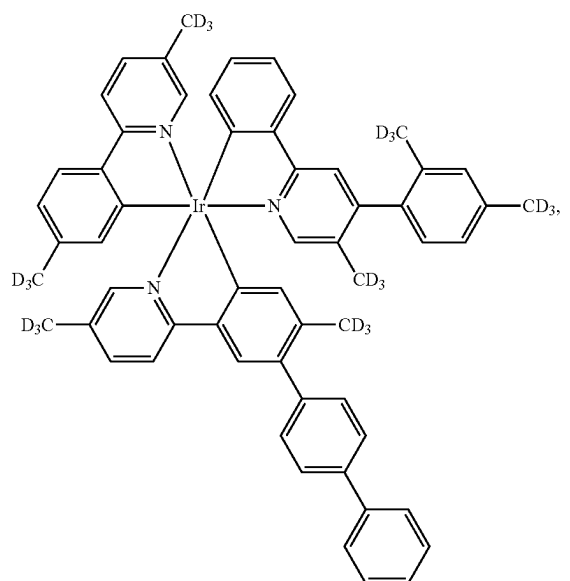
**270**

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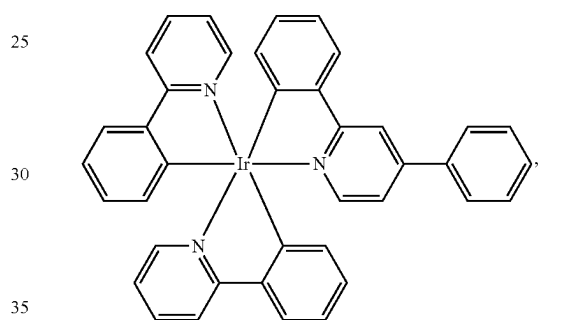
CC-1



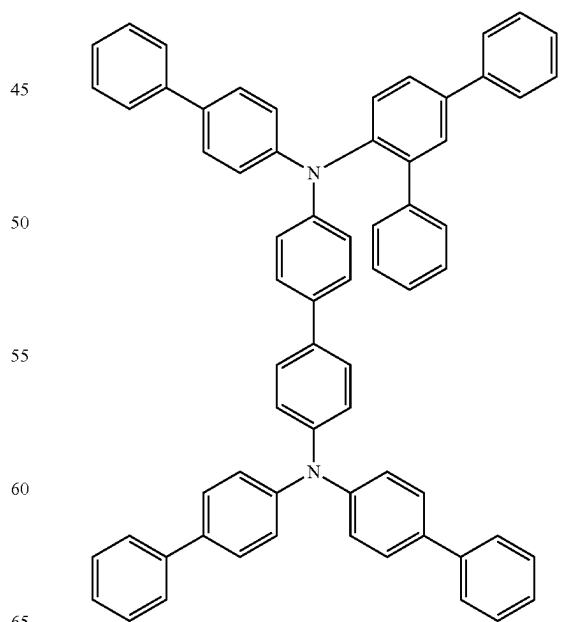
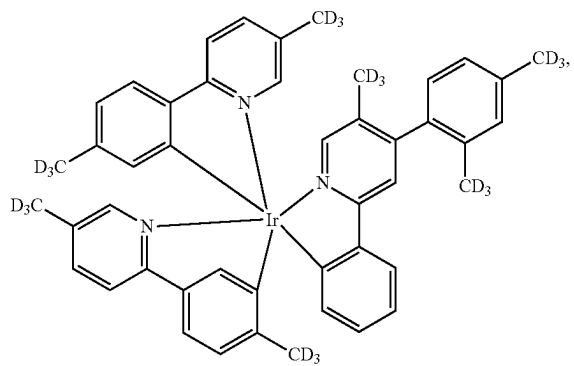
Compound 438



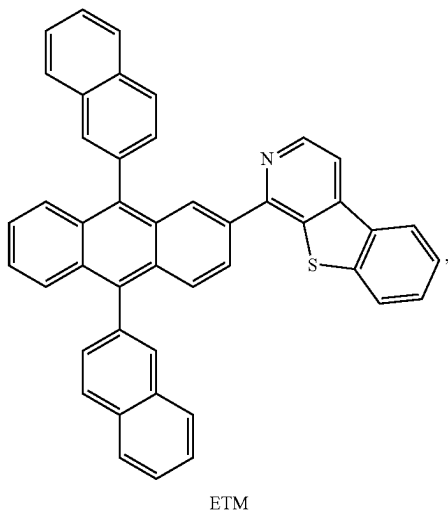
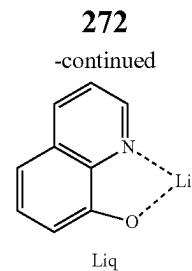
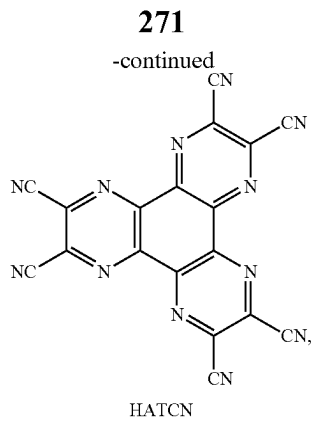
CC-2



Comparative example



HTM



Upon fabrication the devices have been measured for EL, JVL, and lifetime tested at DC 80 mA/cm². Device performance is shown in Table 3, voltage, LE, EQE, PE, and LT_{97%} are all normalized to the comparative compound.

TABLE 2

schematic device structure		
Layer	Material	Thickness [Å]
Anode	ITO	800
HIL	HAT-CN	100
HTL	HTM	450
Green EML	H1:H2: example dopant	400
ETL	Liq:ETM 40%	350
EIL	Liq	10
Cathode	Al	1,000

TABLE 3

Device performance										
Emitter [12%]	1931 CIE				At 10 mA/cm ²				at 80 mA/cm ² *	
	λ		max	FWHM	Voltage	LE	EQE	PE	L ₀	LT _{97%}
	x	y								
Comparative example	0.319	0.624	521	73	1.00	1.00	1.00	1.00	46,497	1.00
Compound 438	0.315	0.628	519	71	1.02	1.04	1.03	1.02	46,542	1.70
Compound 437	0.313	0.628	518	71	0.99	1.12	1.12	1.14	51,738	3.00

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Comparing compounds 437 and 438 with the comparative example; the efficiency of both compound 437 and 438 are higher than the comparative example. Presumably compound 437 and compound 438 have higher horizontal emitting dipole orientation than the comparative example. Elongated and planar substituents with high electrostatic potential enlarge the interacting surface region between Ir complex and host molecules, resulting in stacking Ir complexes parallel to the film surface and increasing the out coupling efficiency. Moreover, the $LT_{97}\%$ at 80 mA/cm^2 of both compound 437 and compound 438 is greater than that of the comparative example, indicating that the elongated substituents not only increase the efficiency but also increase the stability of the complexes in device.

Provided in Table 4 below is a summary of the device data recorded at 9000 nits for the device examples, the EQE value is normalized to Device C-2.

TABLE 4

Device ID	Dopant	Color	EQE (%)
Device 3	Compound 161	Yellow	1.24
Device C-1	CC-1	Yellow	1.10
Device C-2	CC-2	Yellow	1.00

The data in Table 4 show that the device using the inventive compound as the emitter achieves the same color but higher efficiency in comparison with the comparative examples. It is noted that the only difference between the inventive compound and the comparative compound (CC-1) is that the inventive compound has a phenyl moiety replacing one of the protons in the comparative compounds, which increases the distance between the terminal atoms in one direction across the Ir metal center. The device results show that the larger aspect ratio of the emitter molecule seems to be critical in achieving higher device efficiency.

Combination with Other Materials

The materials described herein as useful for a particular layer in an organic light emitting device may be used in combination with a wide variety of other materials present in the device. For example, emissive dopants disclosed herein may be used in conjunction with a wide variety of hosts, transport layers, blocking layers, injection layers, electrodes and other layers that may be present. The materials described or referred to below are non-limiting examples of materials that may be useful in combination with the compounds disclosed herein, and one of skill in the art can readily consult the literature to identify other materials that may be useful in combination.

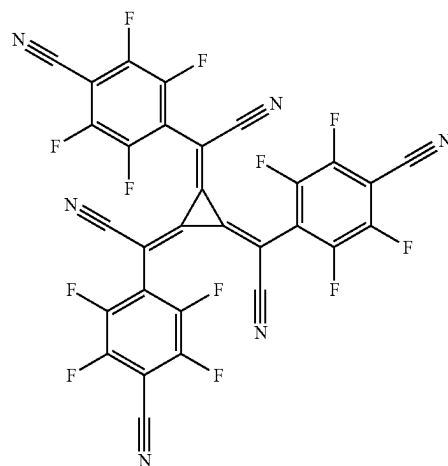
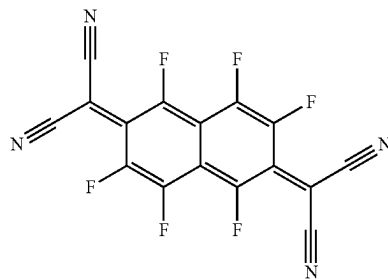
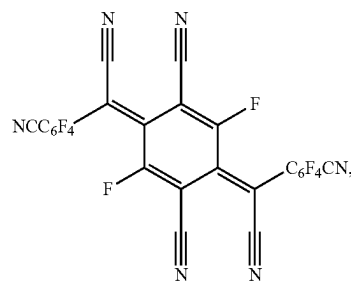
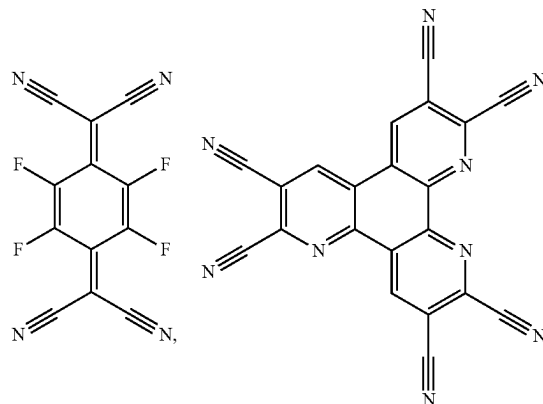
Conductivity Dopants:

A charge transport layer can be doped with conductivity dopants to substantially alter its density of charge carriers, which will in turn alter its conductivity. The conductivity is increased by generating charge carriers in the matrix material, and depending on the type of dopant, a change in the Fermi level of the semiconductor may also be achieved. Hole-transporting layer can be doped by p-type conductivity dopants and n-type conductivity dopants are used in the electron-transporting layer.

Non-limiting examples of the conductivity dopants that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: EP01617493, EP01968131, EP2020694, EP2684932, US20050139810,

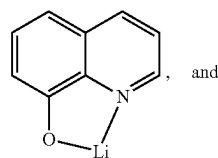
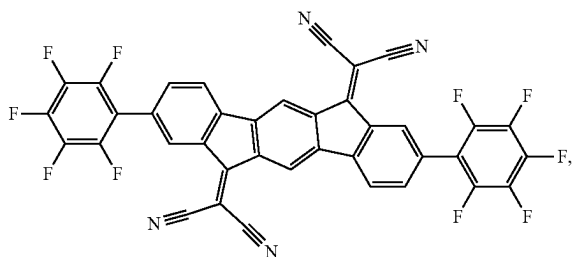
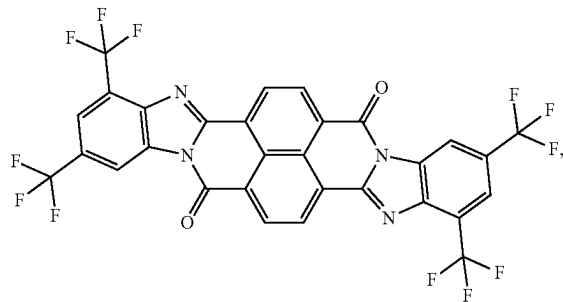
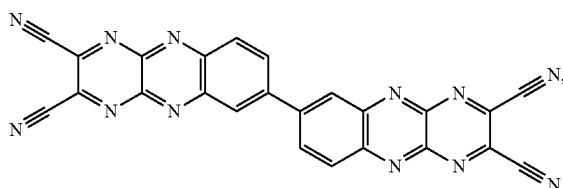
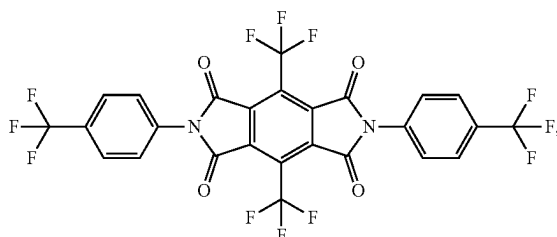
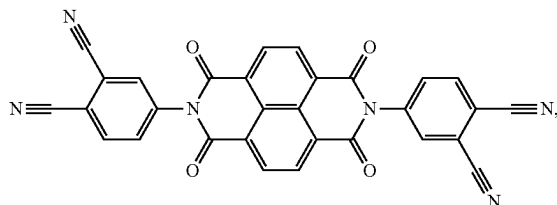
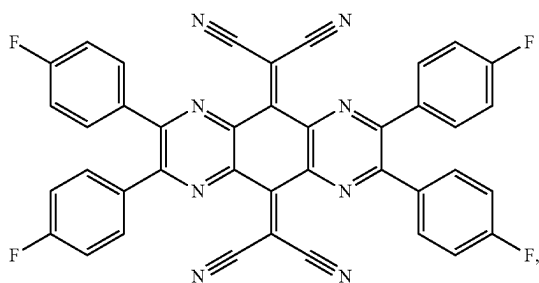
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US20070160905, US20090167167, US2010288362,
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US2015060804 and US2012146012.



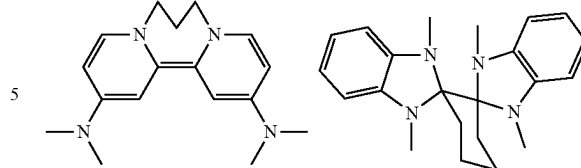
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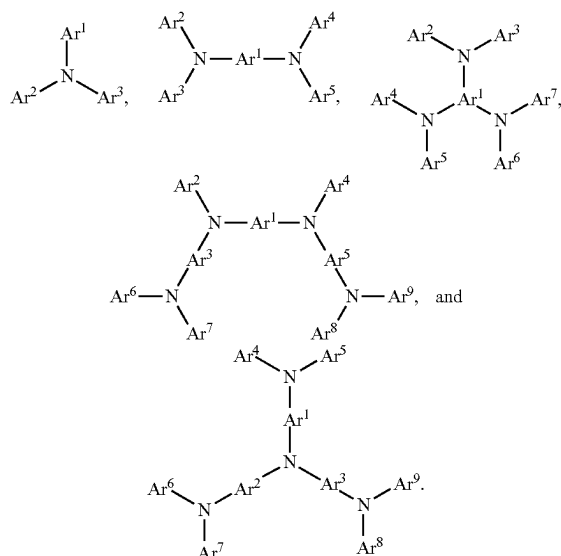
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HIL/HTL:

A hole injecting/transporting material to be used in the present invention is not particularly limited, and any compound may be used as long as the compound is typically used as a hole injecting/transporting material. Examples of the material include, but are not limited to: a phthalocyanine or porphyrin derivative; an aromatic amine derivative; an indolocarbazole derivative; a polymer containing fluorohydrocarbon; a polymer with conductivity dopants; a conducting polymer, such as PEDOT/PSS; a self-assembly monomer derived from compounds such as phosphonic acid and silane derivatives; a metal oxide derivative, such as MoO_x; a p-type semiconducting organic compound, such as 1,4,5,8,9,12-Hexaazatriphenylenehexacarbonitrile; a metal complex, and a cross-linkable compounds.

Examples of aromatic amine derivatives used in HIL or HTL include, but not limit to the following general structures:

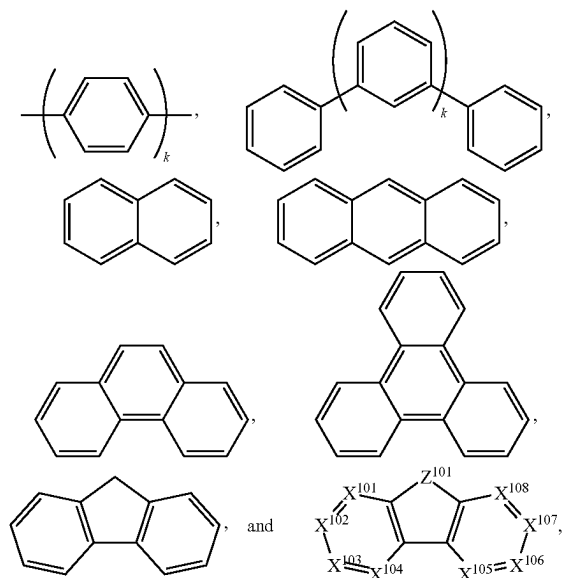


Each of Ar¹ to Ar⁹ is selected from the group consisting of aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, and azulene; the group consisting of aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolidopyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xan-

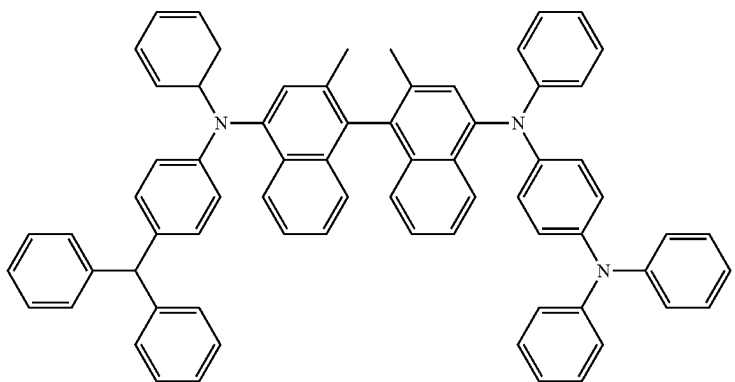
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thene, acridine, phenazine, phenothiazine, phenoxazine, benzofuropyridine, furodipyridine, benzothienopyridine, thienodipyridine, benzoselenophenopyridine, and selenophenodipyridine; and the group consisting of 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Each Ar may be unsubstituted or may be substituted by a substituent selected from the group consisting of deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

In one aspect, Ar¹ to Ar⁹ is independently selected from the group consisting of:

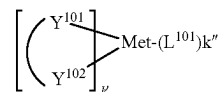


wherein k is an integer from 1 to 20; X¹⁰¹ to X¹⁰⁸ is C (including CH) or N; Z¹⁰¹ is NAr¹, O, or S; Ar¹ has the same group defined above.



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Examples of metal complexes used in HIL or HTL include, but are not limited to the following general formula:



wherein Met is a metal, which can have an atomic weight greater than 40; (Y¹⁰¹-Y¹⁰²) is a bidentate ligand, Y¹⁰¹ and Y¹⁰² are independently selected from C, N, O, P, and S; L¹⁰¹ is an ancillary ligand; k' is an integer value from 1 to the maximum number of ligands that may be attached to the metal; and k'+k'' is the maximum number of ligands that may be attached to the metal.

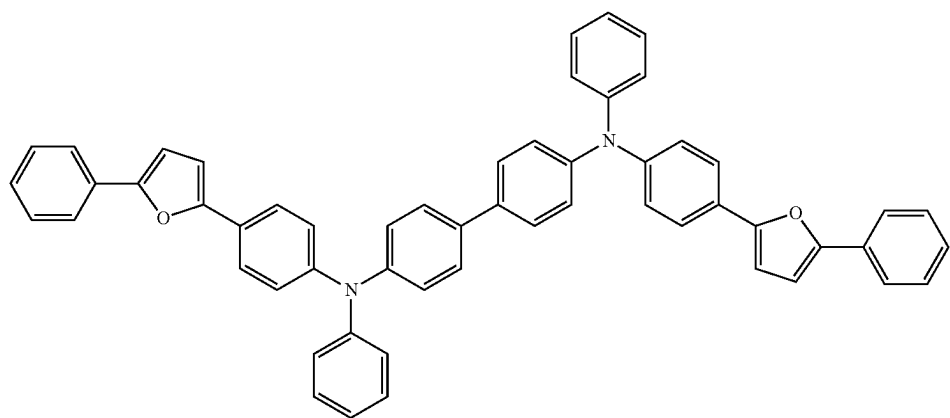
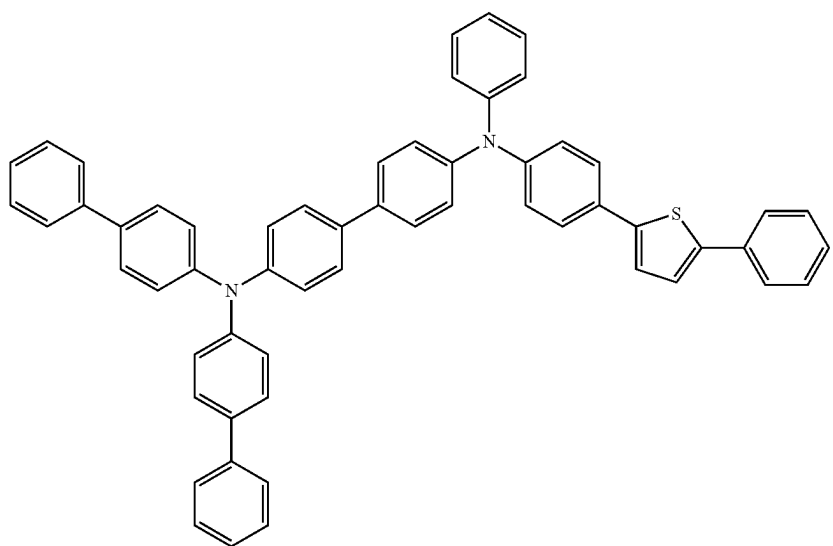
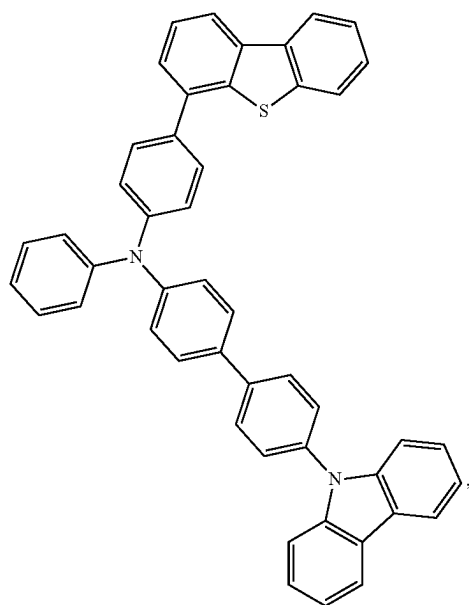
In one aspect, (Y¹⁰¹-Y¹⁰²) is a 2-phenylpyridine derivative. In another aspect, (Y¹⁰¹-Y¹⁰²) is a carbene ligand. In another aspect, Met is selected from Ir, Pt, Os, and Zn. In a further aspect, the metal complex has a smallest oxidation potential in solution vs. Fc⁺/Fc couple less than about 0.6 V.

Non-limiting examples of the HIL and HTL materials that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: CN102702075, DE102012005215, EP01624500, EP01698613, EP01806334, EP01930964, EP01972613, EP01997799, EP02011790, EP02055700, EP02055701, EP1725079, EP2085382, EP2660300, EP650955, JP07-073529, JP2005112765, JP2007091719, JP2008021687, JP2014-009196, KR20110088898, KR20130077473, TW201139402, U.S. Ser. No. 06/517,957, US20020158242, US20030162053, US20050123751, US20060182993, US20060240279, US20070145888, US20070181874, US20070278938, US20080014464, US20080091025, US20080106190, US20080124572, US20080145707, US20080220265, US20080233434, US20080303417, US2008107919, US20090115320, US20090167161, US2009066235, US2011007385, US20110163302, US2011240968, US2011278551, US2012205642, US2013241401, US20140117329, US2014183517, U.S. Pat. Nos. 5,061,569, 5,639,914, WO05075451, WO07125714, WO08023550, WO08023759, WO2009145016, WO2010061824, WO2011075644, WO2012177006, WO2013018530, WO2013039073, WO2013087142, WO2013118812, WO2013120577, WO2013157367, WO2013175747, WO2014002873, WO2014015935, WO2014015937, WO2014030872, WO2014030921, WO2014034791, WO2014104514, WO2014157018.

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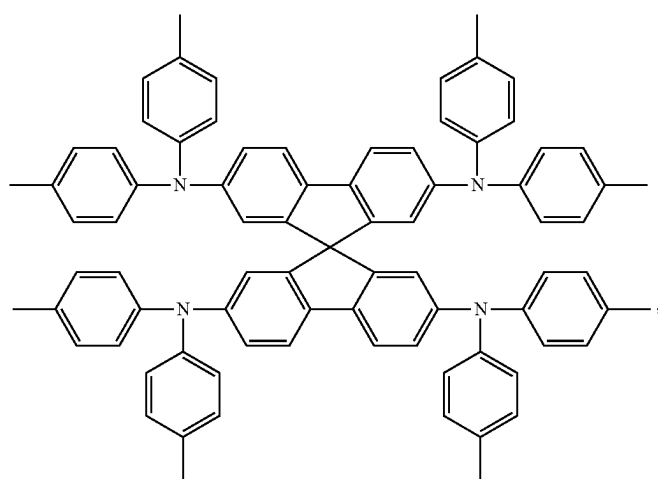
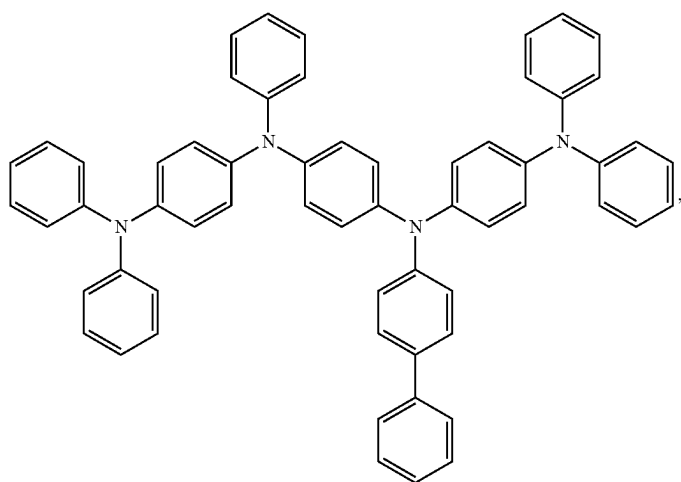
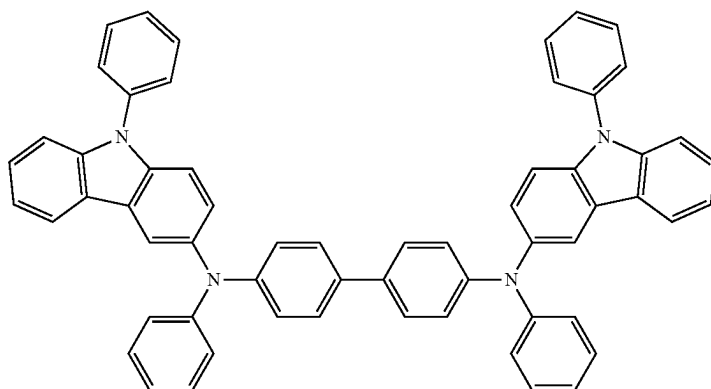
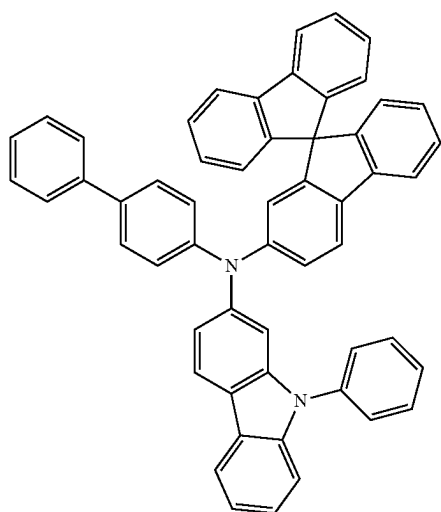
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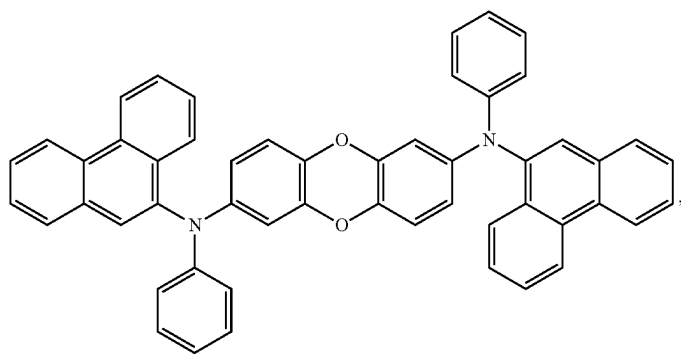
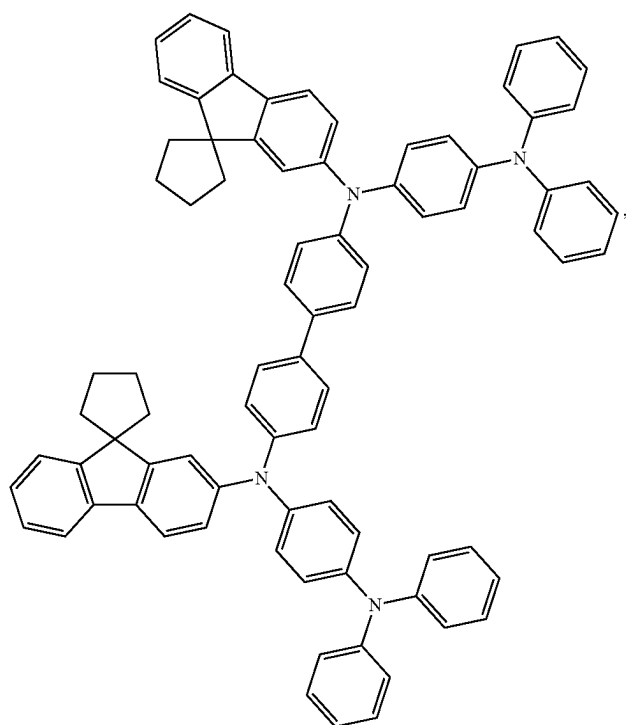
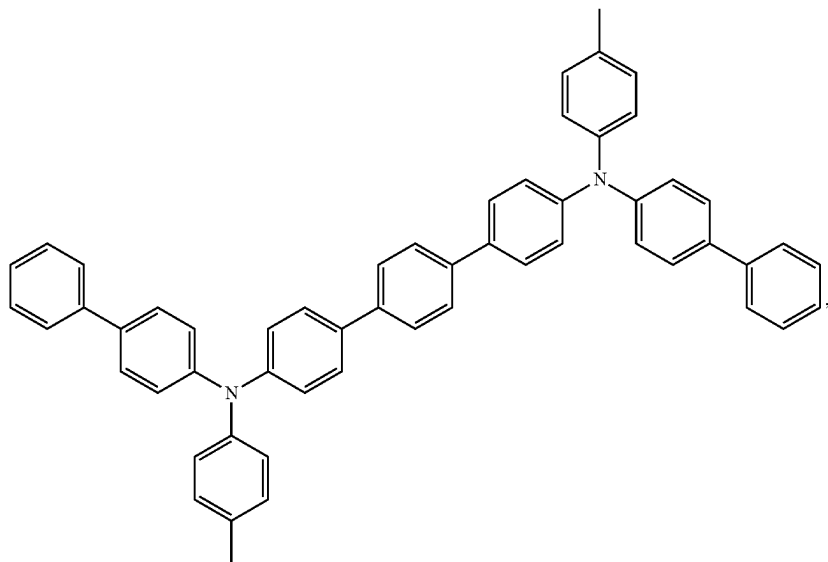
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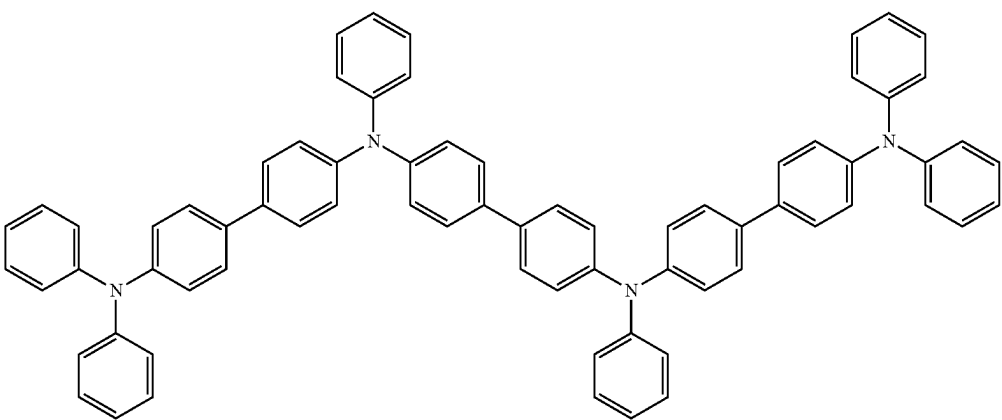
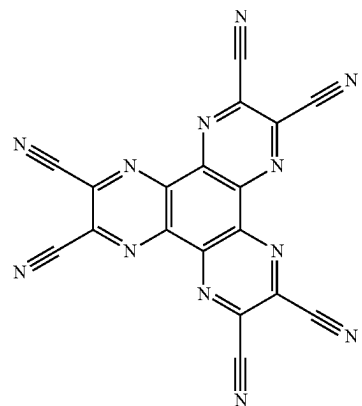
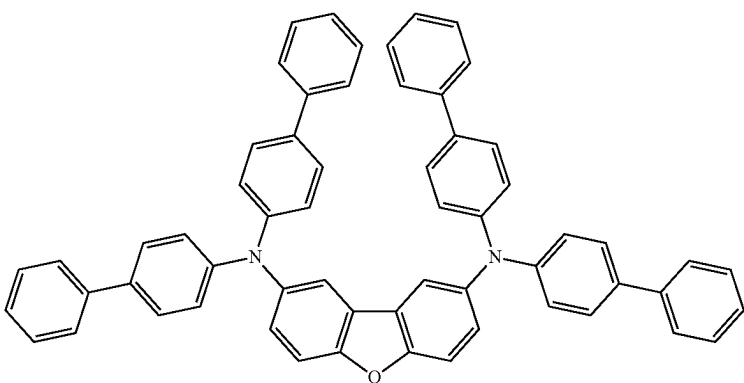
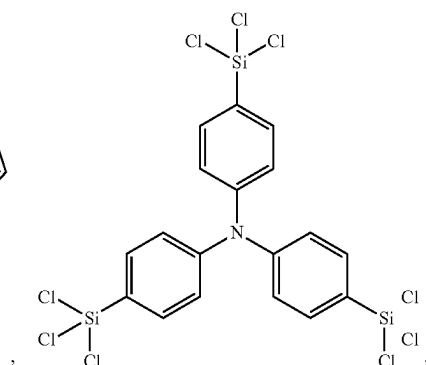
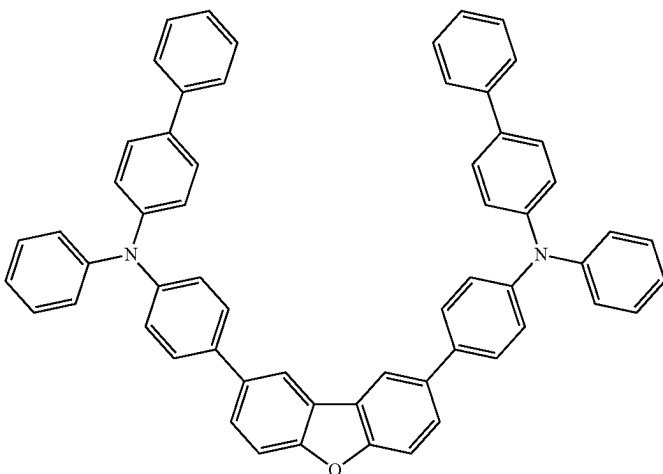
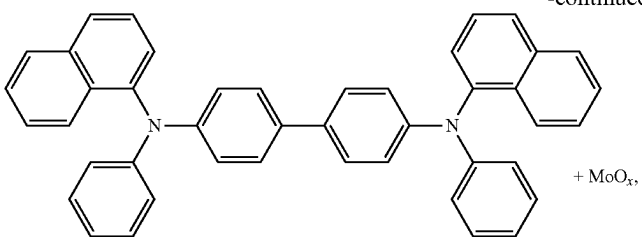
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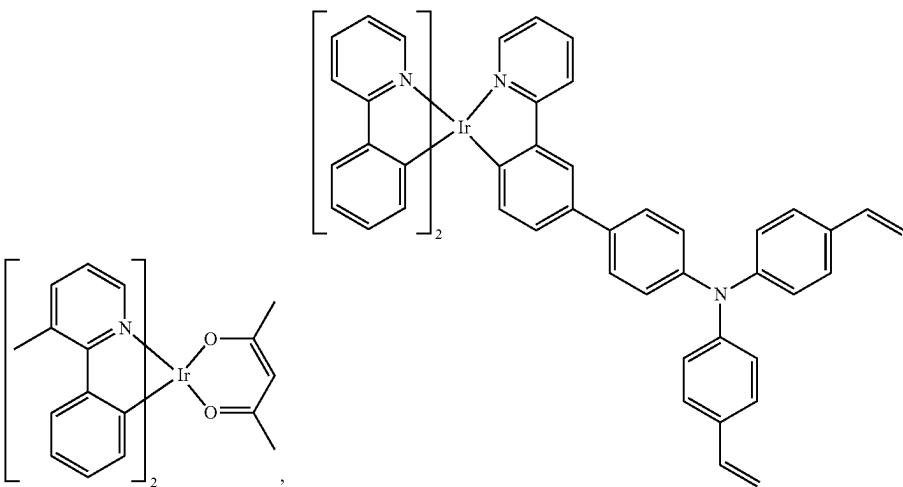
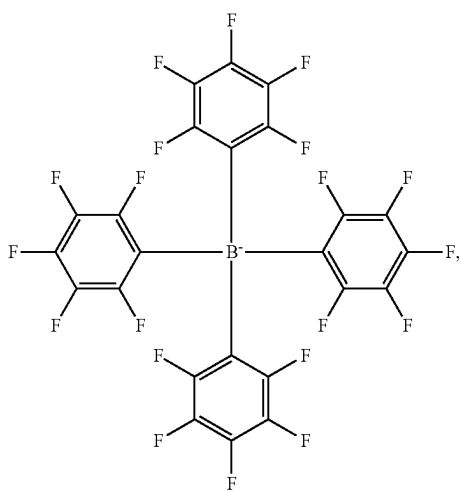
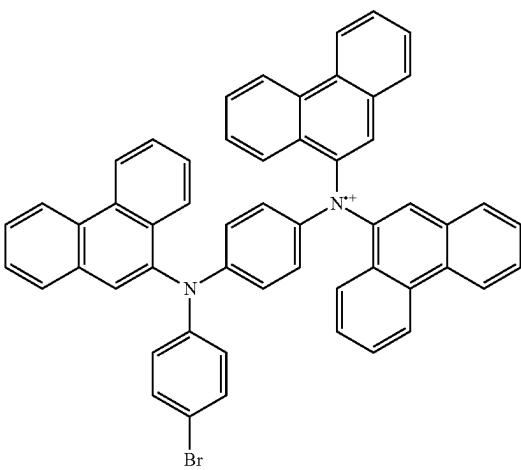
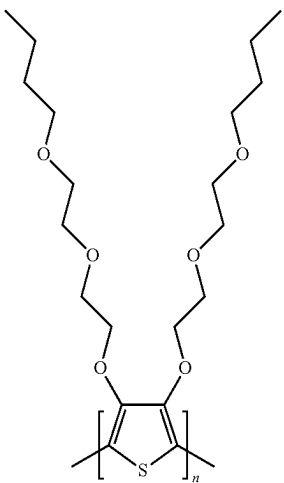
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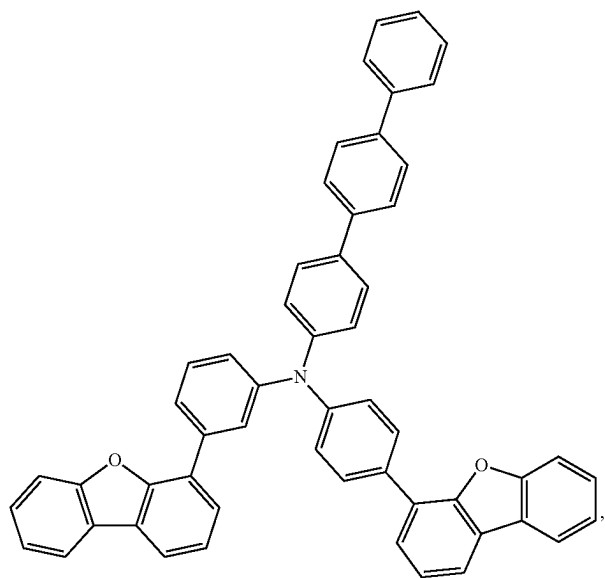
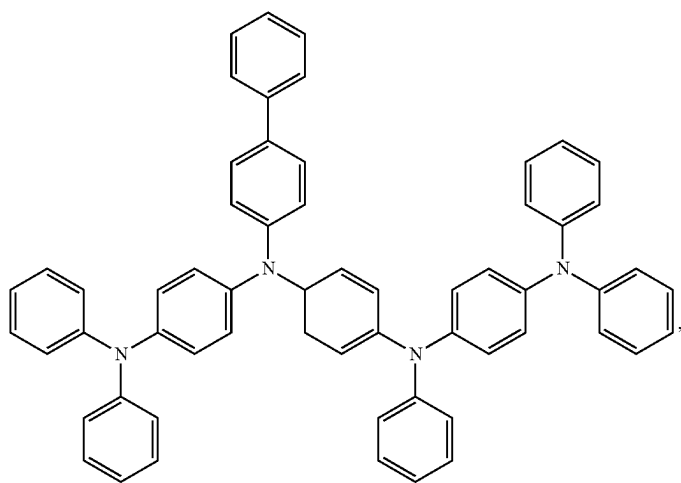
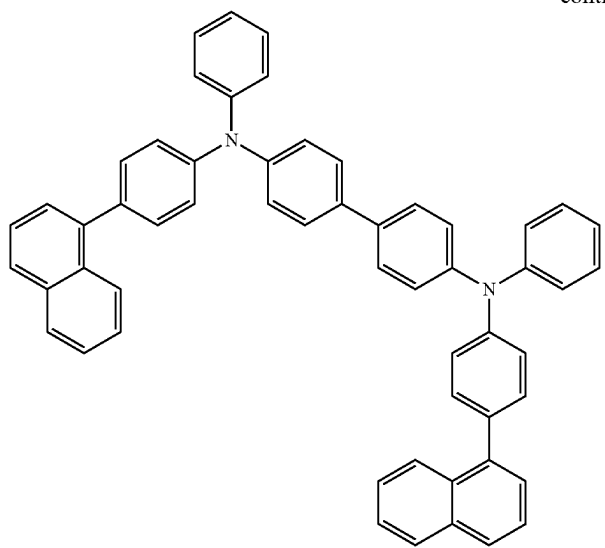
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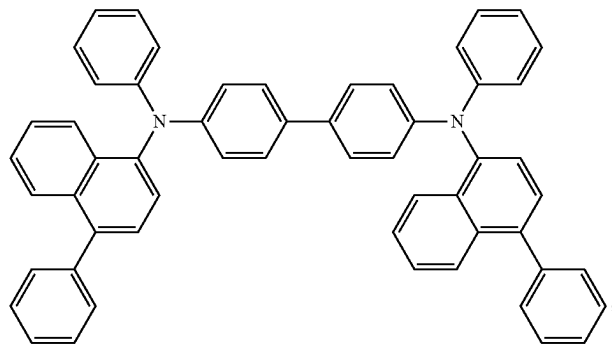
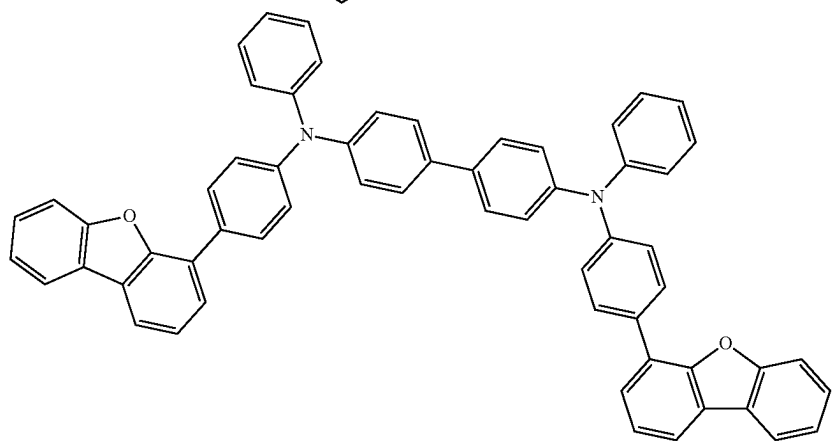
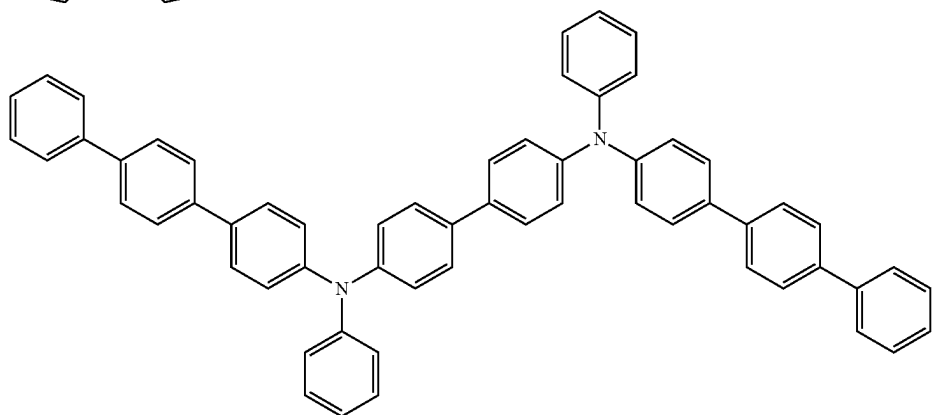
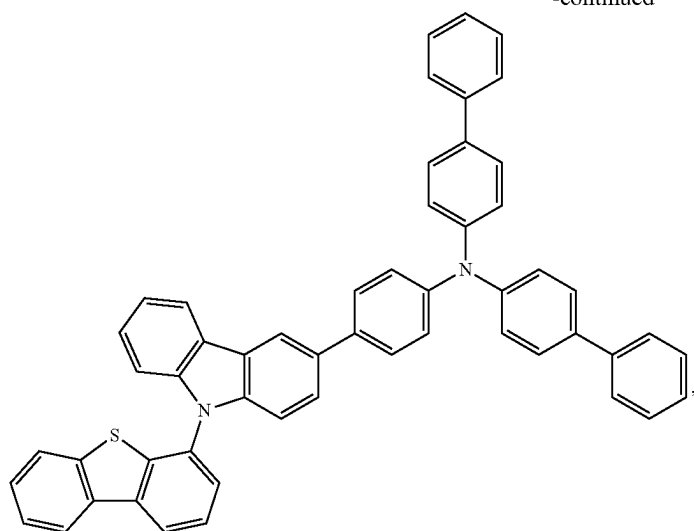
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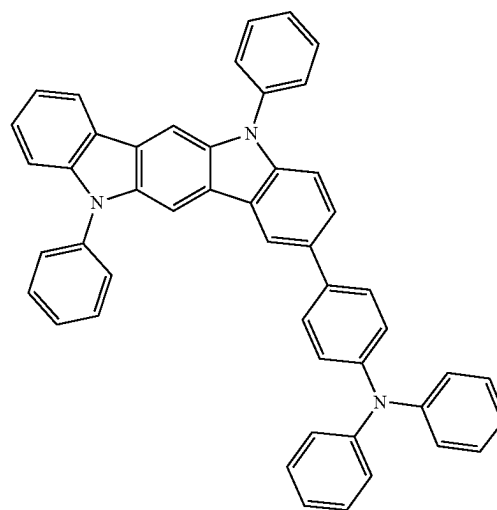
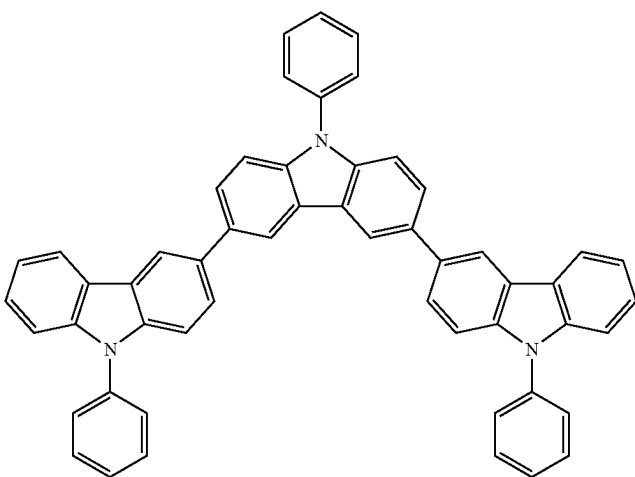
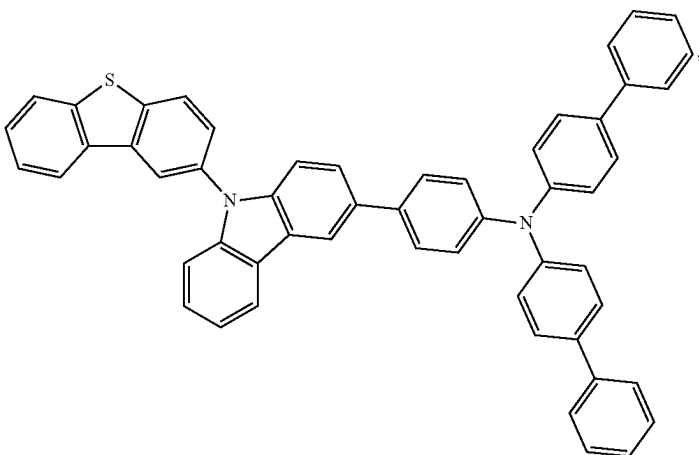
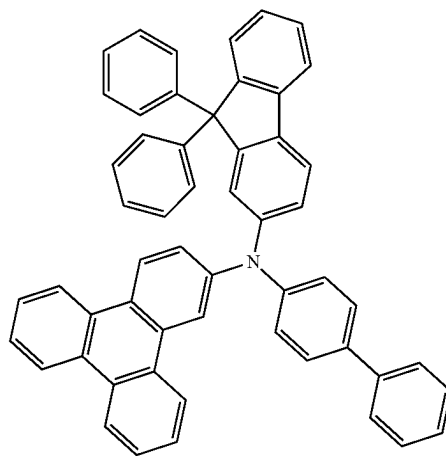
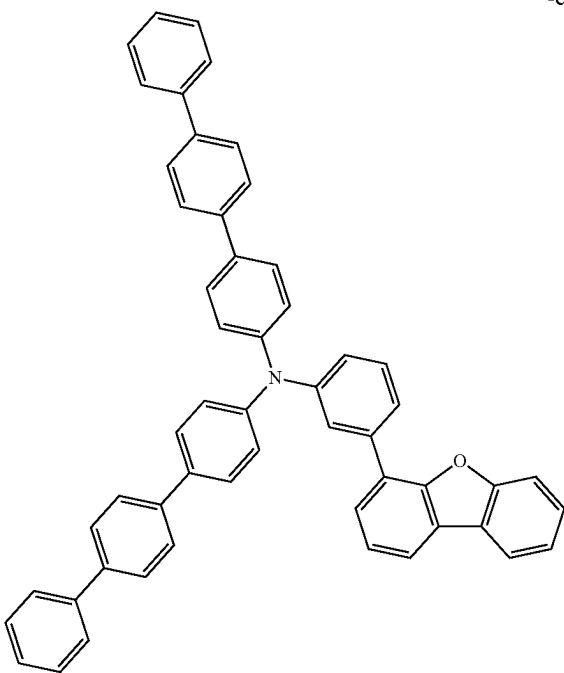
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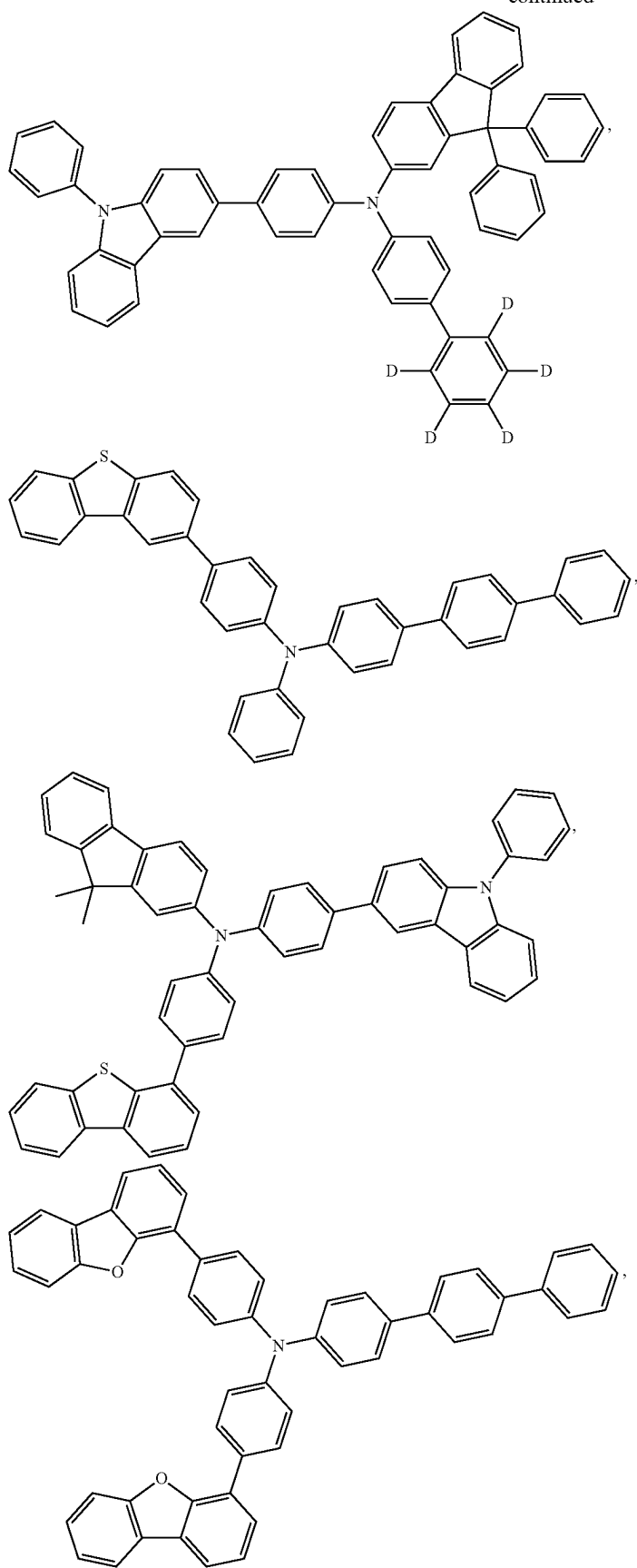
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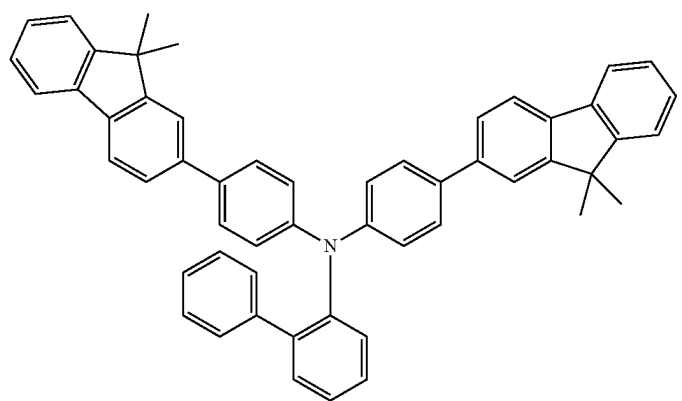
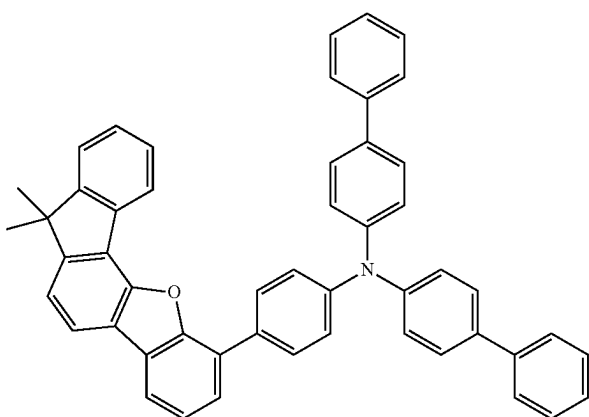
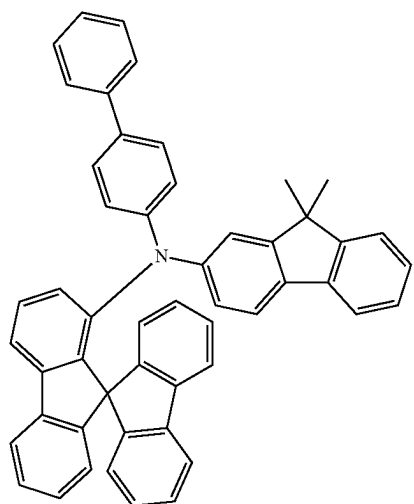
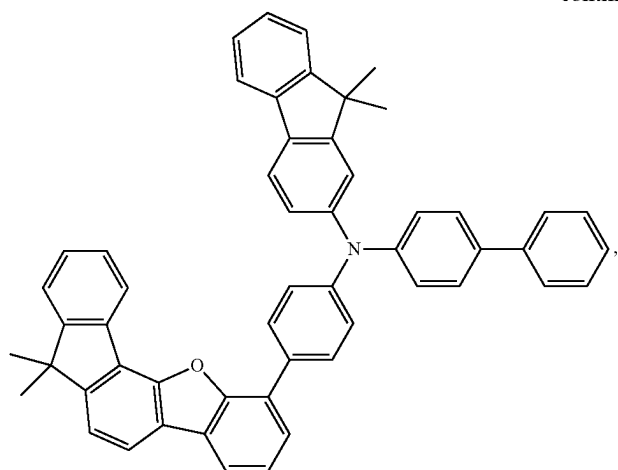
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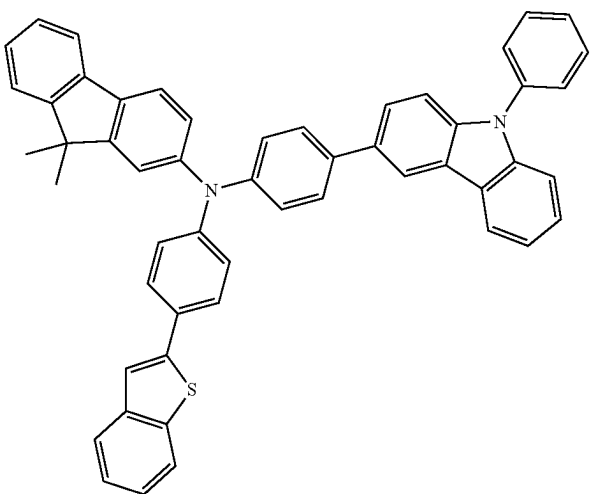
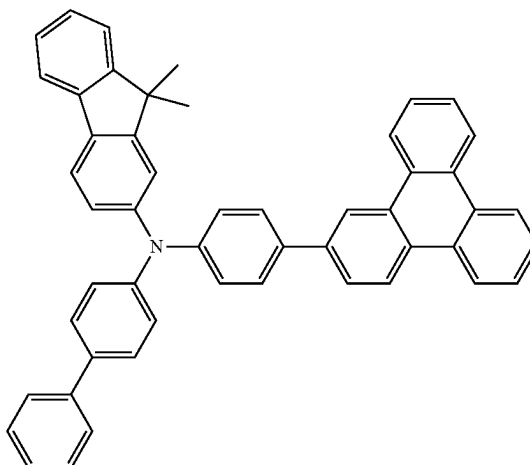
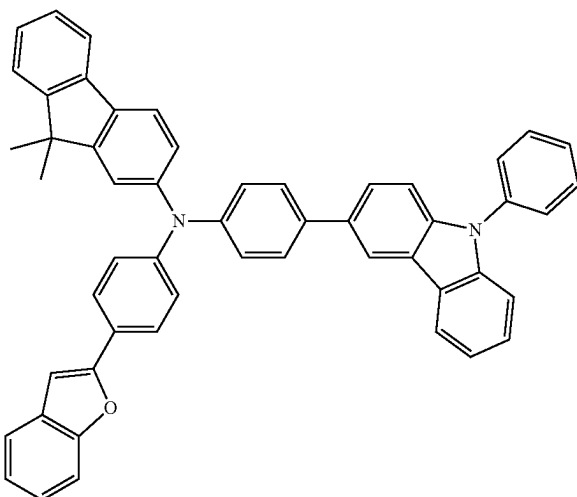
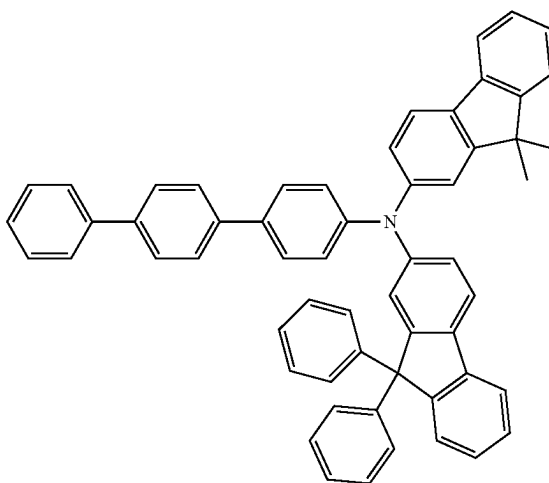
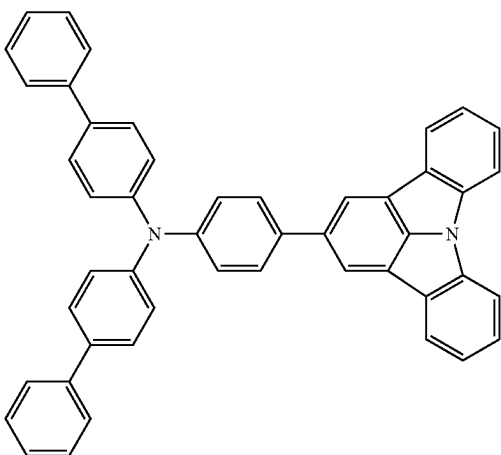
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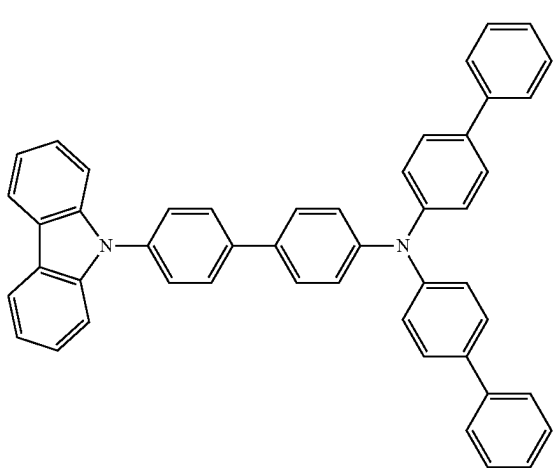
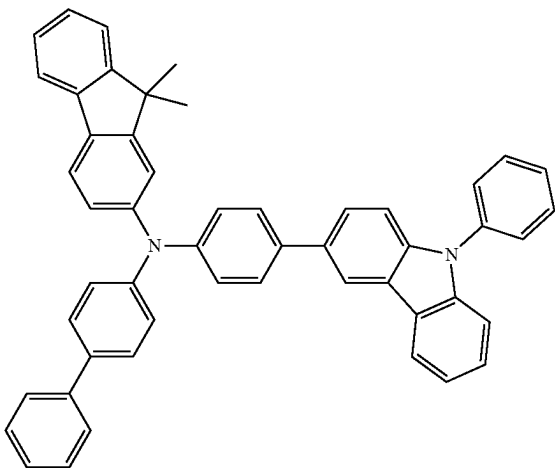
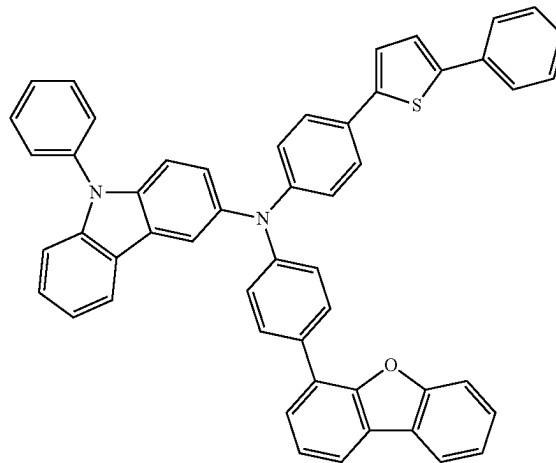
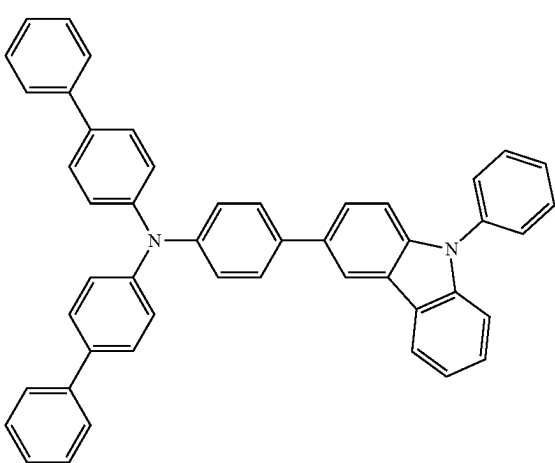
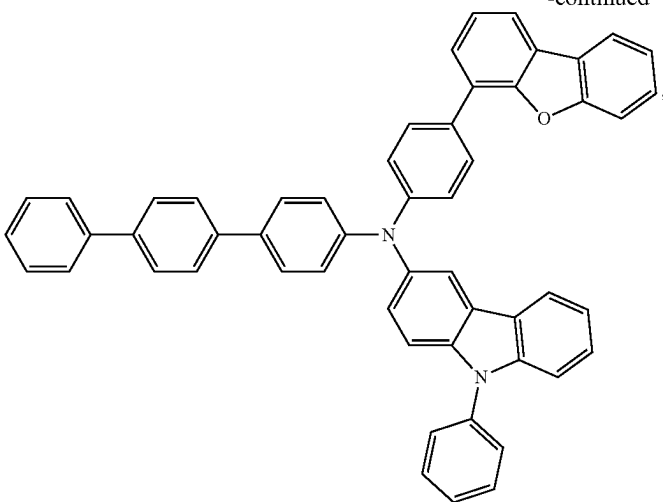
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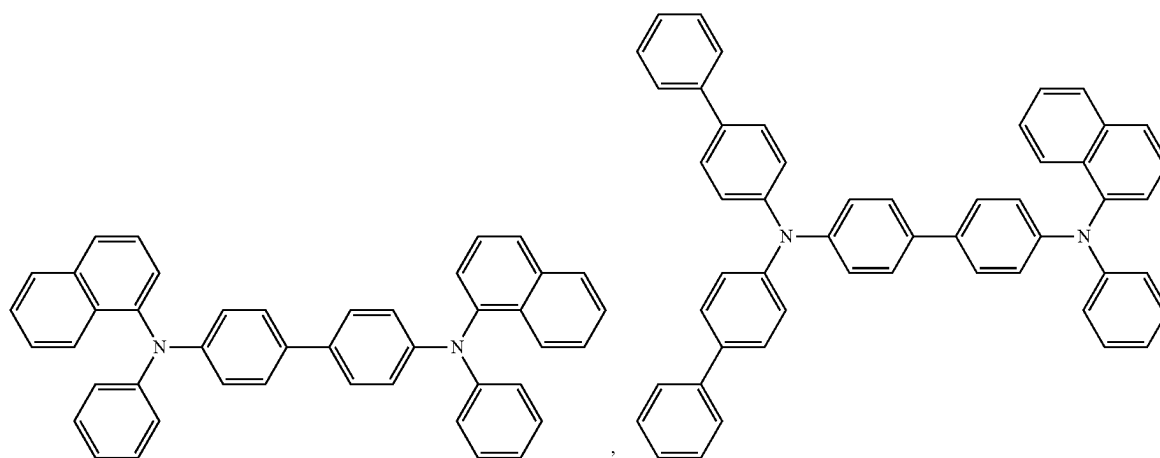
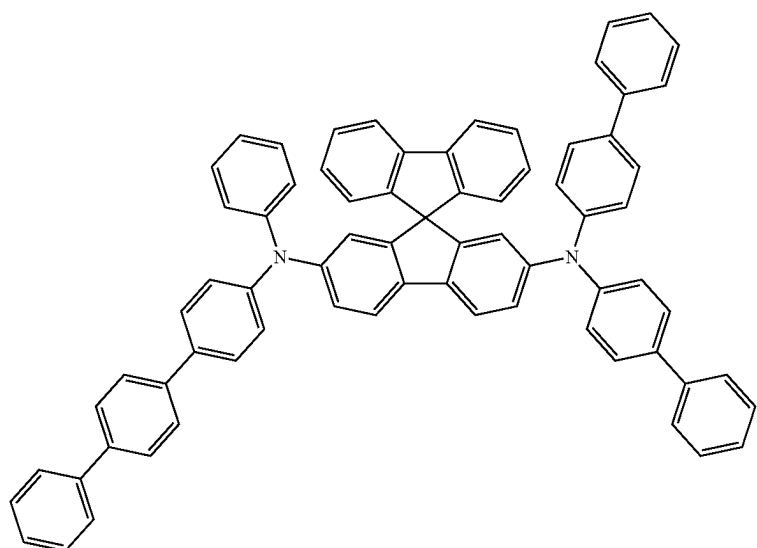
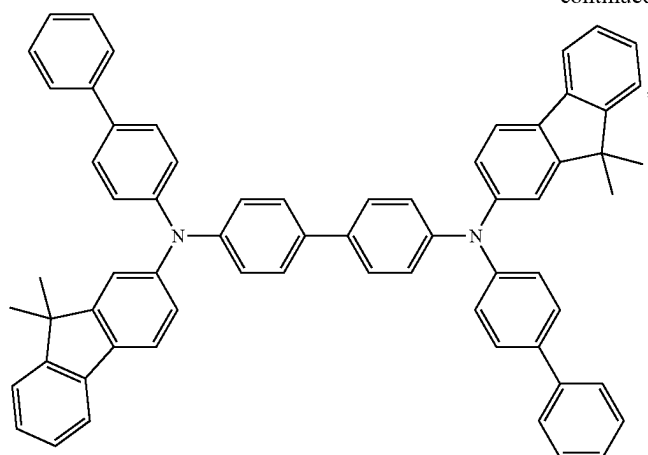
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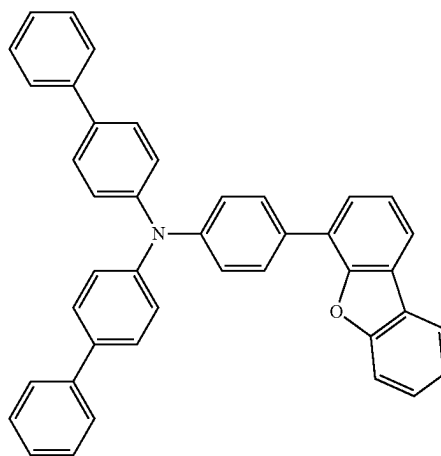
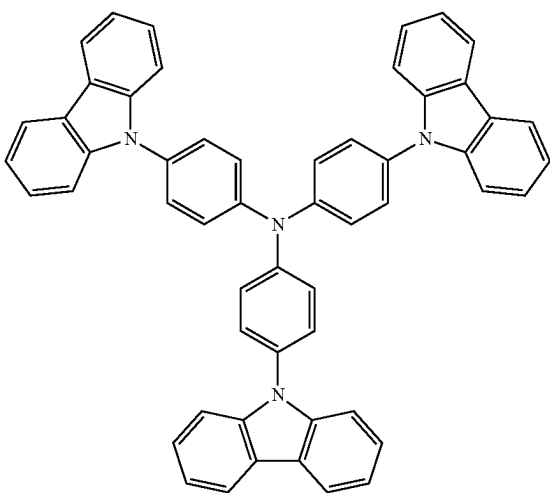
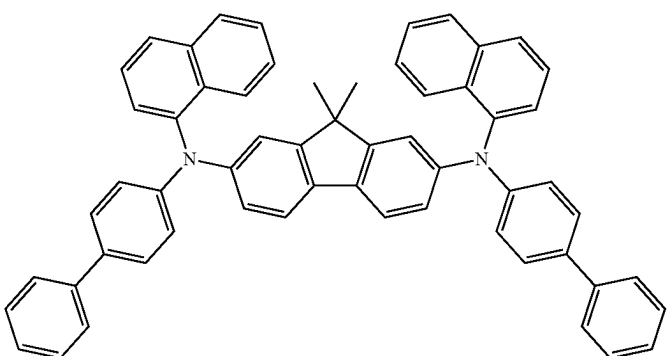
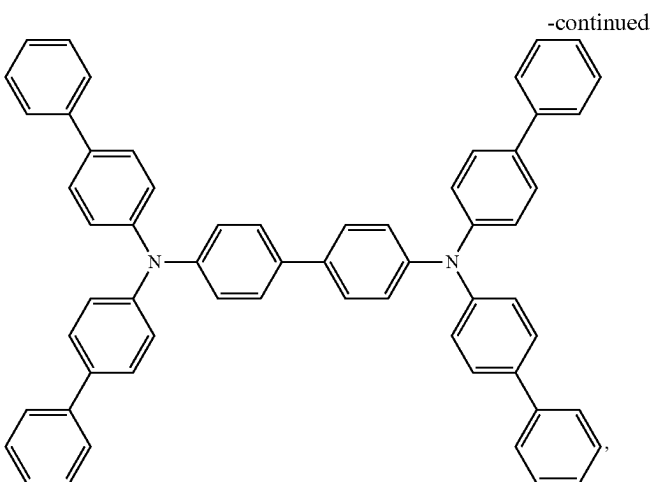
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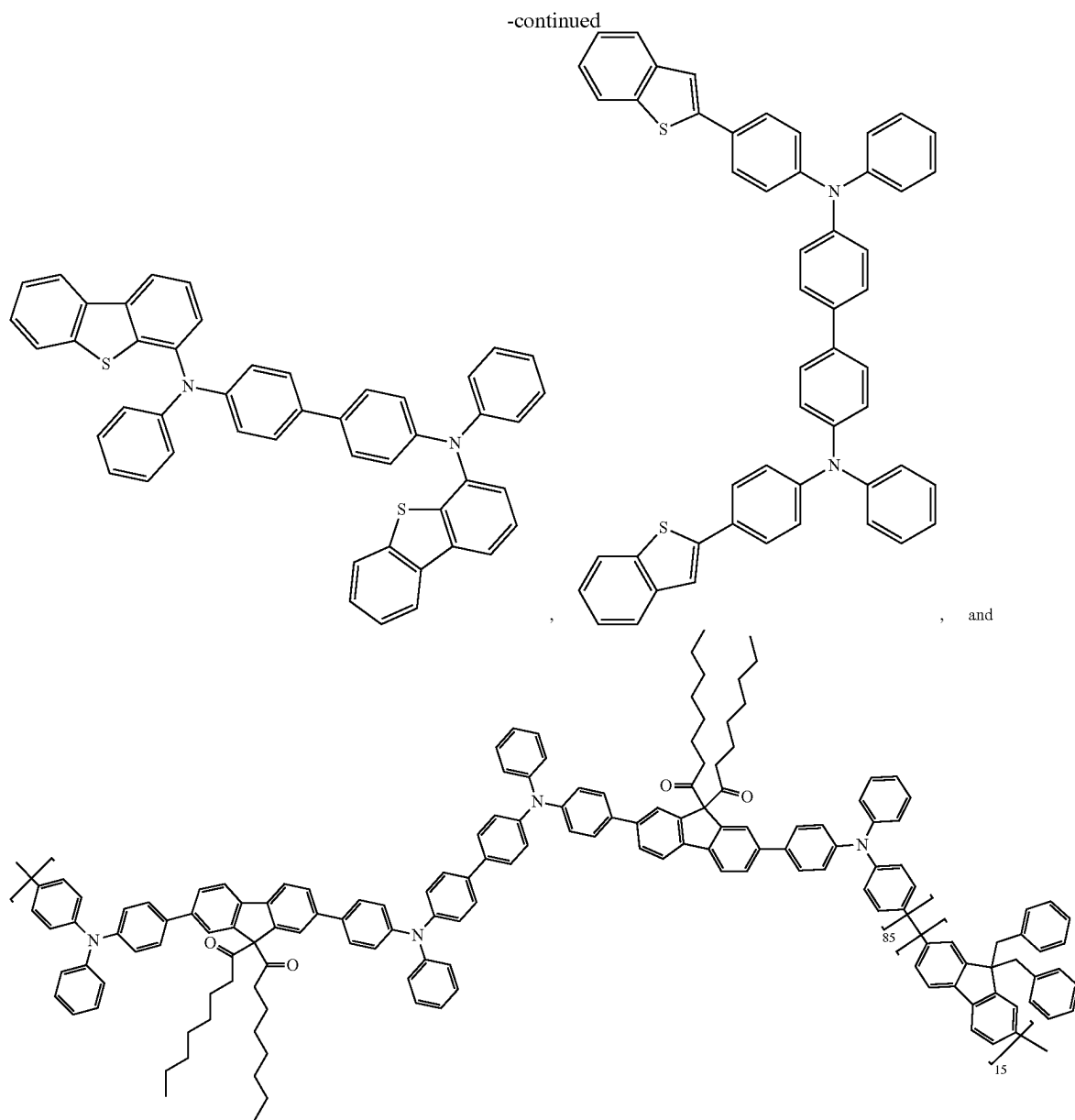
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EBL:

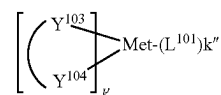
An electron blocking layer (EBL) may be used to reduce the number of electrons and/or excitons that leave the emissive layer. The presence of such a blocking layer in a device may result in substantially higher efficiencies, and/or longer lifetime, as compared to a similar device lacking a blocking layer. Also, a blocking layer may be used to confine emission to a desired region of an OLED. In some embodiments, the EBL material has a higher LUMO (closer to the vacuum level) and/or higher triplet energy than the emitter closest to the EBL interface. In some embodiments, the EBL material has a higher LUMO (closer to the vacuum level) and/or higher triplet energy than one or more of the hosts closest to the EBL interface. In one aspect, the compound used in EBL contains the same molecule or the same functional groups used as one of the hosts described below.

Host:

The light emitting layer of the organic EL device of the present invention preferably contains at least a metal com-

plex as light emitting material, and may contain a host material using the metal complex as a dopant material. Examples of the host material are not particularly limited, and any metal complexes or organic compounds may be used as long as the triplet energy of the host is larger than that of the dopant. Any host material may be used with any dopant so long as the triplet criteria is satisfied.

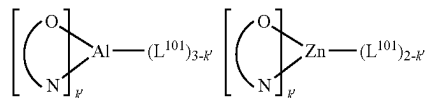
Examples of metal complexes used as host are preferred to have the following general formula:



wherein Met is a metal; $(Y^{103}-Y^{104})$ is a bidentate ligand, Y^{103} and Y^{104} are independently selected from C, N, O, P, and S; L^{101} is another ligand; k' is an integer value

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from 1 to the maximum number of ligands that may be attached to the metal; and $k'+k''$ is the maximum number of ligands that may be attached to the metal. In one aspect, the metal complexes are:

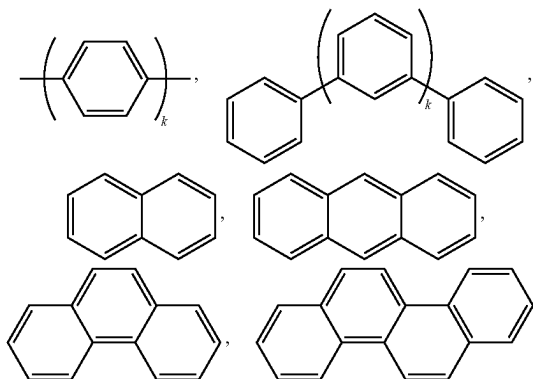


wherein (O—N) is a bidentate ligand, having metal coordinated to atoms O and N.

In another aspect, Met is selected from Ir and Pt. In a further aspect, $(\text{Y}^{103}, \text{Y}^{104})$ is a carbene ligand.

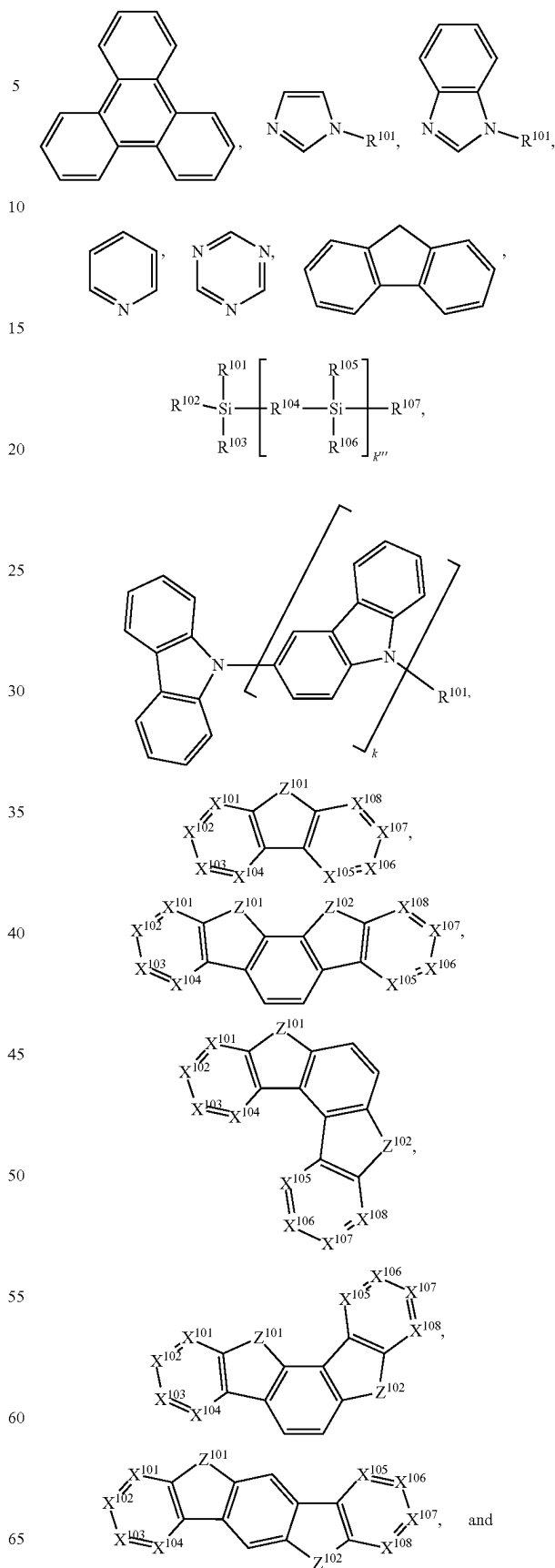
Examples of other organic compounds used as host are selected from the group consisting of aromatic hydrocarbon cyclic compounds such as benzene, biphenyl, triphenyl, triphenylene, tetraphenylene, naphthalene, anthracene, phenalene, phenanthrene, fluorene, pyrene, chrysene, perylene, and azulene; the group consisting of aromatic heterocyclic compounds such as dibenzothiophene, dibenzofuran, dibenzoselenophene, furan, thiophene, benzofuran, benzothiophene, benzoselenophene, carbazole, indolocarbazole, pyridylindole, pyrrolodipyridine, pyrazole, imidazole, triazole, oxazole, thiazole, oxadiazole, oxatriazole, dioxazole, thiadiazole, pyridine, pyridazine, pyrimidine, pyrazine, triazine, oxazine, oxathiazine, oxadiazine, indole, benzimidazole, indazole, indoxazine, benzoxazole, benzisoxazole, benzothiazole, quinoline, isoquinoline, cinnoline, quinazoline, quinoxaline, naphthyridine, phthalazine, pteridine, xanthene, acridine, phenazine, phenothiazine, phenoxazine, benzofuropyridine, furodipyridine, benzothienopyridine, thienodipyridine, benzoselenophenopyridine, and selenophenodipyridine; and the group consisting of 2 to 10 cyclic structural units which are groups of the same type or different types selected from the aromatic hydrocarbon cyclic group and the aromatic heterocyclic group and are bonded to each other directly or via at least one of oxygen atom, nitrogen atom, sulfur atom, silicon atom, phosphorus atom, boron atom, chain structural unit and the aliphatic cyclic group. Each option within each group may be unsubstituted or may be substituted by a substituent selected from the group consisting of deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof.

In one aspect, the host compound contains at least one of the following groups in the molecule:

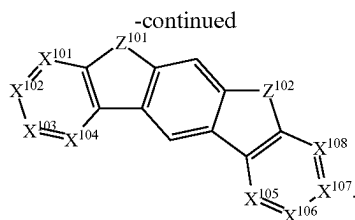


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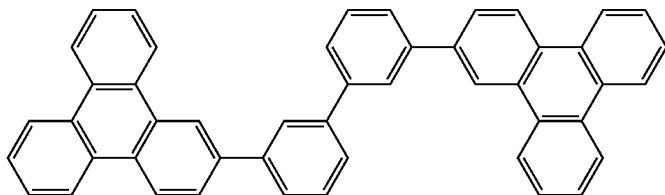
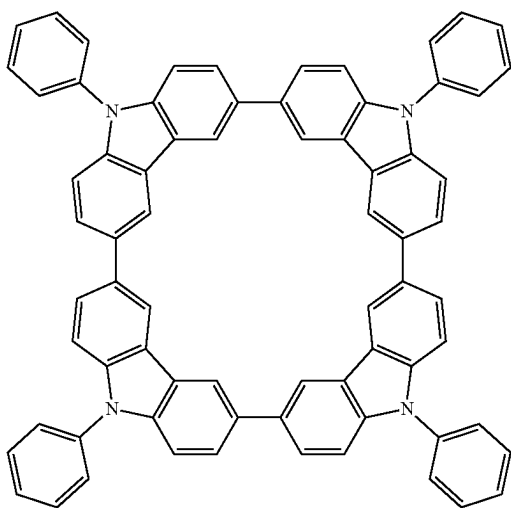
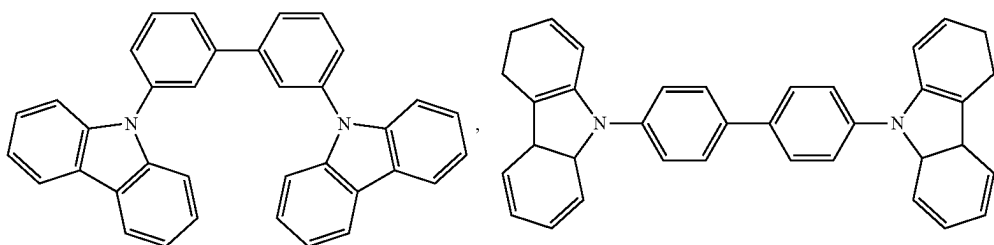
wherein each of R^{101} to R^{107} is independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, and when it is aryl or heteroaryl, it has the similar definition as Ar's mentioned above. k is an integer from 0 to 20 or 1 to 20; k'' is an integer from 0 to 20. X^{101} to X^{108} is selected from C (including CH) or N.

Z^{101} and Z^{102} is selected from NR^{101} , O, or S.

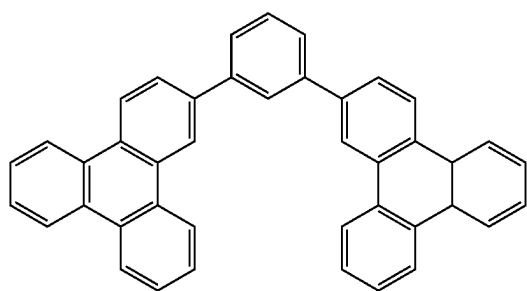
312

Non-limiting examples of the host materials that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: EP2034538, EP2034538A,

- | | | | |
|----|----------------|--------------------------|----------------|
| 5 | EP2757608, | JP2007254297, | KR20100079458, |
| | KR20120088644, | KR20120129733, | KR20130115564, |
| | TW201329200, | US20030175553, | US20050238919, |
| | US20060280965, | US20090017330, | US20090030202, |
| | US20090167162, | US20090302743, | US20090309488, |
| 10 | US20100012931, | US20100084966, | US20100187984, |
| | US2010187984, | US2012075273, | US2012126221, |
| | US2013009543, | US2013105787, | US2013175519, |
| | US2014001446, | US20140183503, | US20140225088, |
| | US2014034914, | U.S. Pat. No. 7,154,114, | WO2001039234, |
| 15 | WO2004093207, | WO2005014551, | WO2005089025, |
| | WO2006072002, | WO2006114966, | WO2007063754, |
| | WO2008056746, | WO2009003898, | WO2009021126, |
| | WO2009063833, | WO2009066778, | WO2009066779, |
| | WO2009086028, | WO2010056066, | WO2010107244, |
| 20 | WO2011081423, | WO2011081431, | WO2011086863, |
| | WO2012128298, | WO2012133644, | WO2012133649, |
| | WO2013024872, | WO2013035275, | WO2013081315, |
| | WO2013191404, | WO2014142472, | |

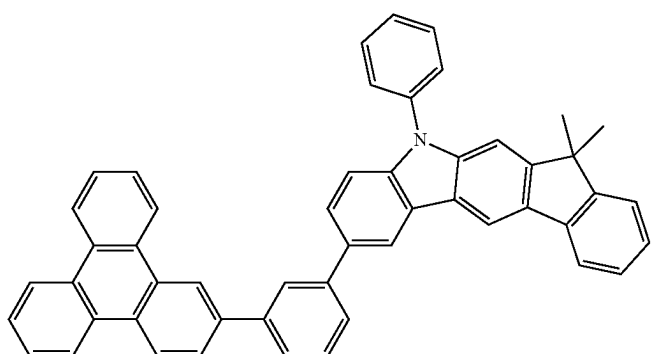
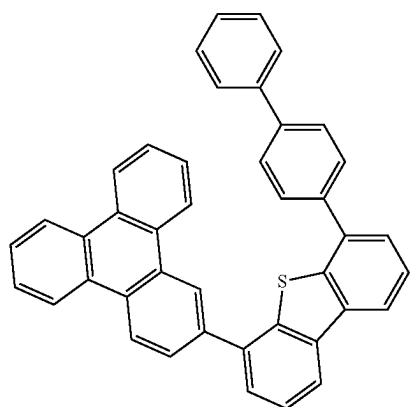
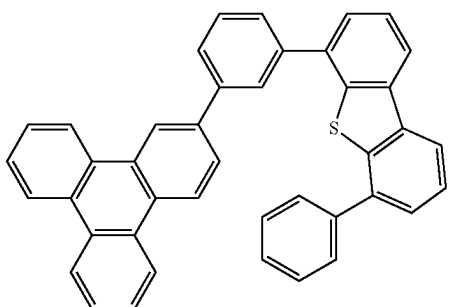
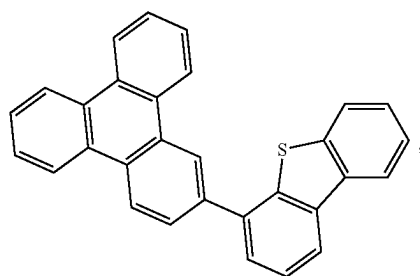
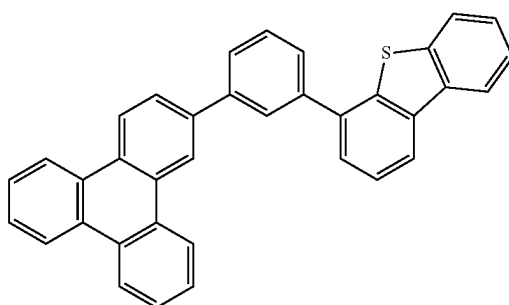
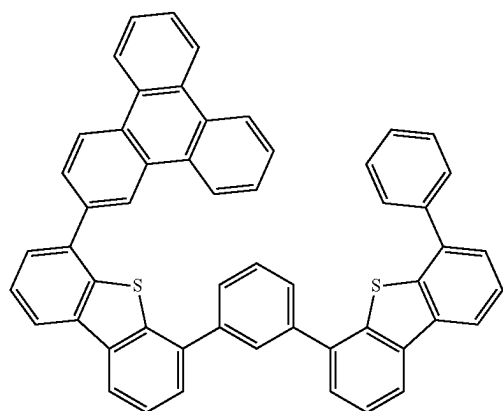
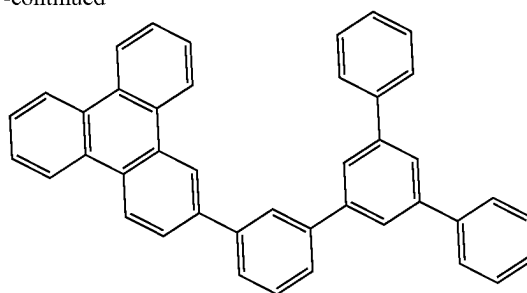


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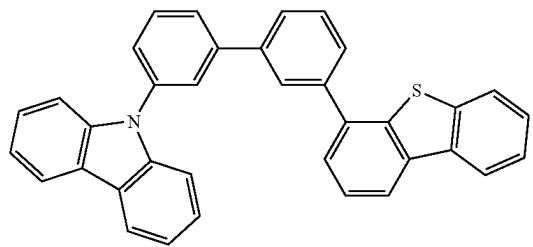
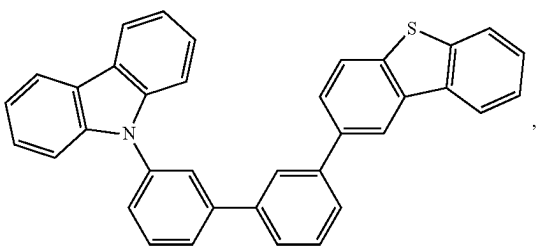
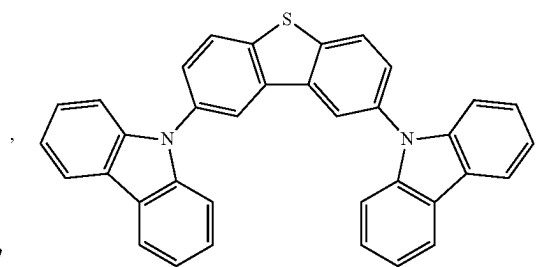
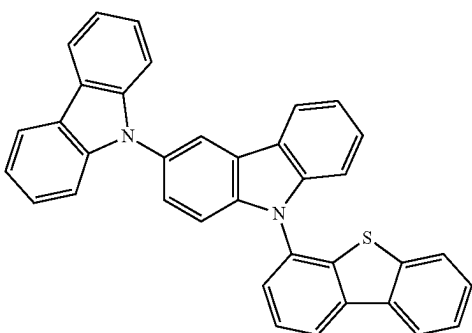
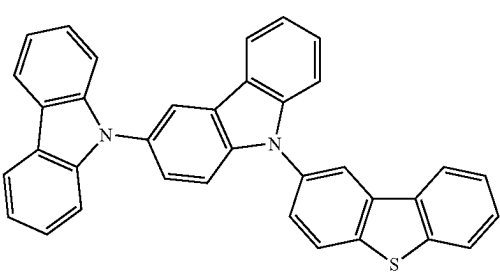
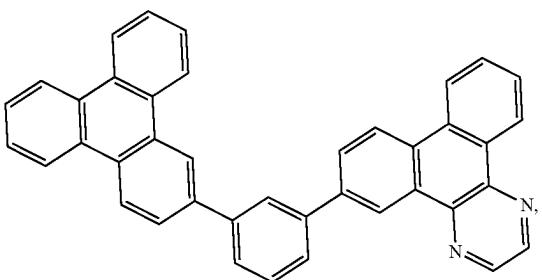
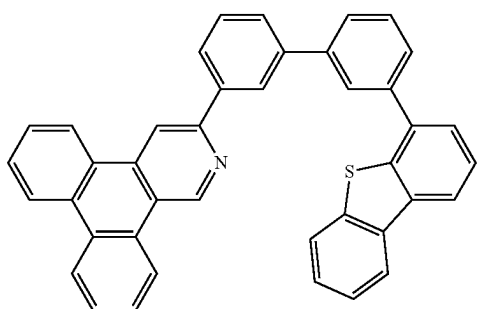
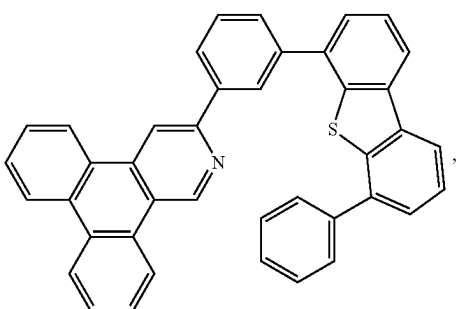
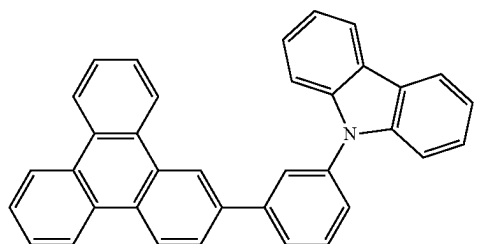
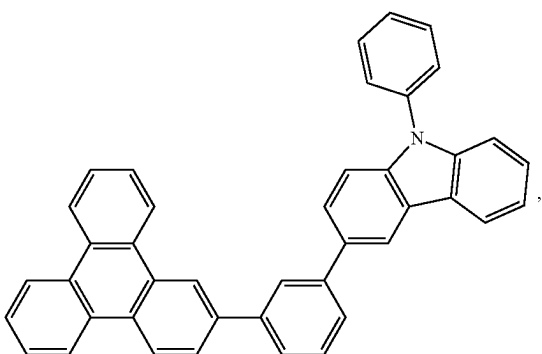
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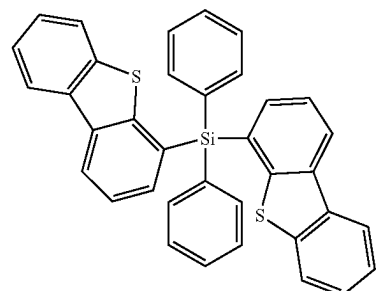
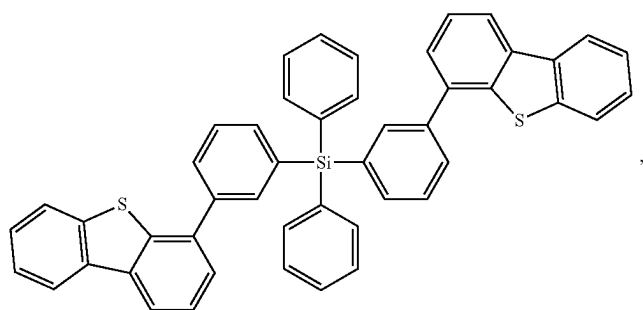
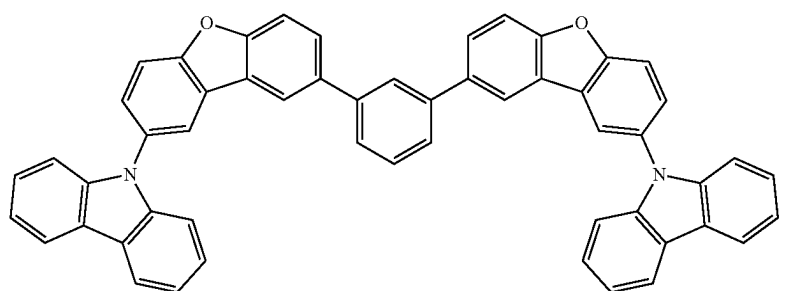
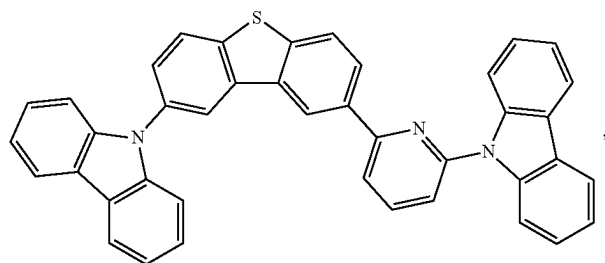
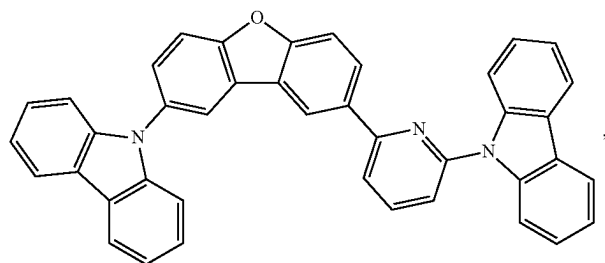
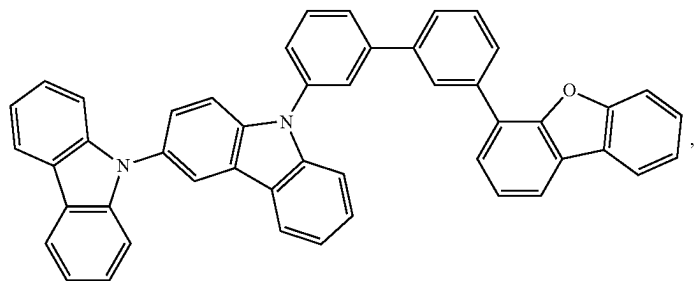
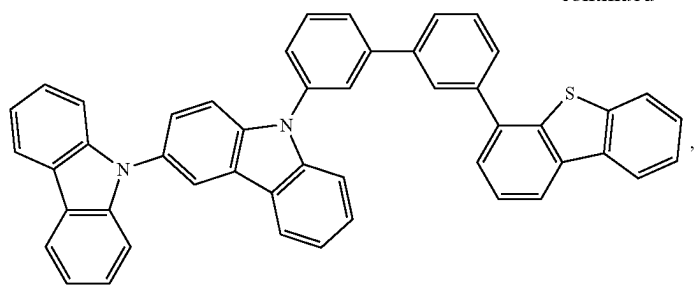
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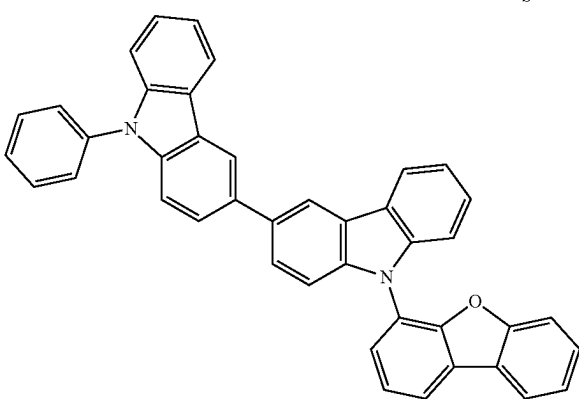
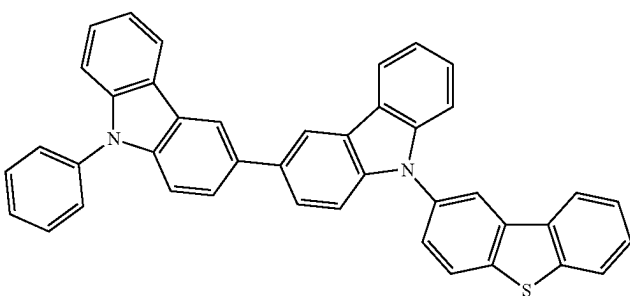
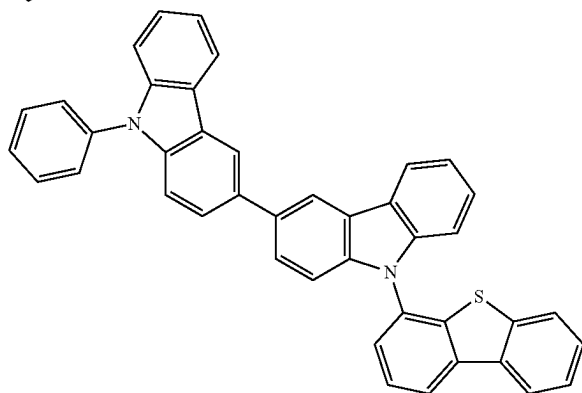
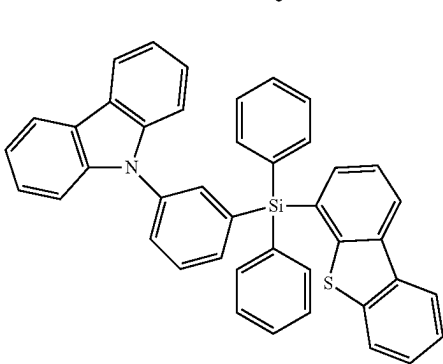
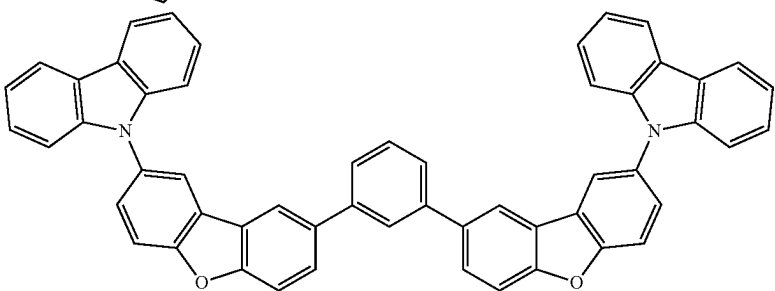
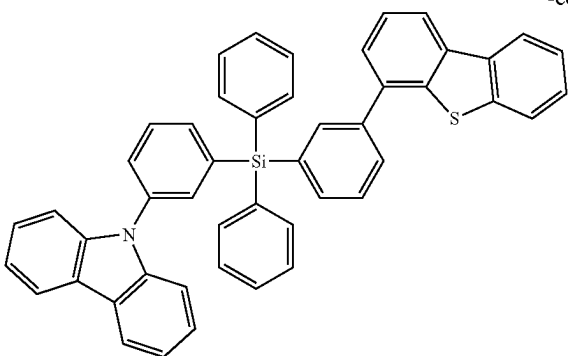
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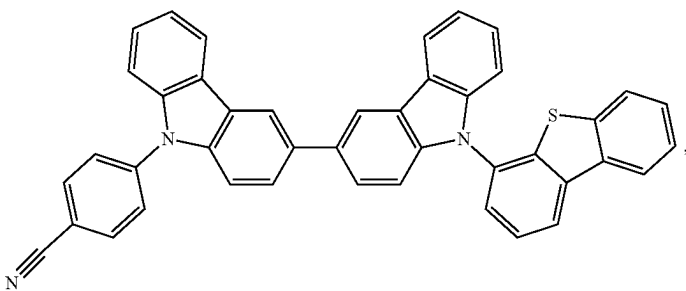
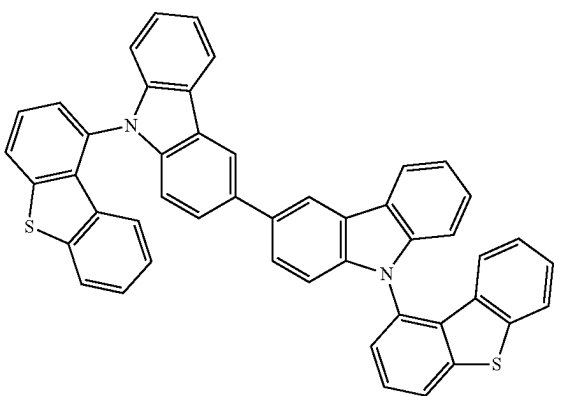
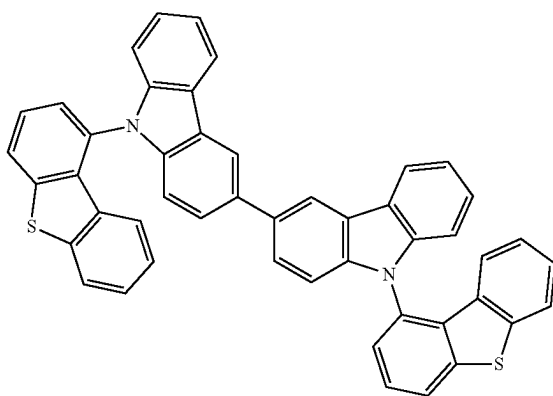
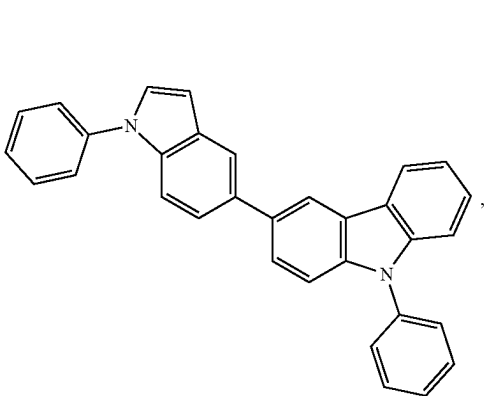
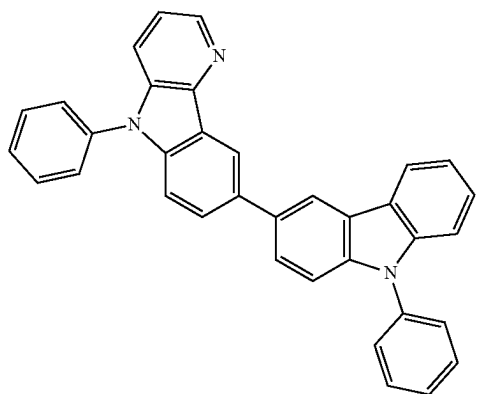
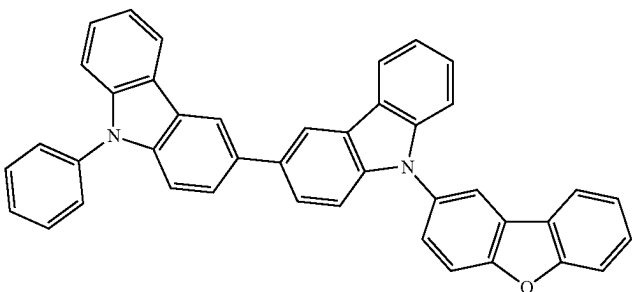
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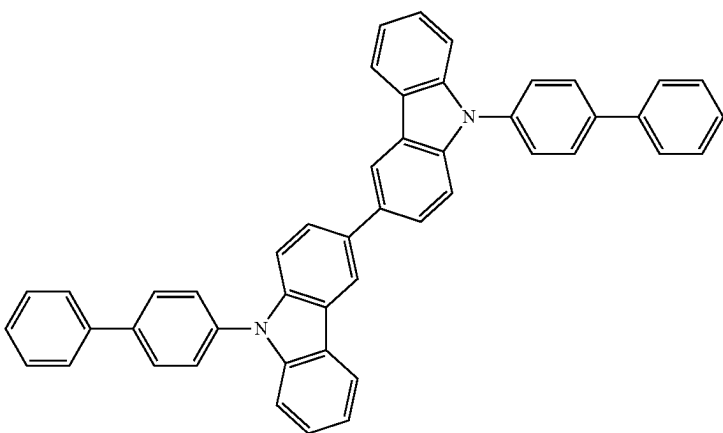
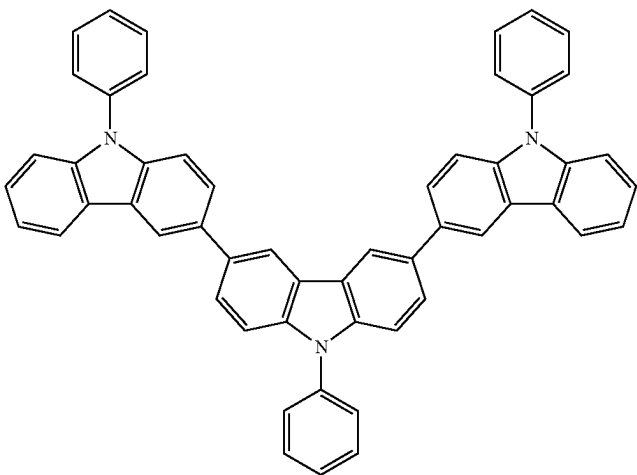
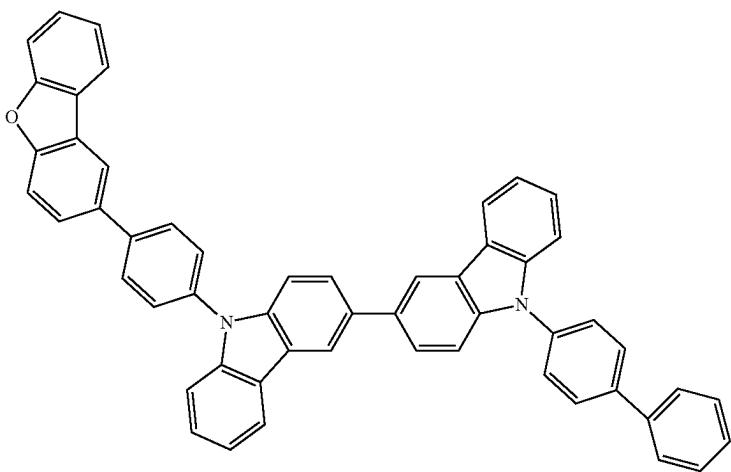
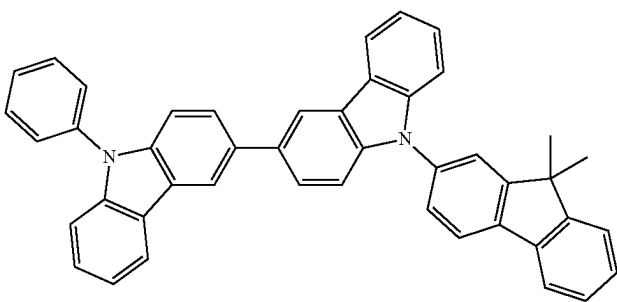
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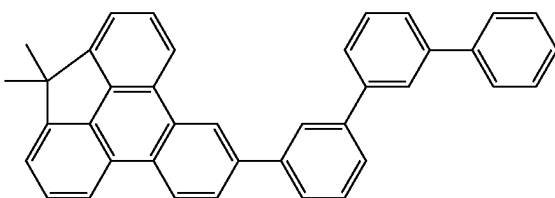
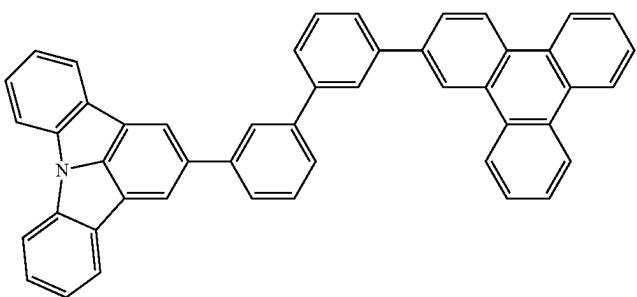
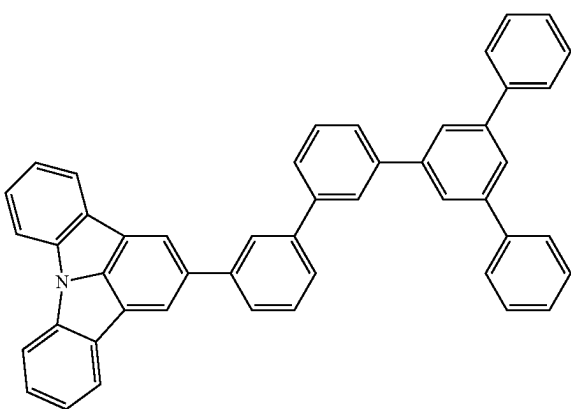
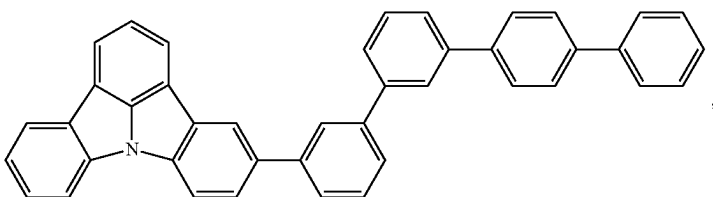
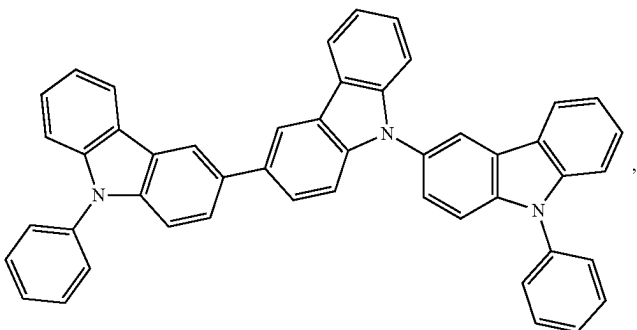
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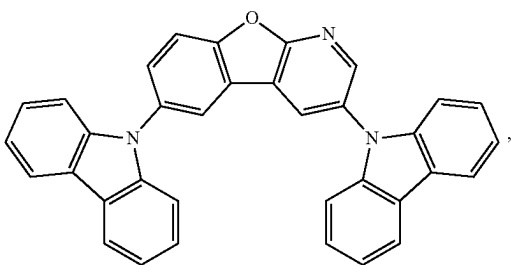
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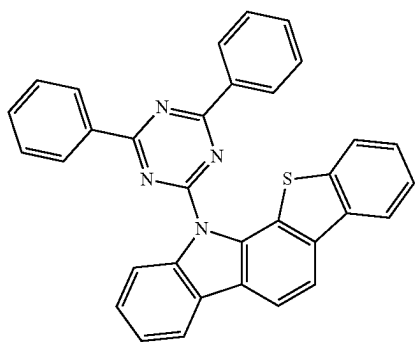
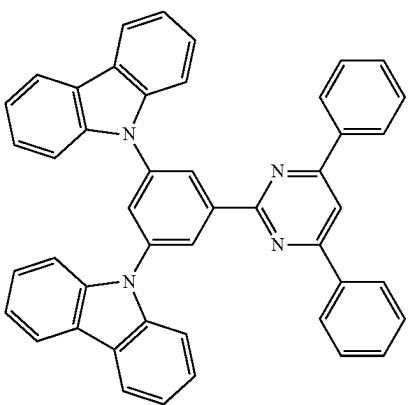
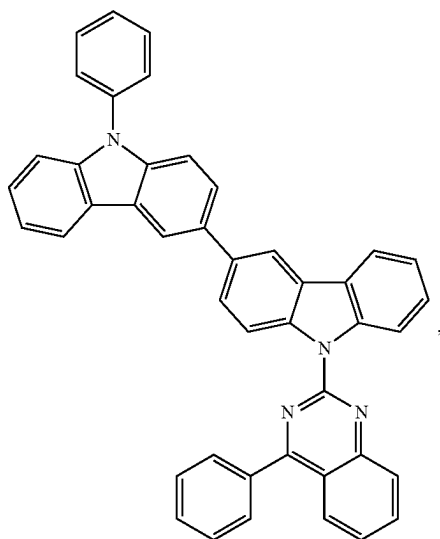
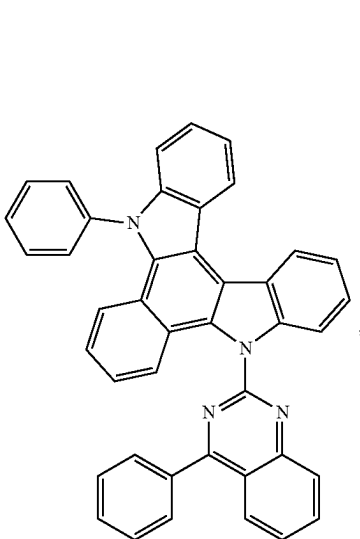
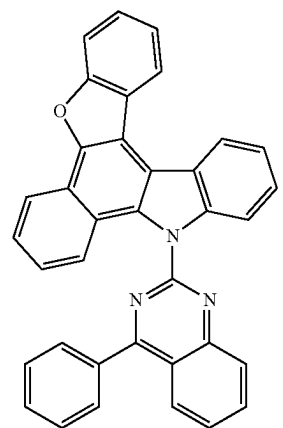
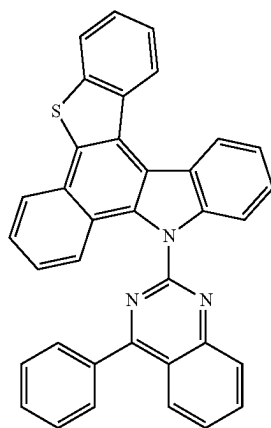
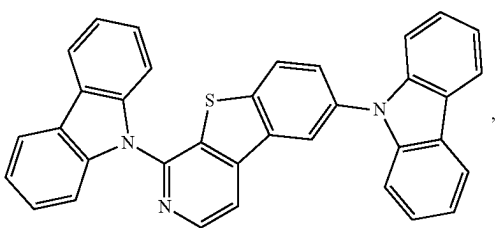
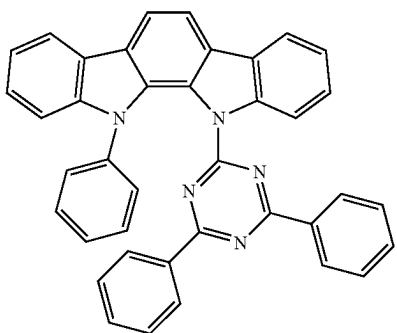


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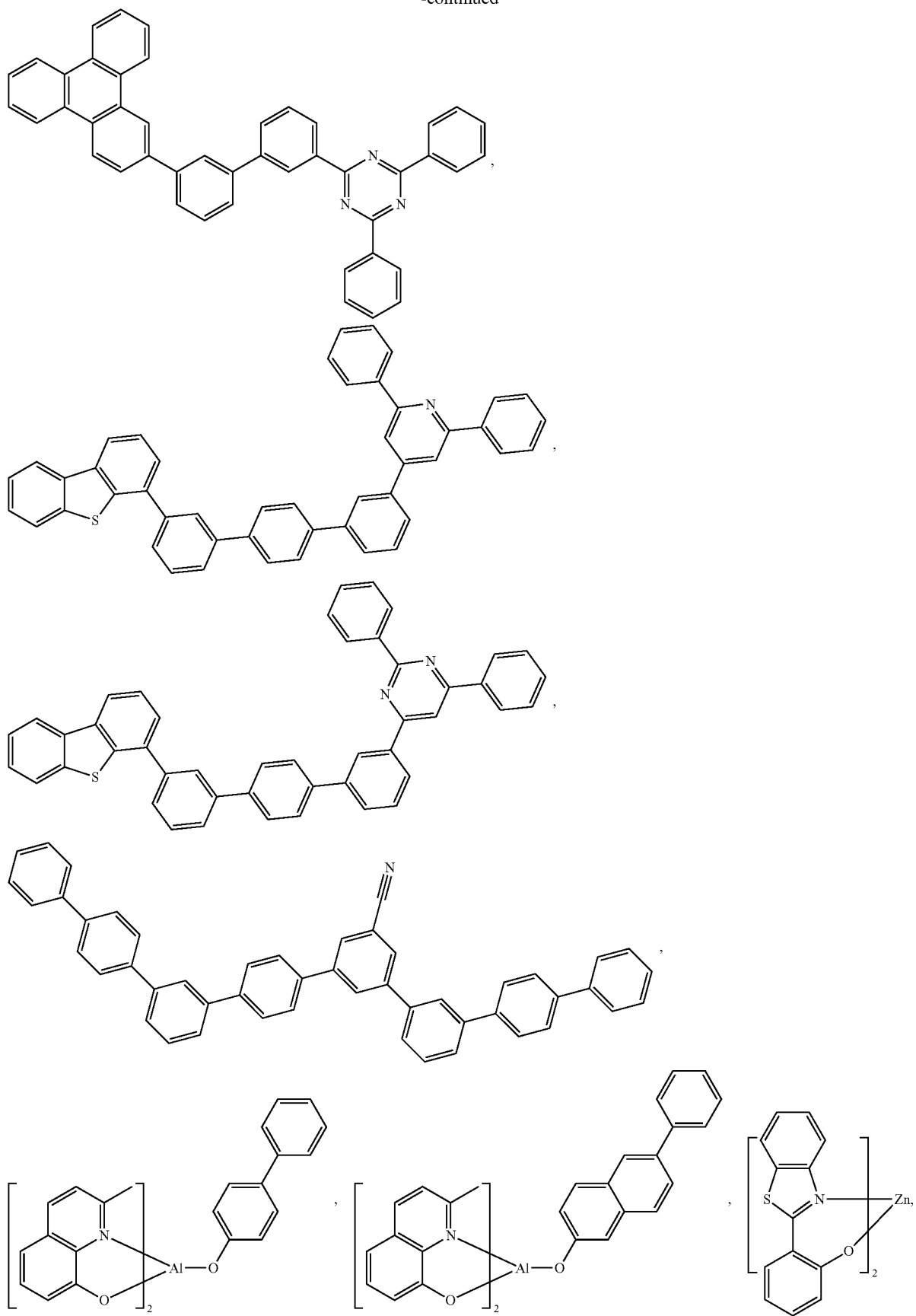
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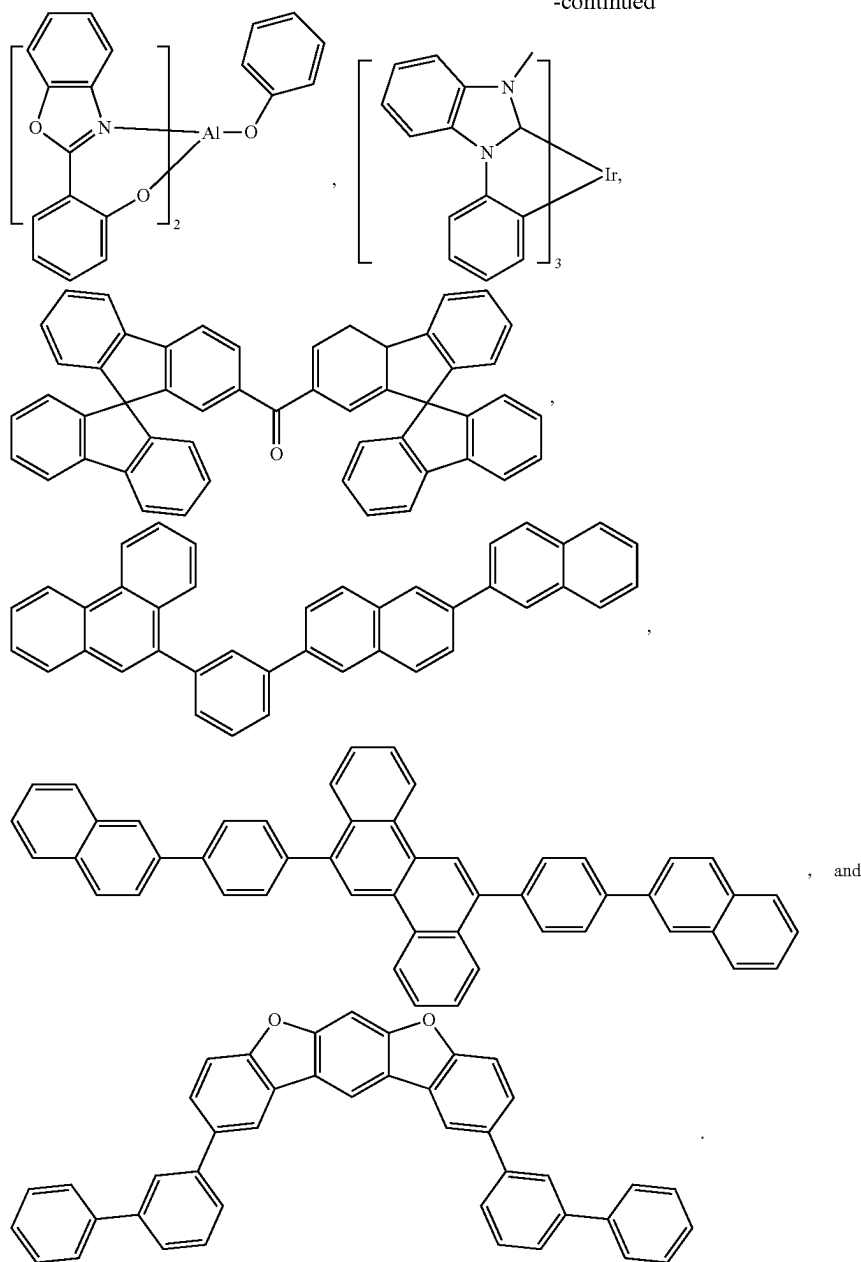
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Additional Emitters:

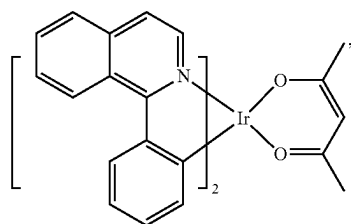
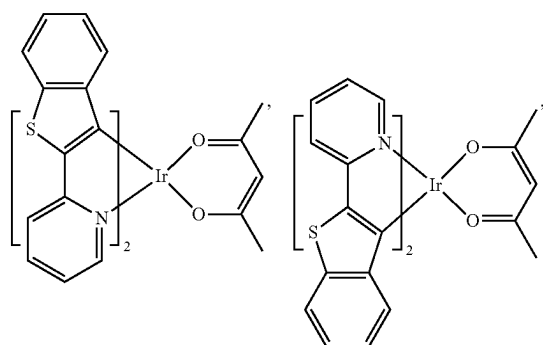
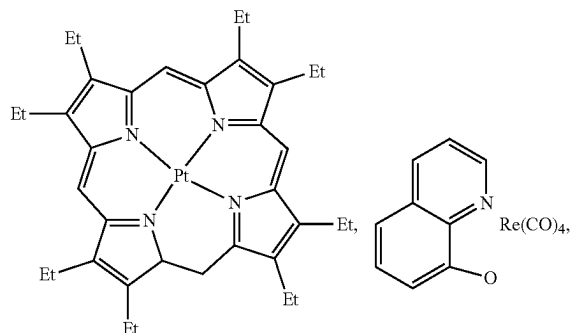
One or more additional emitter dopants may be used in conjunction with the compound of the present disclosure. Examples of the additional emitter dopants are not particularly limited, and any compounds may be used as long as the compounds are typically used as emitter materials. Examples of suitable emitter materials include, but are not limited to, compounds which can produce emissions via phosphorescence, fluorescence, thermally activated delayed fluorescence, i.e., TADF (also referred to as E-type delayed fluorescence), triplet-triplet annihilation, or combinations of these processes.

Non-limiting examples of the emitter materials that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: CN103694277, CN1696137,

EB01238981, EP01239526, EP01961743, EP1239526, EP1244155, EP1642951, EP1647554, EP1841834, EP1841834B, EP2062907, EP2730583, JP2012074444, JP2013110263, JP4478555, KR1020090133652, KR20120032054, KR20130043460, TW201332980, U.S. Ser. No. 06/699,599, U.S. Ser. No. 06/916,554, US20010019782, US20020034656, US20030068526, US20030072964, US20030138657, US20050123788, US20050244673, US2005123791, US2005260449, US20060008670, US20060065890, US20060127696, US20060134459, US20060134462, US20060202194, US20060251923, US20070034863, US20070087321, US20070103060, US20070111026, US20070190359, US20070231600, US2007034863, US2007104979, US2007104980, US2007138437, US2007224450, US2007278936, US20080020237, US20080233410,

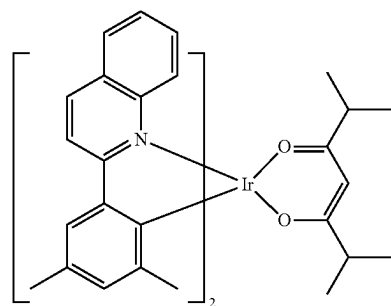
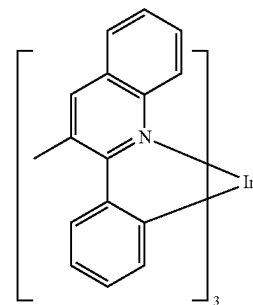
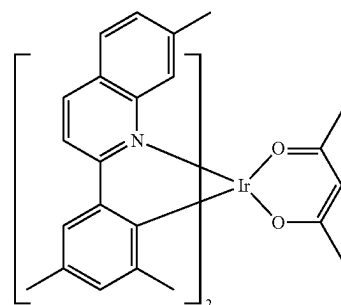
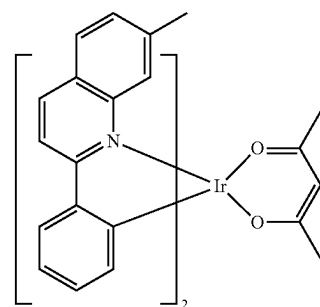
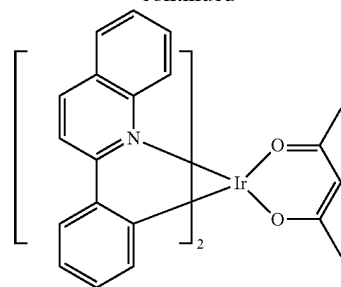
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US20080261076, US20080297033, US200805851,
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 US2011285275, US2012292601, US20130146848, 10
 US2013033172, US2013165653, US2013181190,
 US2013334521, US20140246656, US2014103305, U.S.
 Pat. Nos. 6,303,238, 6,413,656, 6,653,654, 6,670,645,
 6,687,266, 6,835,469, 6,921,915, 7,279,704, 7,332,232,
 7,378,162, 7,534,505, 7,675,228, 7,728,137, 7,740,957, 15
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 WO06081973, WO06121811, WO07018067,
 WO07108362, WO07115970, WO07115981,
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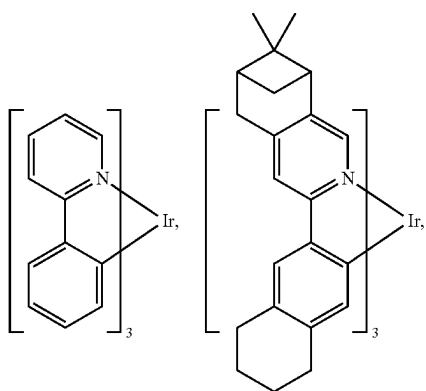
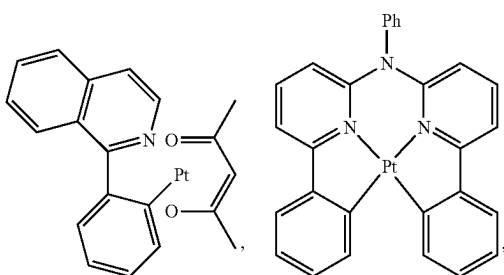
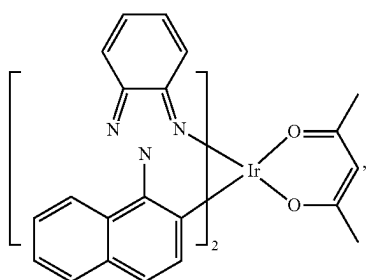
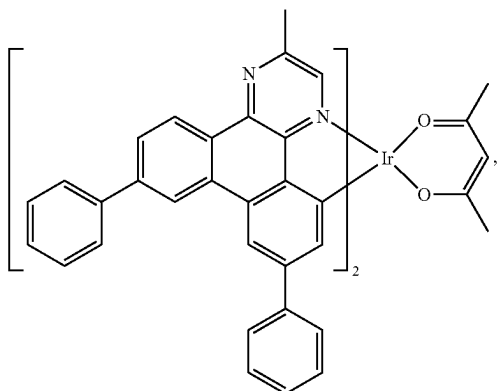
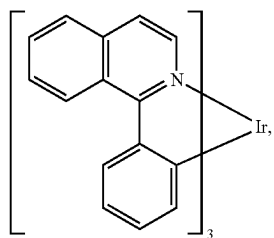
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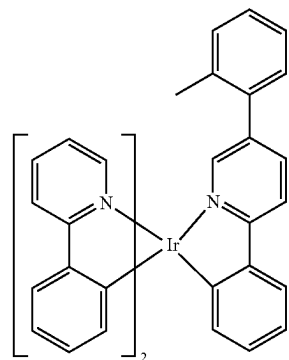
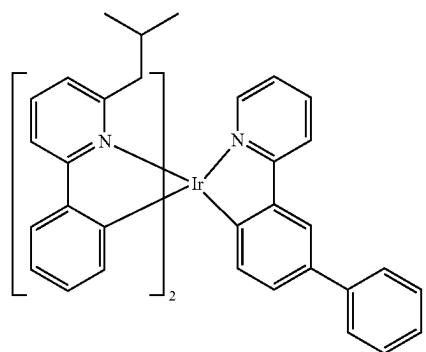
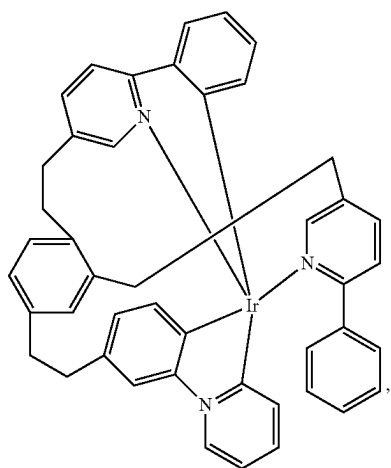
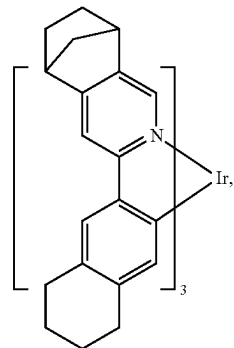
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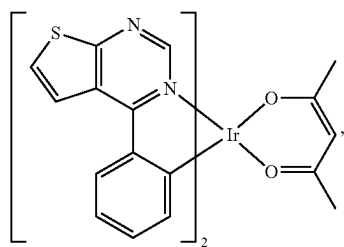
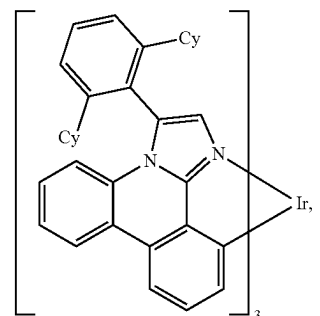
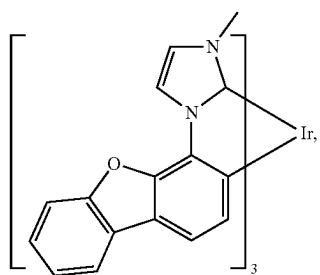
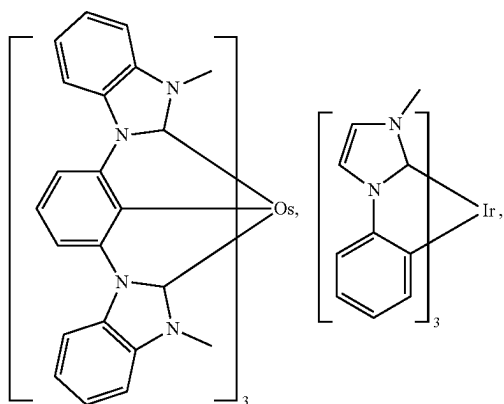
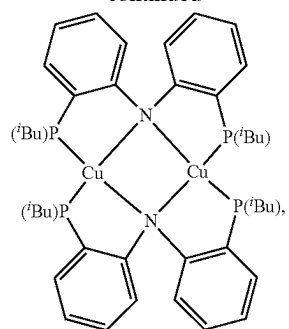
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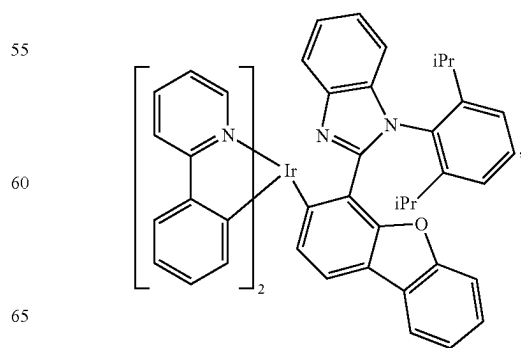
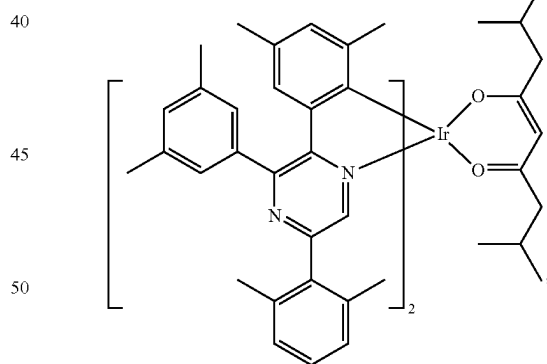
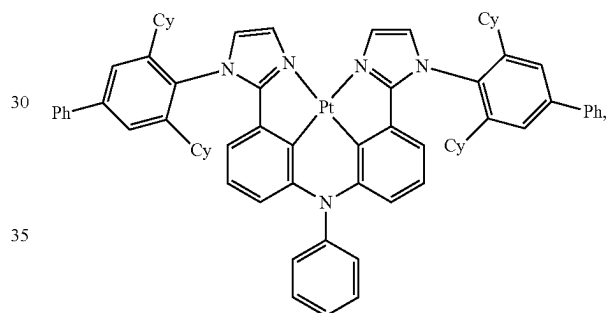
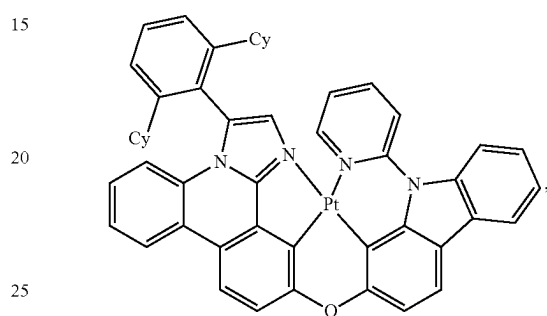
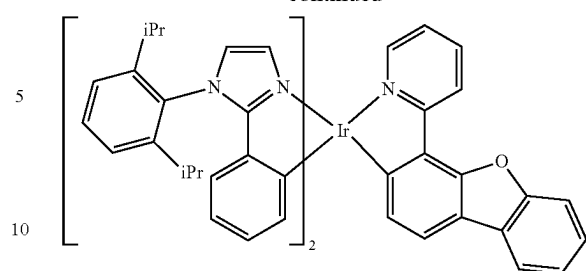


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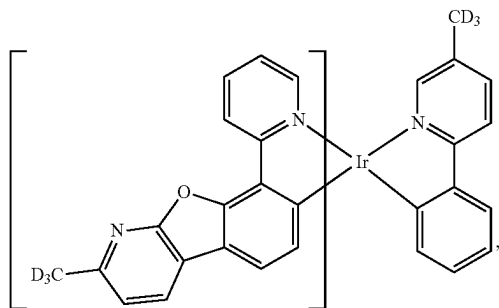
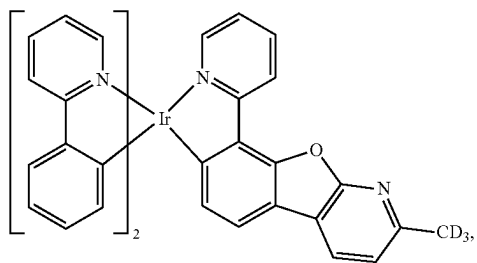
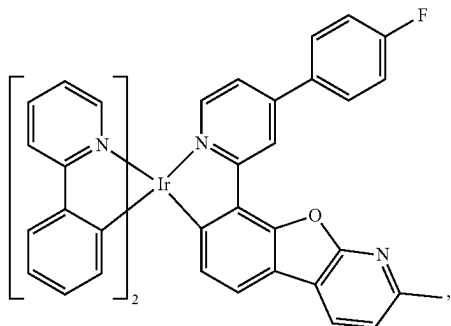
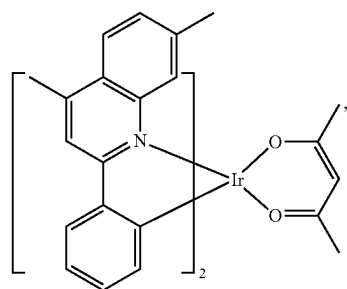
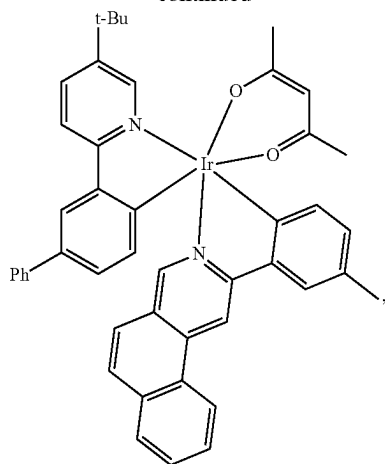
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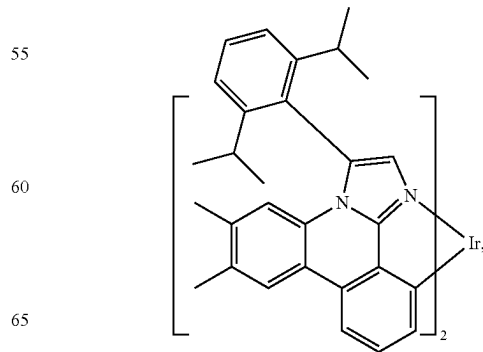
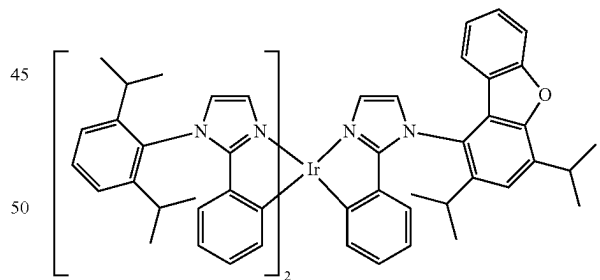
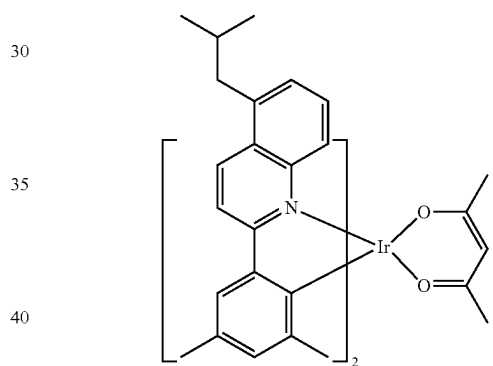
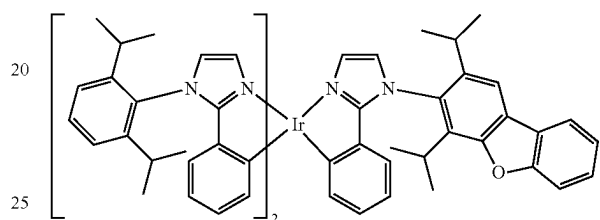
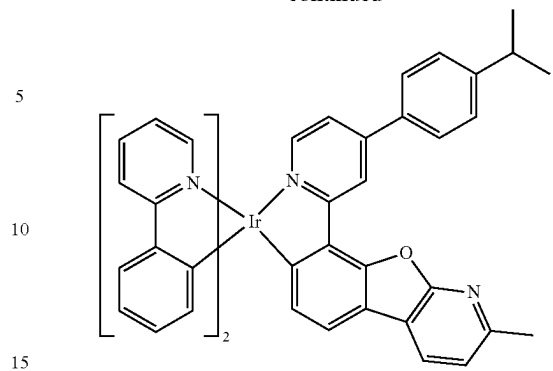


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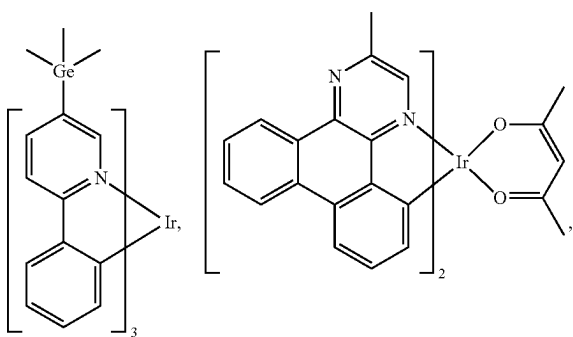
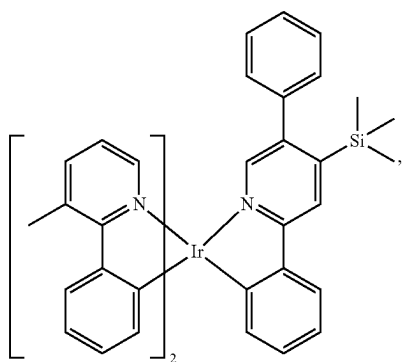
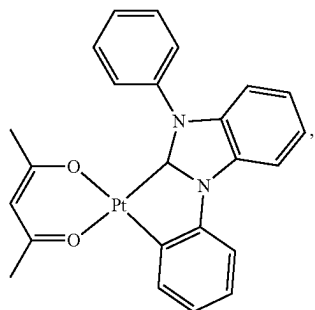
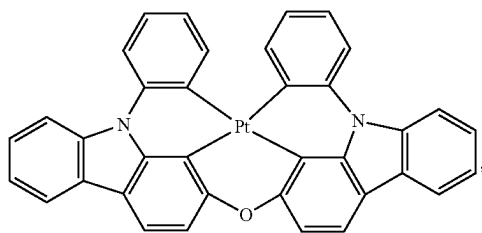
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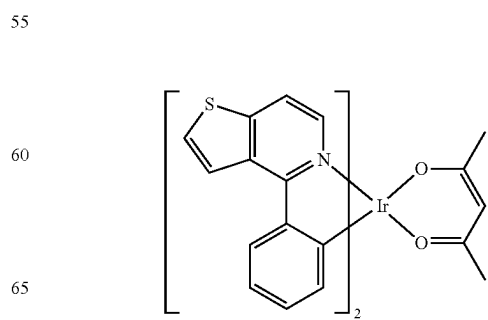
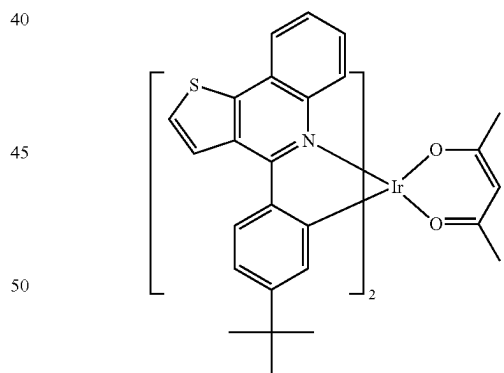
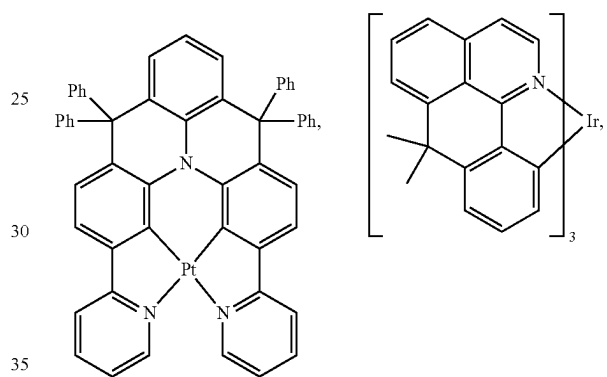
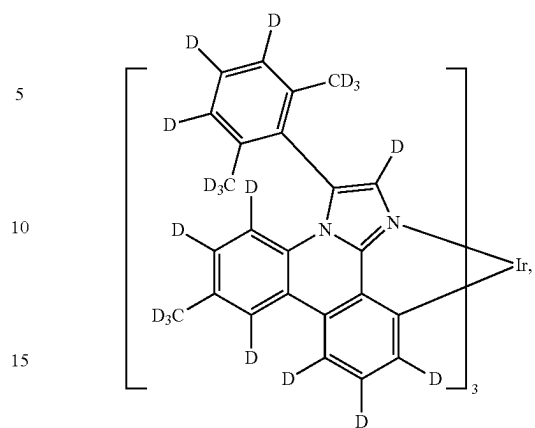


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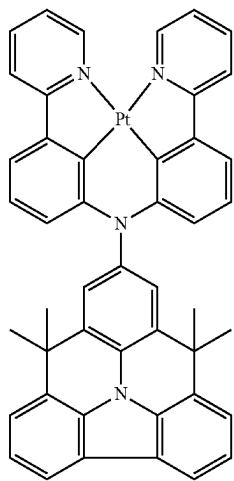
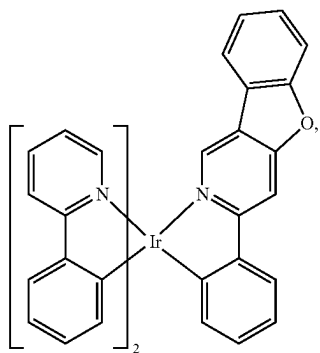
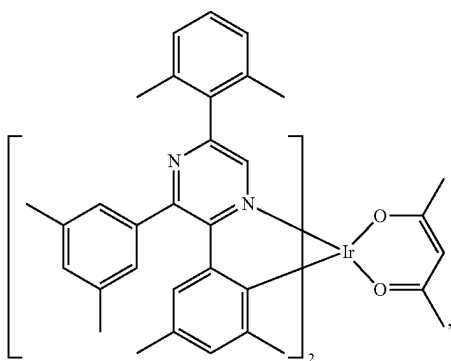
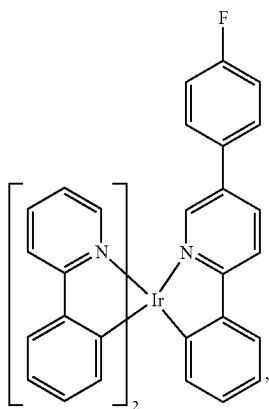
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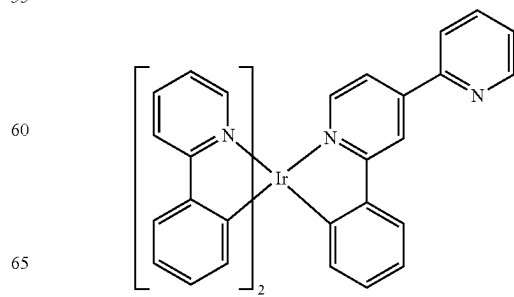
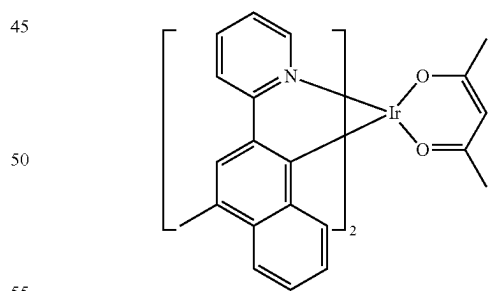
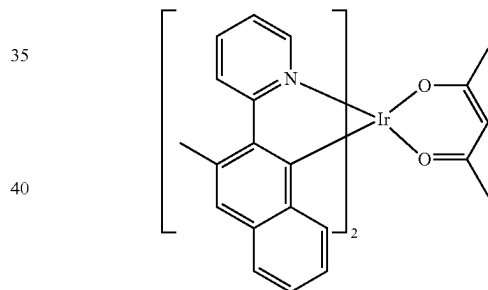
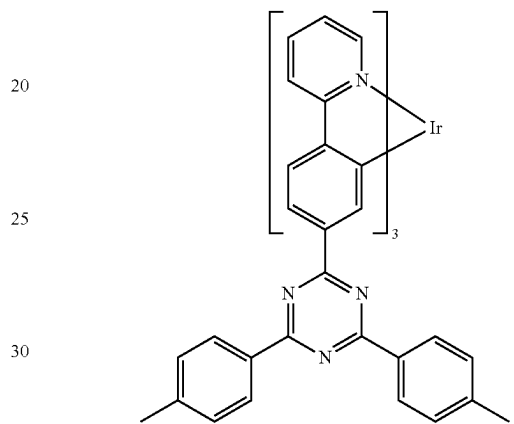
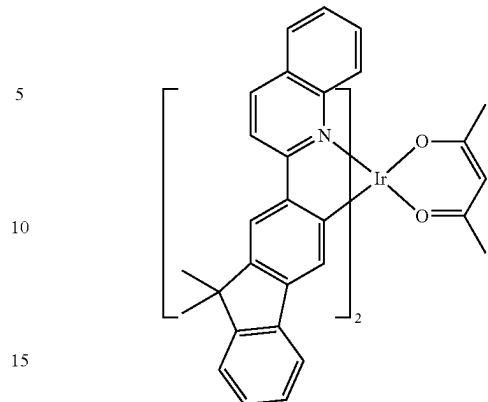


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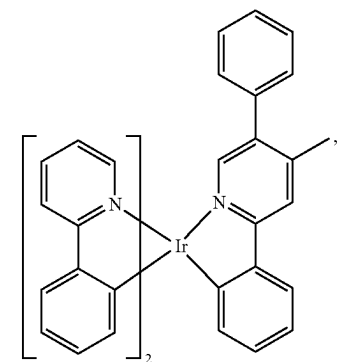
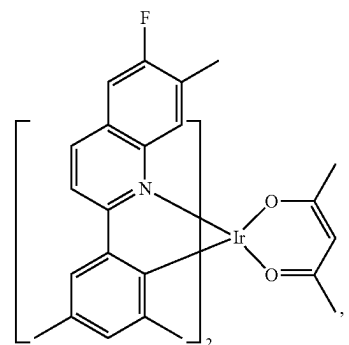
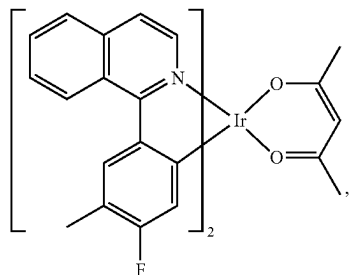
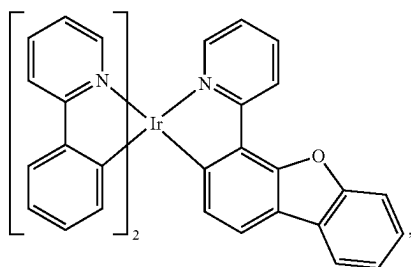
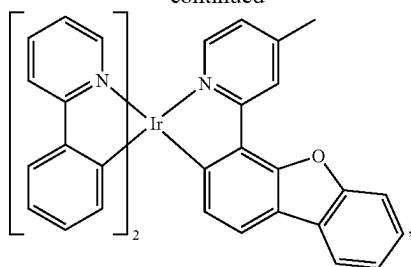
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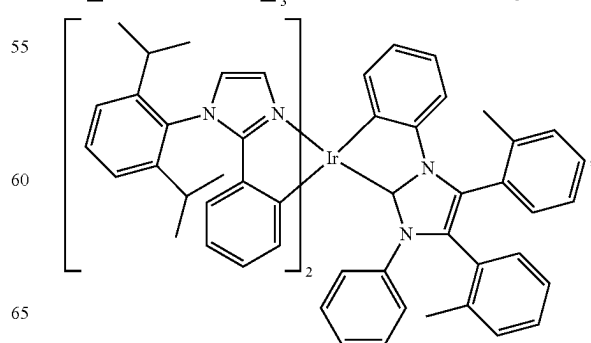
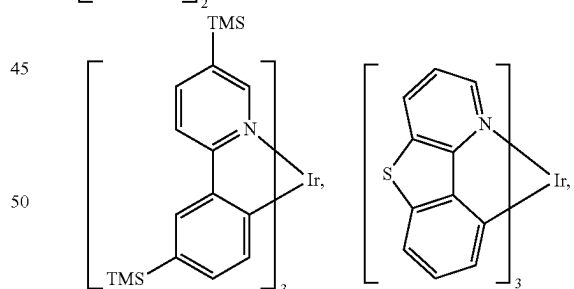
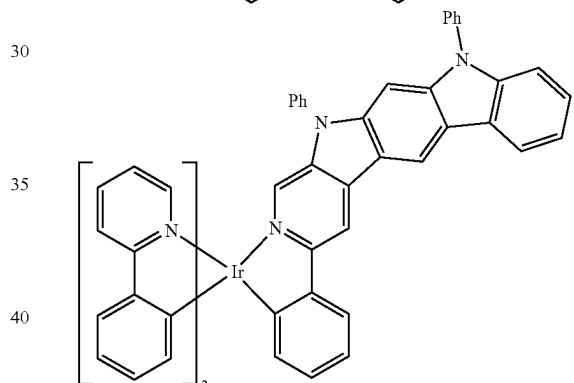
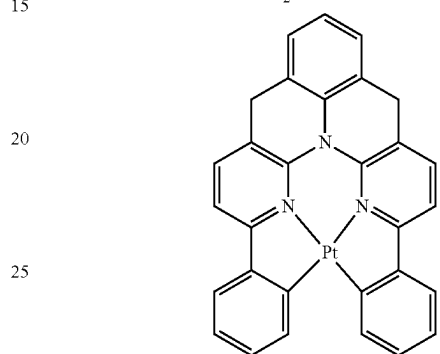
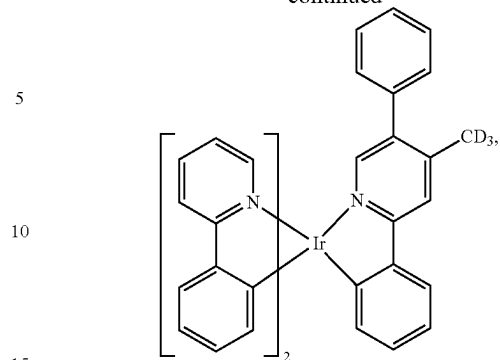


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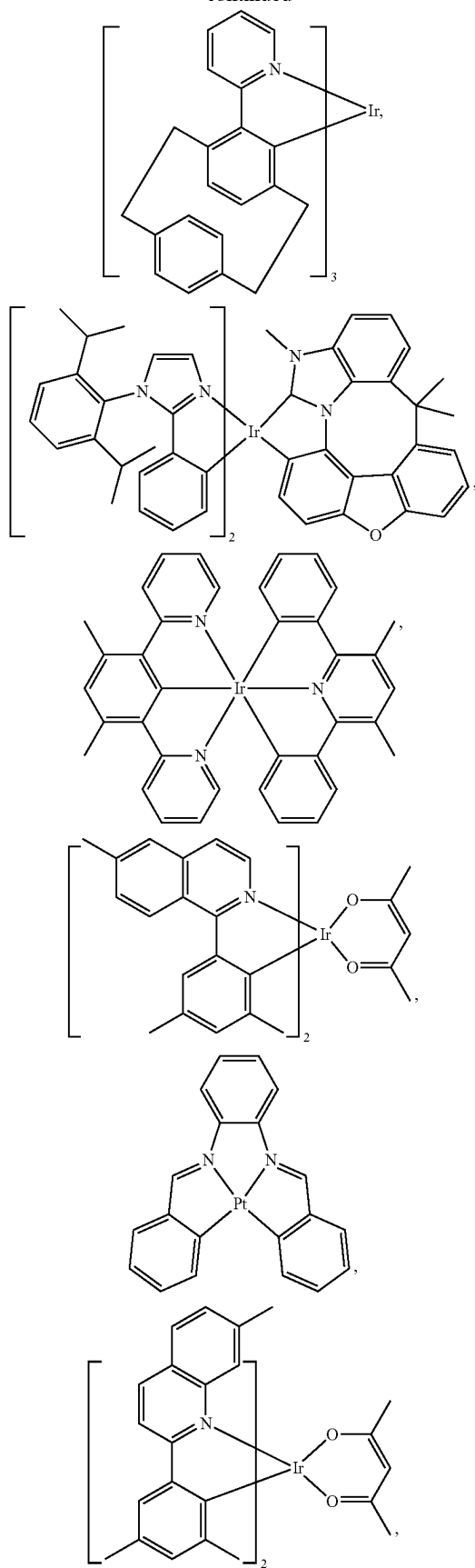
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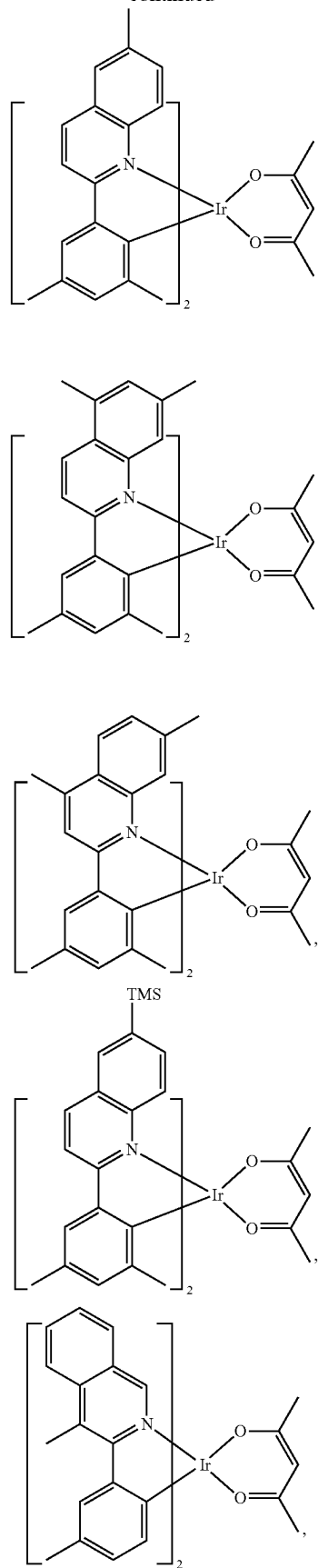
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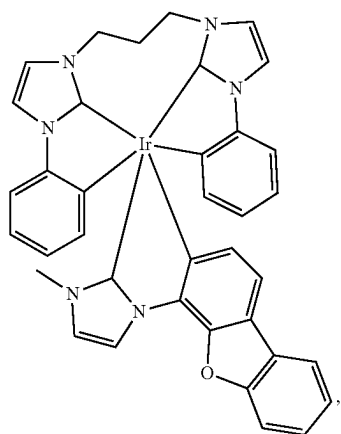
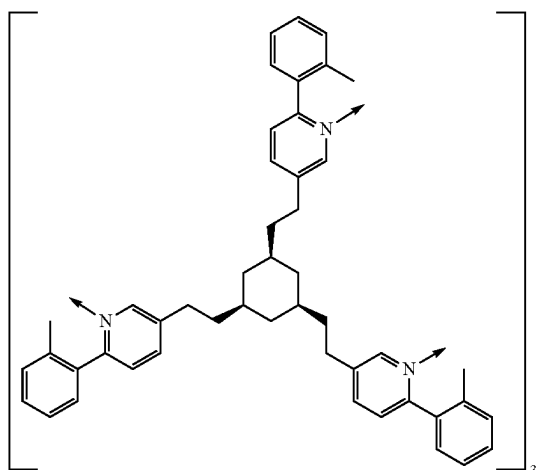
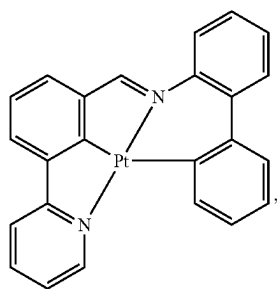
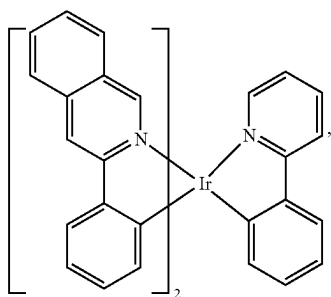
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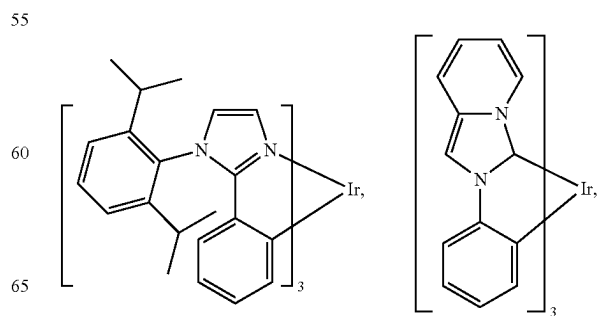
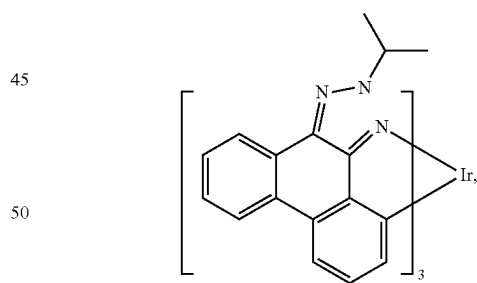
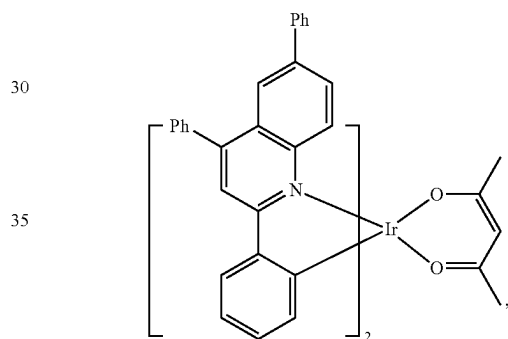
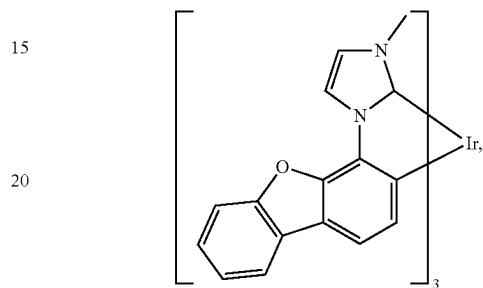
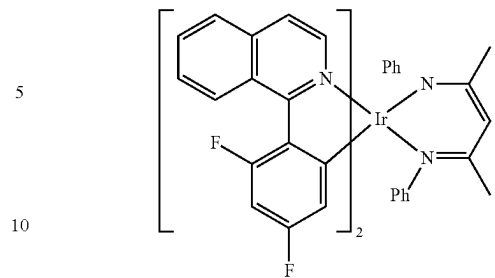


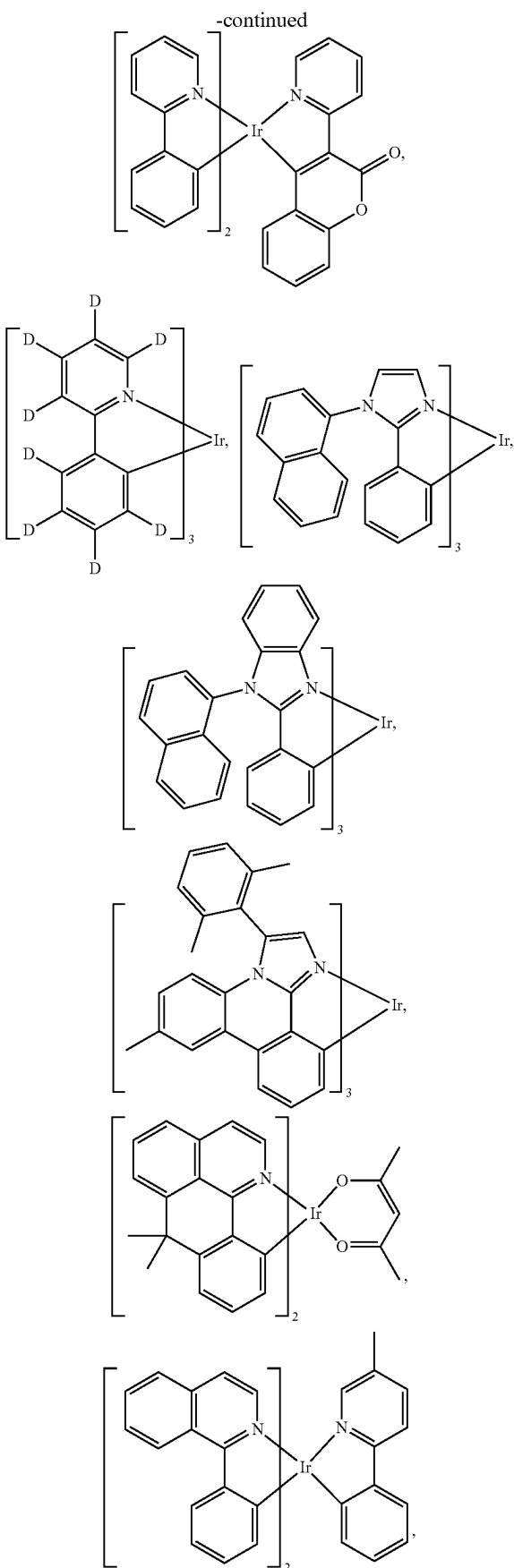
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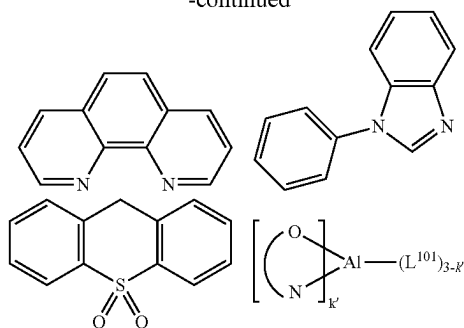
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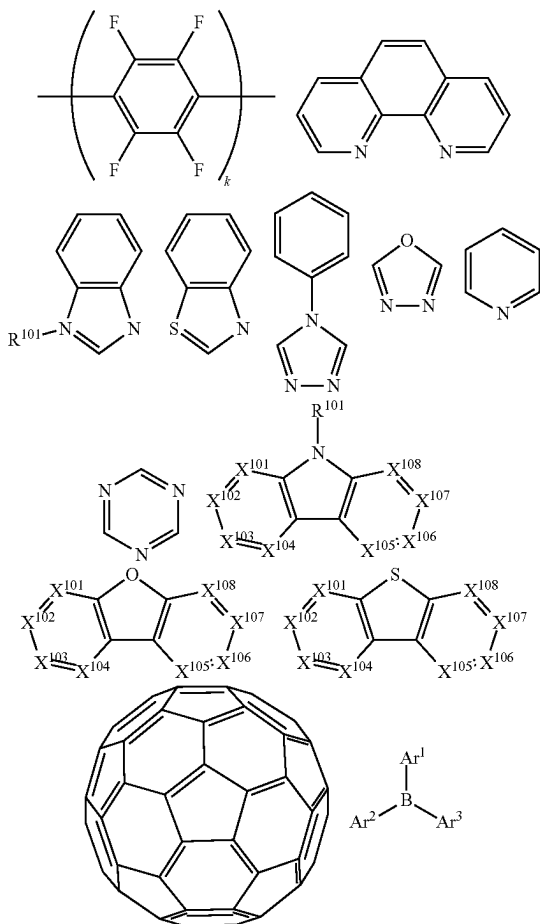


wherein k is an integer from 1 to 20; L¹⁰¹ is an another ligand, k' is an integer from 1 to 3.

ETI:

Electron transport layer (ETL) may include a material capable of transporting electrons. Electron transport layer may be intrinsic (undoped), or doped. Doping may be used to enhance conductivity. Examples of the ETL material are not particularly limited, and any metal complexes or organic compounds may be used as long as they are typically used to transport electrons.

In one aspect, compound used in ETL contains at least one of the following groups in the molecule:

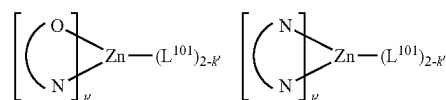
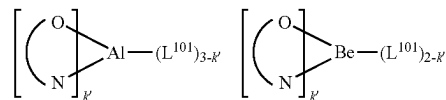


wherein R¹⁰¹ is selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, aryl-

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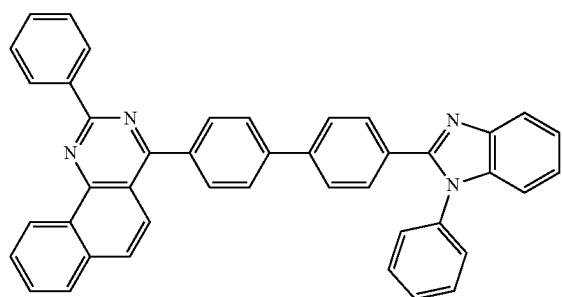
alkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof, when it is aryl or heteroaryl, it has the similar definition as Ar's mentioned above. Ar¹ to Ar³ has the similar definition as Ar's mentioned above. k is an integer from 1 to 20. X¹⁰¹ to X¹⁰⁸ is selected from C (including CH) or N.

10 In another aspect, the metal complexes used in ETL contains, but not limit to the following general formula:



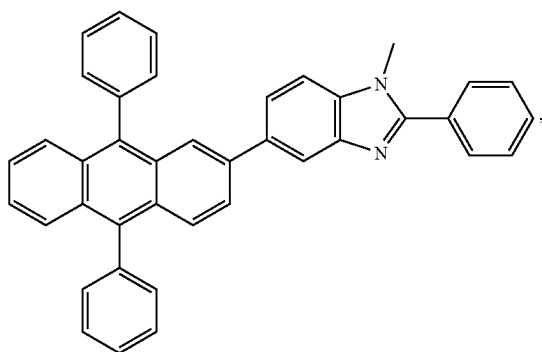
30 wherein (O—N) or (N—N) is a bidentate ligand, having
metal coordinated to atoms O, N or N, N; L^{101} is
another ligand; k' is an integer value from 1 to the
35 maximum number of ligands that may be attached to
the metal.

Non-limiting examples of the ETL materials that may be used in an OLED in combination with materials disclosed herein are exemplified below together with references that disclose those materials: CN103508940, EP01602648, 40 EP01734038, EP01956007, JP2004-022334, JP2005149918, JP2005-268199, KR0117693, KR20130108183, US20040036077, US20070104977, US2007018155, US20090101870, US20090115316, US20090140637, US20090179554, US2009218940, 45 US2010108990, US2011156017, US2011210320, US2012193612, US2012214993, US2014014925, US2014014927, US20140284580, U.S. Pat. Nos. 6,656,612, 8,415,031, WO2003060956, WO2007111263, 50 WO2009148269, WO2010067894, WO2010072300, WO2011074770, WO2011105373, WO2013079217, WO2013145667, WO2013180376, WO2014104499, WO2014104535.

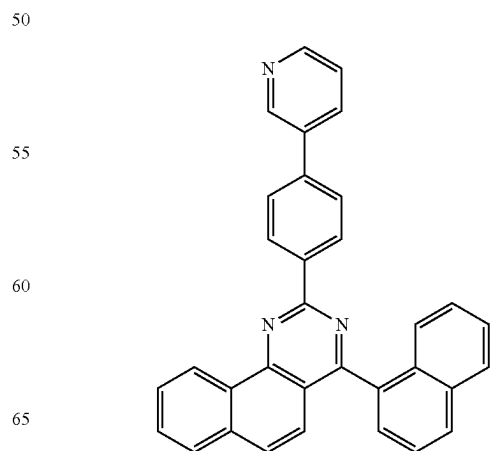
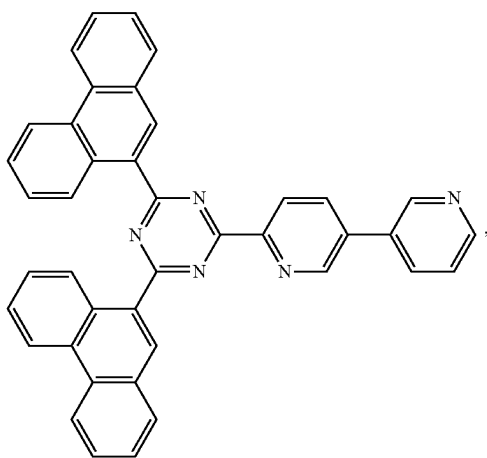
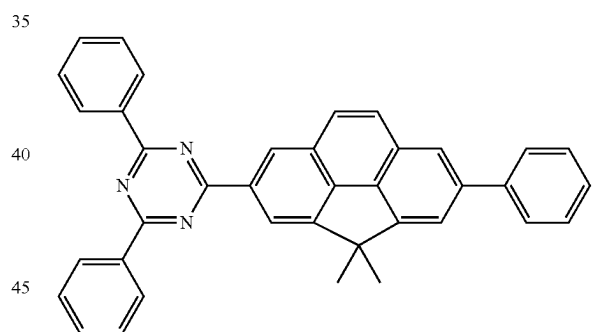
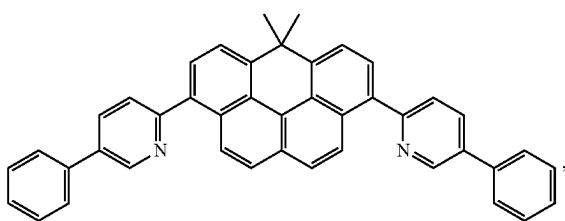
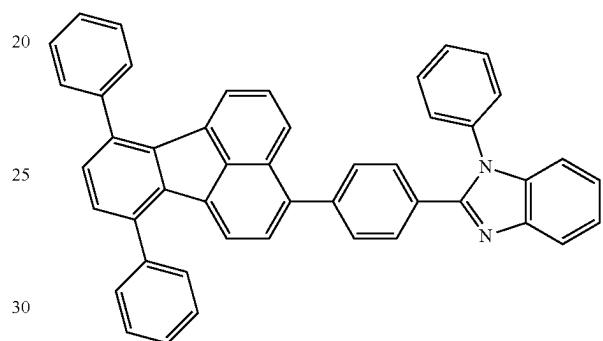
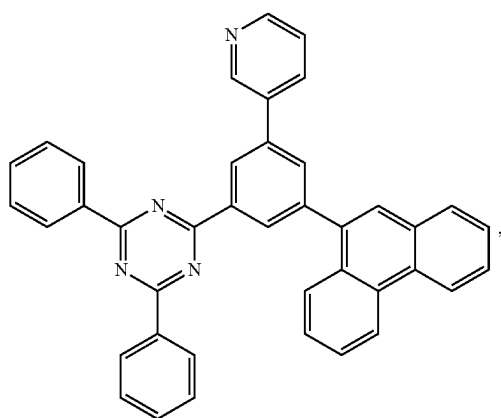
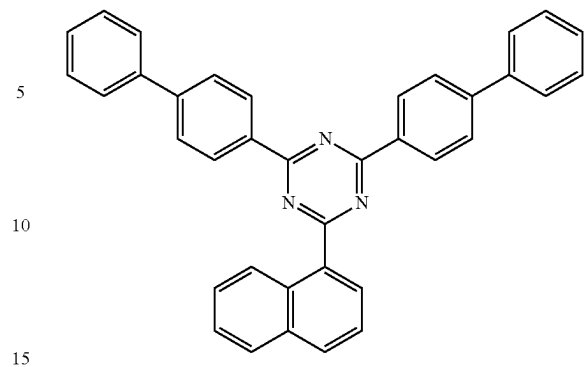


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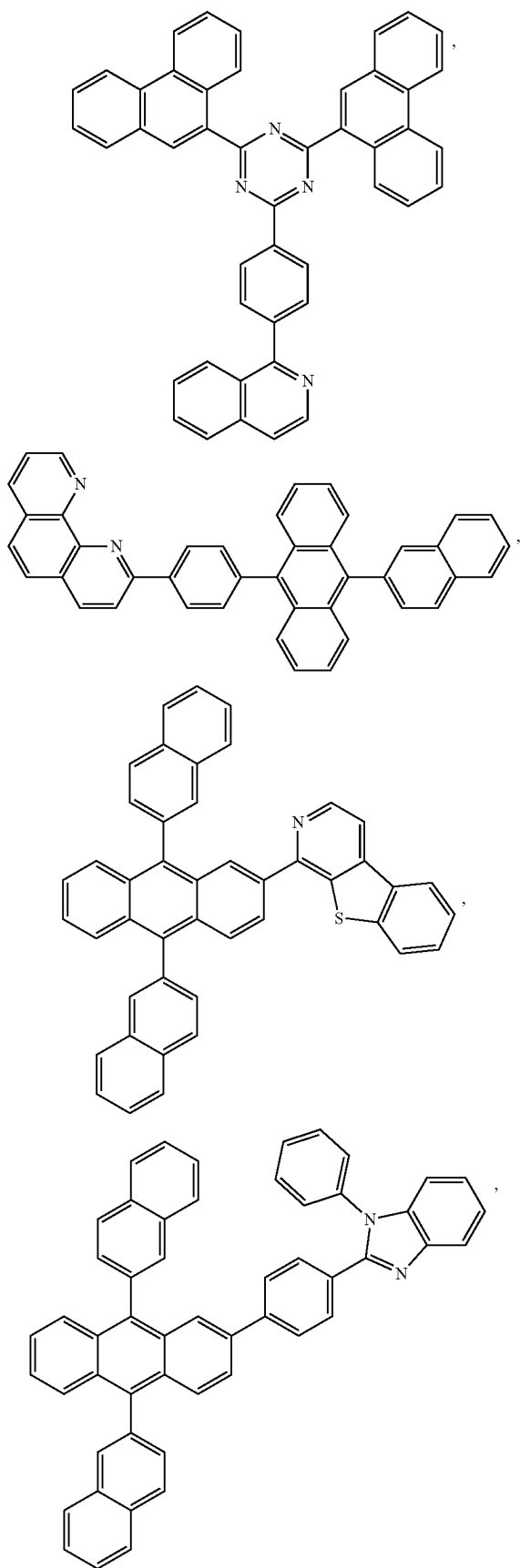
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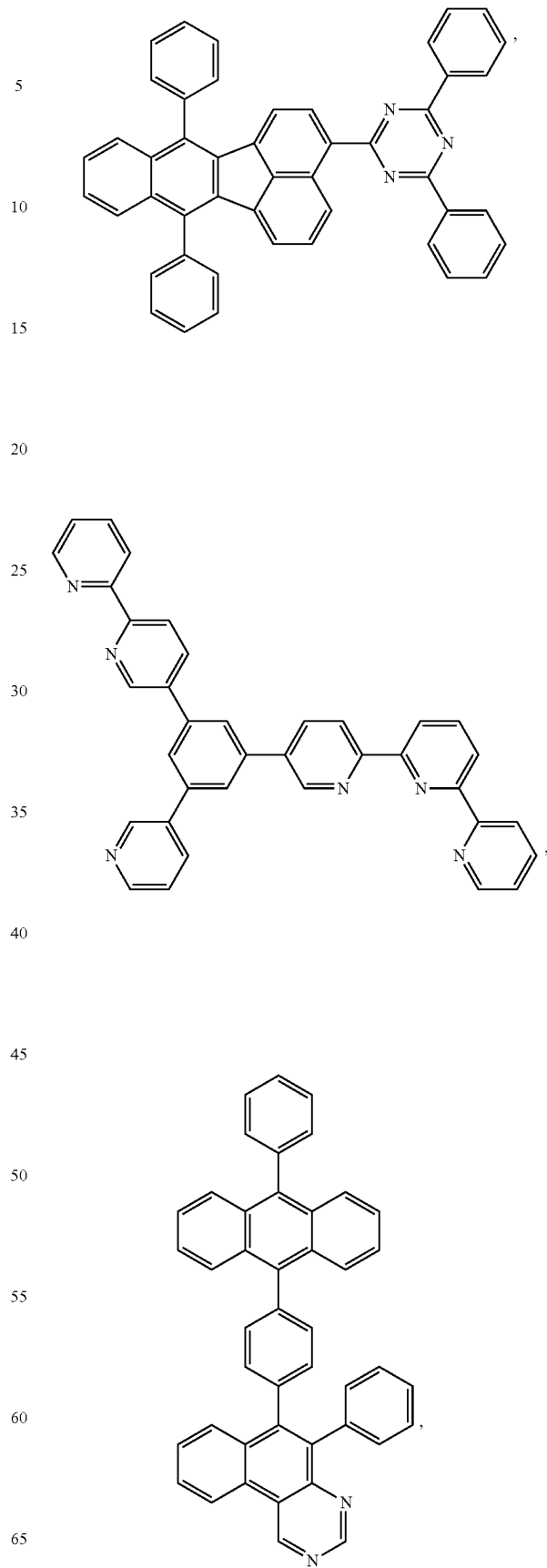


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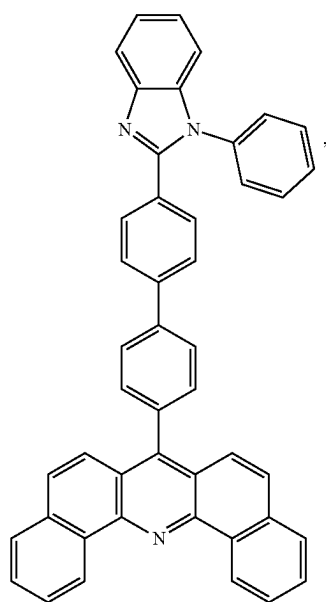
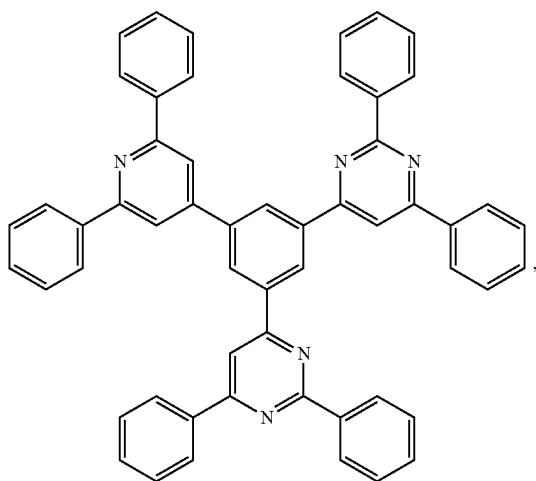
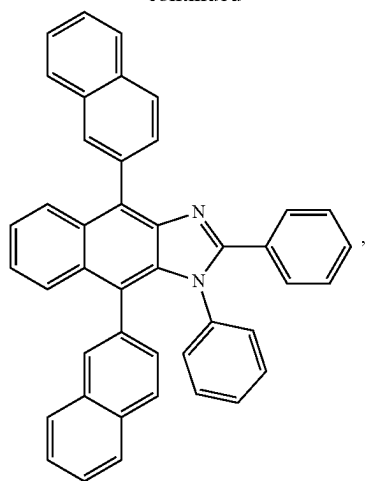
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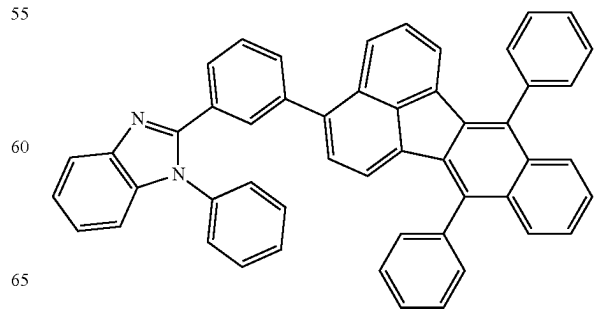
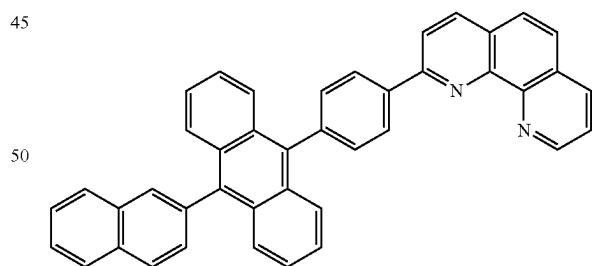
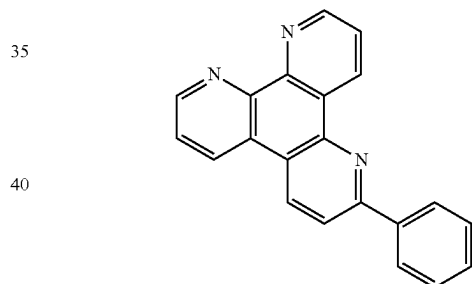
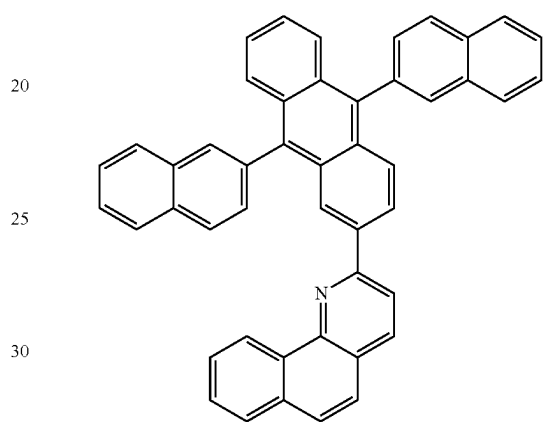
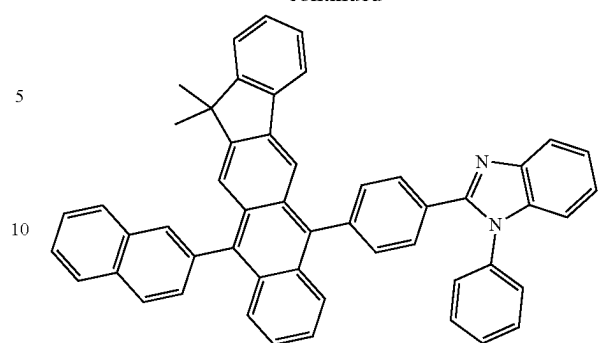


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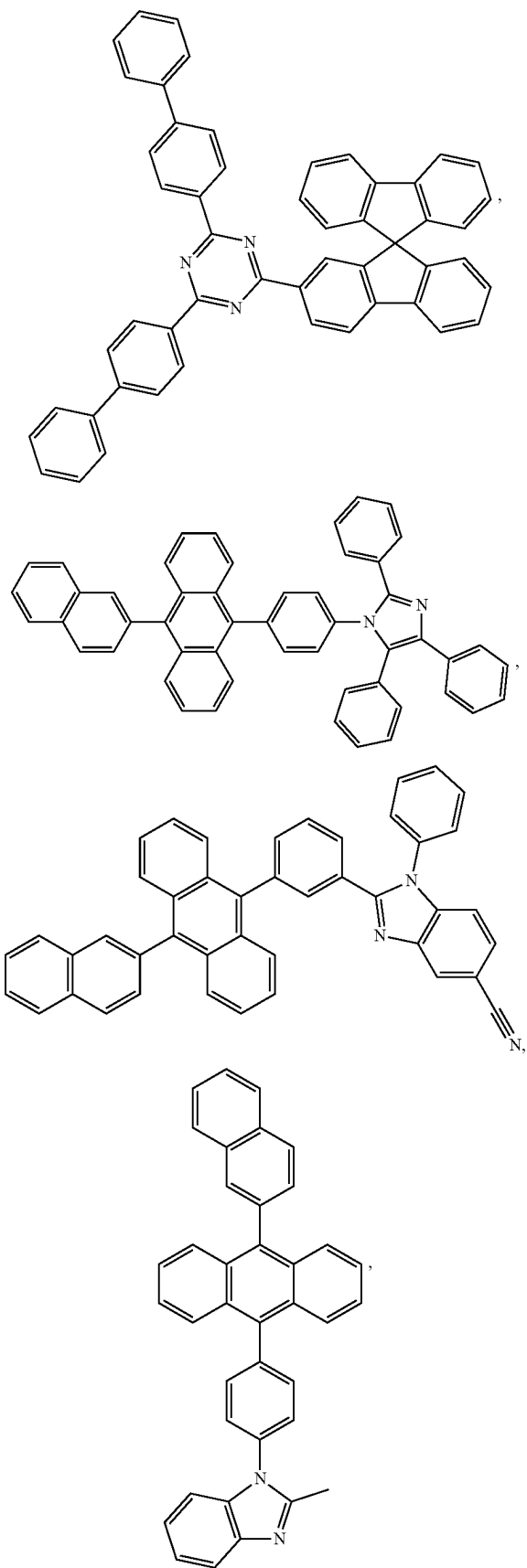
**362**

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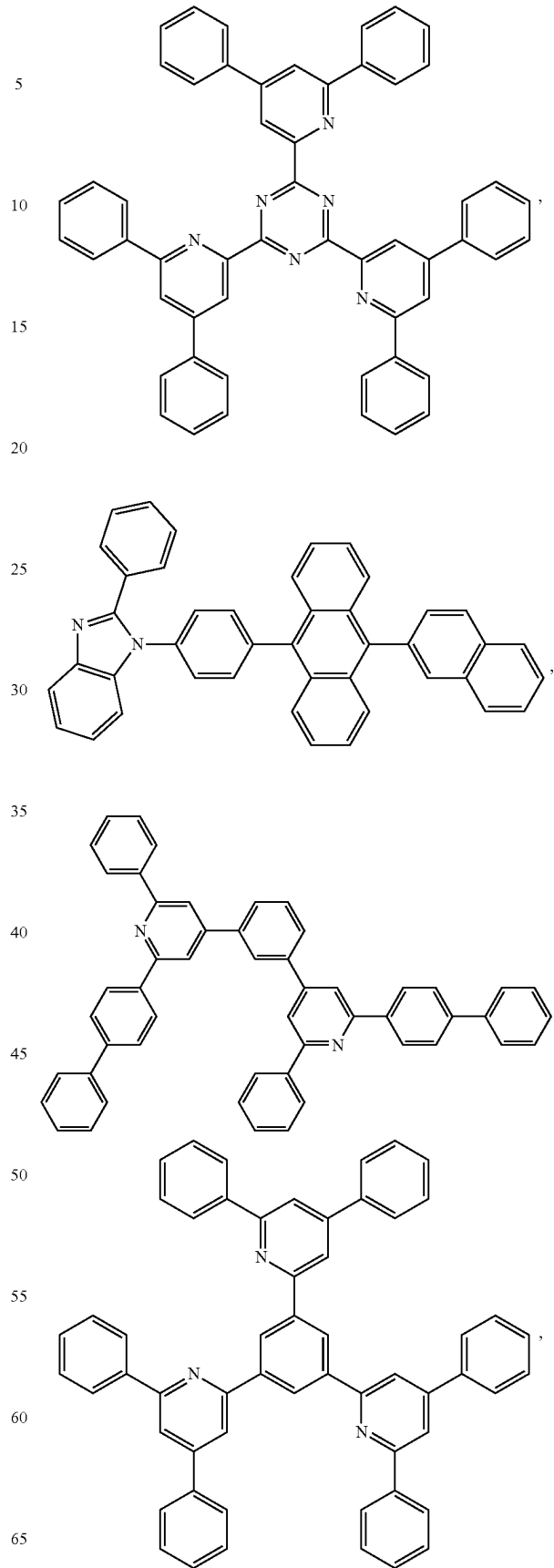


363

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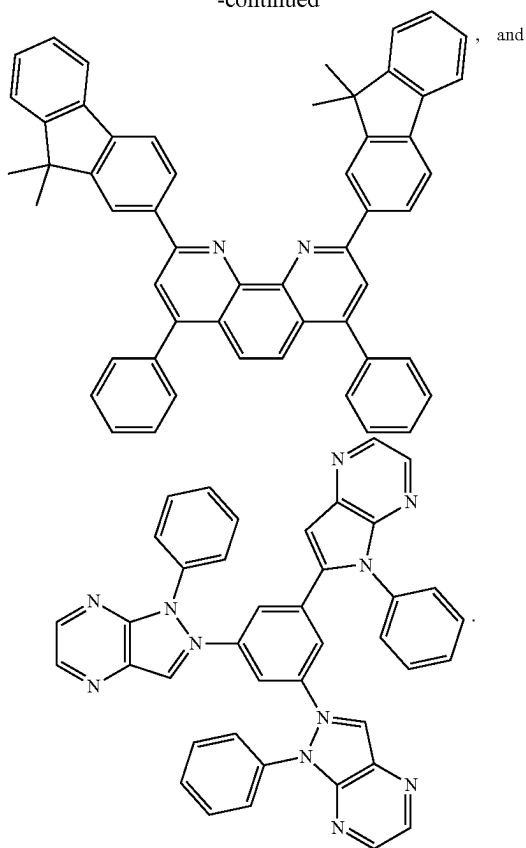
**364**

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Charge Generation Layer (CGL)

In tandem or stacked OLEDs, the CGL plays an essential role in the performance, which is composed of an n-doped layer and a p-doped layer for injection of electrons and holes, respectively. Electrons and holes are supplied from the CGL and electrodes. The consumed electrons and holes in the CGL are refilled by the electrons and holes injected from the cathode and anode, respectively; then, the bipolar currents reach a steady state gradually. Typical CGL materials include n and p conductivity dopants used in the transport layers.

In any above-mentioned compounds used in each layer of the OLED device, the hydrogen atoms can be partially or fully deuterated. Thus, any specifically listed substituent, such as, without limitation, methyl, phenyl, pyridyl, etc. may be undeuterated, partially deuterated, and fully deuterated versions thereof. Similarly, classes of substituents such as, without limitation, alkyl, aryl, cycloalkyl, heteroaryl, etc. also may be undeuterated, partially deuterated, and fully deuterated versions thereof.

It is understood that the various embodiments described herein are by way of example only, and are not intended to limit the scope of the invention. For example, many of the materials and structures described herein may be substituted with other materials and structures without deviating from the spirit of the invention. The present invention as claimed may therefore include variations from the particular examples and preferred embodiments described herein, as will be apparent to one of skill in the art. It is understood that various theories as to why the invention works are not intended to be limiting.

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We claim:

1. A compound having a formula of Formula III,

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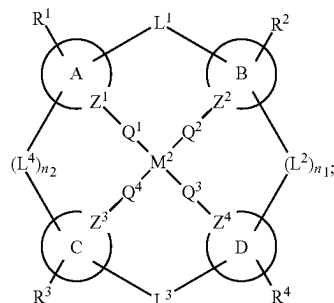
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Formula III



wherein rings A, B, C, and D are each a 5 or 6-membered carbocyclic or heterocyclic ring;

wherein L¹ and L³ are each independently selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof;

wherein n₁ and n₂ each independently is 0 or 1;

wherein, if n₁ or n₂ is 1, then L² or L⁴ is selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof; and

wherein, if n₁ or n₂ is 0, L² or L⁴ is not present;

Q¹, Q², Q³ and Q⁴ are each independently selected from the group consisting of direct bond and oxygen;

wherein Z¹, Z², Z³, and Z⁴ are each independently carbon or nitrogen;

when any of Z¹, Z², Z³ and Z⁴ is nitrogen, the Q¹, Q², Q³ and Q⁴ attached thereto is a direct bond;

ring A is trans to ring D, ring B is trans to ring C in a square-planar coordination configuration;

wherein R¹, R², R³, and R⁴ each represents mono to the maximum possible number of substitution, or no substitution;

wherein M² is Pt, and the compound comprises at least one Pt-carbene bond;

wherein a fourth distance is the distance between the atom in R¹ that is the farthest away from M² to the atom in R⁴ that is the farthest away from M²;

wherein a fifth distance is the distance between the atom in R² that is the farthest away from M² to the atom in R³ that is the farthest away from M²;

wherein the fourth distance is longer than the fifth distance by at least 1.5 Å;

wherein R, R', R¹, R², R³, and R⁴ are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof;

wherein any two substituents among R, R', R¹, R², R³, and R⁴ are optionally joined or fused into a ring; and

wherein at least one of the following is true:

(i) at least one of L¹ to L³ is a C—C direct bond;

(ii) each of Q¹ to Q⁴ is a direct bond; or

(iii) at least two of Q¹ to Q⁴ are O.

2. The compound of claim 1, wherein rings A, B, C, and D are each independently selected from the group consisting of phenyl, pyridine, and imidazole.

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3. The compound of claim 1, wherein rings A and C are phenyl, and rings B and D are selected from the group consisting of pyridine, pyrimidine, imidazole, and pyrazole.

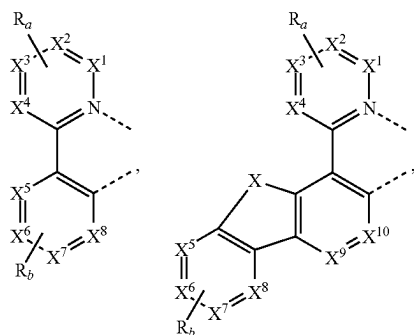
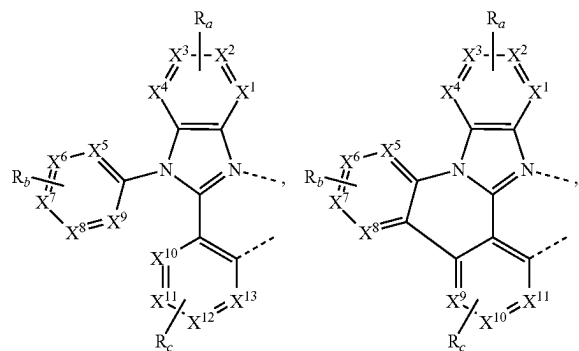
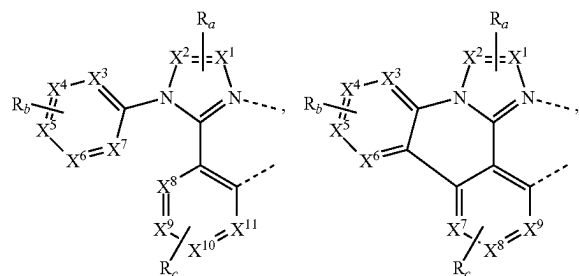
4. The compound of claim 1, wherein rings B and D are selected from the group consisting of pyridine, pyrimidine, imidazole, and pyrazole.

5. The compound of claim 1, wherein R^1 , R^2 , R^3 , and R^4 are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, silyl, aryl, heteroaryl, and combinations thereof.

6. The compound of claim 1, wherein the fourth distance is longer than the fifth distance by at least 4.3 Å.

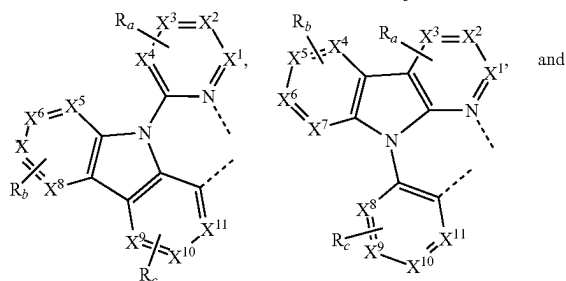
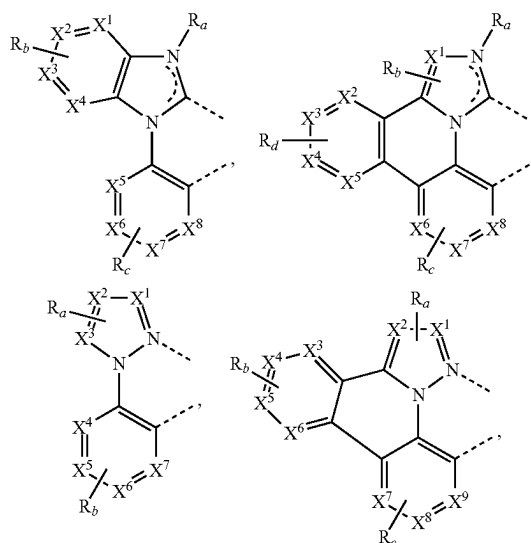
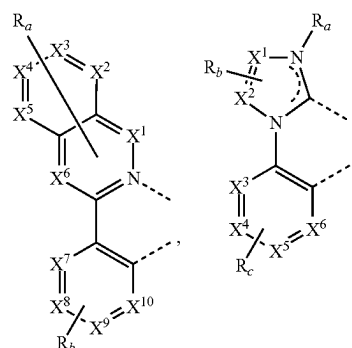
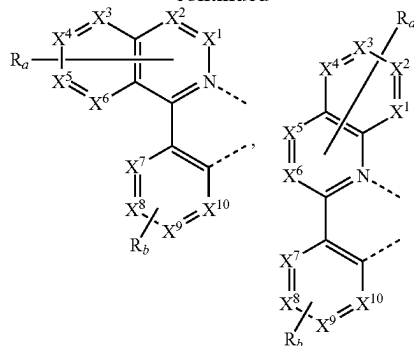
7. The compound of claim 1, wherein at least one of the rings A, B, C, or D is fused by another 5- or 6-membered ring.

8. The compound of claim 1, wherein the bidentate ligands A-B and C-D are each independently selected from the group consisting of:



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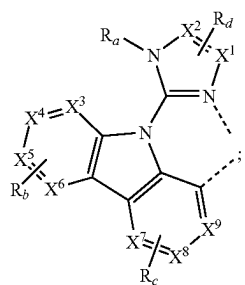
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and

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wherein X^1 to X^{13} are each independently selected from the group consisting of carbon and nitrogen;

wherein X is selected from the group consisting of BR' , NR' , PR' , O, S, Se, $C=O$, $S=O$, SO_2 , $CR'R''$, $SiR'R''$, and $GeR'R''$;

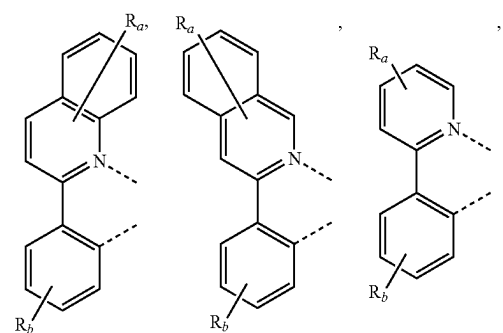
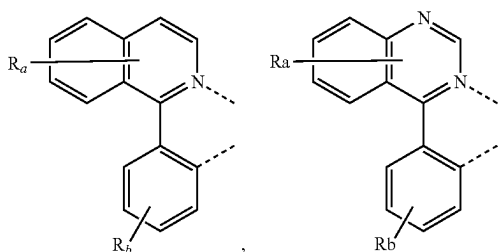
wherein R' and R'' are optionally fused or joined to form a ring;

wherein R_a , R_b , R_c , and R_d each represents from mono substitution to the maximum possible number of substitution, or no substitution;

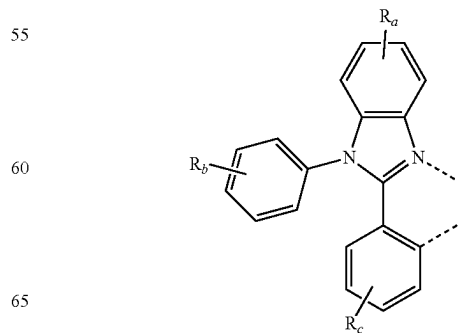
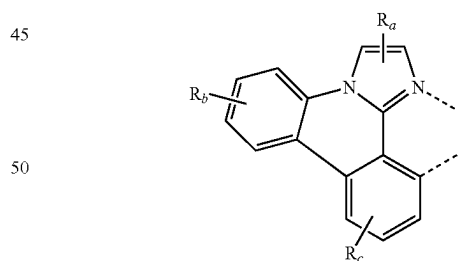
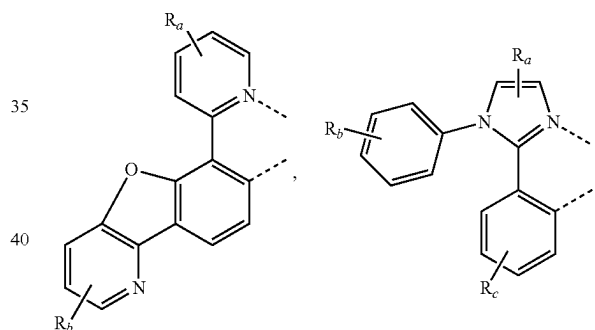
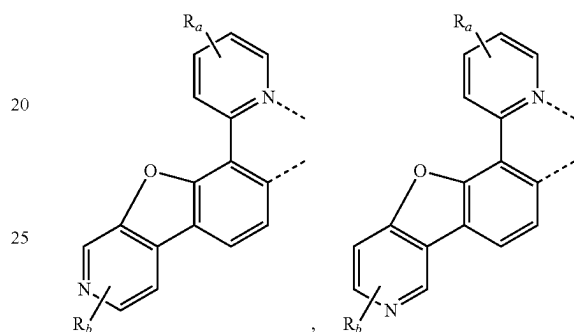
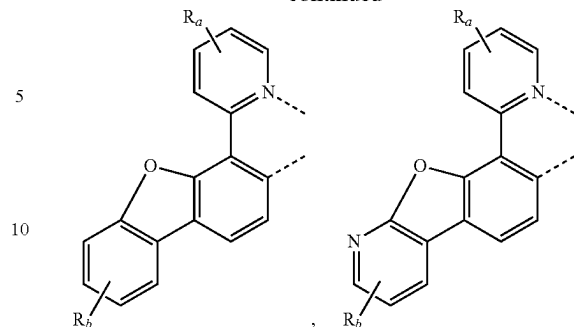
wherein R' , R'' , R_a , R_b , R_c , and R_d are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and

wherein any two substituents among R_a , R_b , R_c , and R_d are optionally fused or joined to form a ring.

9. The compound of claim 8, wherein the bidentate ligands A-B and C-D are each independently selected from the group consisting of:

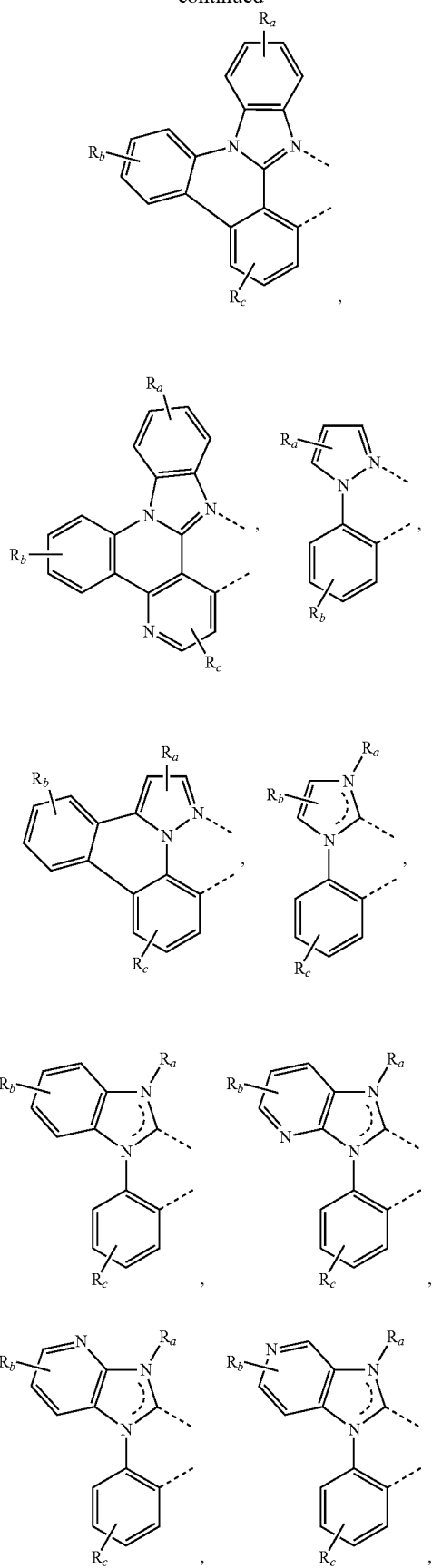
**370**

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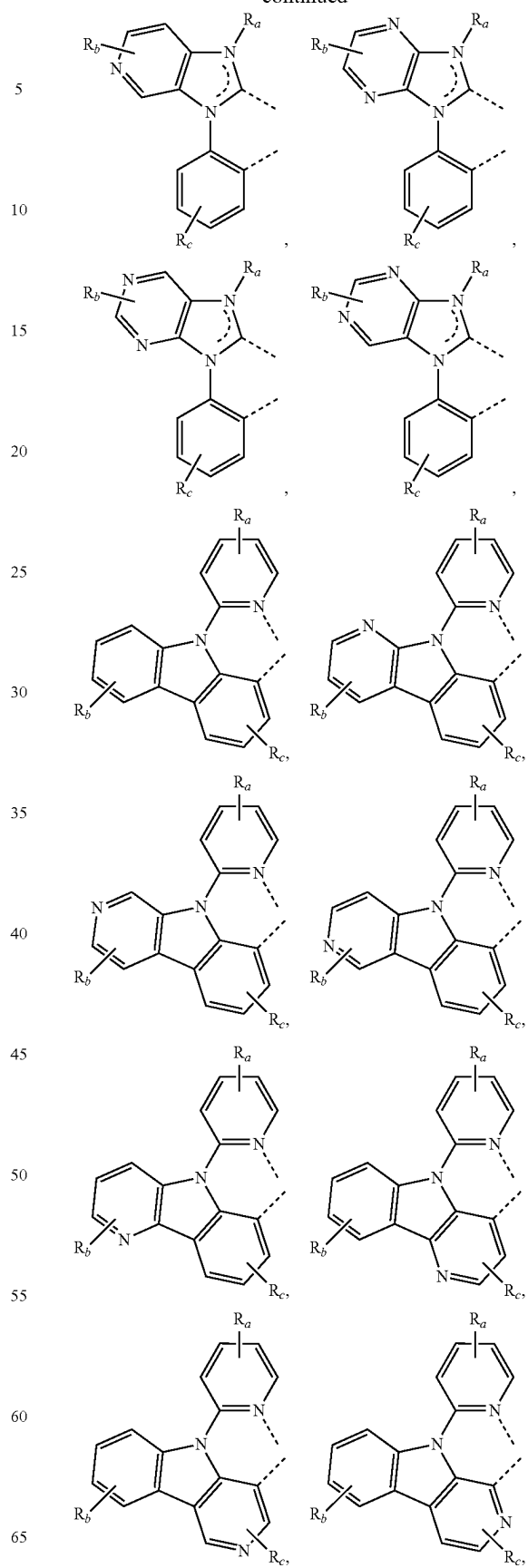


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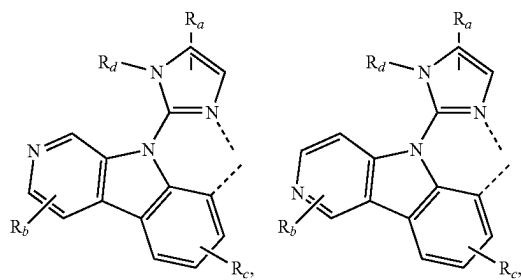
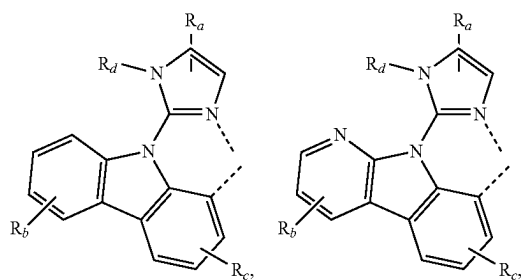
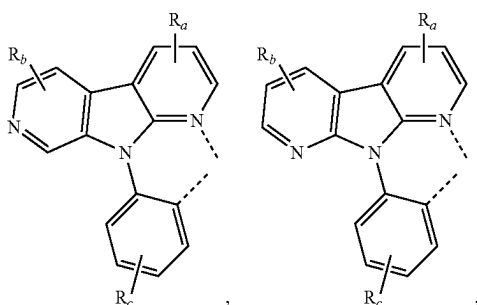
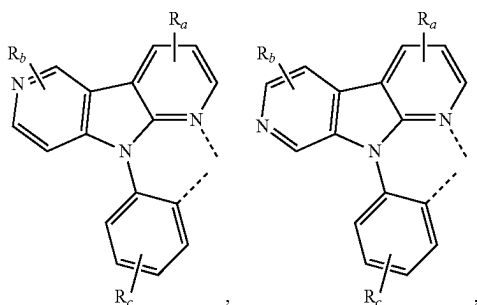
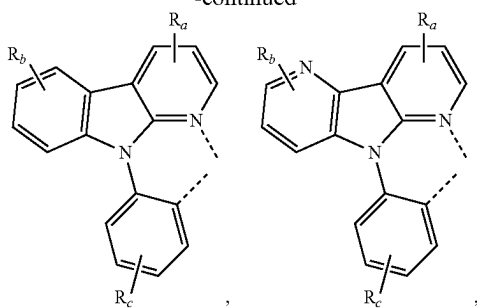
**372**

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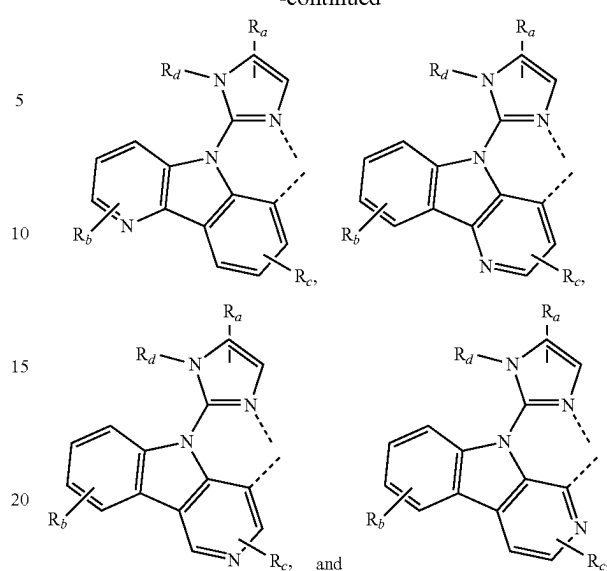
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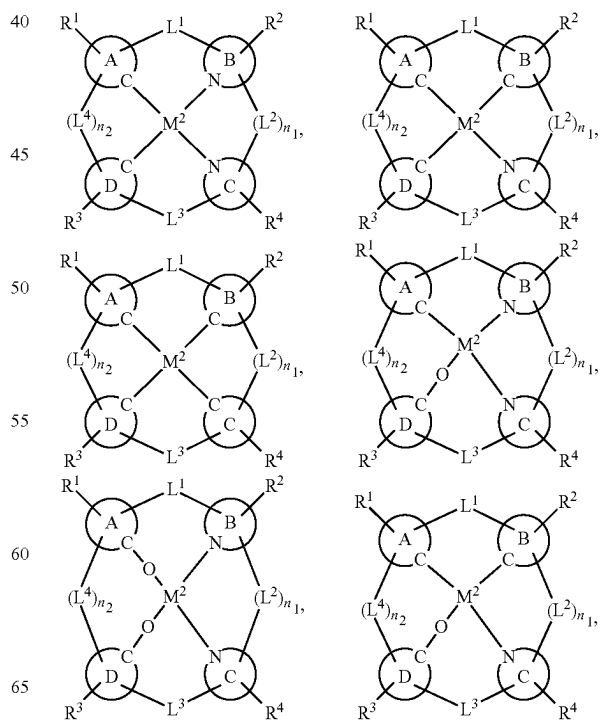
10. The compound of claim 1, wherein each of Q^1 , Q^2 , Q^3 , and Q^4 is a direct bond.

11. The compound of claim 1, wherein two of Z^1 , Z^2 , Z^3 , and Z^4 are carbon atoms, and the remaining two of Z^1 , Z^2 , Z^3 , and Z^4 are nitrogen atoms;

wherein three of Z^1 , Z^2 , Z^3 , and Z^4 are carbon atoms, and the remaining one of Z^1 , Z^2 , Z^3 , and Z^4 is a nitrogen atom; or

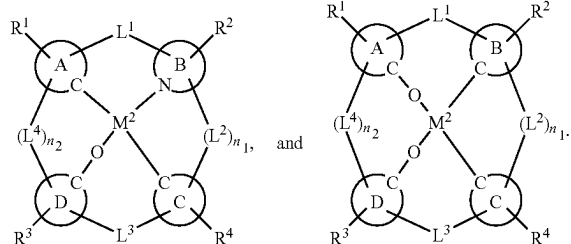
wherein each of Z^1 , Z^2 , Z^3 , and Z^4 are carbon atoms.

12. The compound of claim 1, wherein the compound is selected from the group consisting of:

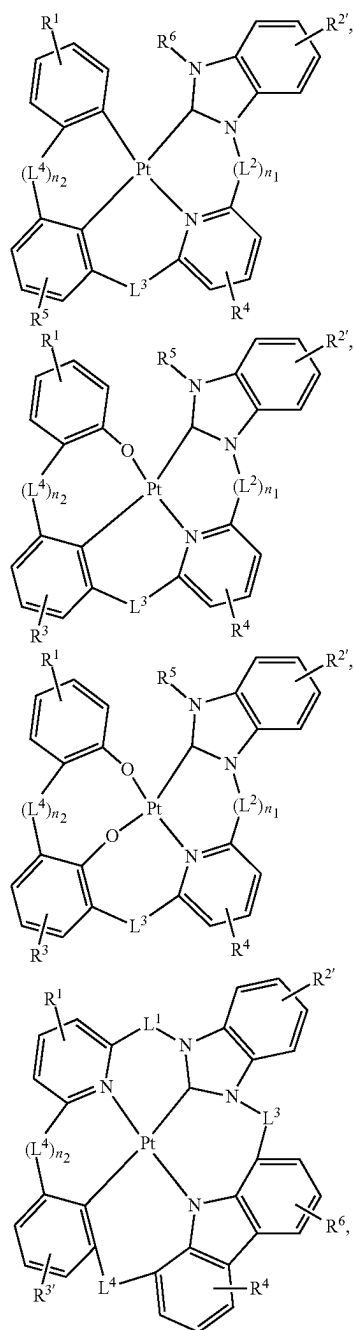


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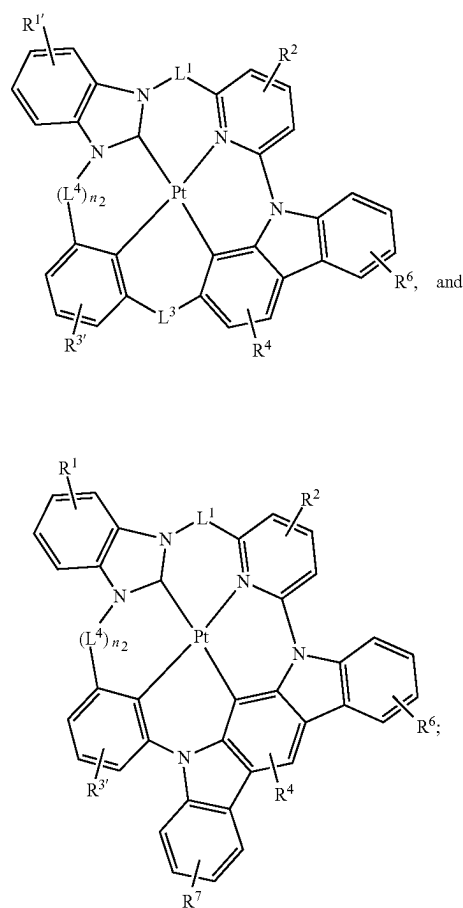
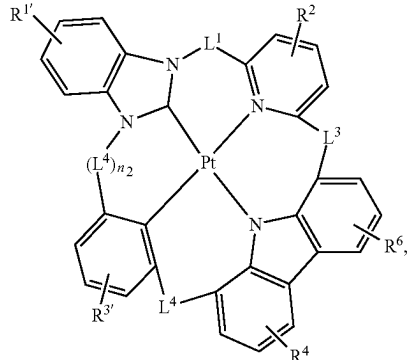
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13. The compound of claim 1, wherein the compound is selected from the group consisting of:

**376**

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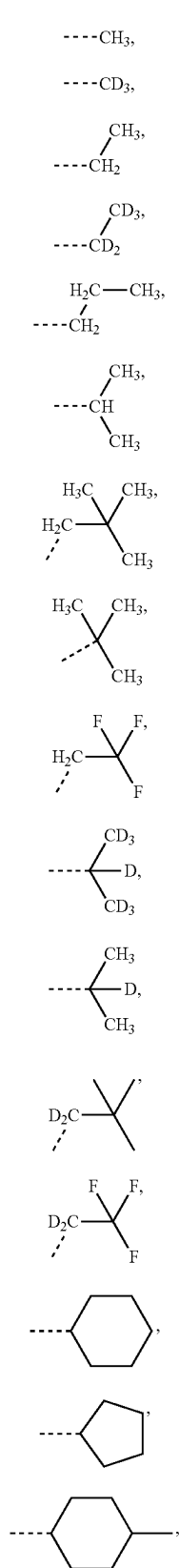
wherein R^5 , R^6 , R^7 , $R^{2'}$, and $R^{3'}$ each represents mono to the maximum possible number of substitution, or no substitution;

wherein R^5 , R^6 , R^7 , $R^{2'}$, and $R^{3'}$ are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof; and

wherein any two substituents among R , R' , R^1 , R^2 , R^3 , R^4 , R^5 , R^6 , and R^7 are optionally joined or fused into a ring.

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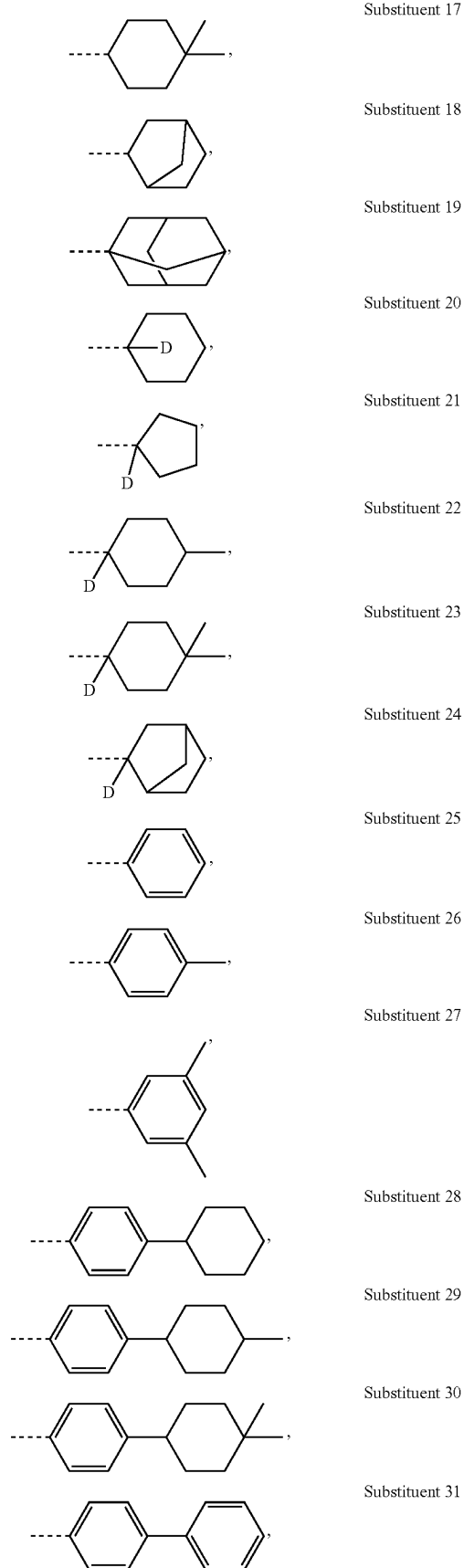
14. The compound of claim 13, wherein at least one R¹ and at least one R⁴ is selected from the group consisting of:



Substituent 1 5
 Substituent 2
 Substituent 3 10
 Substituent 4
 Substituent 5 15
 Substituent 6
 Substituent 7 20
 Substituent 8 25
 Substituent 9 30
 Substituent 10 35
 Substituent 11 40
 Substituent 12 45
 Substituent 13 50
 Substituent 14 55
 Substituent 15 60
 Substituent 16 65

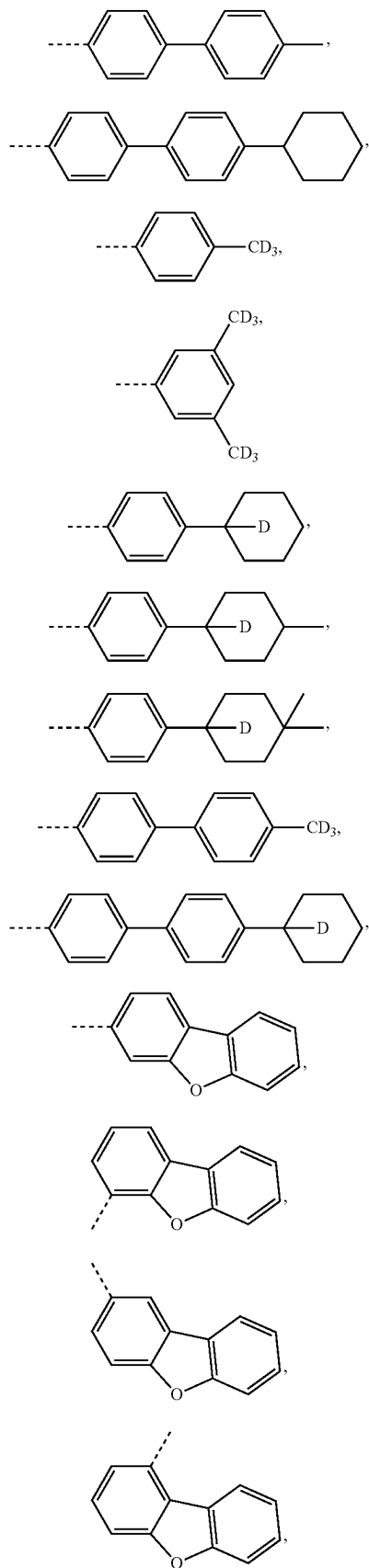
378

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379

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Substituent 32

5

Substituent 33

Substituent 34 10

Substituent 35 15

Substituent 36

Substituent 37 25

Substituent 38 30

Substituent 39 35

Substituent 40 40

Substituent 41 45

Substituent 42 50

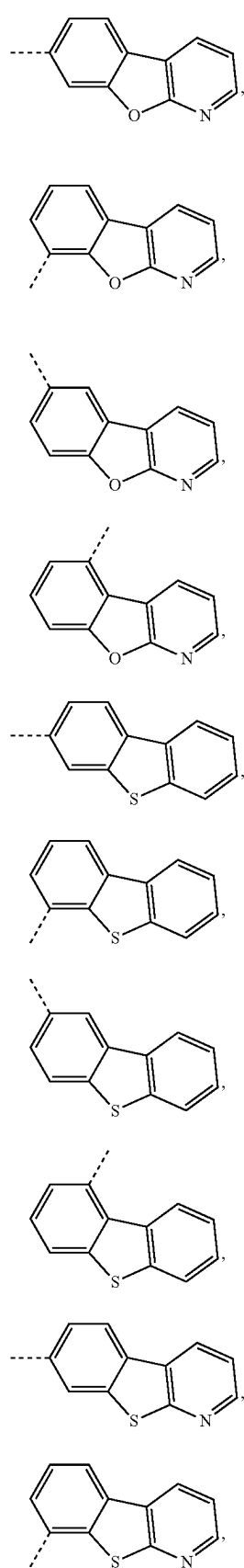
Substituent 43 55

Substituent 44 60

65

380

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Substituent 45

Substituent 46

Substituent 47

Substituent 48

Substituent 49

Substituent 50

Substituent 51

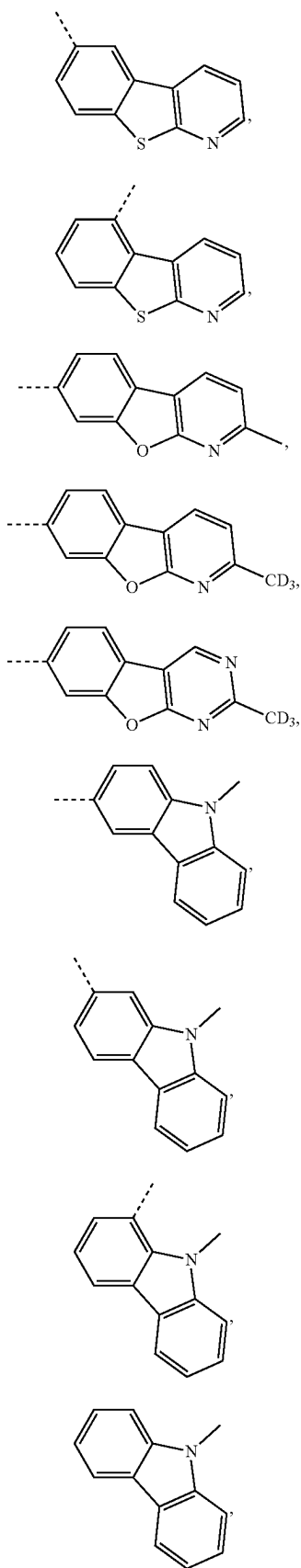
Substituent 52

Substituent 53

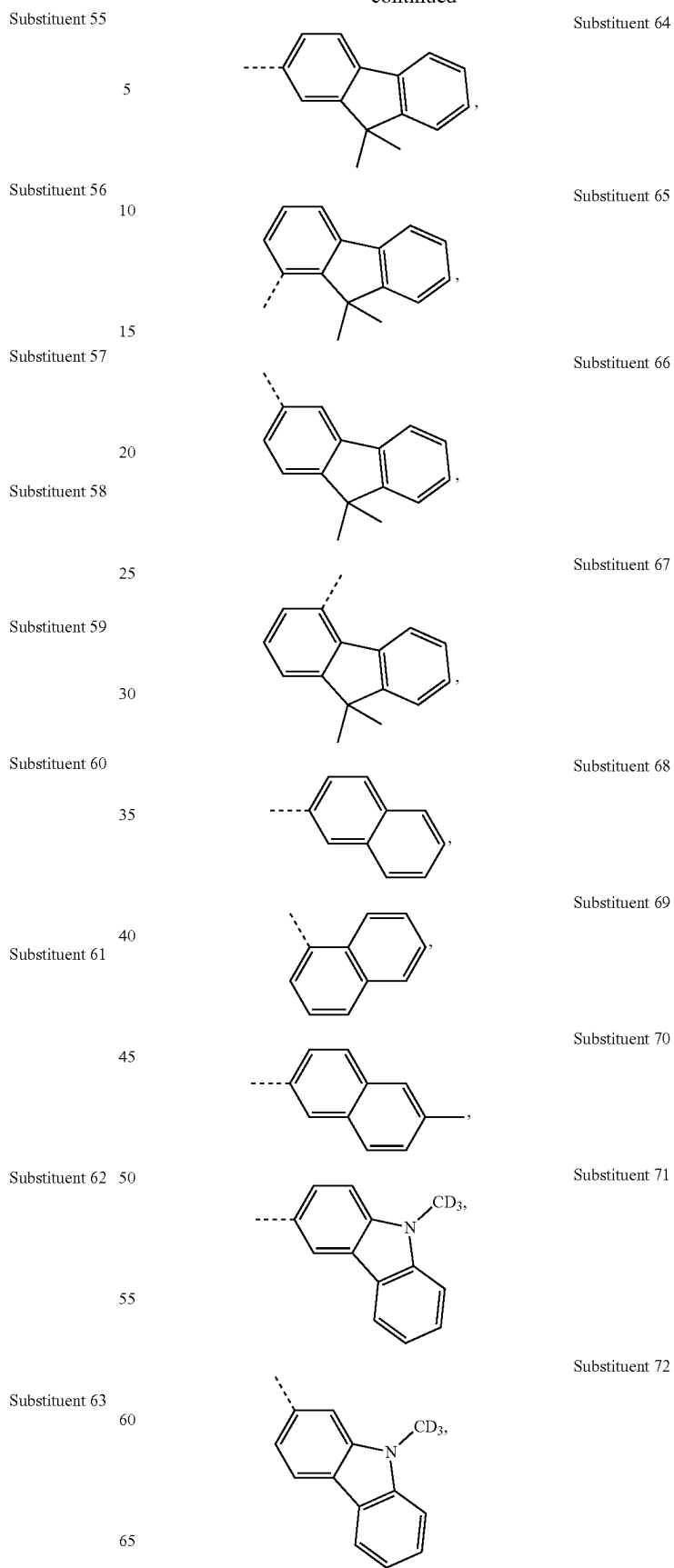
Substituent 54

381

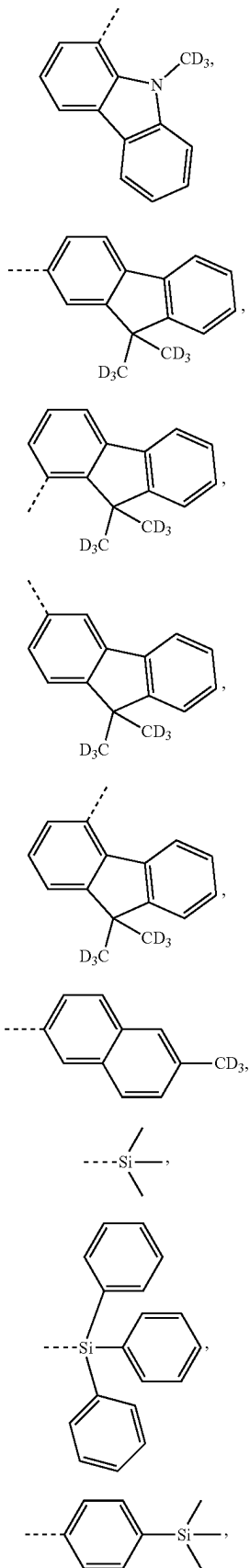
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**382**

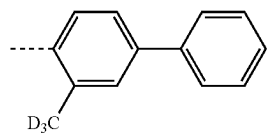
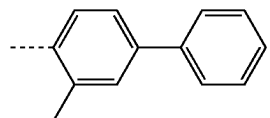
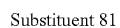
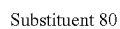
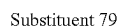
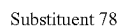
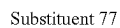
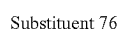
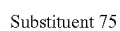
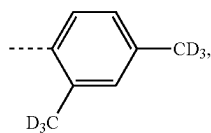
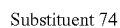
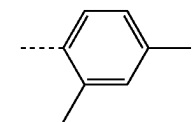
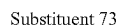
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Substituent 82

Substituent 83

Substituent 84

Substituent 85

Substituent 86

Substituent 87

Substituent 88

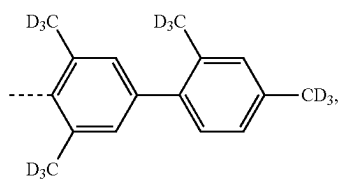
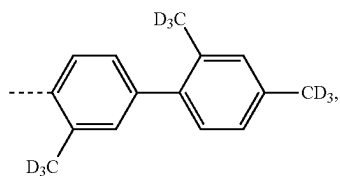
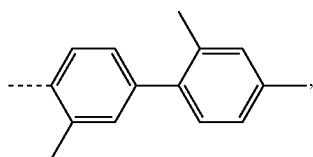
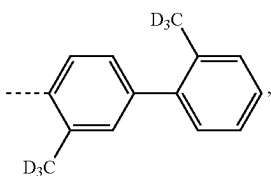
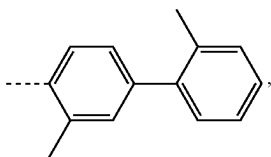
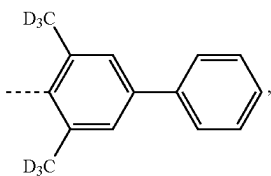
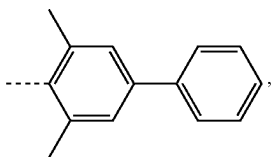
Substituent 89

Substituent 90

Substituent 91

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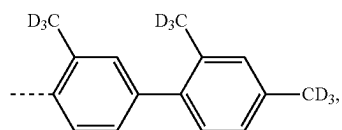
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**386**

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Substituent 92

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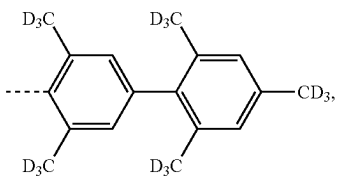


Substituent 99

10

Substituent 93

15

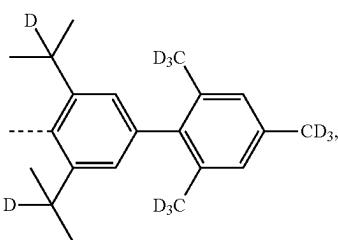


Substituent 100

Substituent 101

Substituent 94

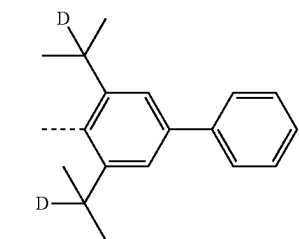
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Substituent 102

Substituent 95

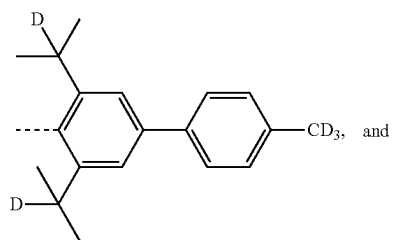
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Substituent 103

Substituent 96

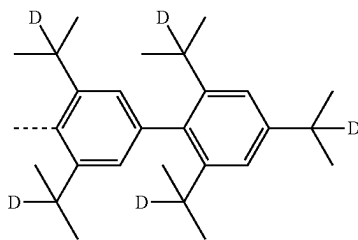
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Substituent 104

Substituent 97

50



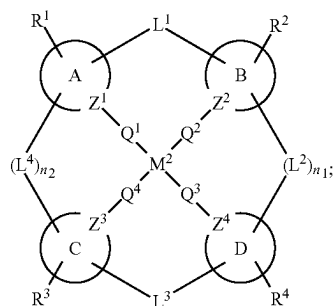
Substituent 98

60

15. An organic light-emitting device (OLED) comprising:
 an anode;
 a cathode; and
 an organic layer, disposed between the anode and the cathode, comprising a compound having a formula of Formula III,

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wherein rings A, B, C, and D are each a 5 or 6-membered carbocyclic or heterocyclic ring;

wherein L^1 and L^3 are each independently selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof;

wherein n_1 and n_2 each independently is 0 or 1;

wherein, if n_1 or n_2 is 1, then L^2 or L^4 is selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof; and

wherein, if n_1 or n_2 is 0, L^2 or L^4 is not present;

Q^1 , Q^2 , Q^3 and Q^4 are each independently selected from the group consisting of direct bond and oxygen;

wherein Z^1 , Z^2 , Z^3 , and Z^4 are each independently carbon or nitrogen;

when any of Z^1 , Z^2 , Z^3 and Z^4 is nitrogen, the Q^1 , Q^2 , Q^3 and Q^4 attached thereto is a direct bond;

ring A is trans to ring D, ring B is trans to ring C in a square-planar coordination configuration;

wherein R^1 , R^2 , R^3 , and R^4 each represents mono to the maximum possible number of substitution, or no substitution;

wherein M^2 is Pt, and the compound comprises at least one Pt-carbene bond;

wherein a fourth distance is the distance between the atom in R^1 that is the farthest away from M^2 to the atom in R^4 that is the farthest away from M^2 ;

wherein a fifth distance is the distance between the atom in R^2 that is the farthest away from M^2 to the atom in R^3 that is the farthest away from M^2 ;

wherein the fourth distance is longer than the fifth distance by at least 1.5 Å;

wherein R, R', R^1 , R^2 , R^3 , and R^4 are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof;

wherein any two substituents among R, R', R^1 , R^2 , R^3 , and R^4 are optionally joined or fused into a ring; and

wherein at least one of the following is true:

(i) at least one of L^1 to L^3 is a C—C direct bond;

(ii) each of Q^1 to Q^4 is a direct bond; or

(iii) at least two of Q^1 to Q^4 are O.

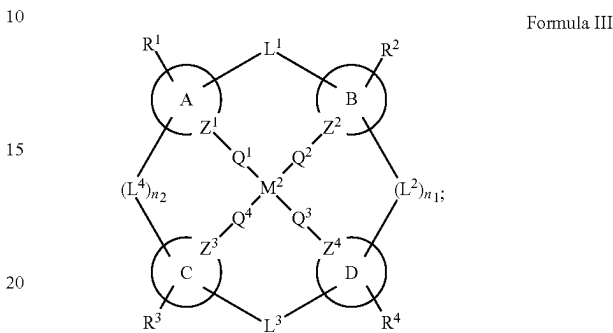
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16. A consumer product comprising an organic light-emitting device comprising:

an anode;

a cathode; and

5 an organic layer, disposed between the anode and the cathode, comprising a compound having a formula of Formula III,



wherein rings A, B, C, and D are each a 5 or 6-membered carbocyclic or heterocyclic ring;

wherein L^1 and L^3 are each independently selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof;

wherein n_1 and n_2 each independently is 0 or 1;

wherein, if n_1 or n_2 is 1, then L^2 or L^4 is selected from the group consisting of a direct bond, BR, NR, PR, O, S, Se, C=O, S=O, SO₂, CRR', SiRR', GeRR', alkyl, and combinations thereof; and

wherein, if n_1 or n_2 is 0, L^2 or L^4 is not present;

Q^1 , Q^2 , Q^3 and Q^4 are each independently selected from the group consisting of direct bond and oxygen;

wherein Z^1 , Z^2 , Z^3 , and Z^4 are each independently carbon or nitrogen;

when any of Z^1 , Z^2 , Z^3 and Z^4 is nitrogen, the Q^1 , Q^2 , Q^3 and Q^4 attached thereto is a direct bond;

ring A is trans to ring D, ring B is trans to ring C in a square-planar coordination configuration;

wherein R^1 , R^2 , R^3 , and R^4 each represents mono to the maximum possible number of substitution, or no substitution;

wherein M^2 is Pt, and the compound comprises at least one Pt-carbene bond;

wherein a fourth distance is the distance between the atom in R^1 that is the farthest away from M^2 to the atom in R^4 that is the farthest away from M^2 ;

wherein a fifth distance is the distance between the atom in R^2 that is the farthest away from M^2 to the atom in R^3 that is the farthest away from M^2 ;

wherein the fourth distance is longer than the fifth distance by at least 1.5 Å;

wherein R, R', R^1 , R^2 , R^3 , and R^4 are each independently selected from the group consisting of hydrogen, deuterium, halide, alkyl, cycloalkyl, heteroalkyl, arylalkyl, alkoxy, aryloxy, amino, silyl, alkenyl, cycloalkenyl, heteroalkenyl, alkynyl, aryl, heteroaryl, acyl, carbonyl, carboxylic acids, ester, nitrile, isonitrile, sulfanyl, sulfinyl, sulfonyl, phosphino, and combinations thereof;

wherein any two substituents among R, R', R^1 , R^2 , R^3 , and R^4 are optionally joined or fused into a ring; and

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wherein at least one of the following is true:

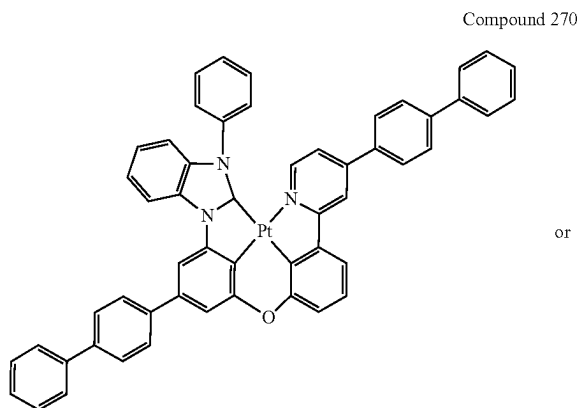
- (i) at least one of L^1 to L^3 is a C—C direct bond;
- (ii) each of Q^1 to Q^4 is a direct bond; or
- (iii) at least two of Q^1 to Q^4 are O.

17. A formulation comprising a compound of claim **1**.

18. The compound of claim **1**, wherein at least one of L^1 to L^3 is a C—C direct bond.

19. The compound of claim **1**, wherein at least two of Q^1 to Q^4 are O.

20. The compound of claim **1**, wherein the compound is

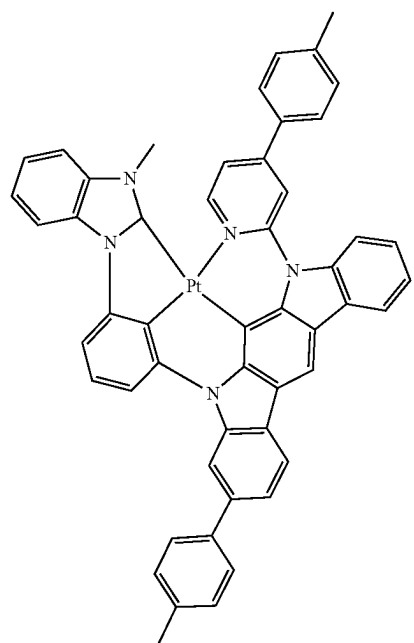


or

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-continued

Compound 296



* * * * *